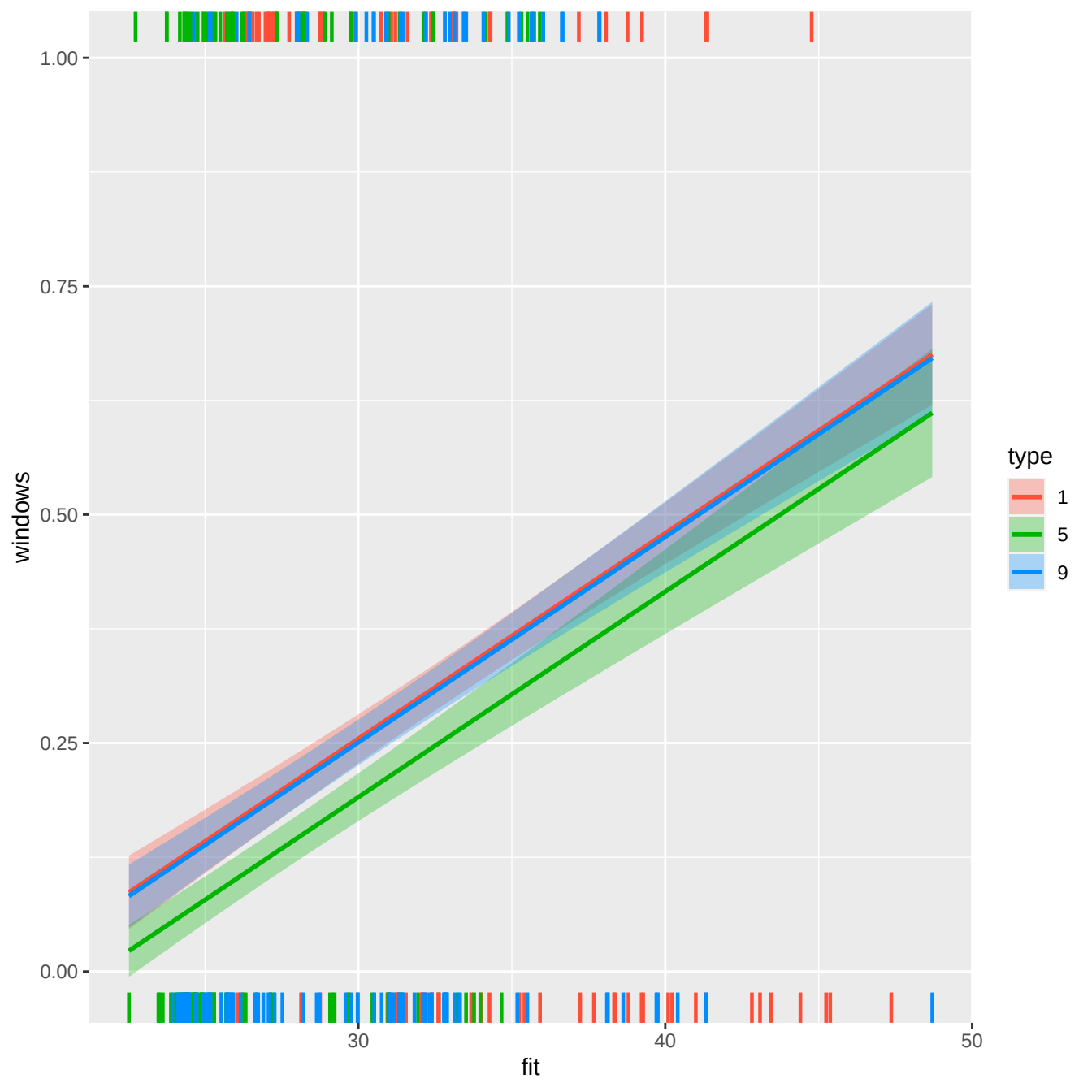
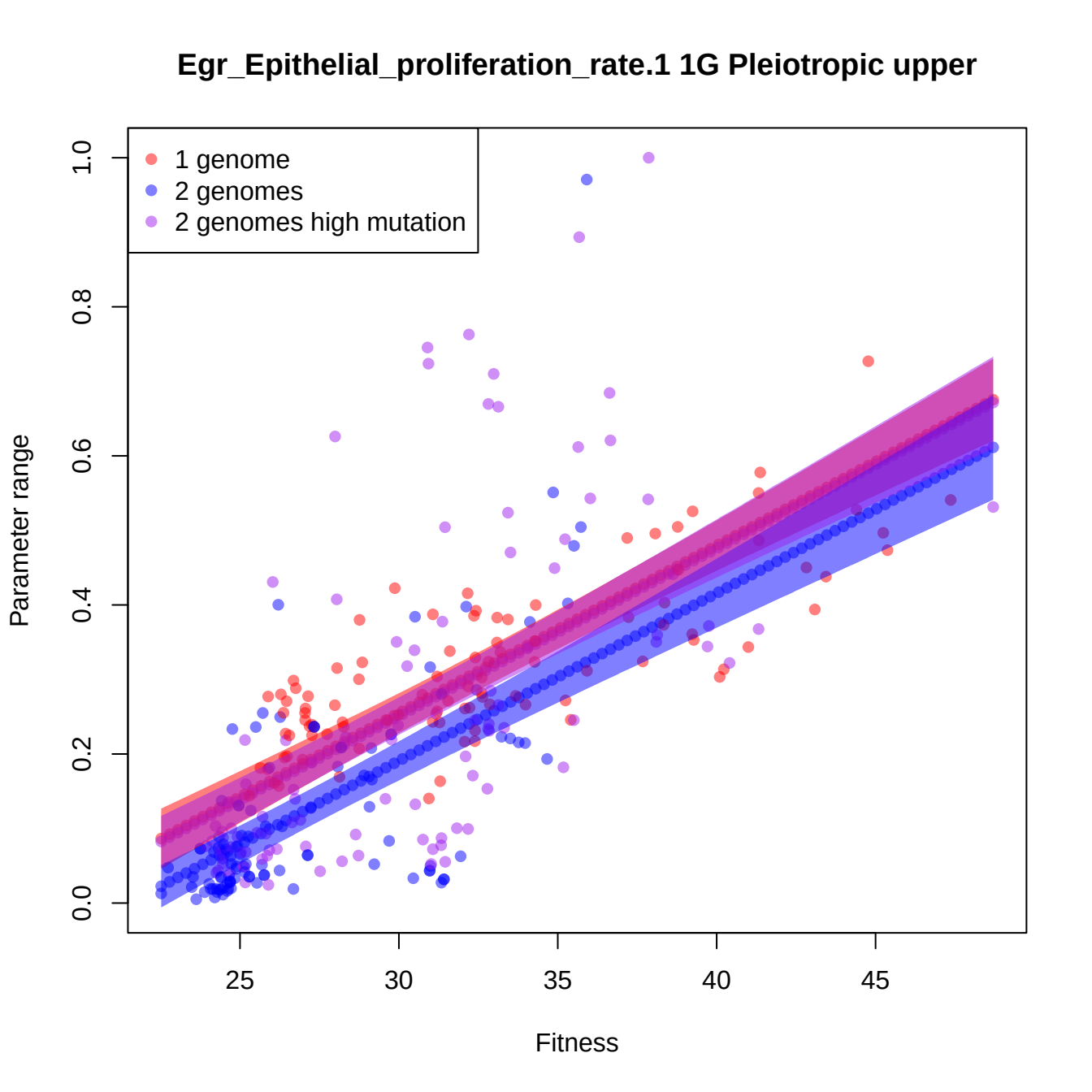
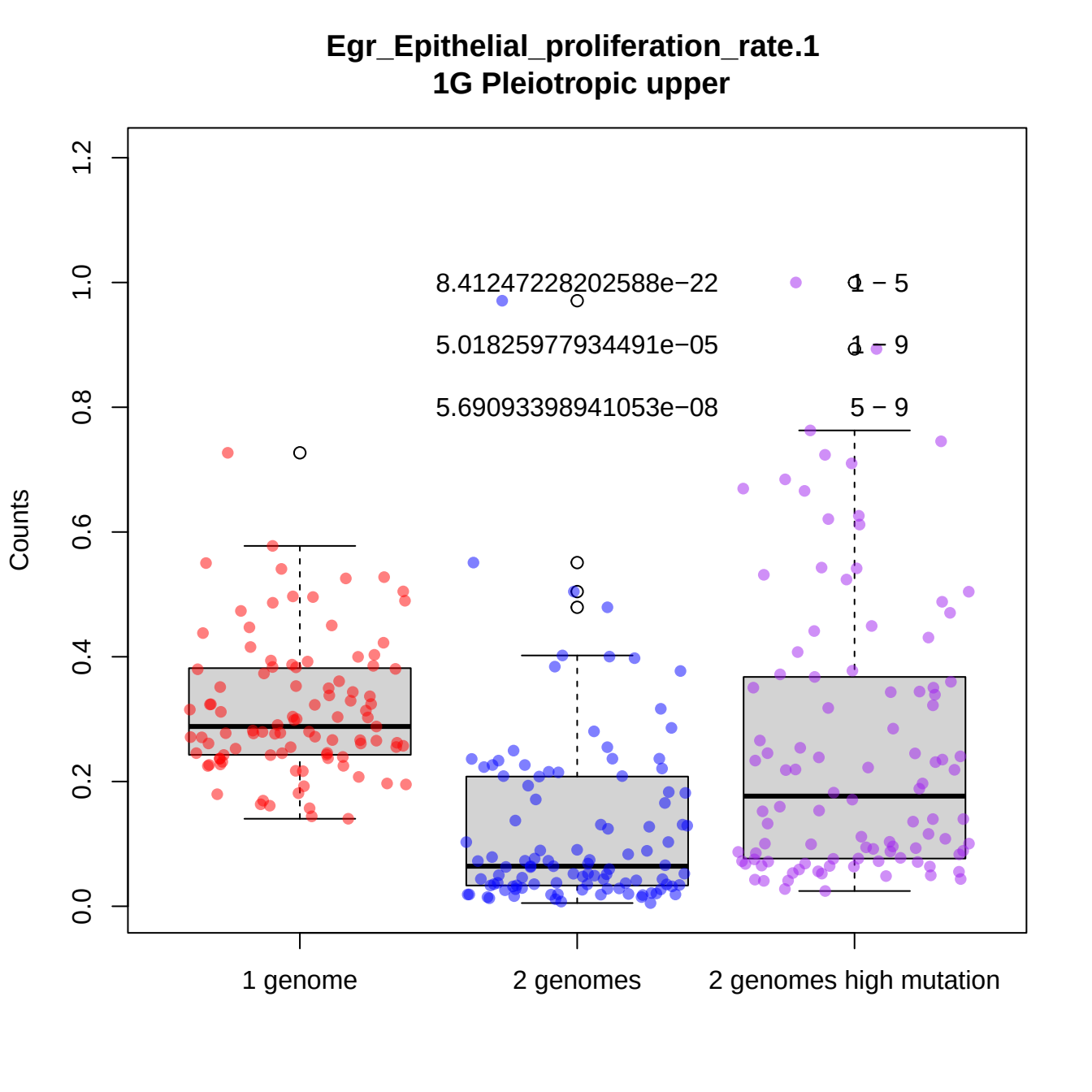
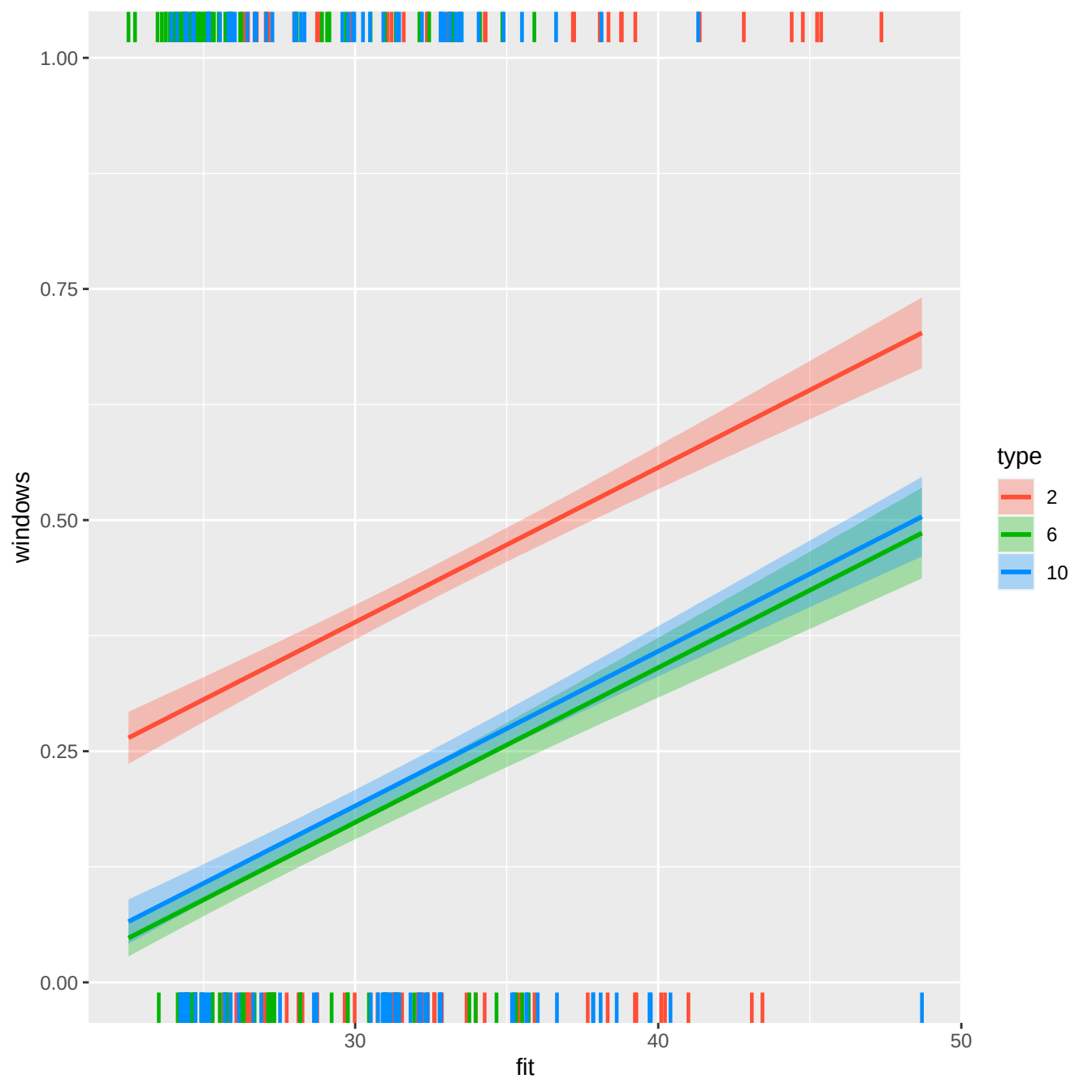
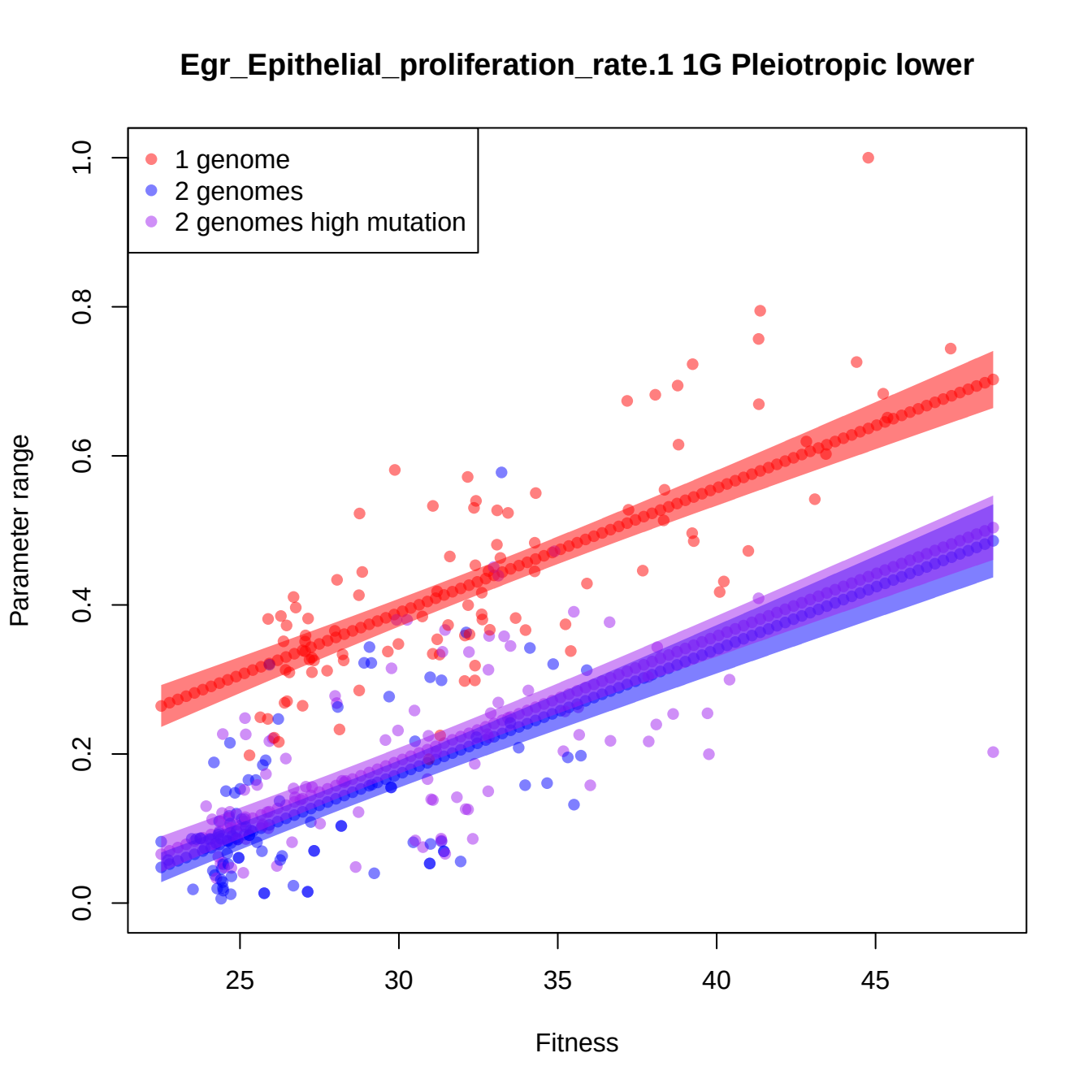
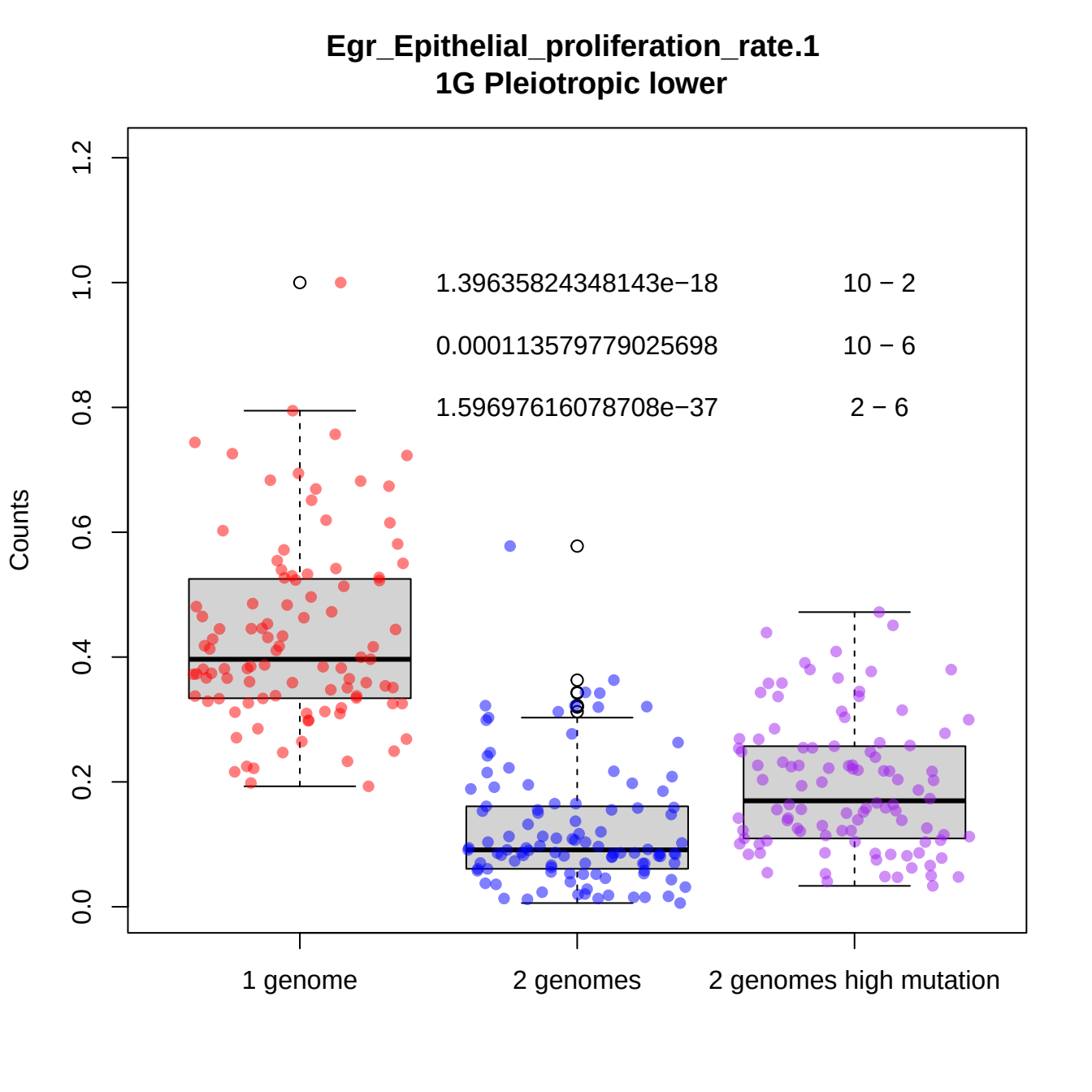

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

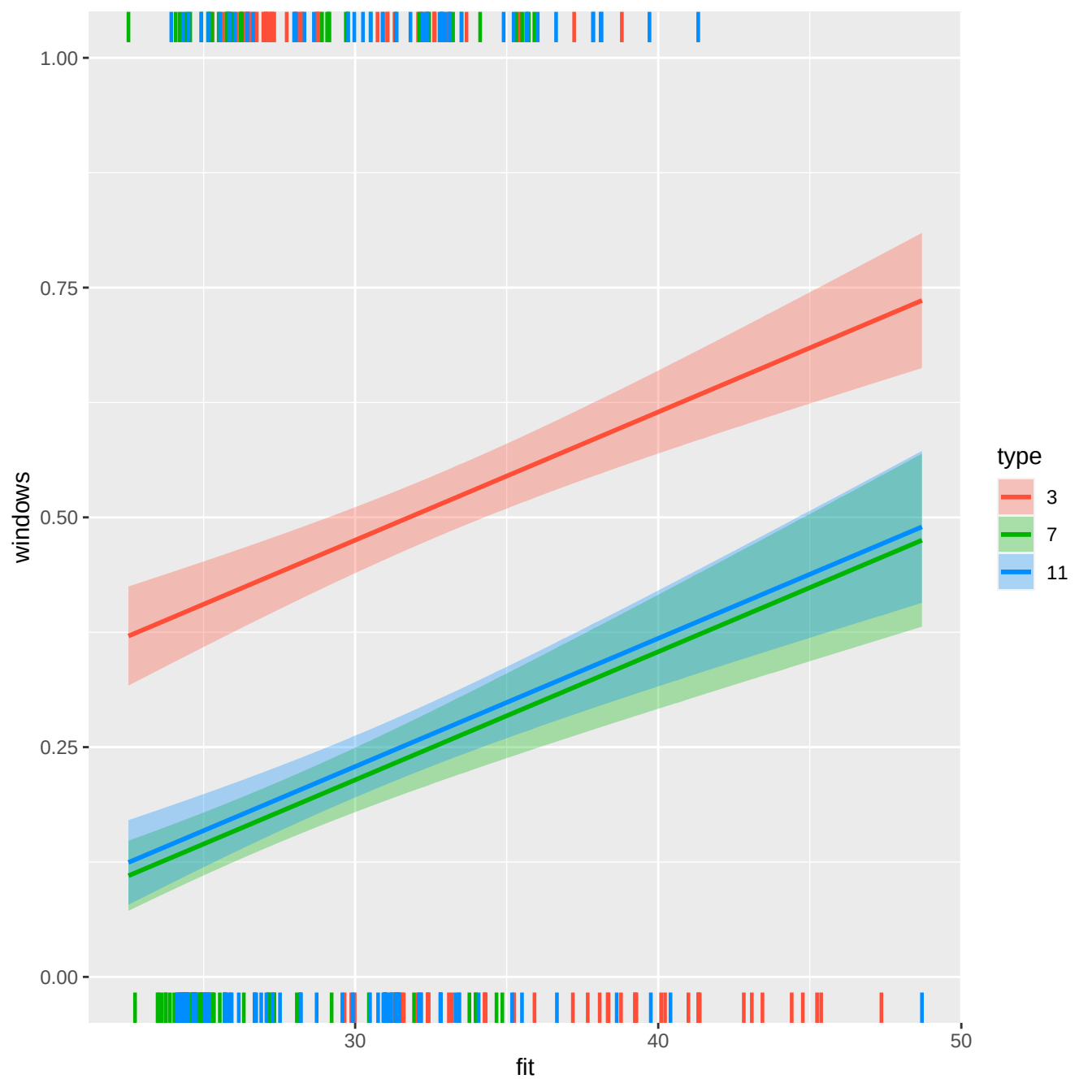

Egr_Epithelial_proliferation_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

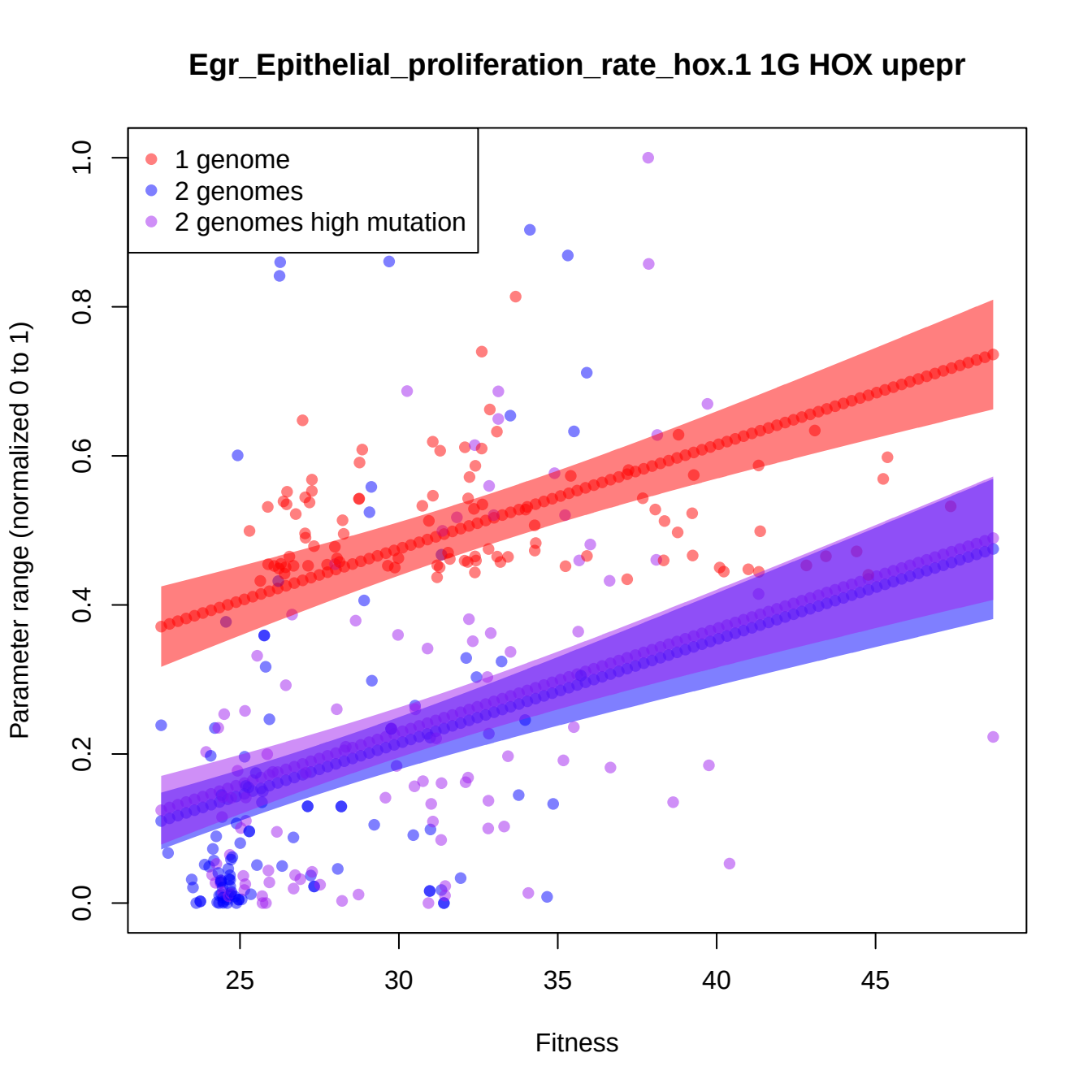

Egr_Epithelial_proliferation_rate.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
8.41247228202588e−22
5.01825977934491e−05
5.69093398941053e−08
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

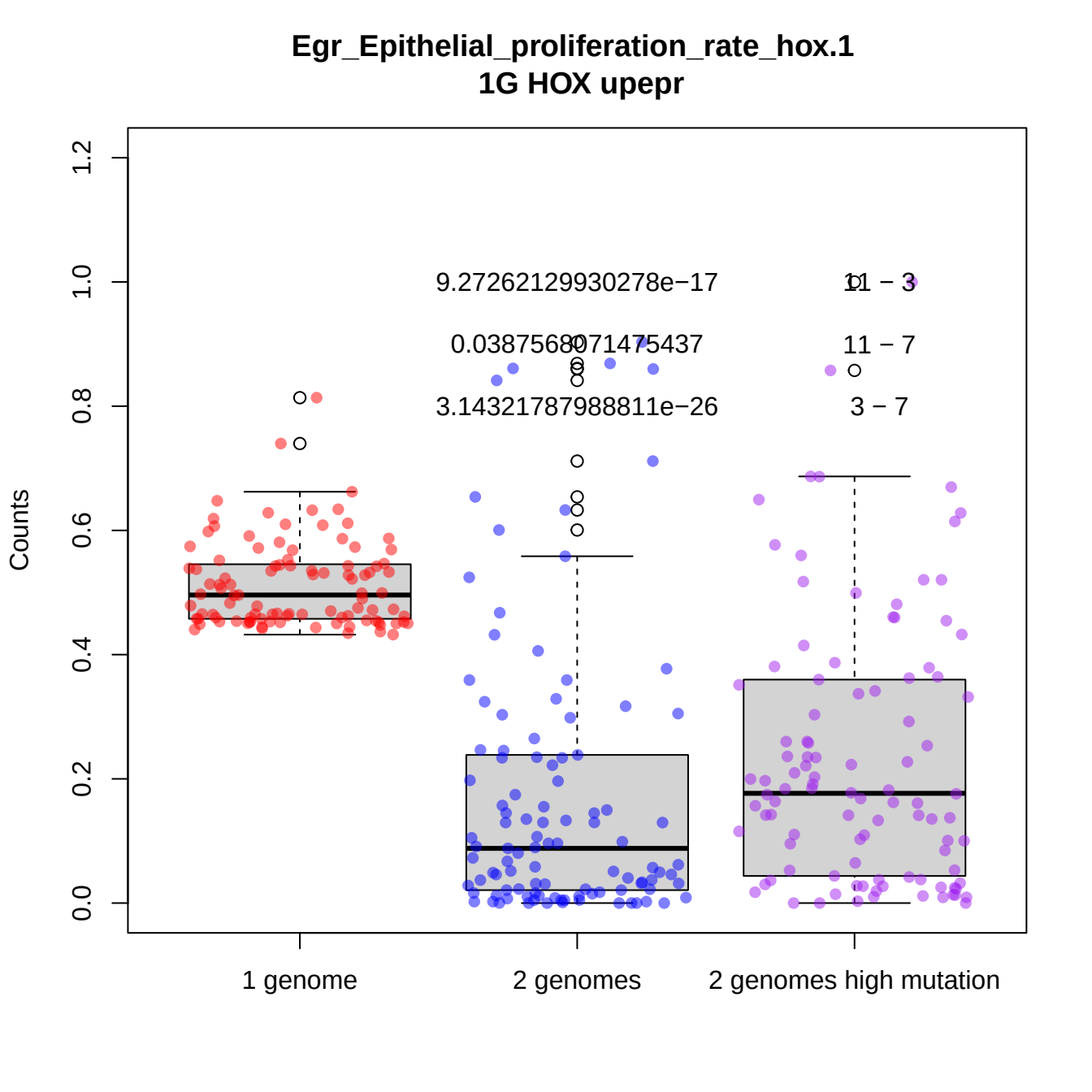

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

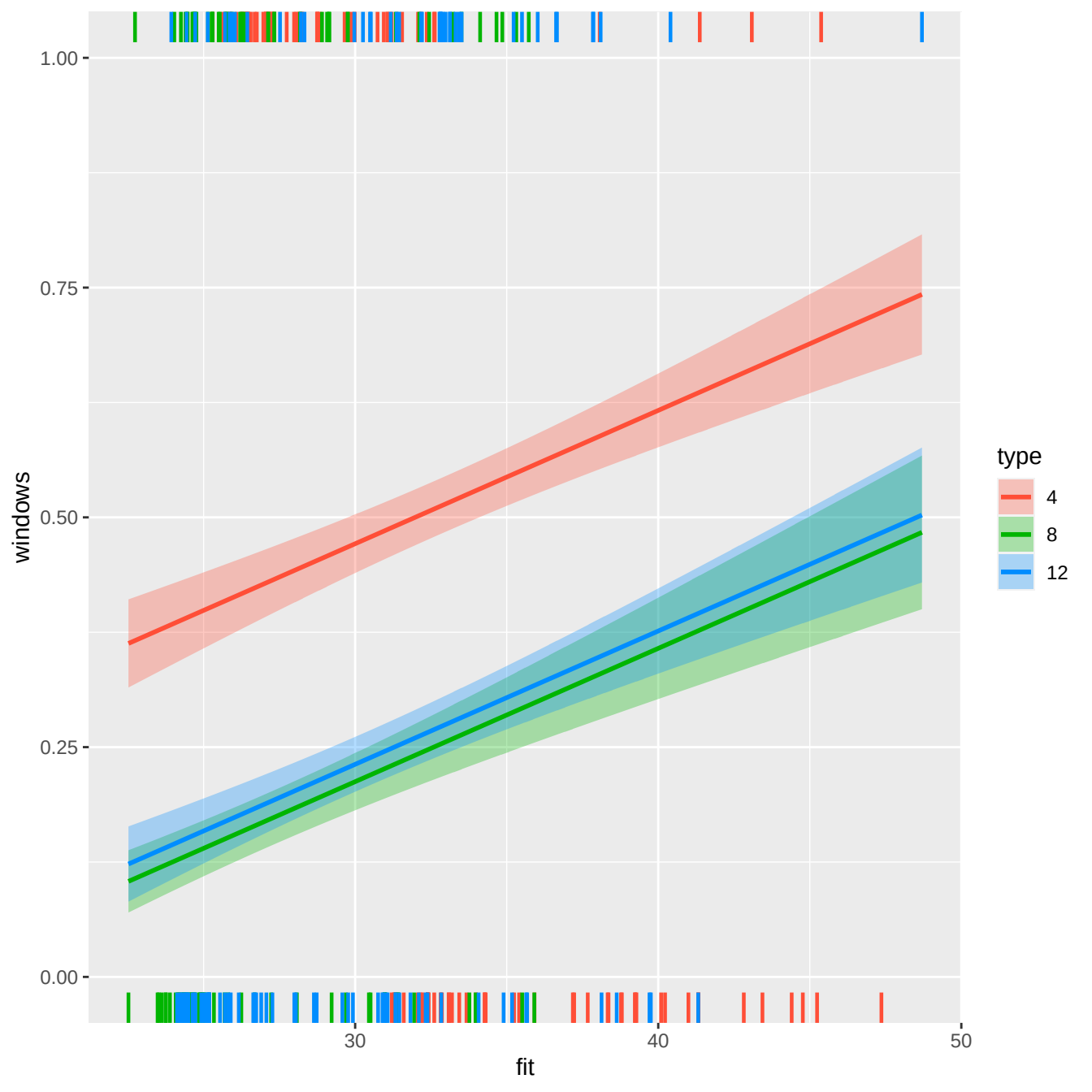

Egr_Epithelial_proliferation_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

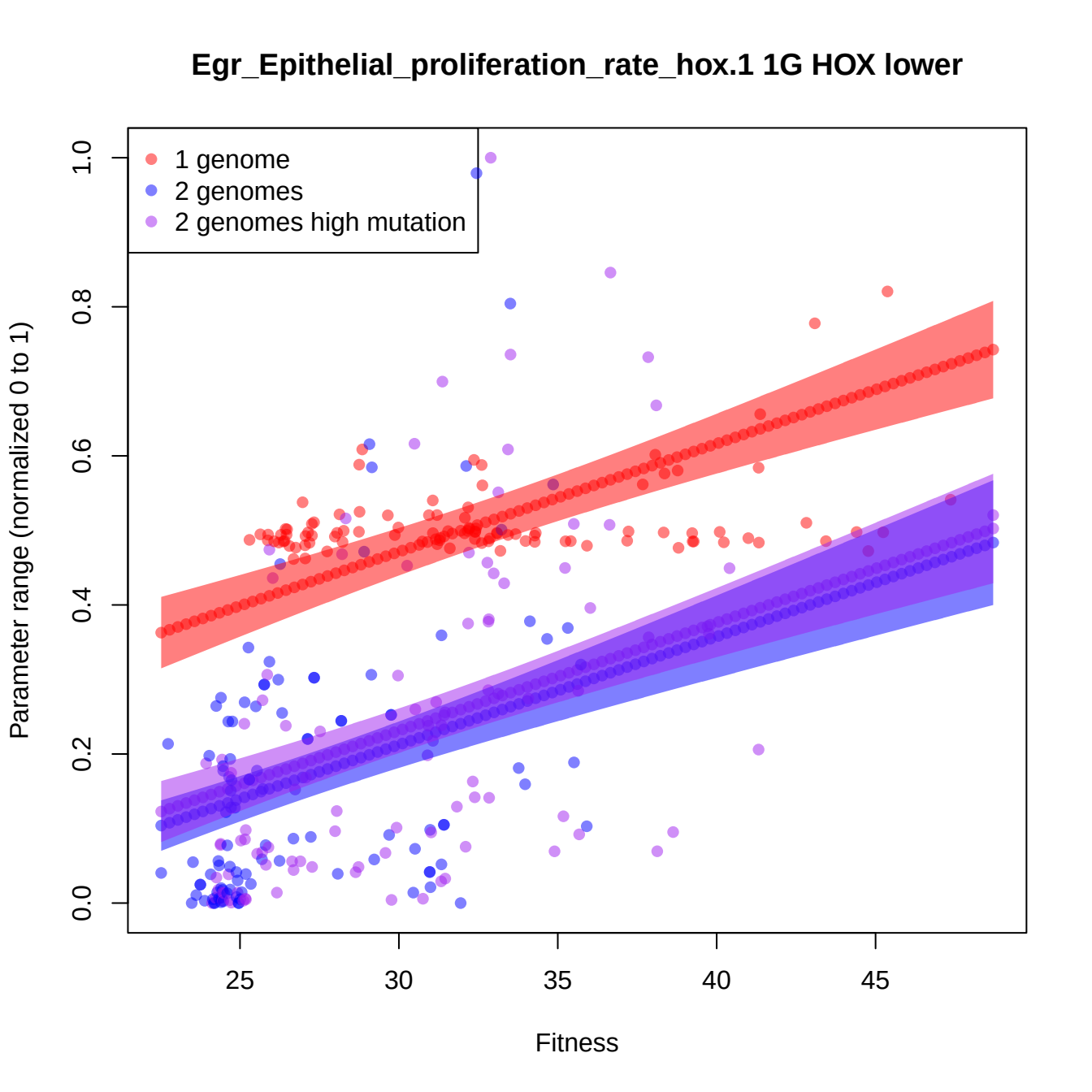

Egr_Epithelial_proliferation_rate.1
1G Pleiotropic lower
1.39635824348143e−18
10 − 2
0.000113579779025698
10 − 6
1.59697616078708e−37
2 − 6
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

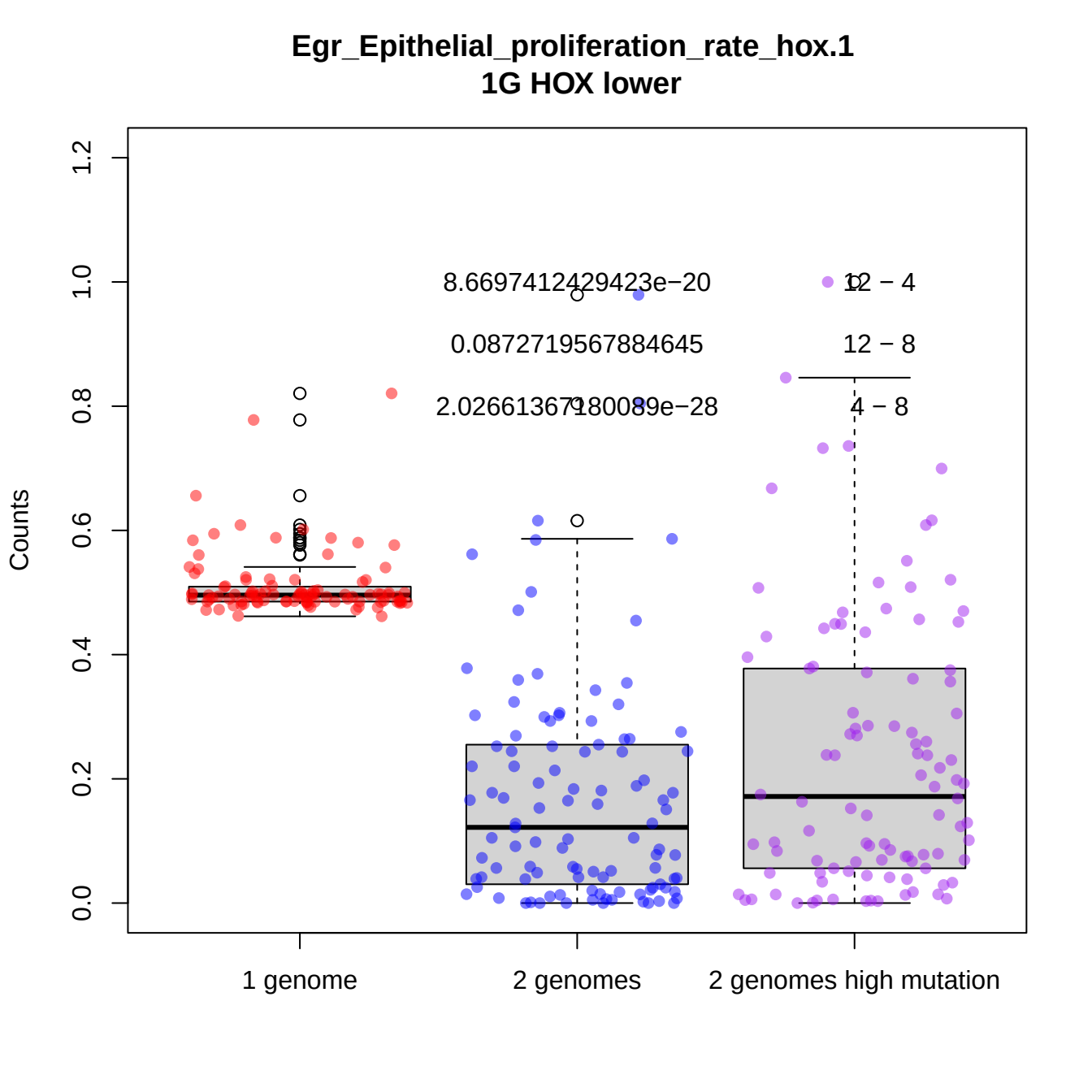

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

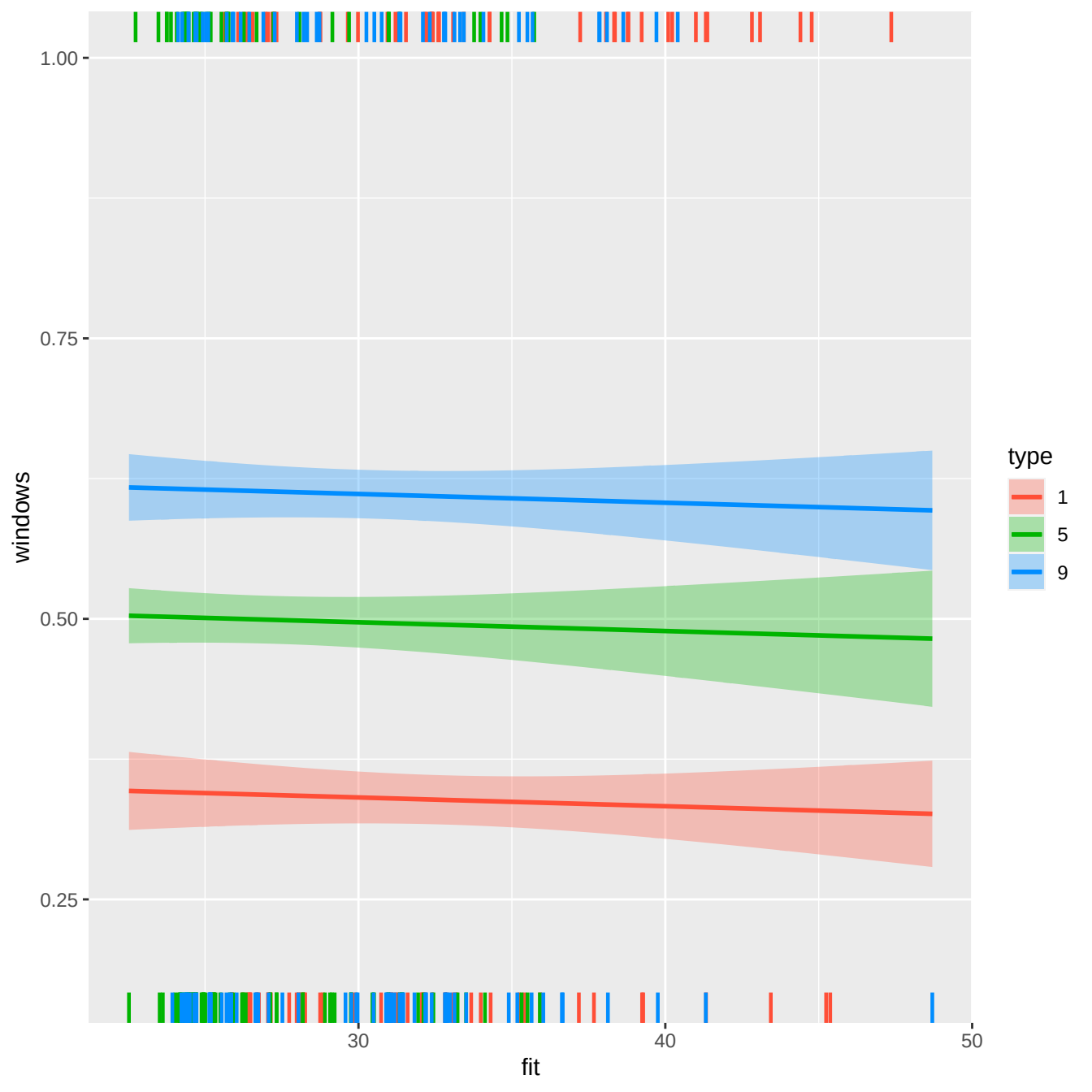

Egr_Epithelial_proliferation_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

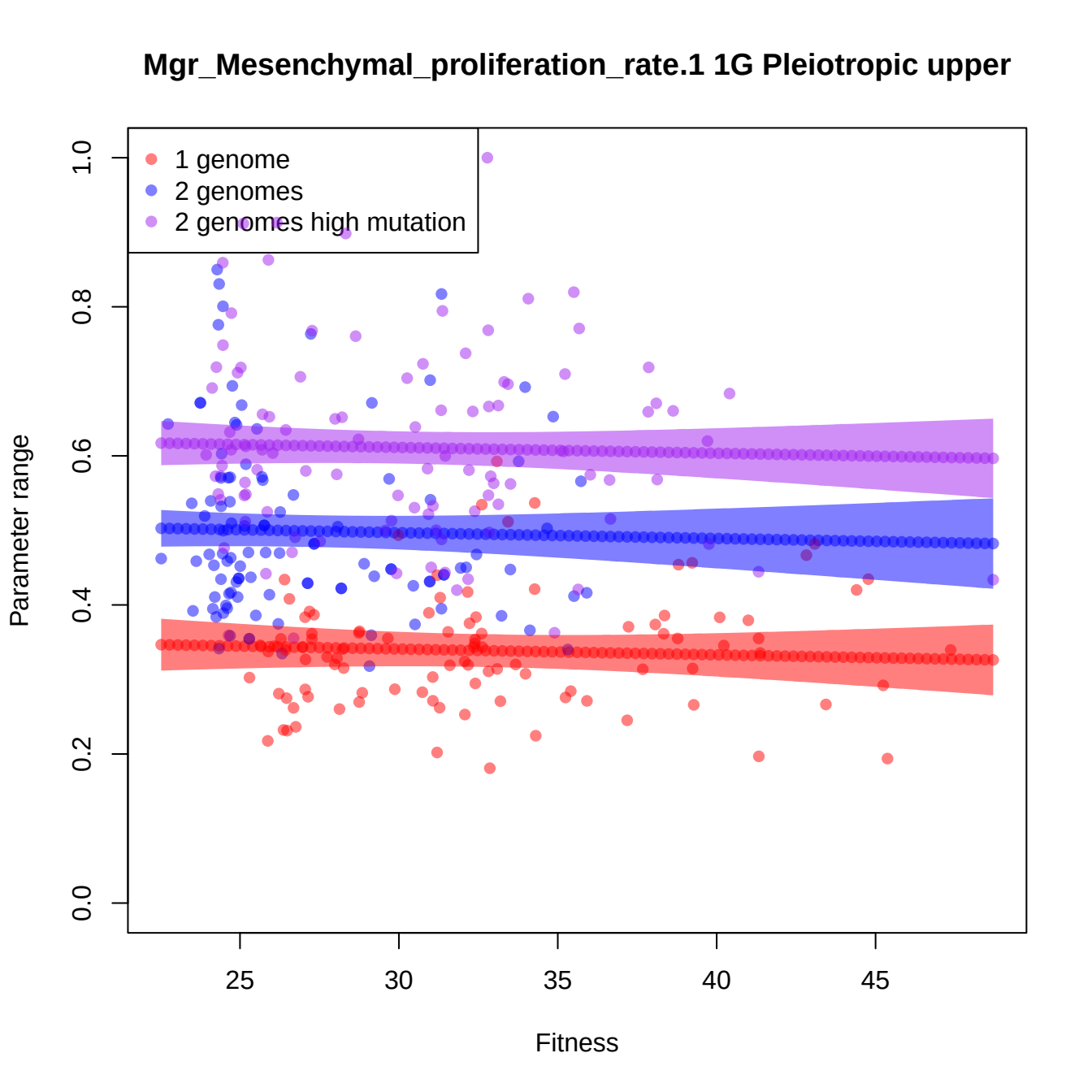

Egr_Epithelial_proliferation_rate_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
9.27262129930278e−17
0.0387568071475437
3.14321787988811e−26
1 genome
2 genomes
2 genomes high mutation

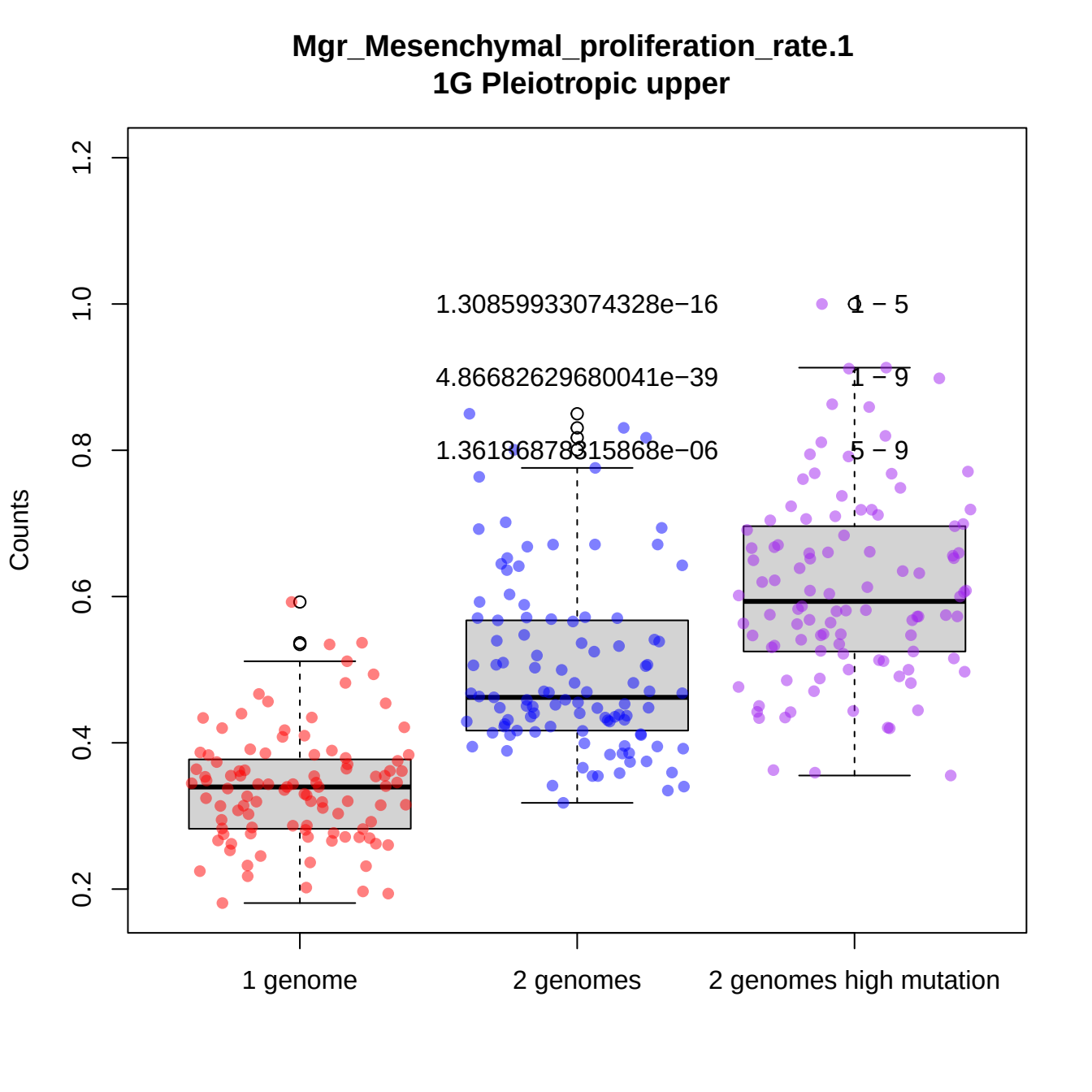

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

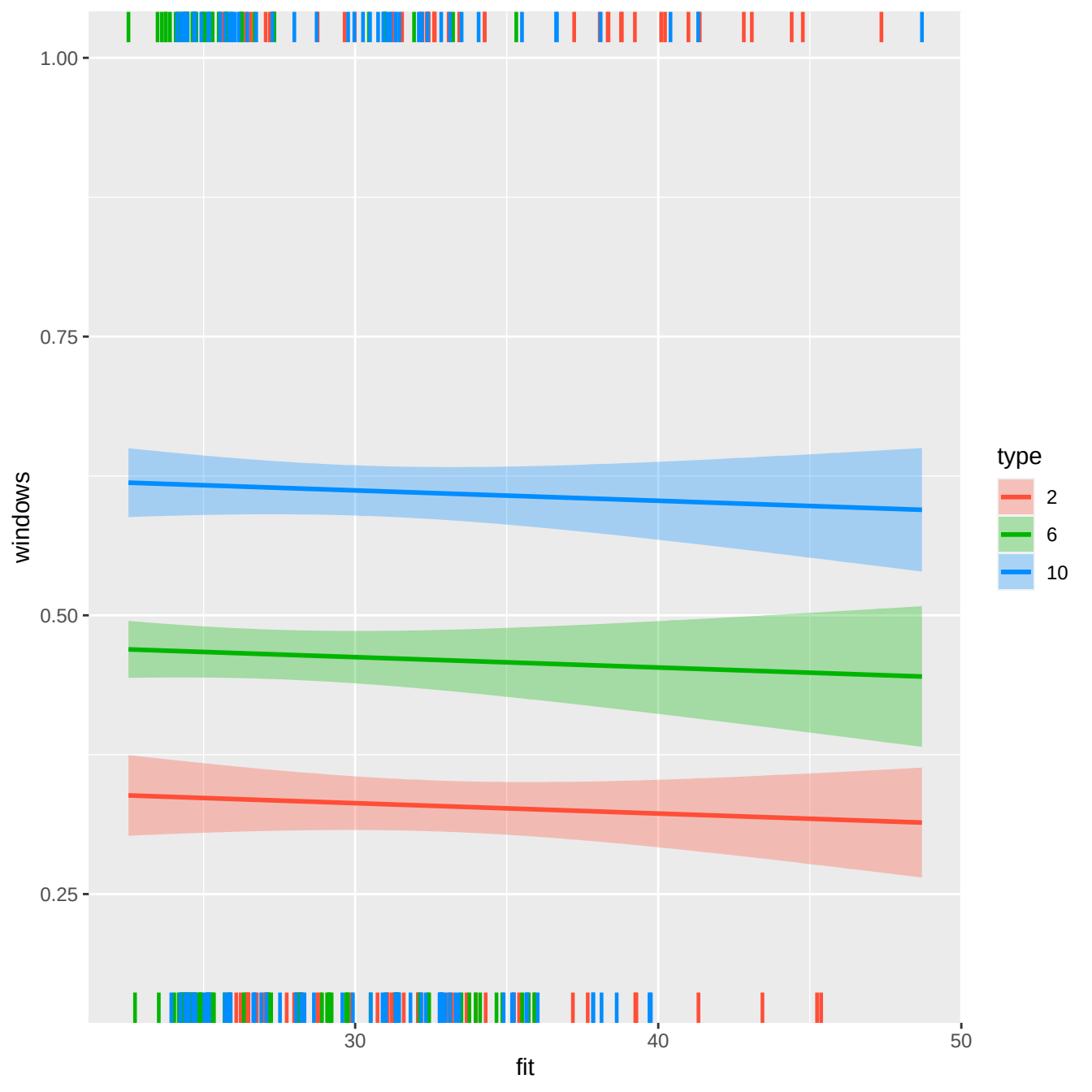

Egr_Epithelial_proliferation_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

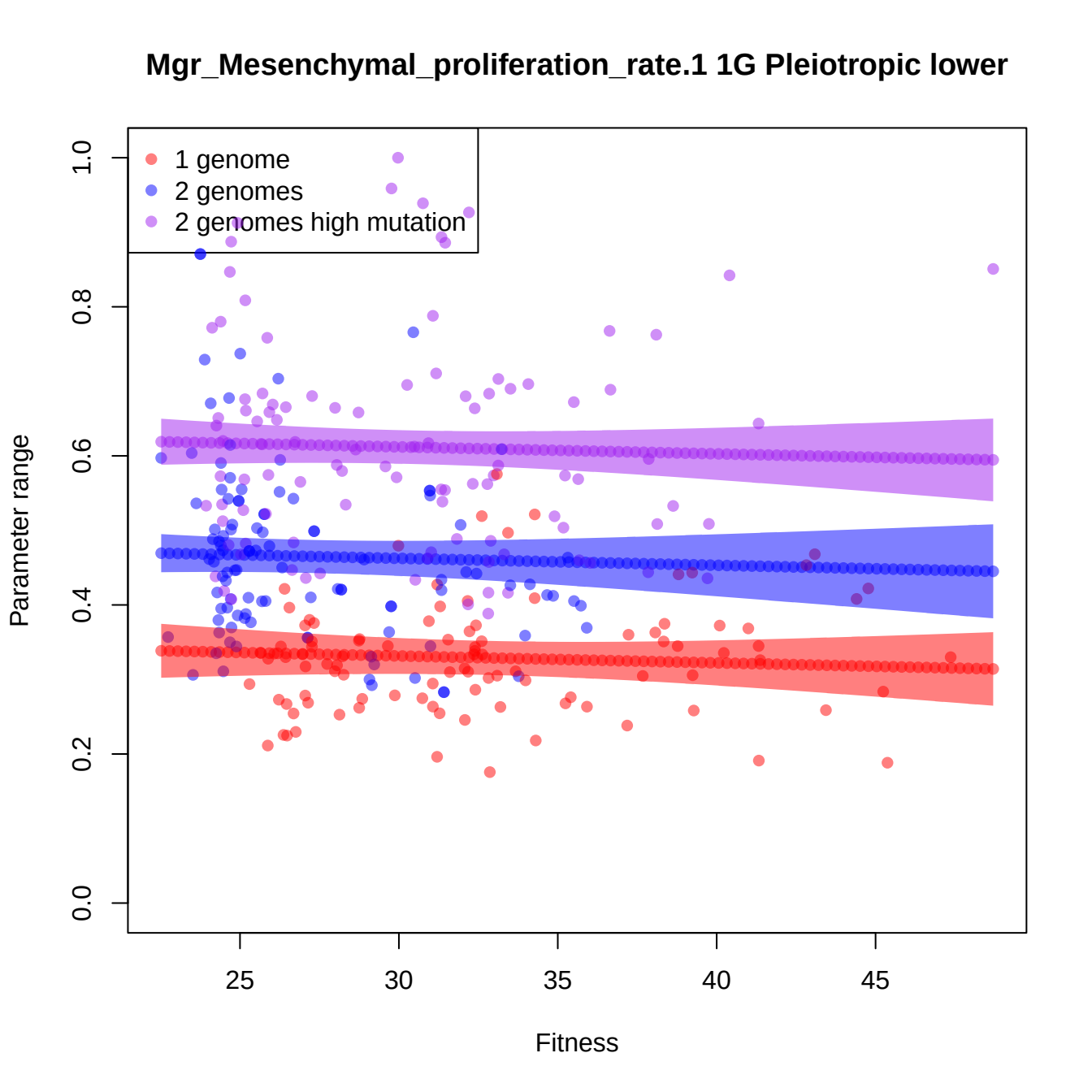

Egr_Epithelial_proliferation_rate_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
8.66974124294 23e−20
0.0872719567884645
2.026613671800 89e−28
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

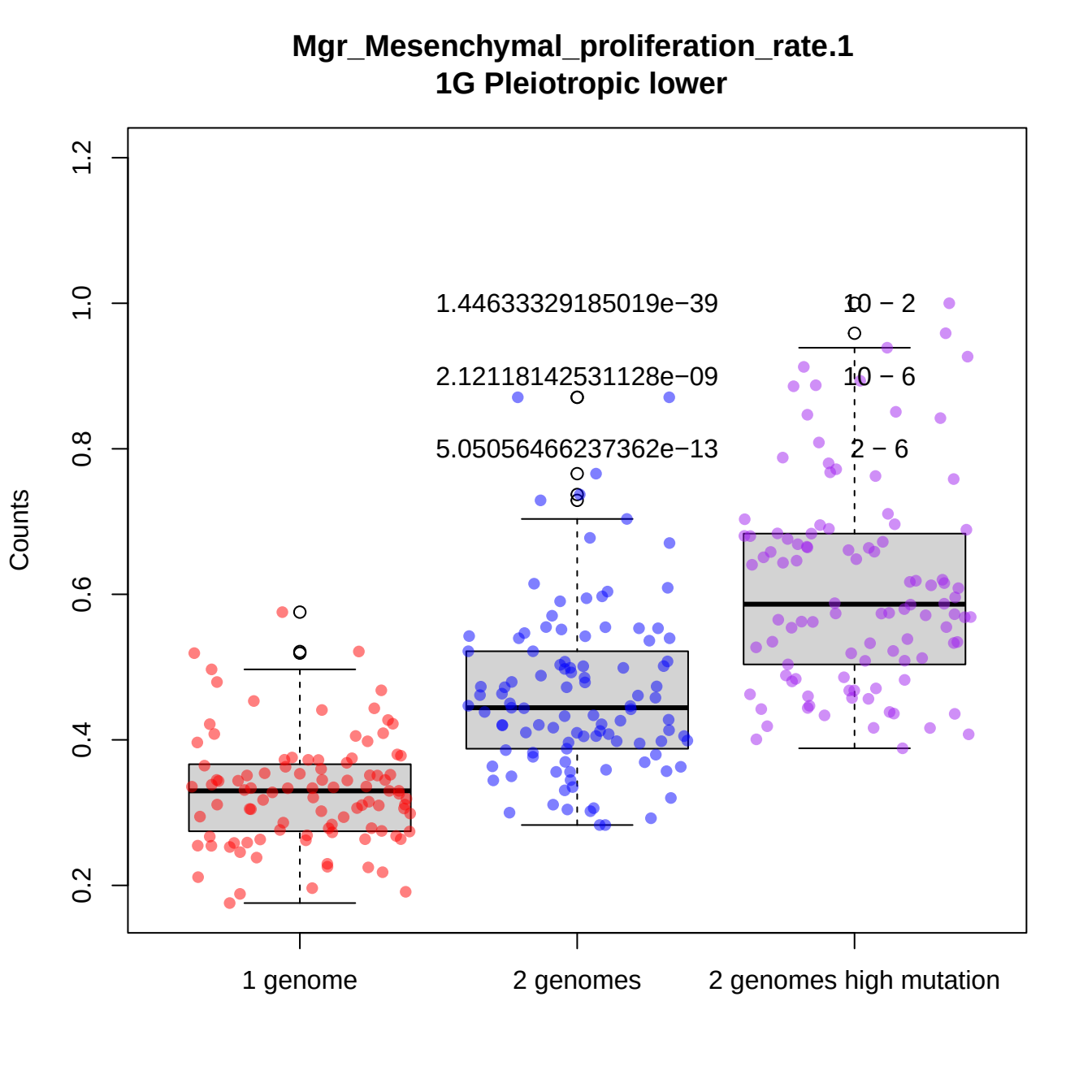

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

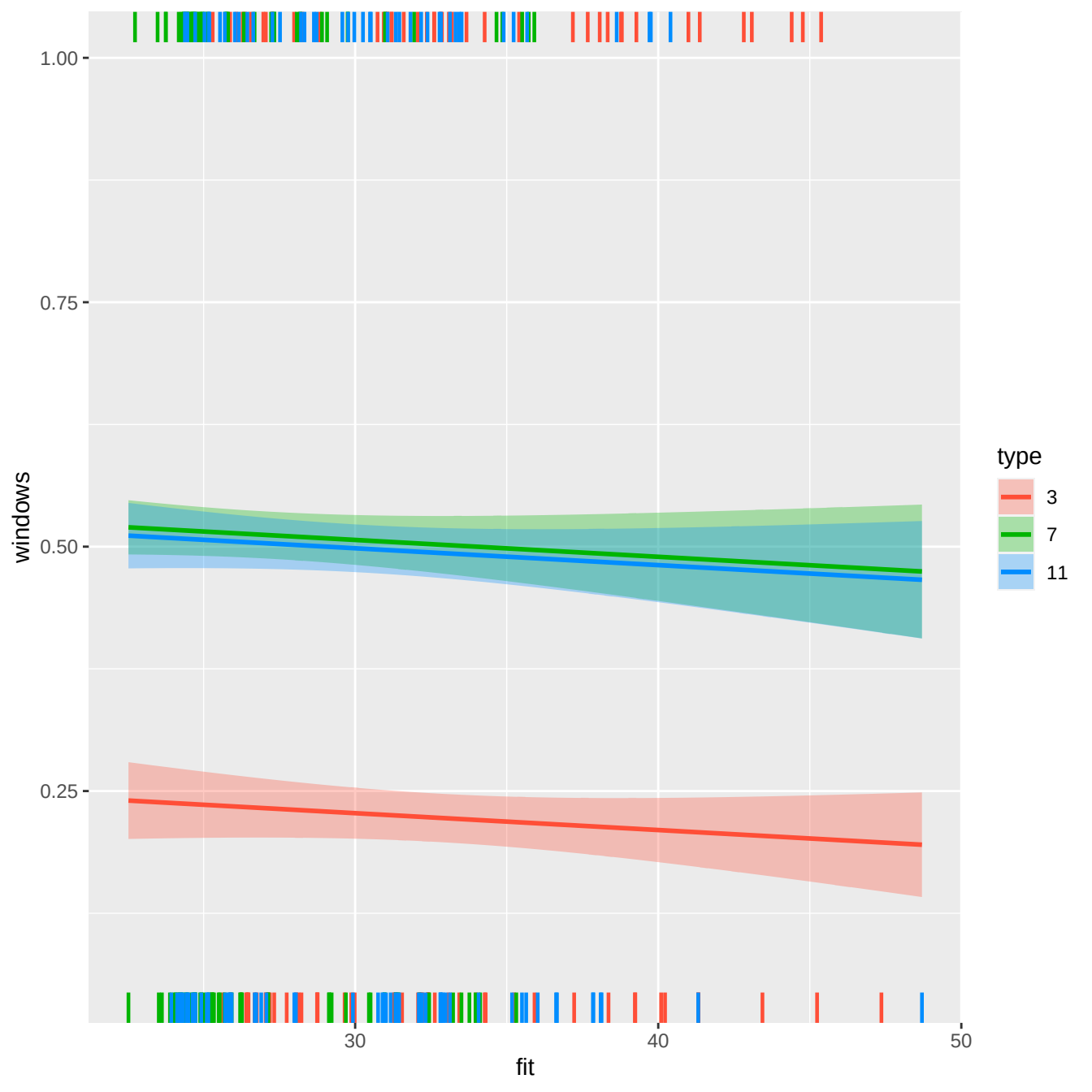

Mgr_Mesenchymal_proliferation_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

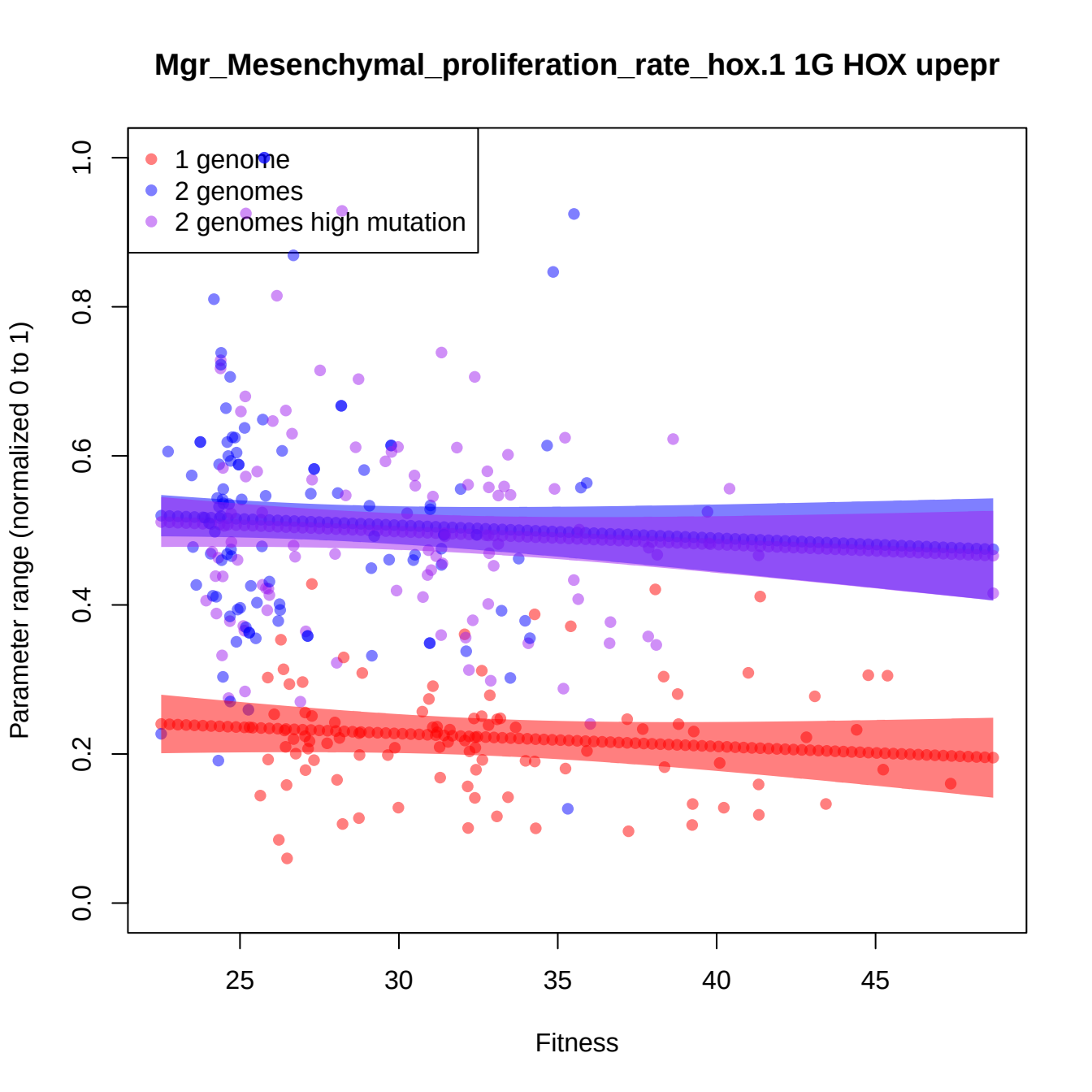

Mgr_Mesenchymal_proliferation_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.30859933074328e−16
4.86682629680041e−39
1.36186878315868e−06
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

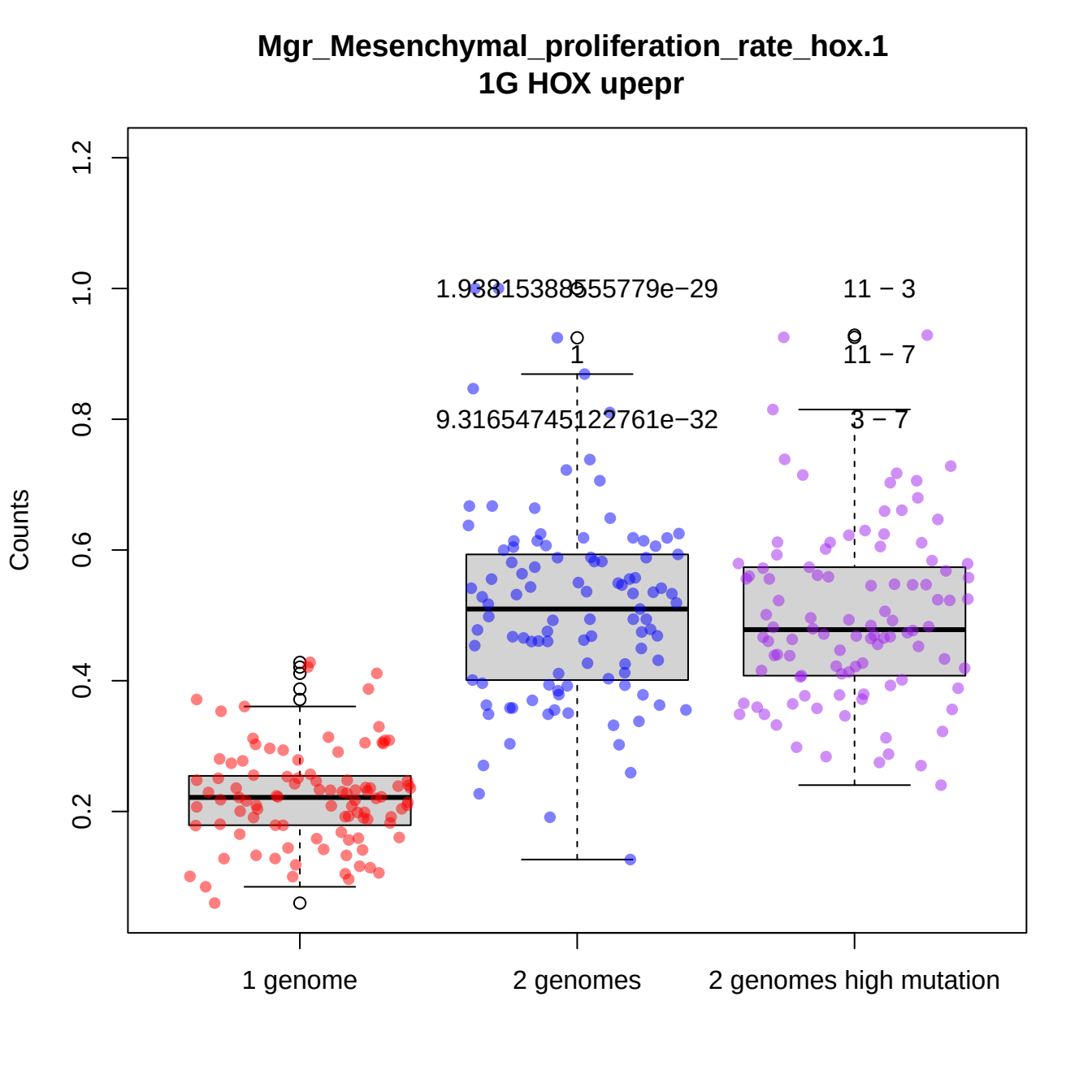

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

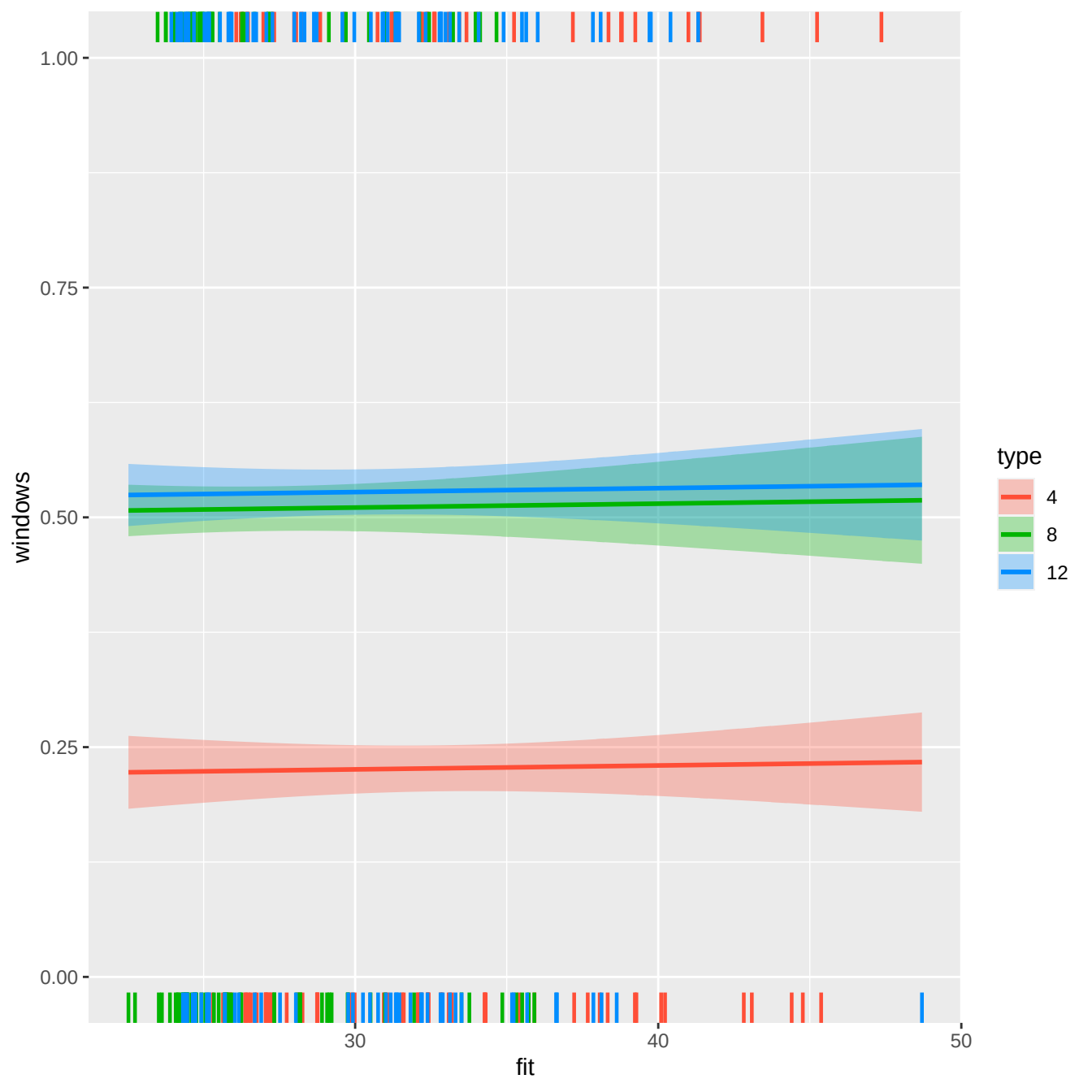

Mgr_Mesenchymal_proliferation_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

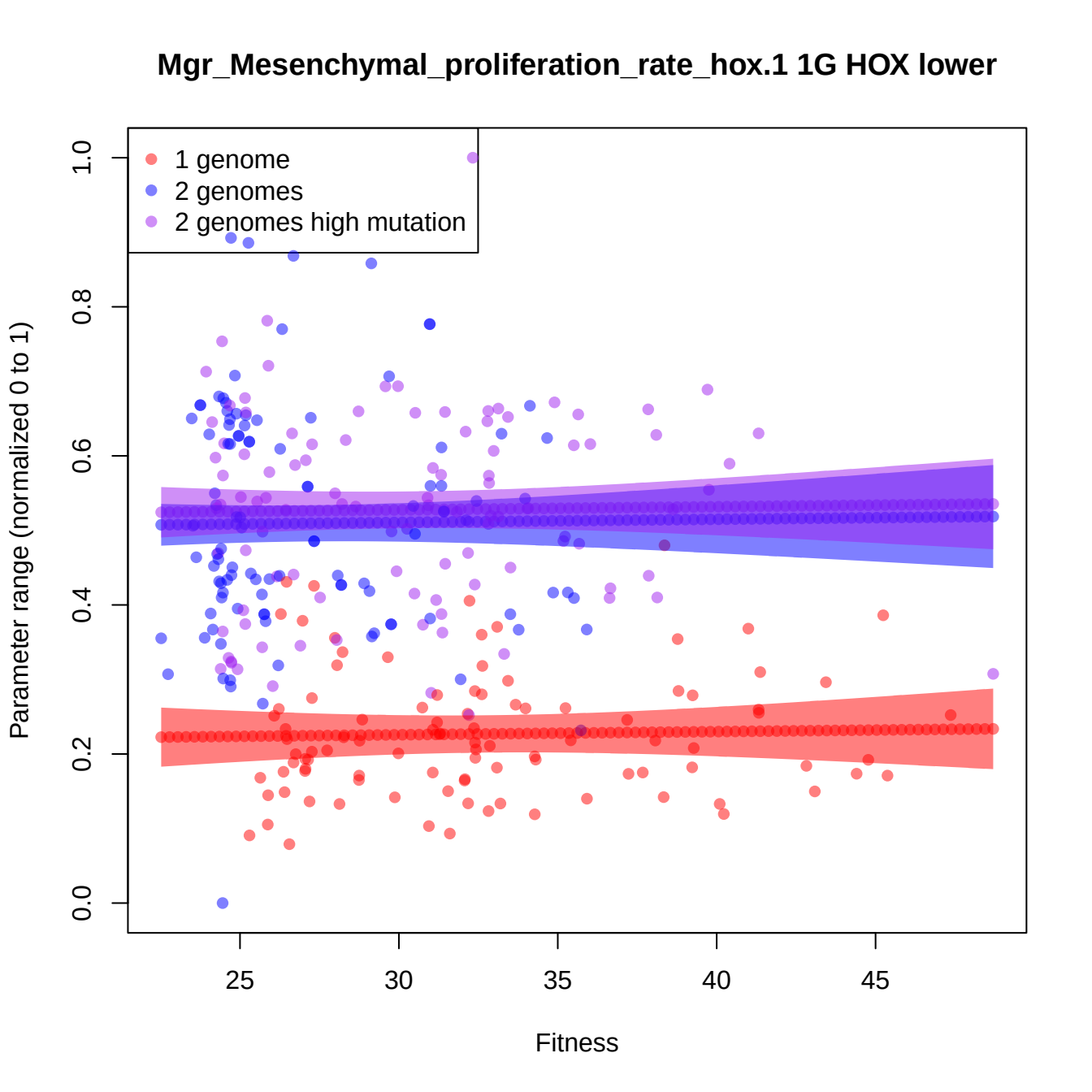

Mgr_Mesenchymal_proliferation_rate.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.44633329185019e−39
2.12118142531128e−09
5.05056466237362e−13
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

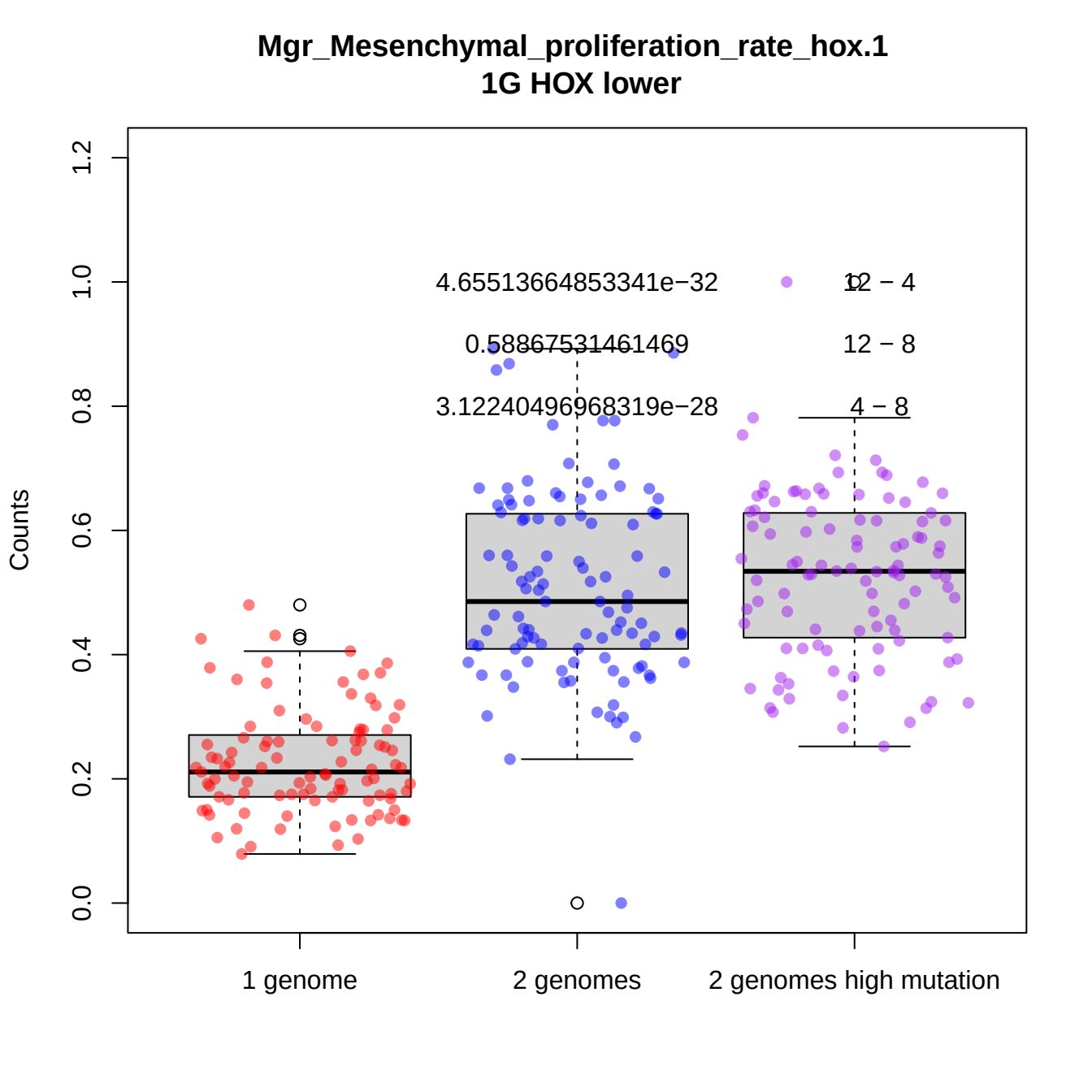

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

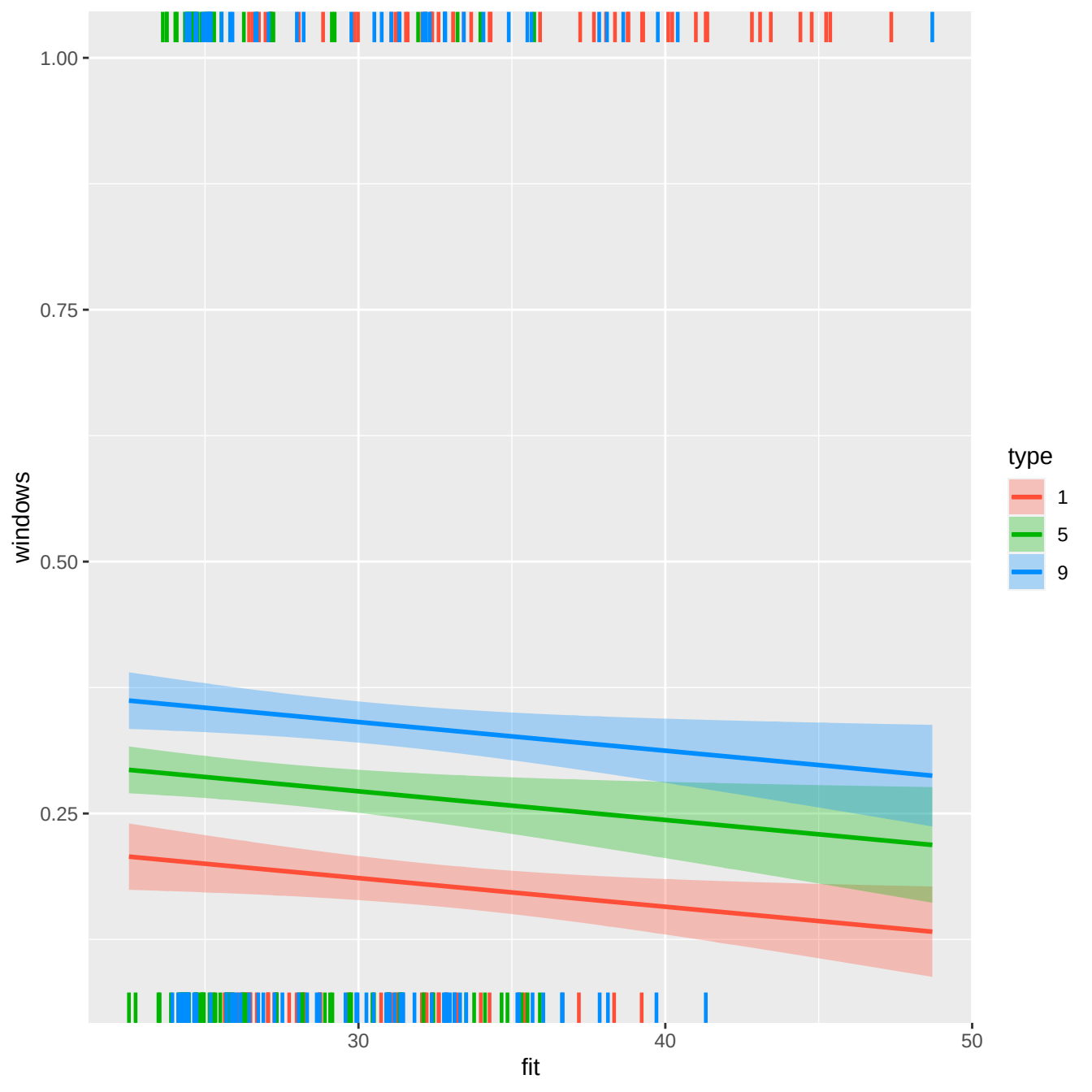

Mgr_Mesenchymal_proliferation_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

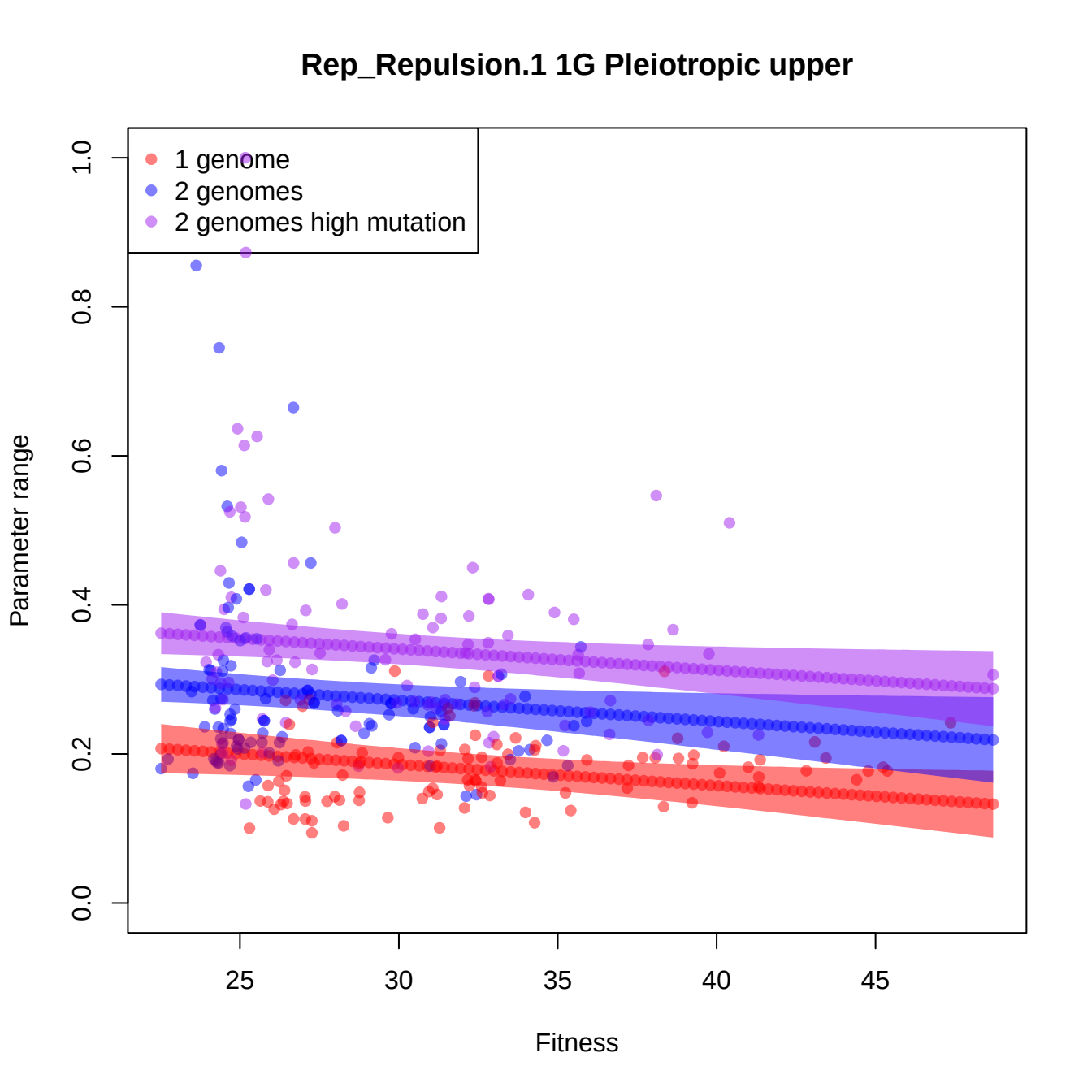

Mgr_Mesenchymal_proliferation_rate_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.938153885557779e−29
1
9.316547451227 61e−32
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

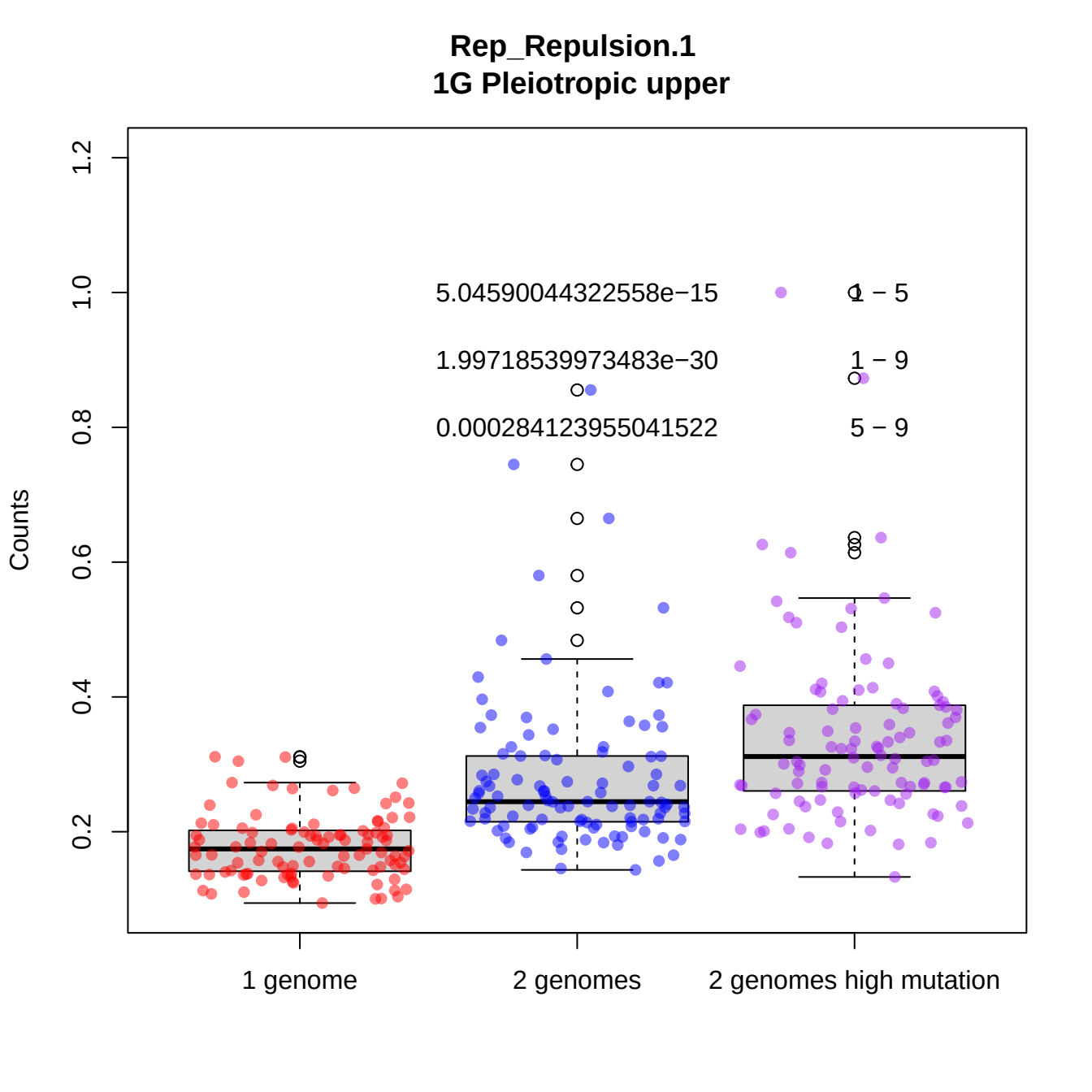

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

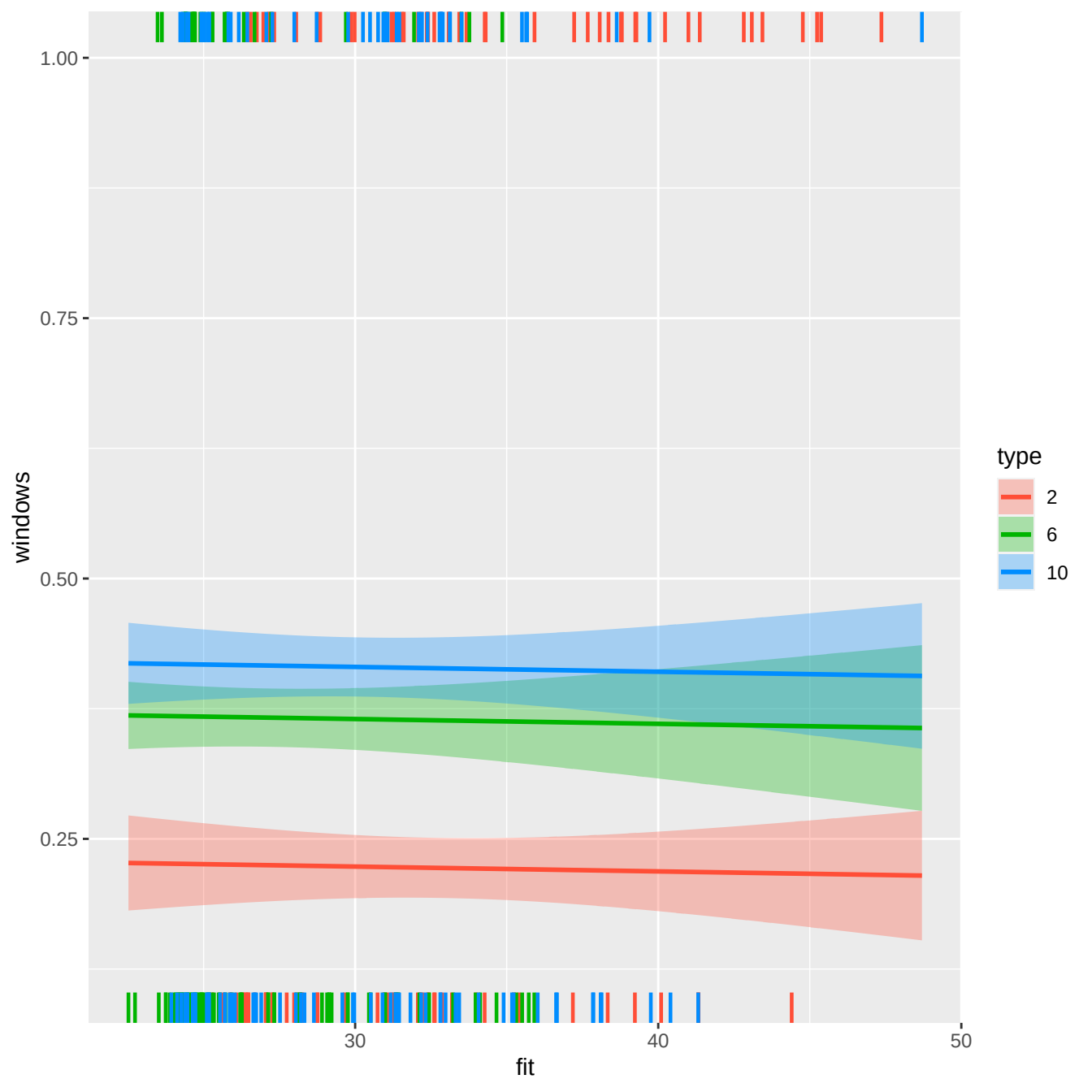

Mgr_Mesenchymal_proliferation_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

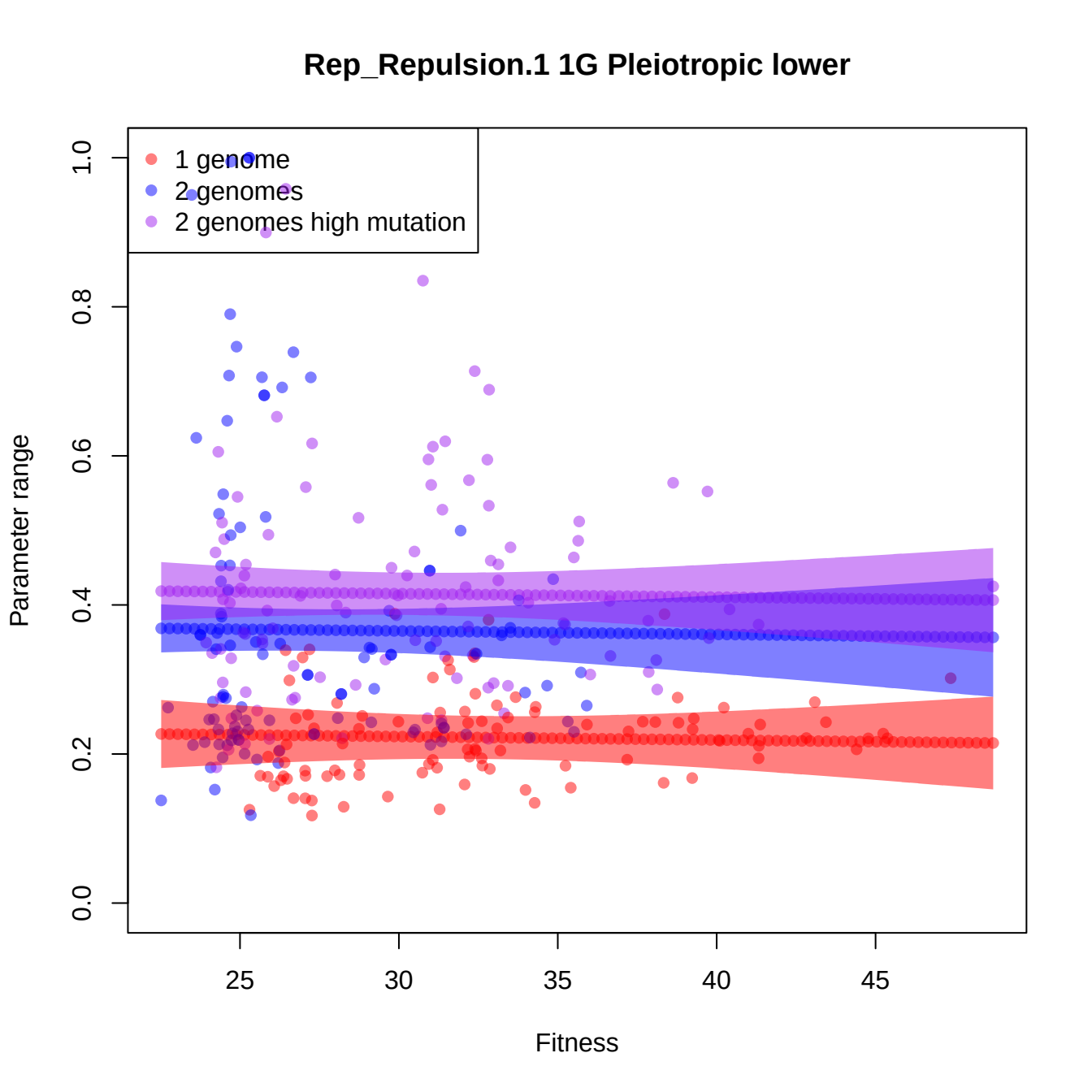

Mgr_Mesenchymal_proliferation_rate_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
4.65513664853341e−32
0.58867531461469
3.12240496968319e−28
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

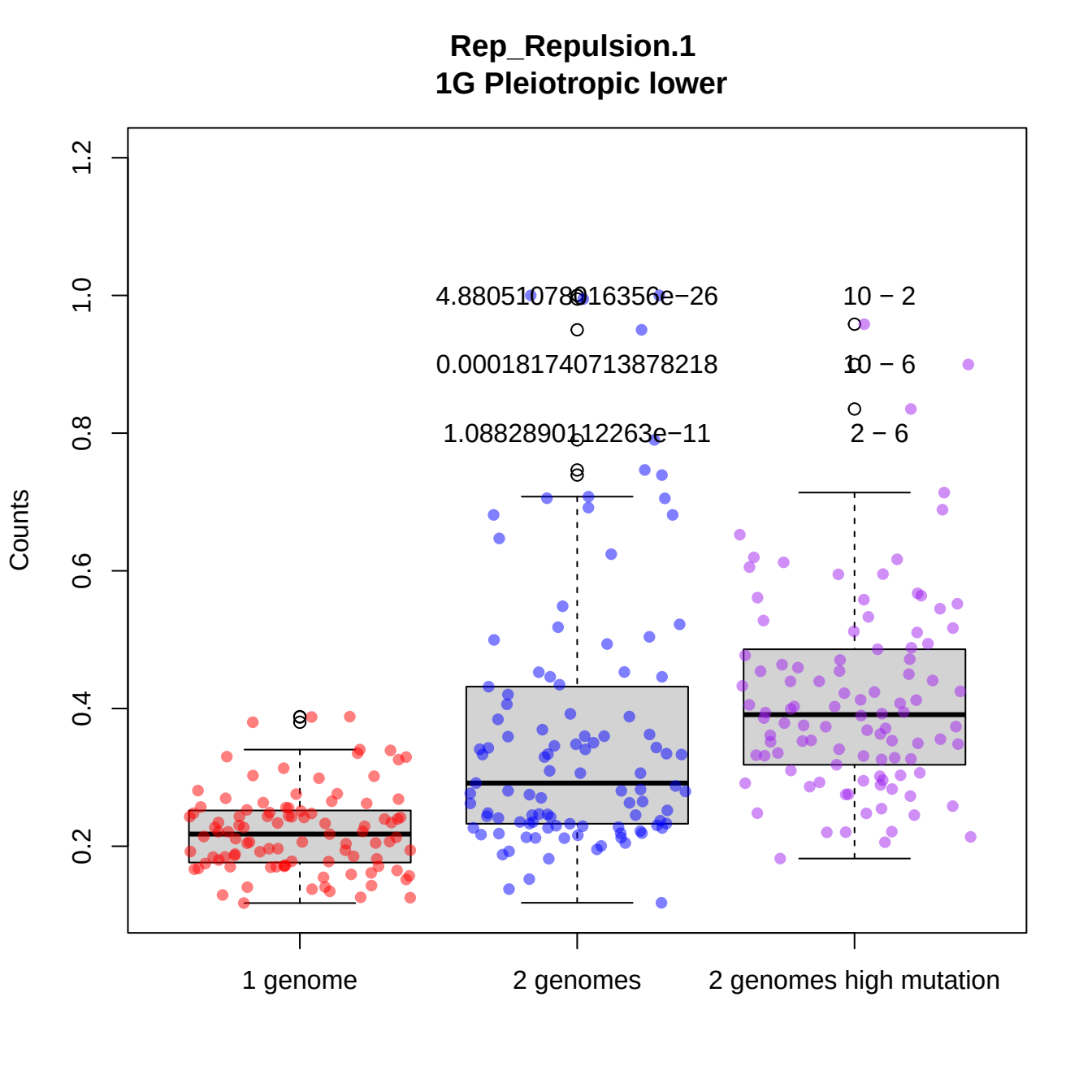

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

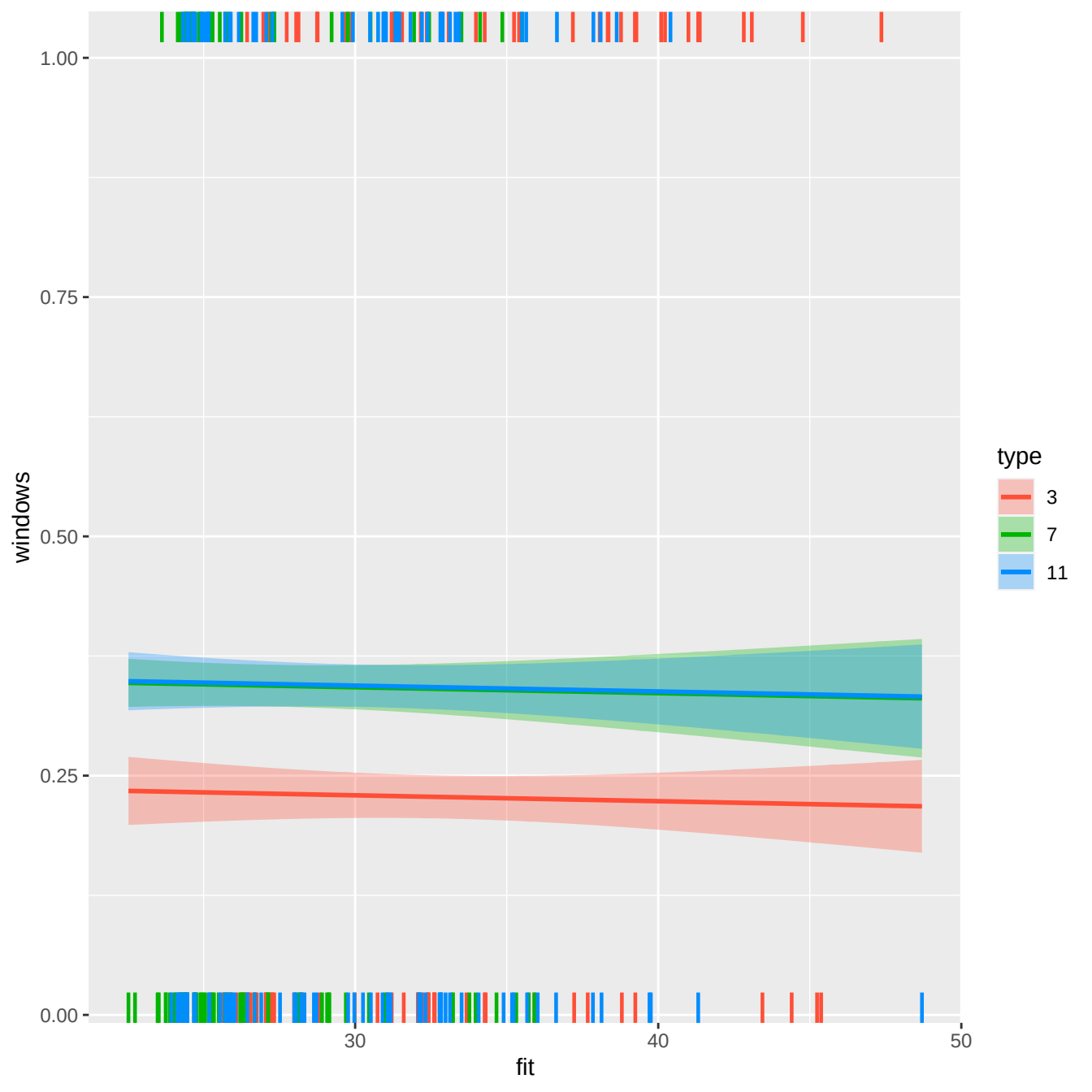

Rep_Repulsion.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

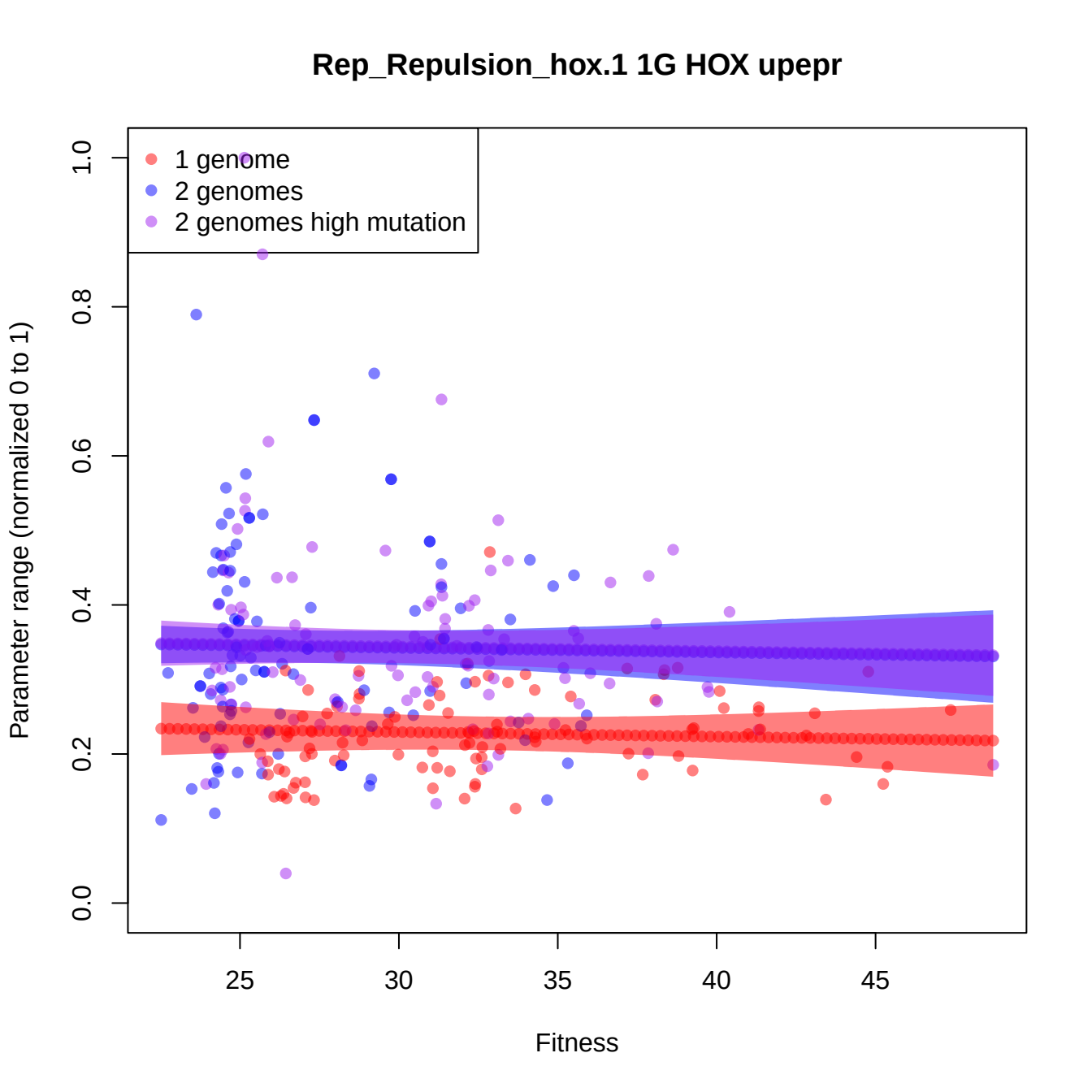

Rep_Repulsion.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
5.04590044322558e−15
1.99718539973483e−30
0.000284123955041522
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

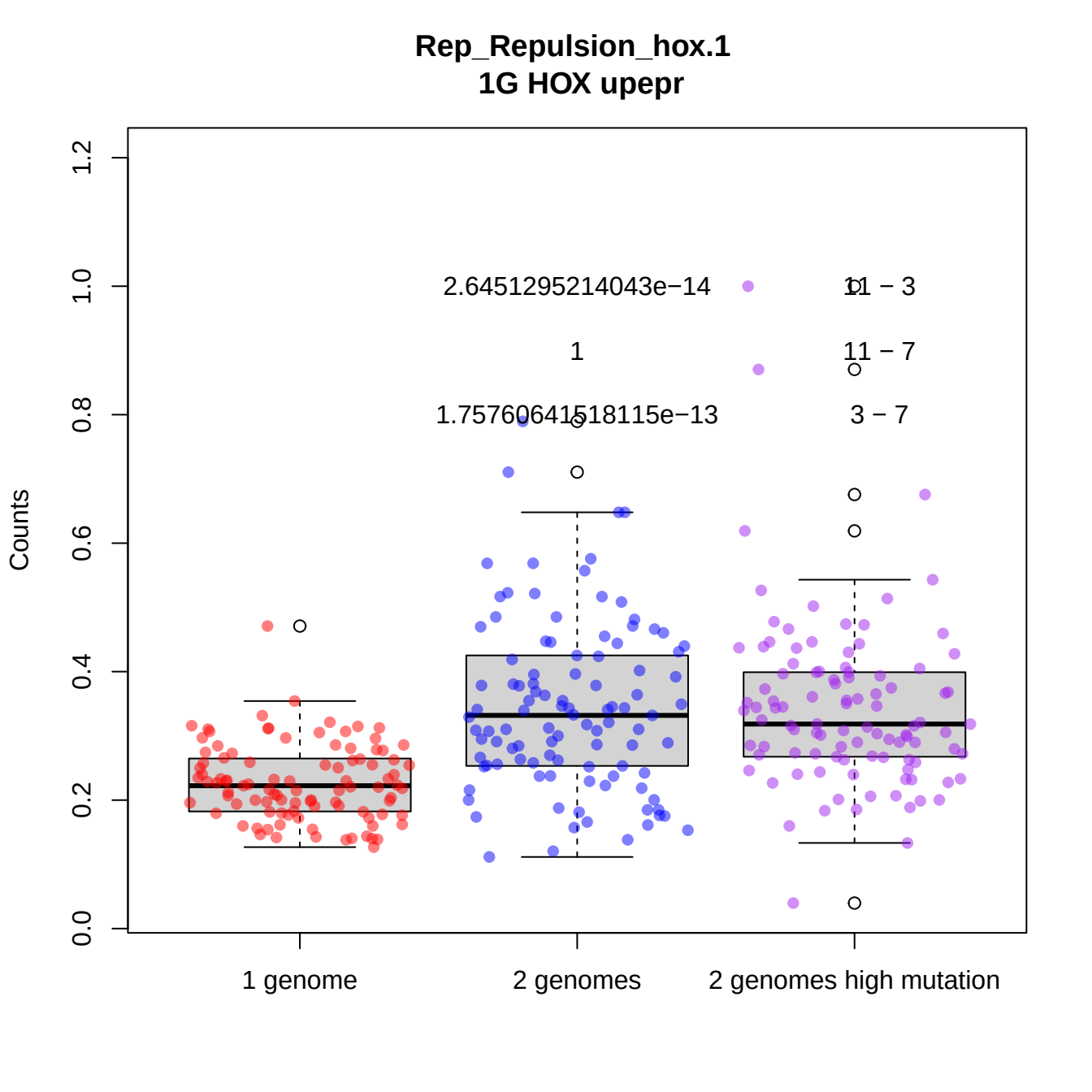

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

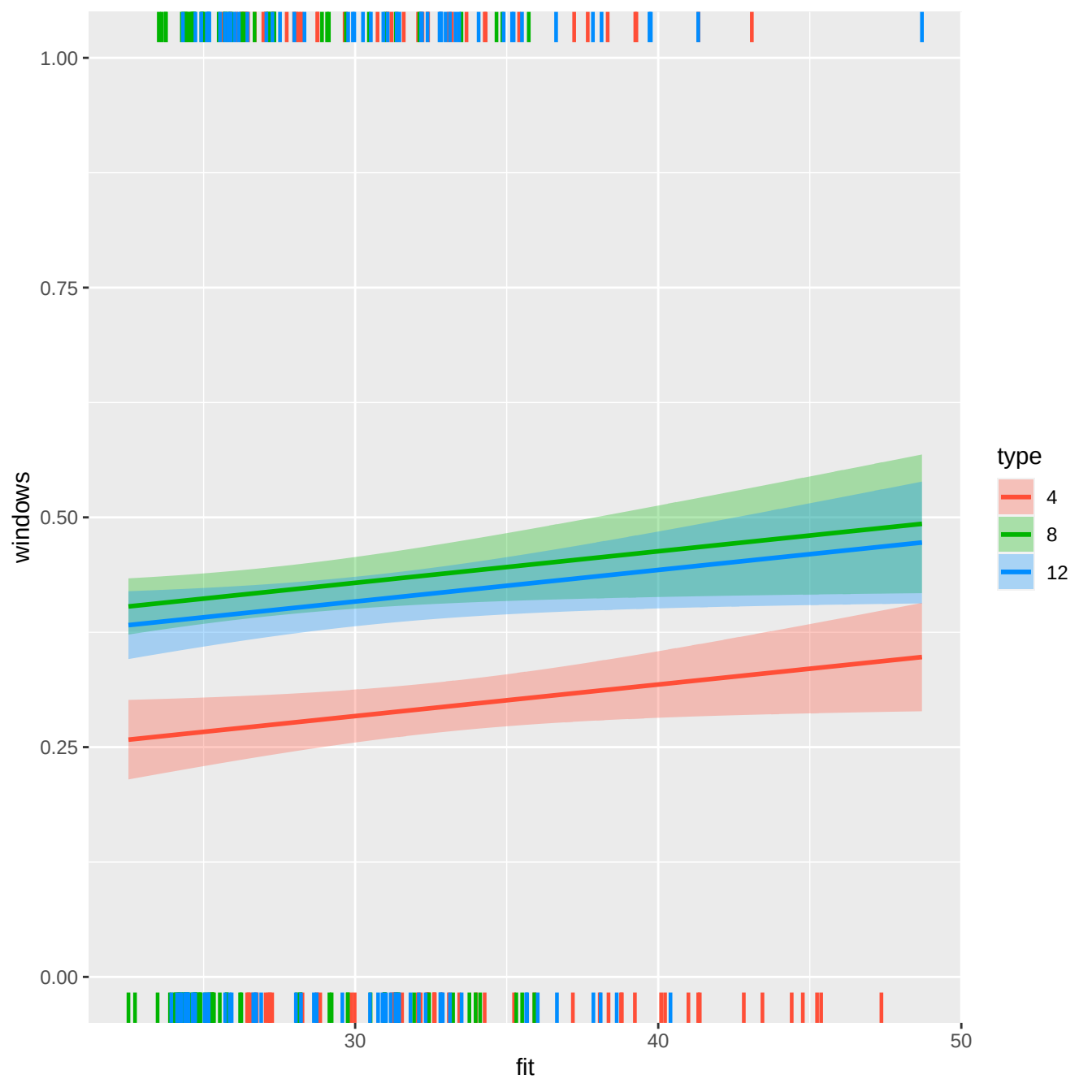

Rep_Repulsion.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

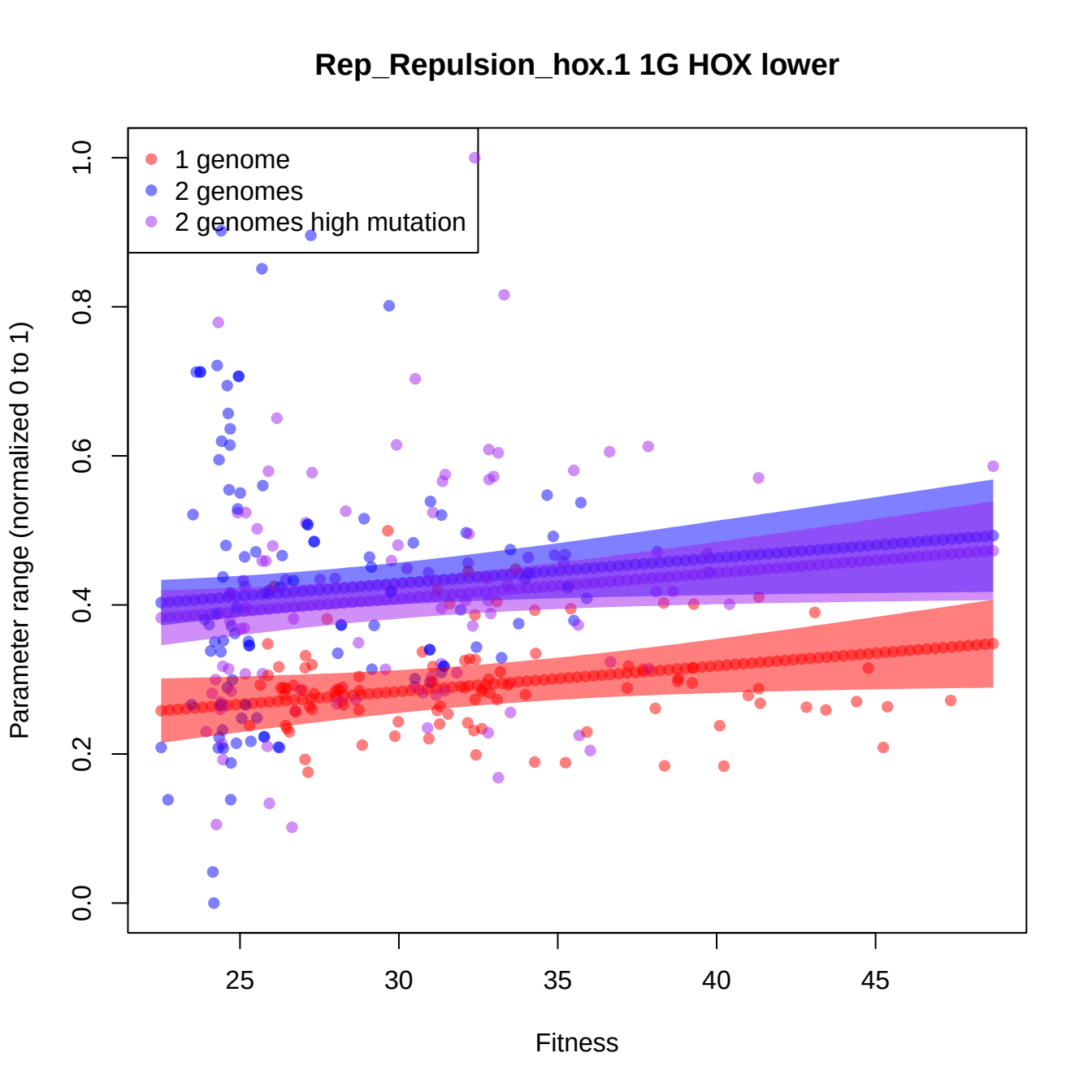

Rep_Repulsion.1
1G Pleiotropic lower
Counts
0.2
0.4
0.6
0.8
1.0
1.2
4.880510780163560e−26
0.000181740713878218
1.088289012263e−11
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

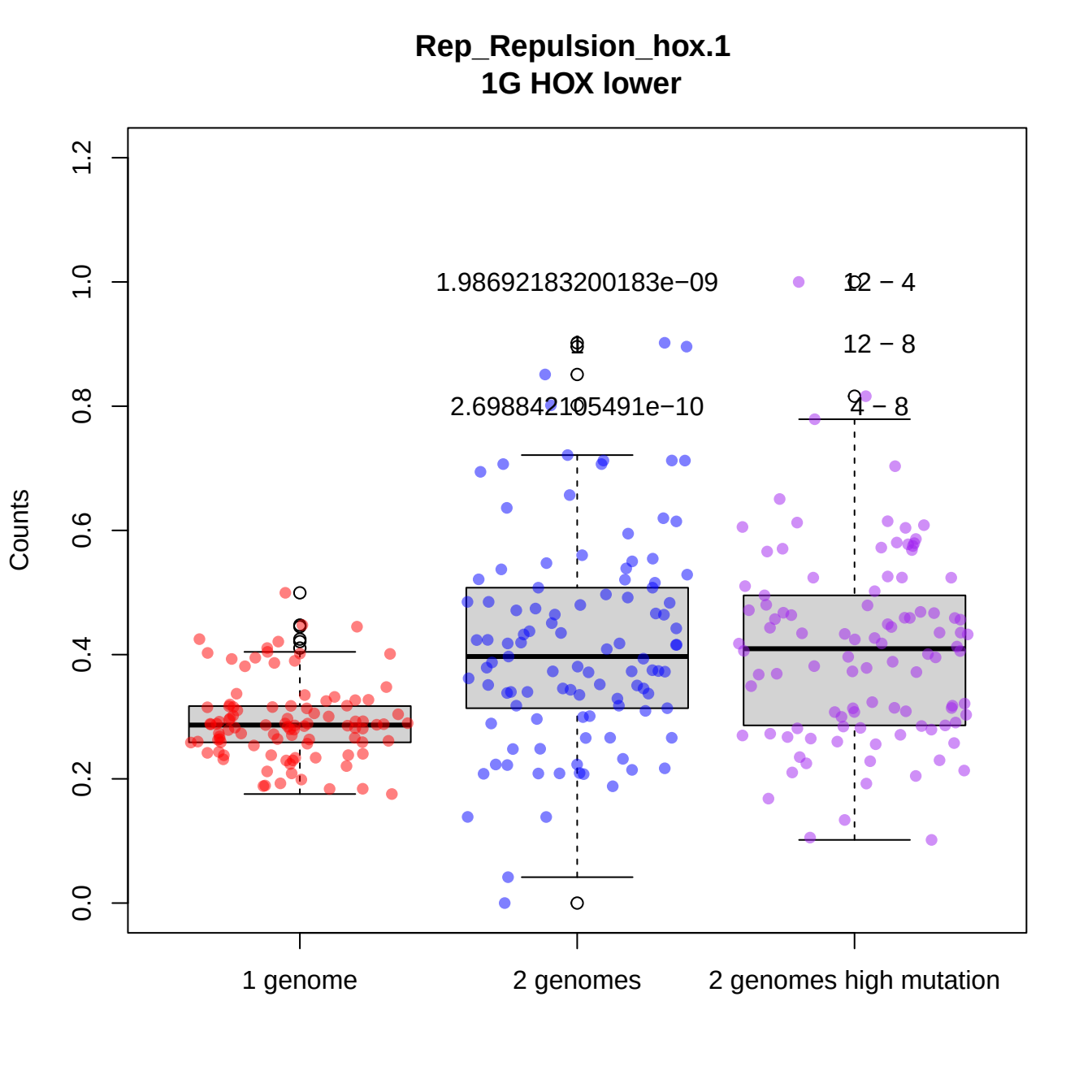

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

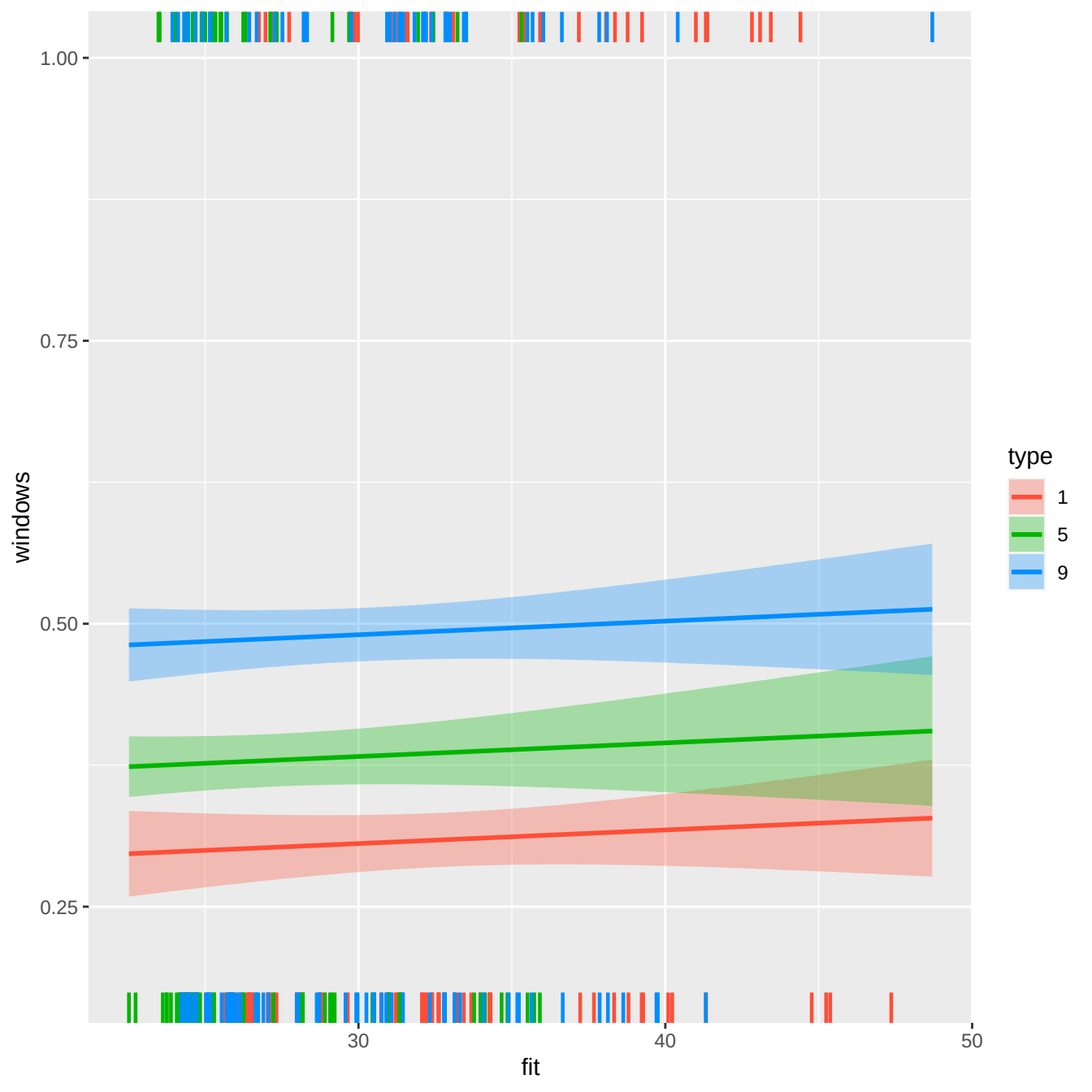

Rep_Repulsion_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

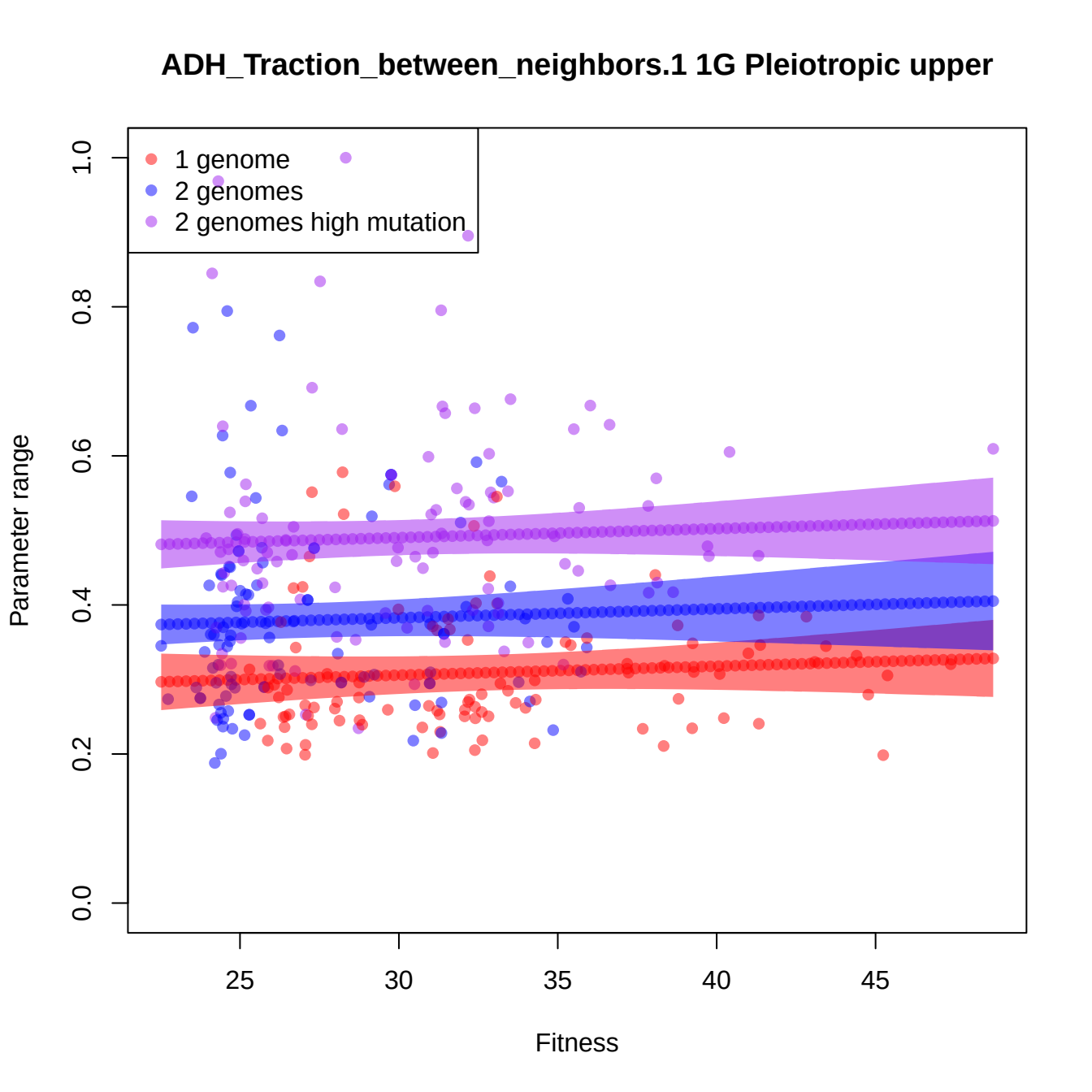

Rep_Repulsion_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.6451295214043e−14
1
1.7576064151811 5e−13
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

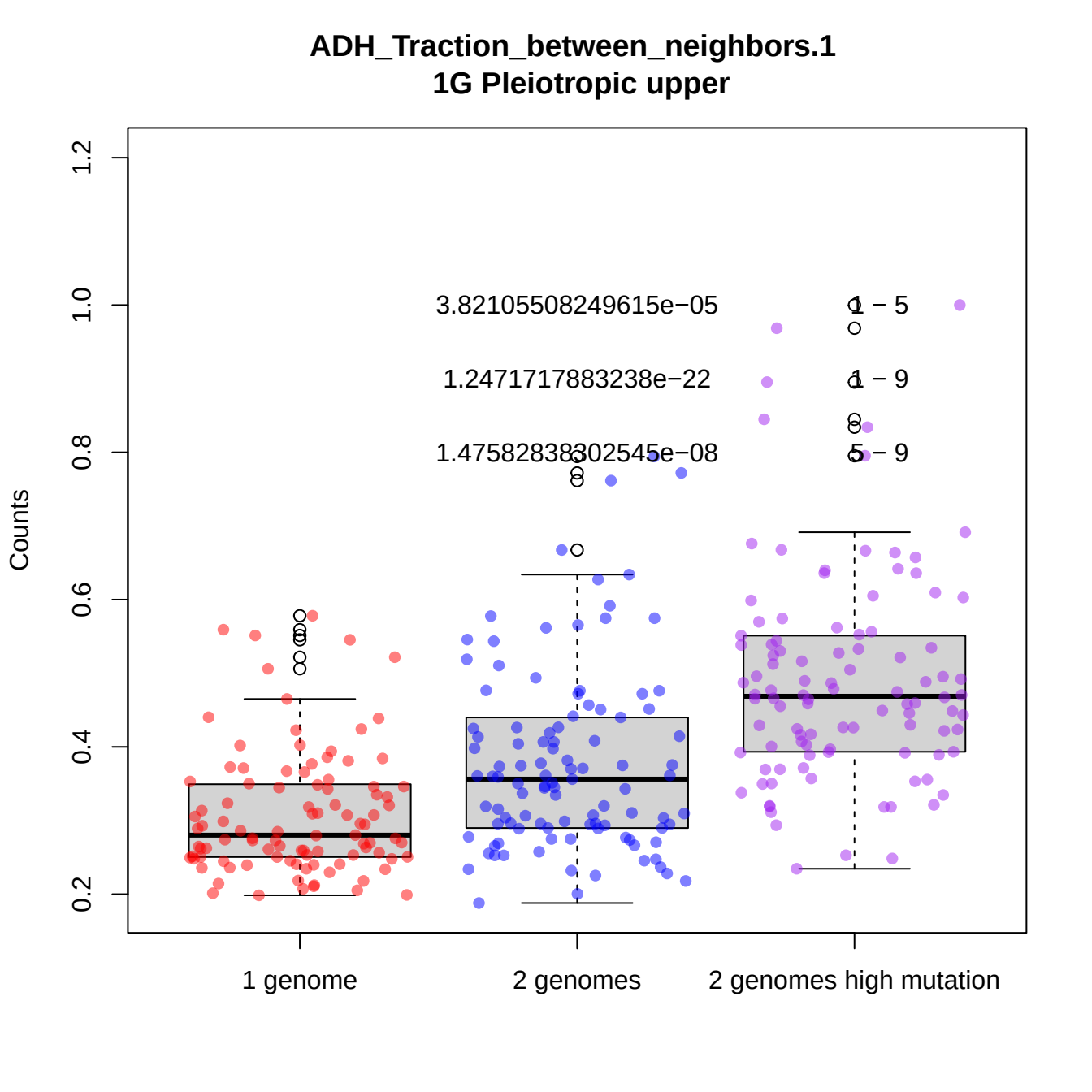

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

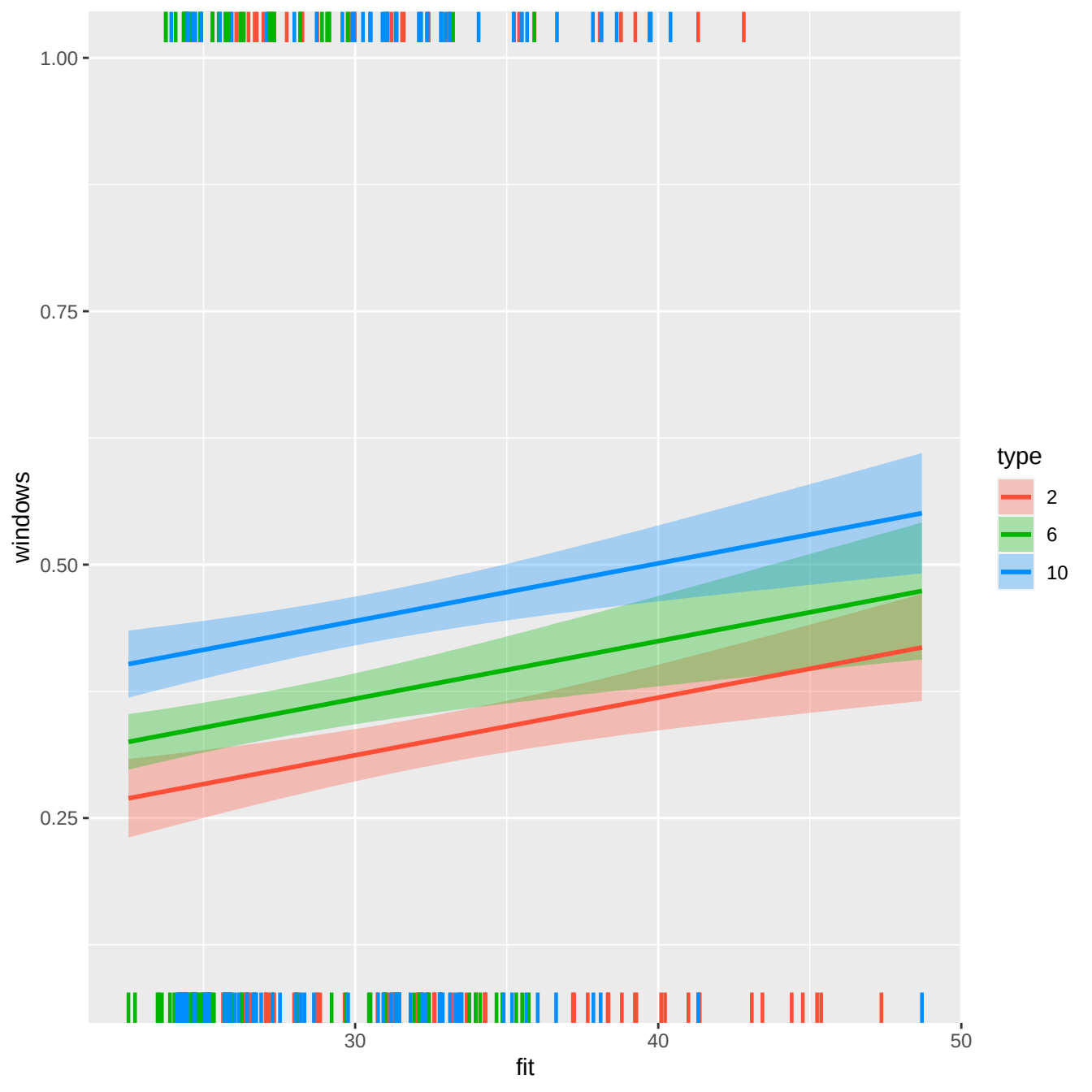

Rep_Repulsion_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

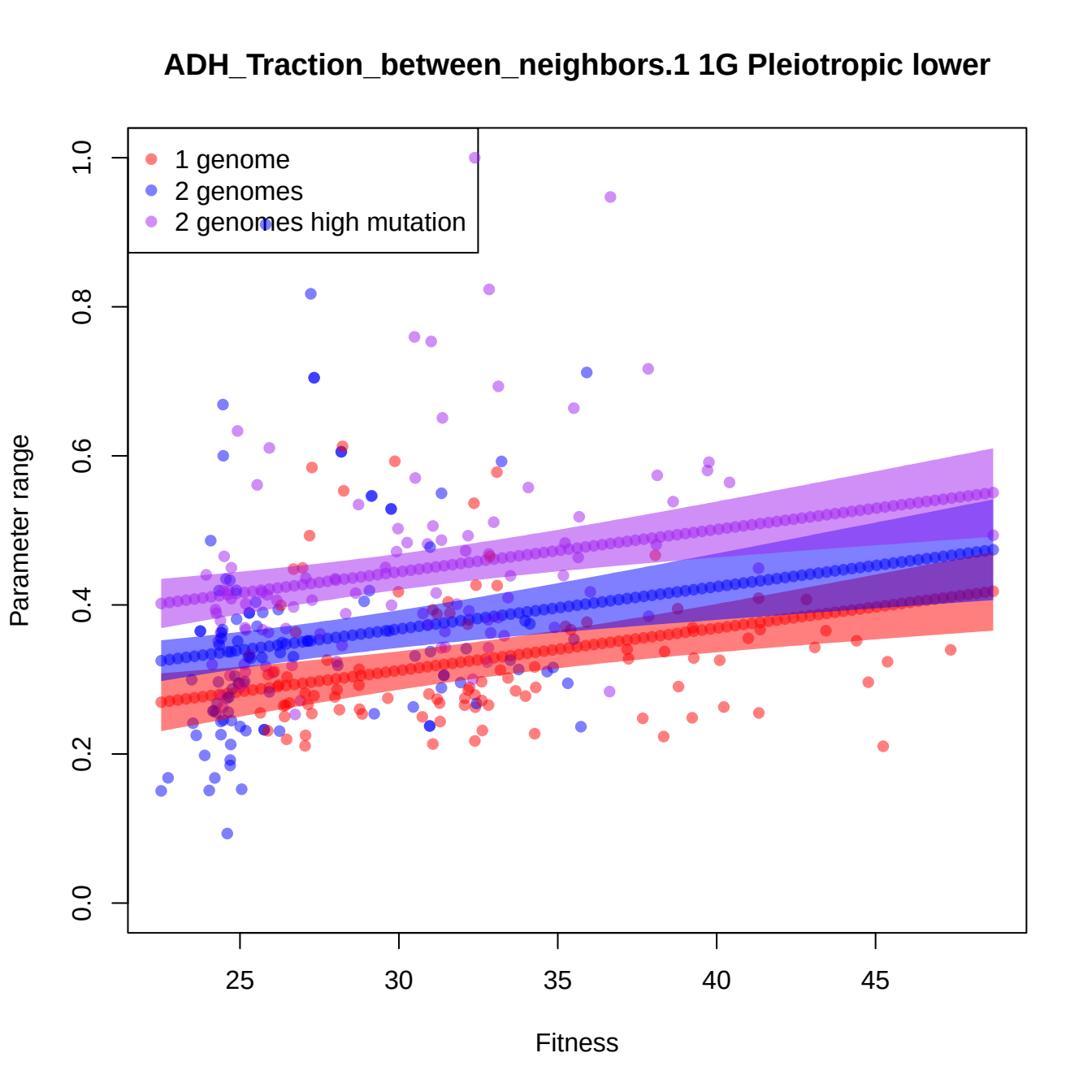

Rep_Repulsion_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.98692183200183e−09
2.69884210549 1e−10
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

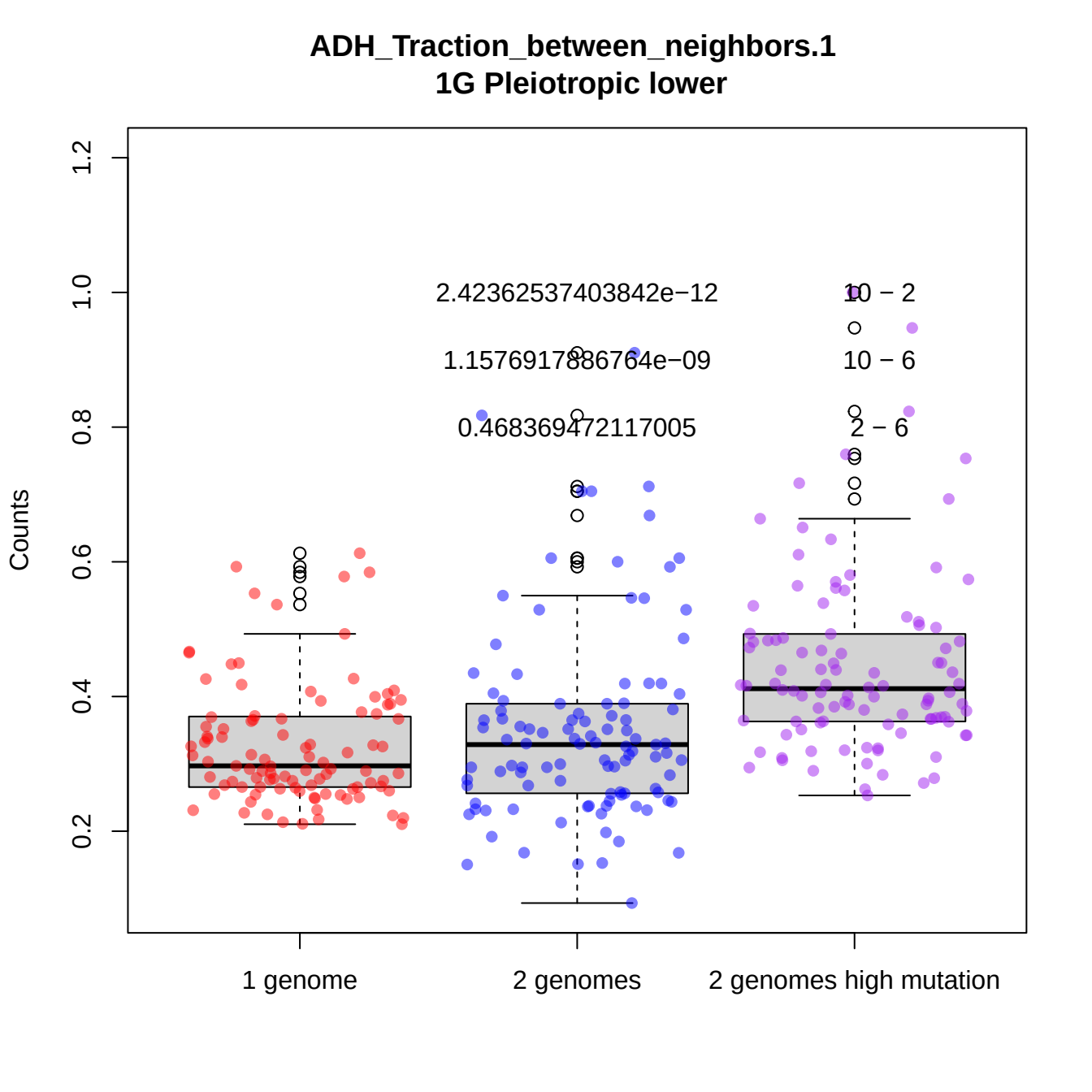

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

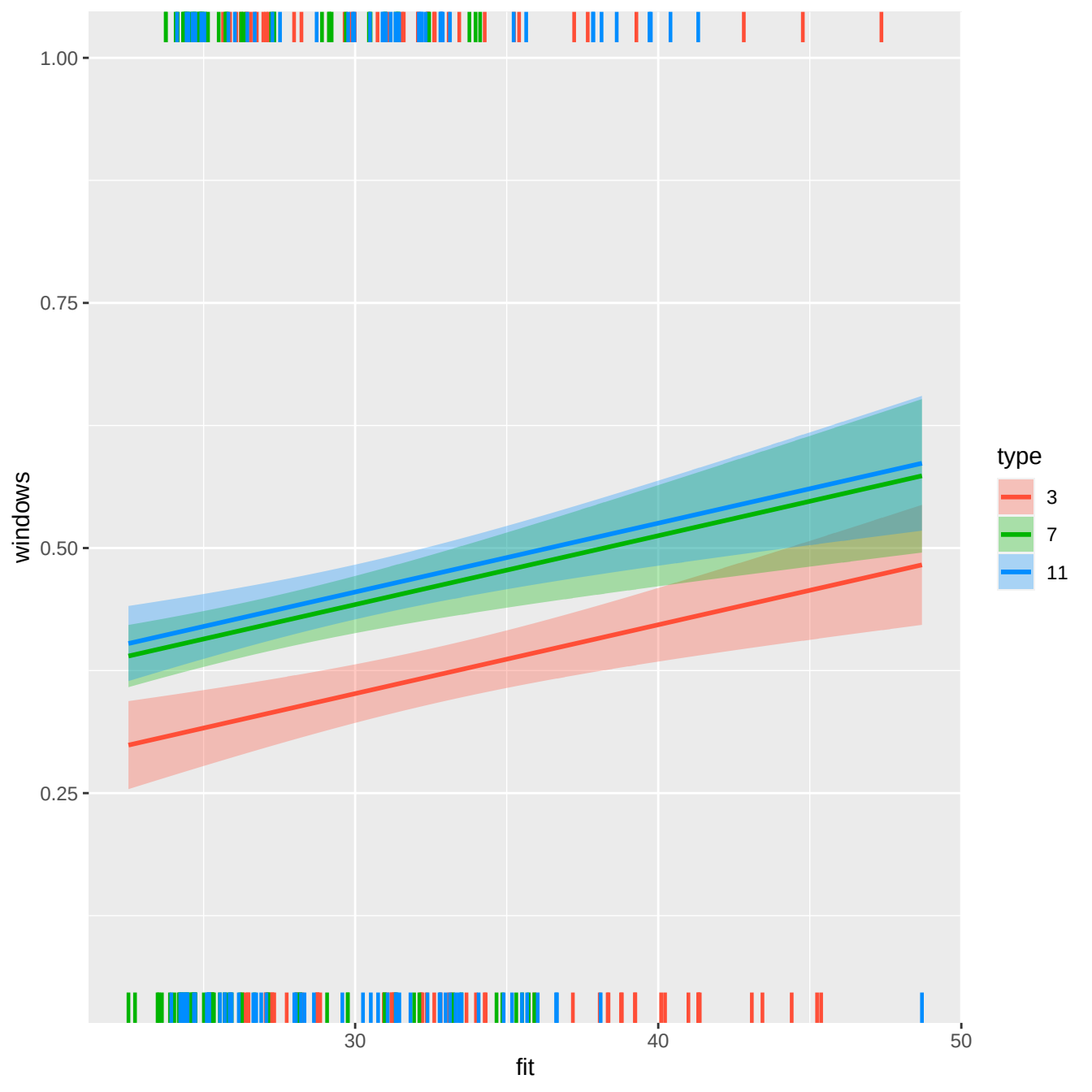

ADH_Traction_between_neighbors.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

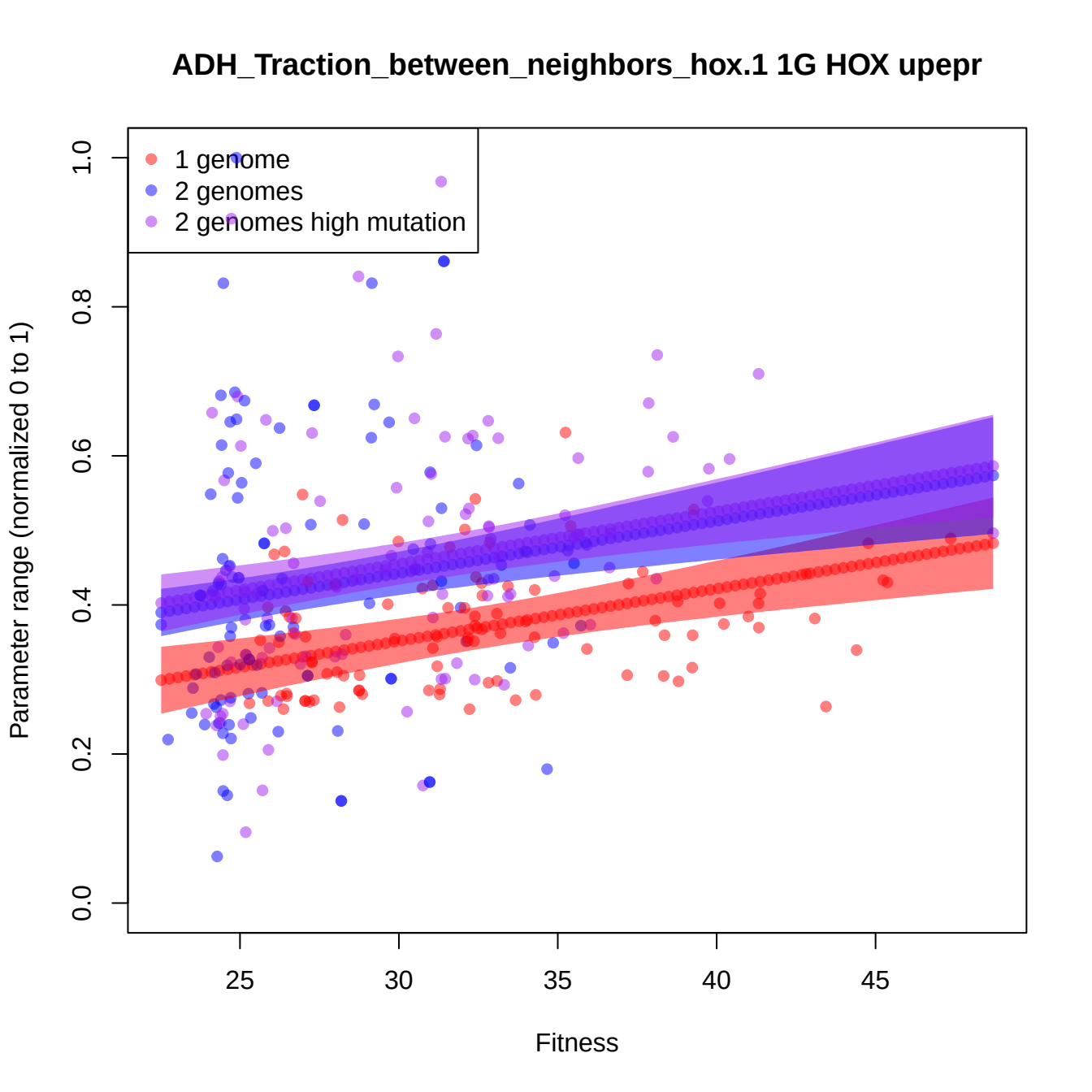

ADH_Traction_between_neighbors.1
1G Pleiotropic upper
Counts
0.2
0.4
0.6
0.8
1.0
1.2
3.82105508249615e−05
1.2471717883238e−22
1.47582838302545e−08
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

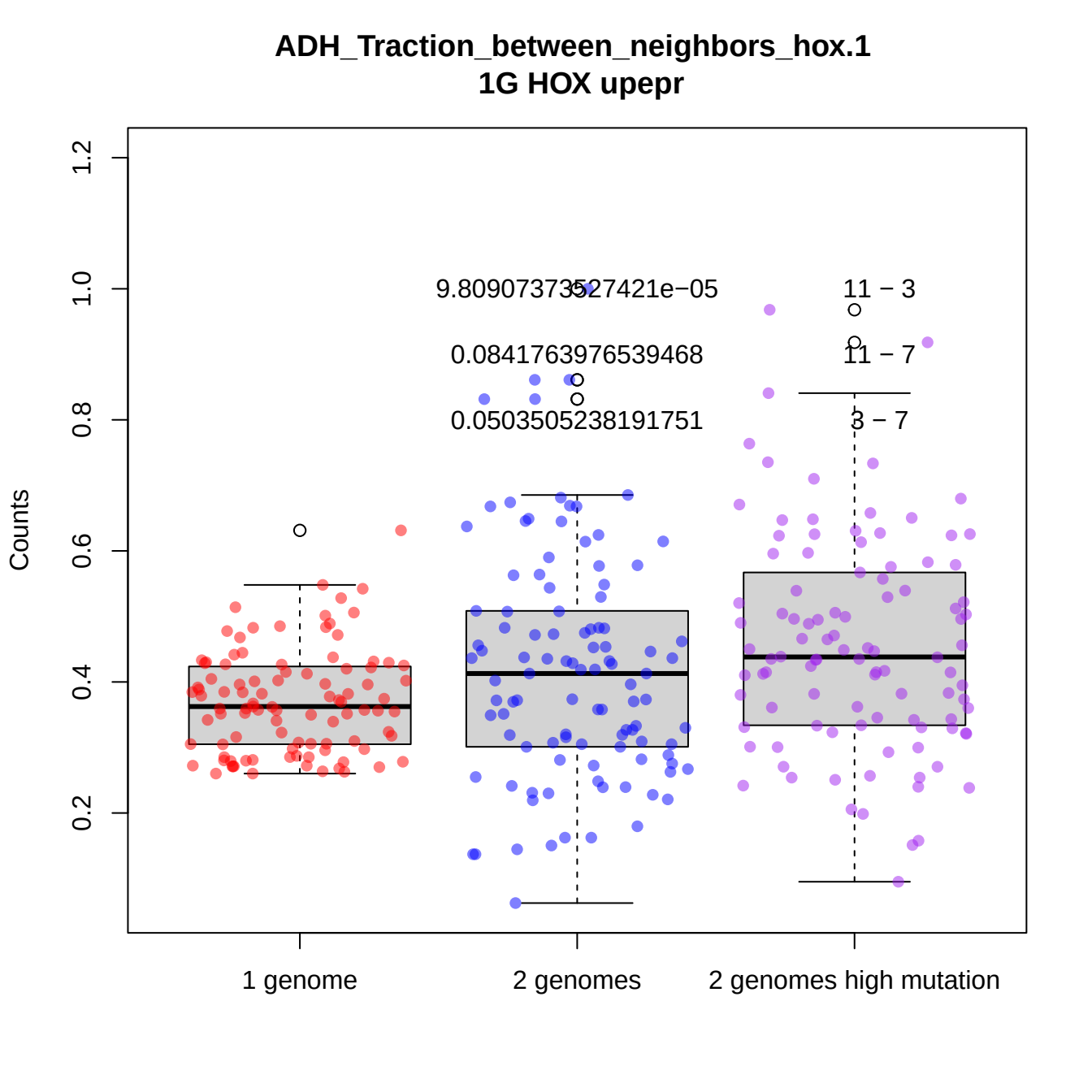

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

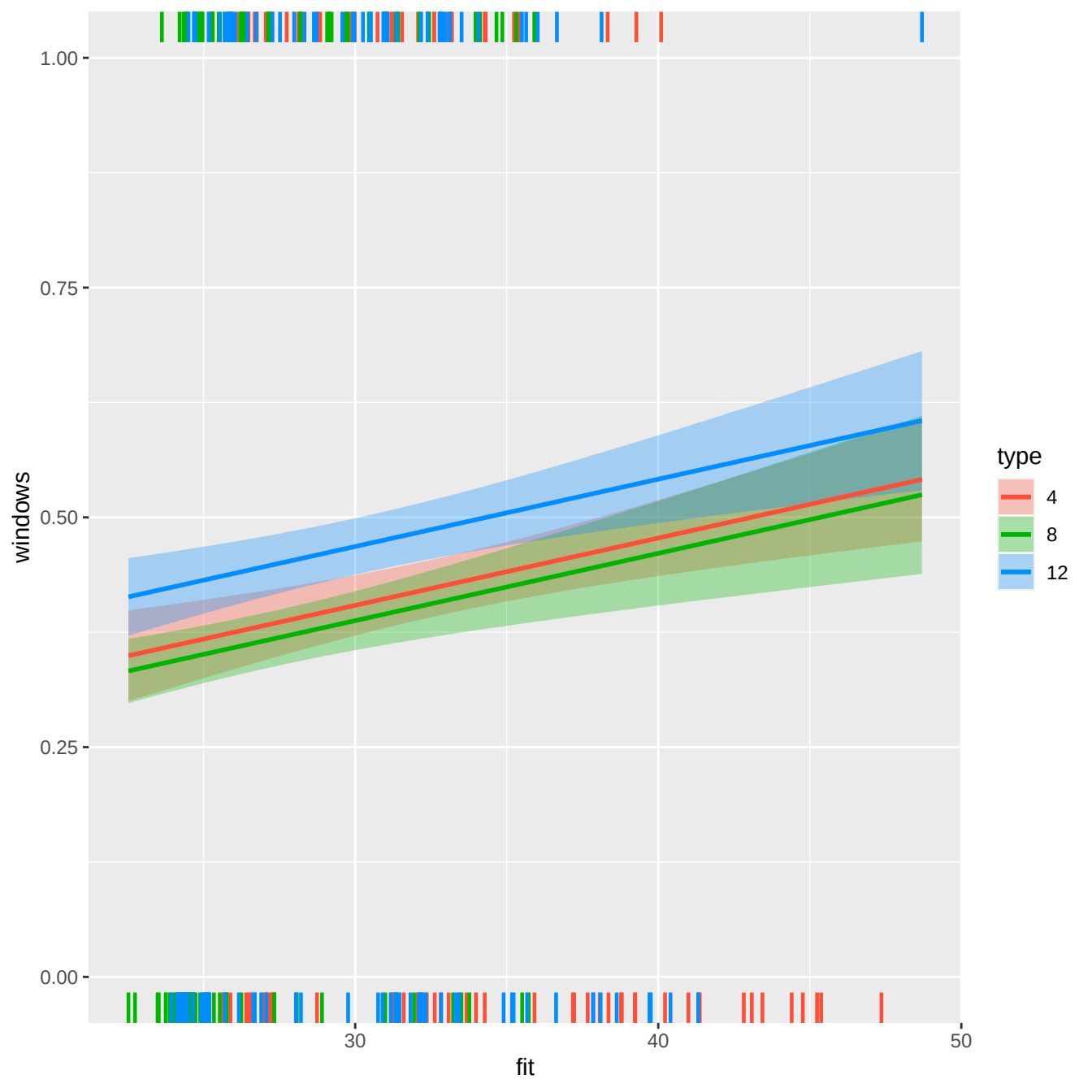

ADH_Traction_between_neighbors.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

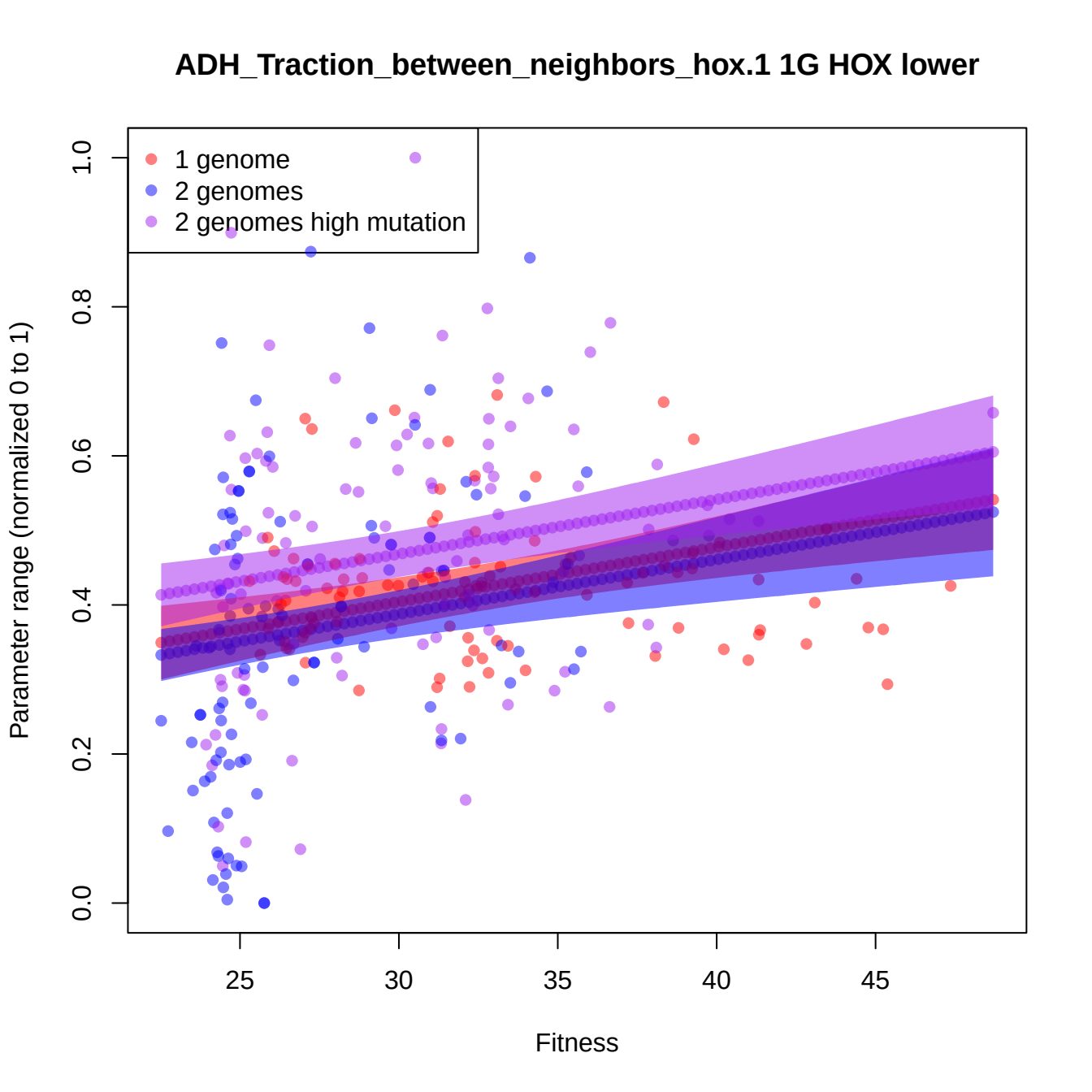

ADH_Traction_between_neighbors.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.42362537403842e−12
1.15769178867 64e−09
0.468369472117005
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

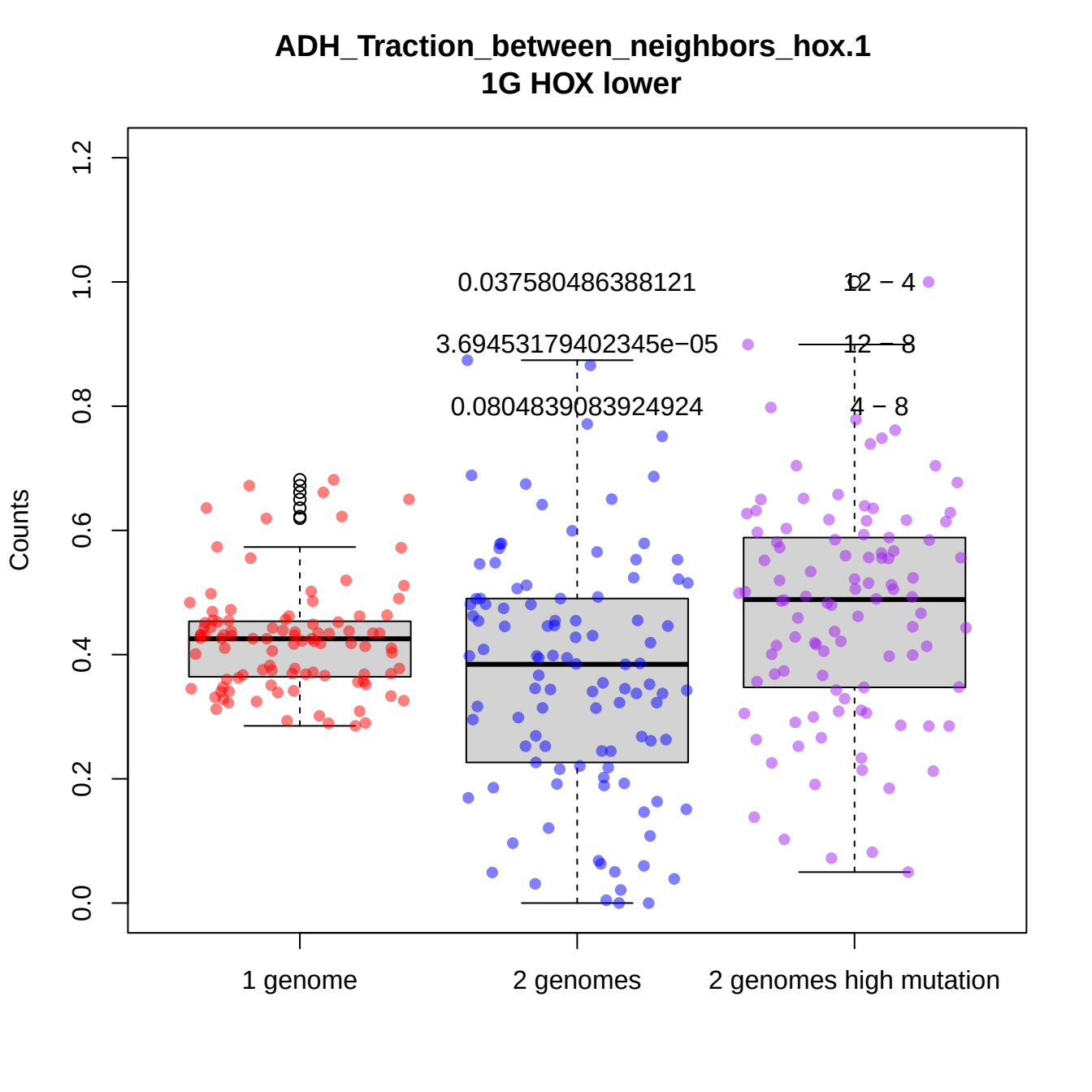

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

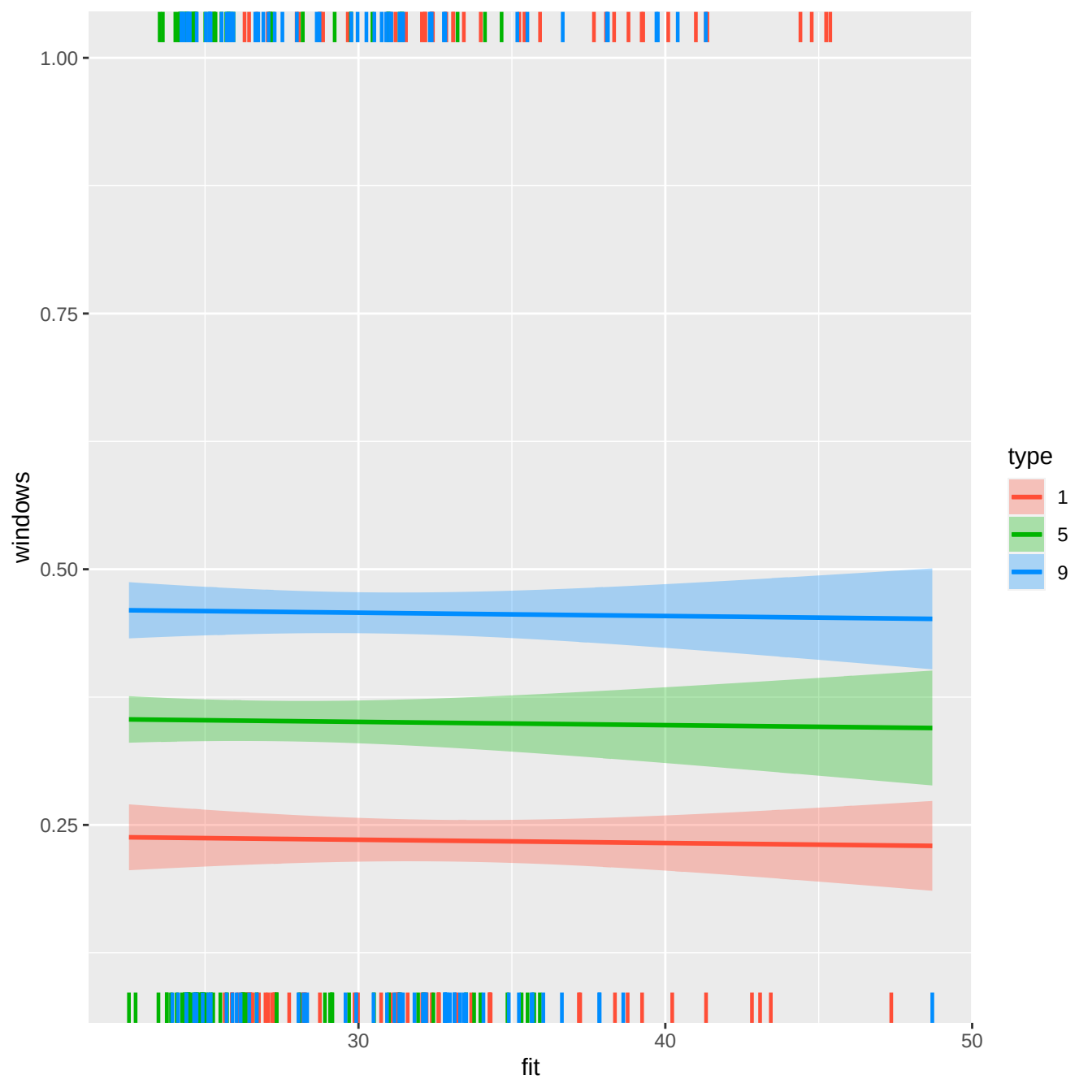

ADH_Traction_between_neighbors_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

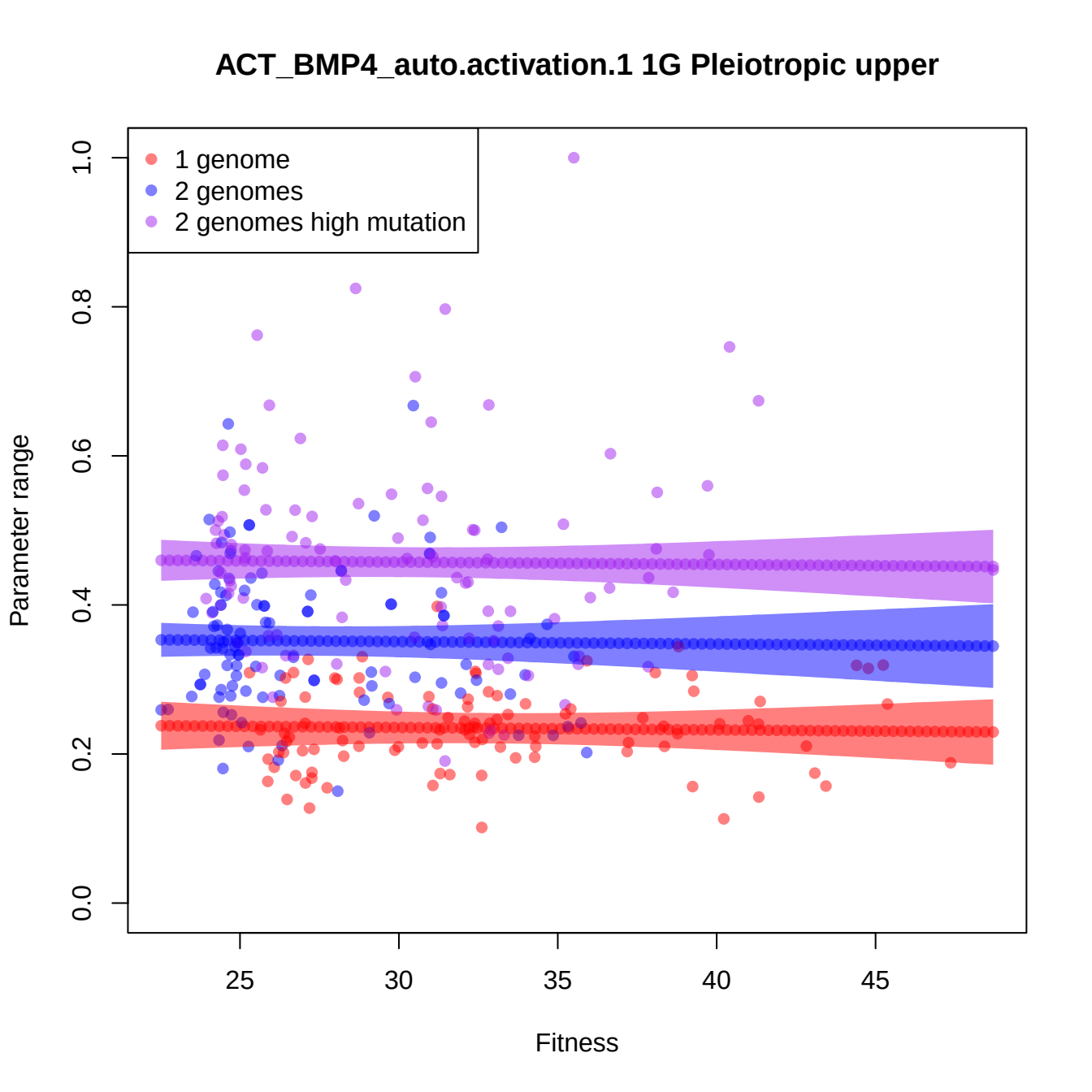

ADH_Traction_between_neighbors_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
9.809073735527421e−05
0.0841763976539468
0.0503505238191751
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

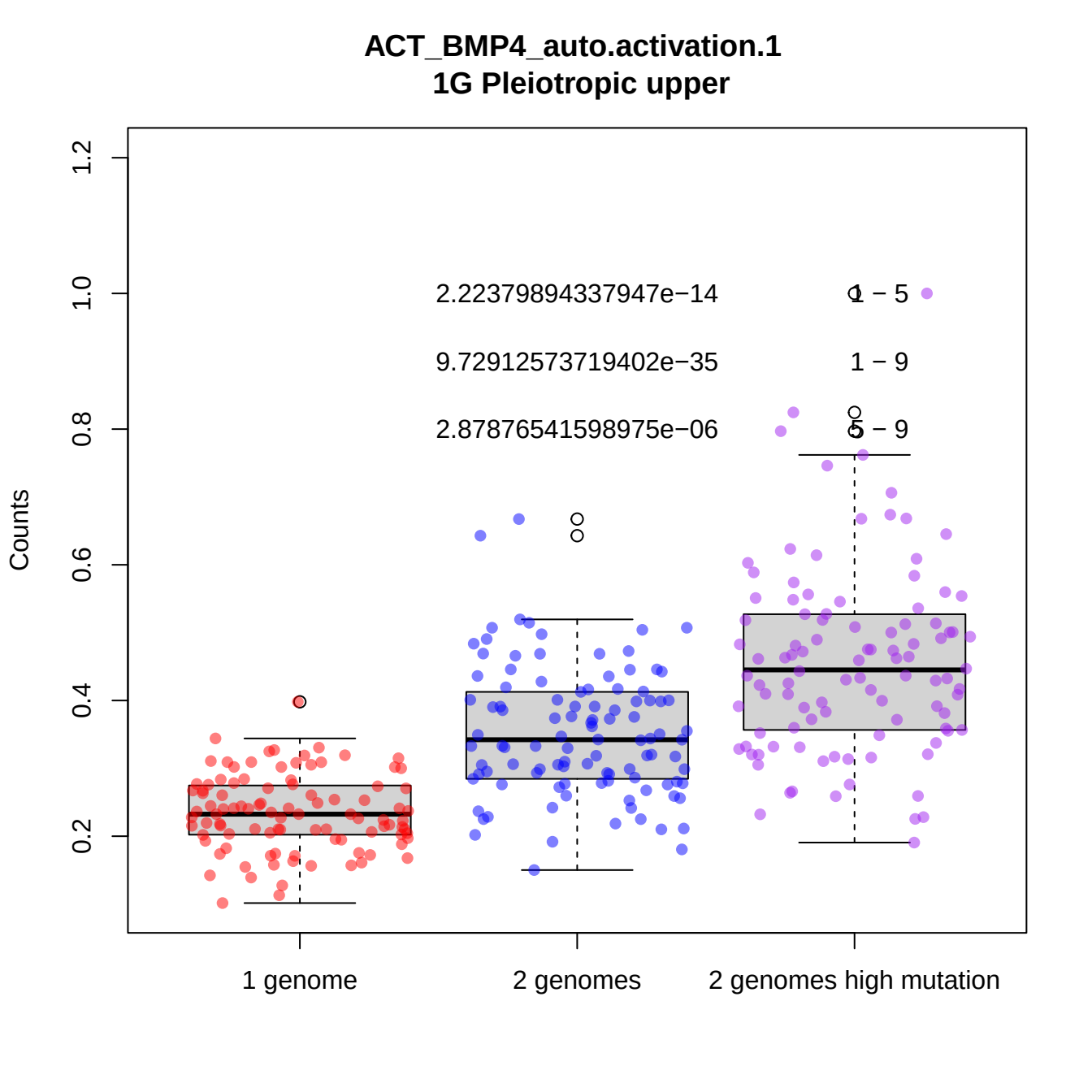

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

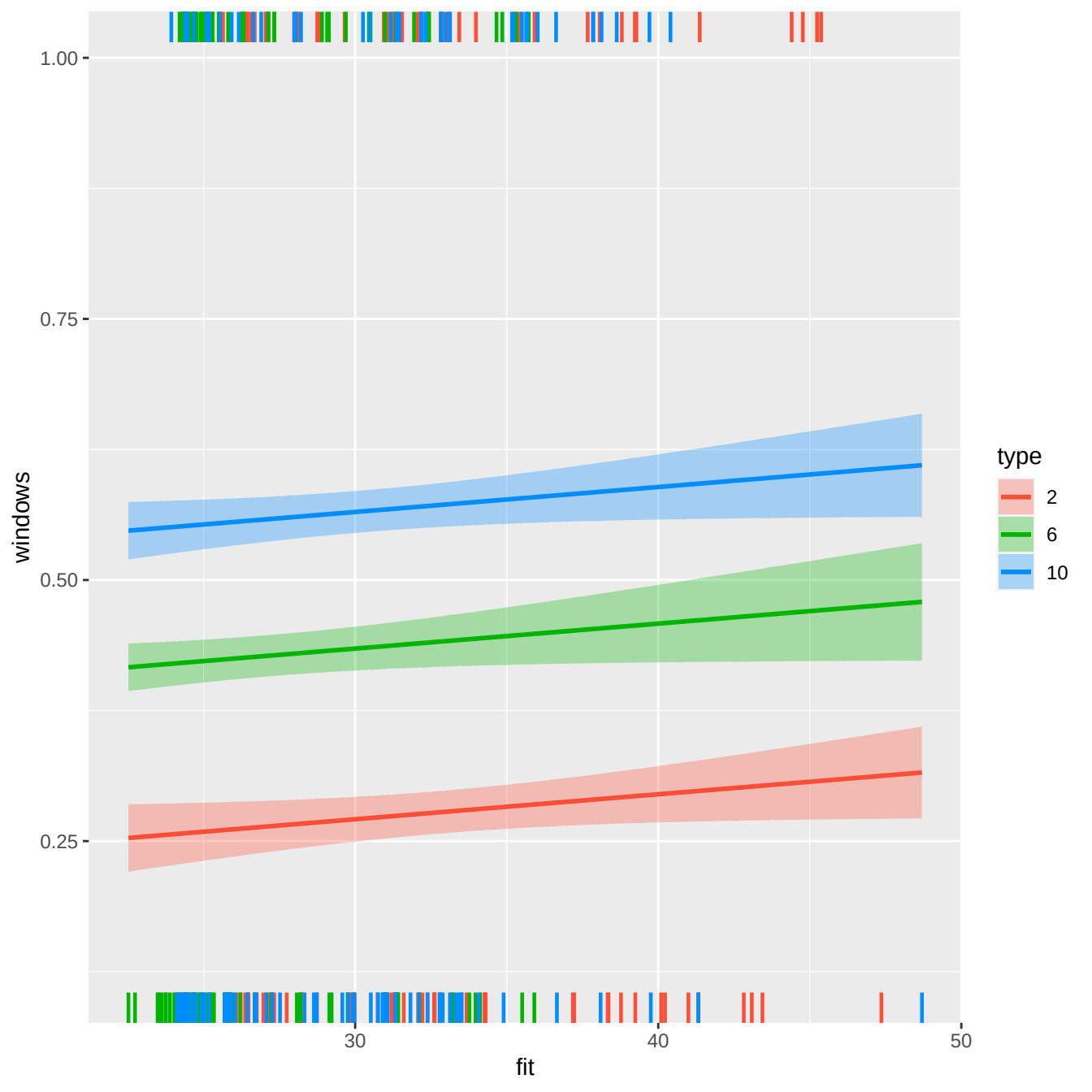

ADH_Traction_between_neighbors_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

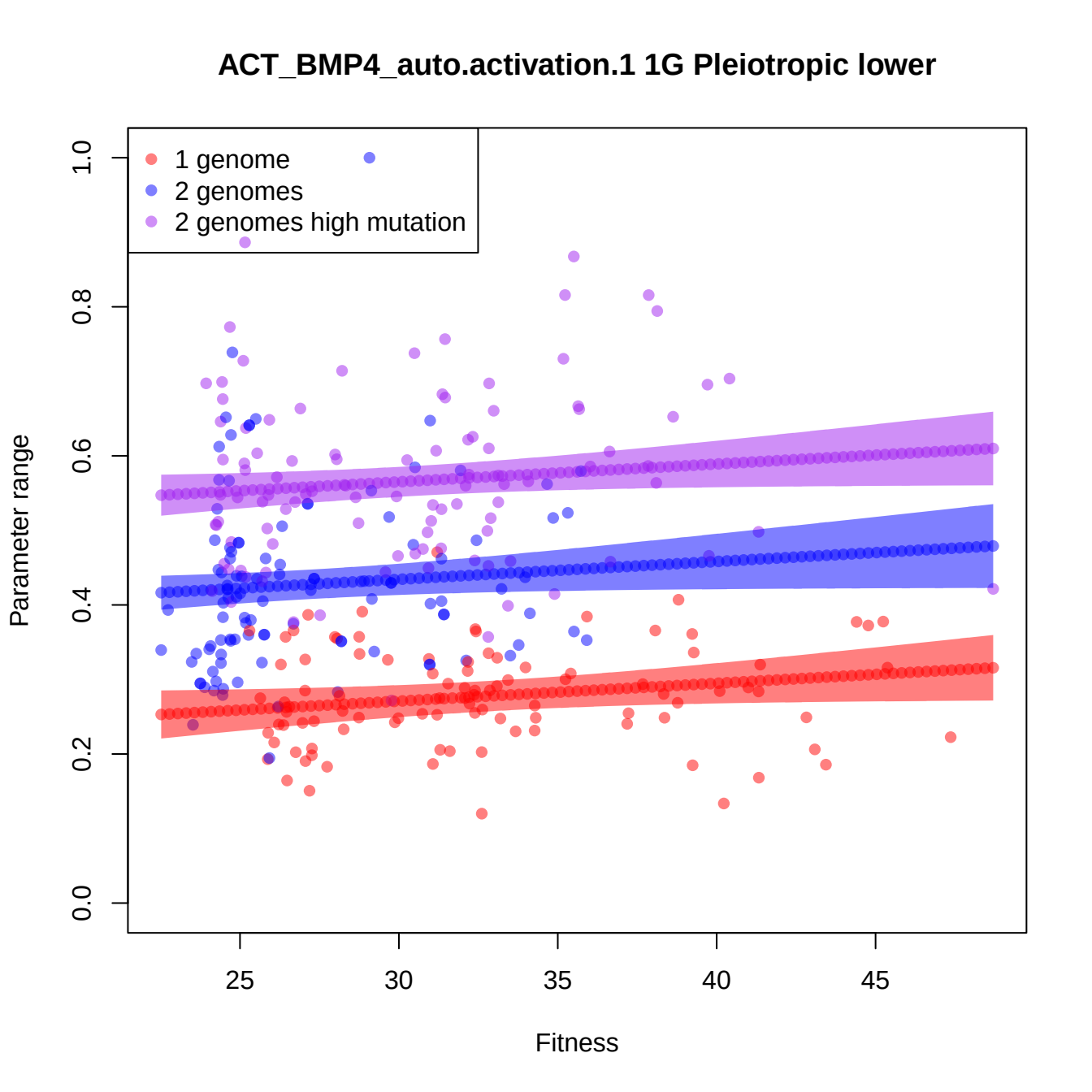

ADH_Traction_between_neighbors_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.037580486388121
3.69453179402345e−05
0.0804839083924924
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

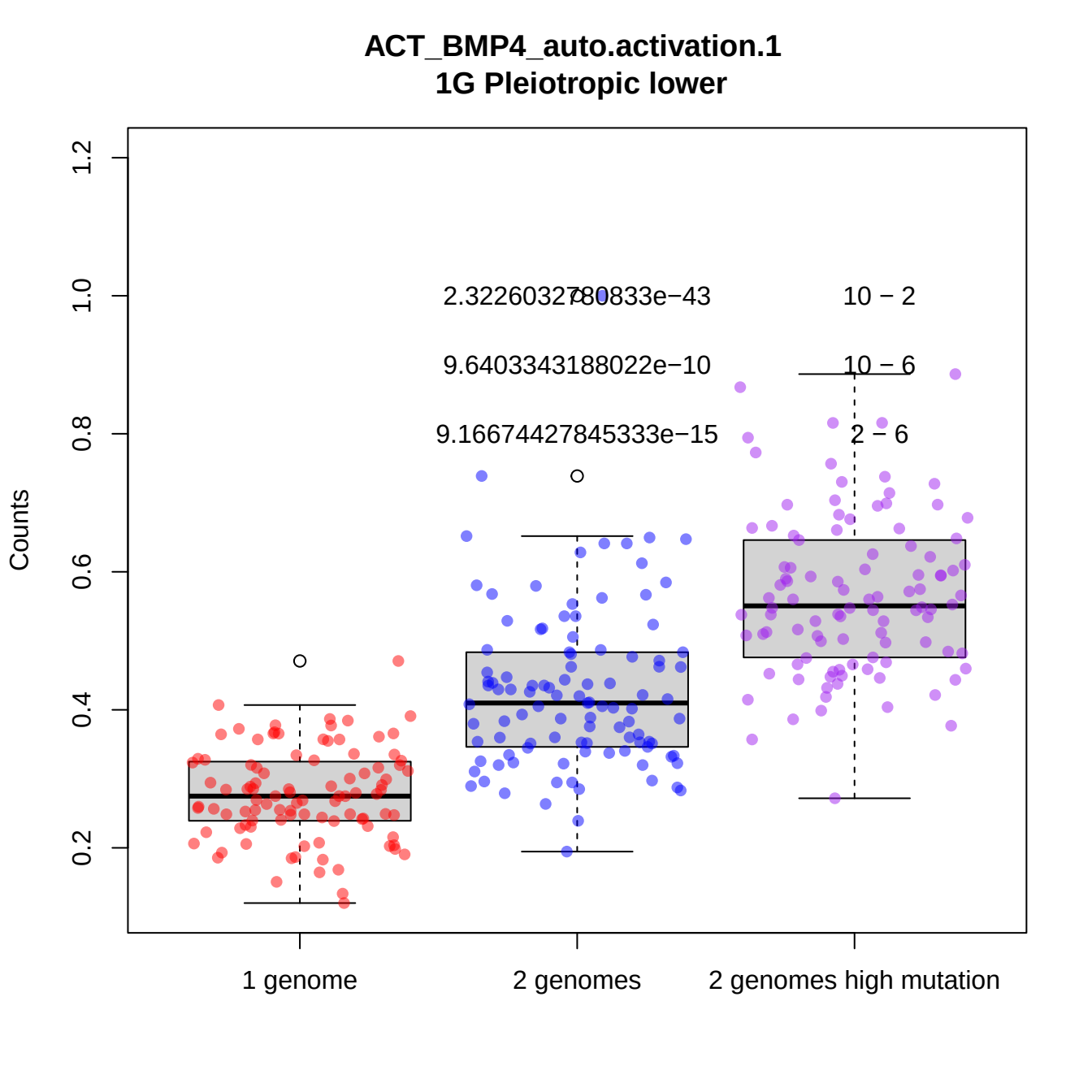

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

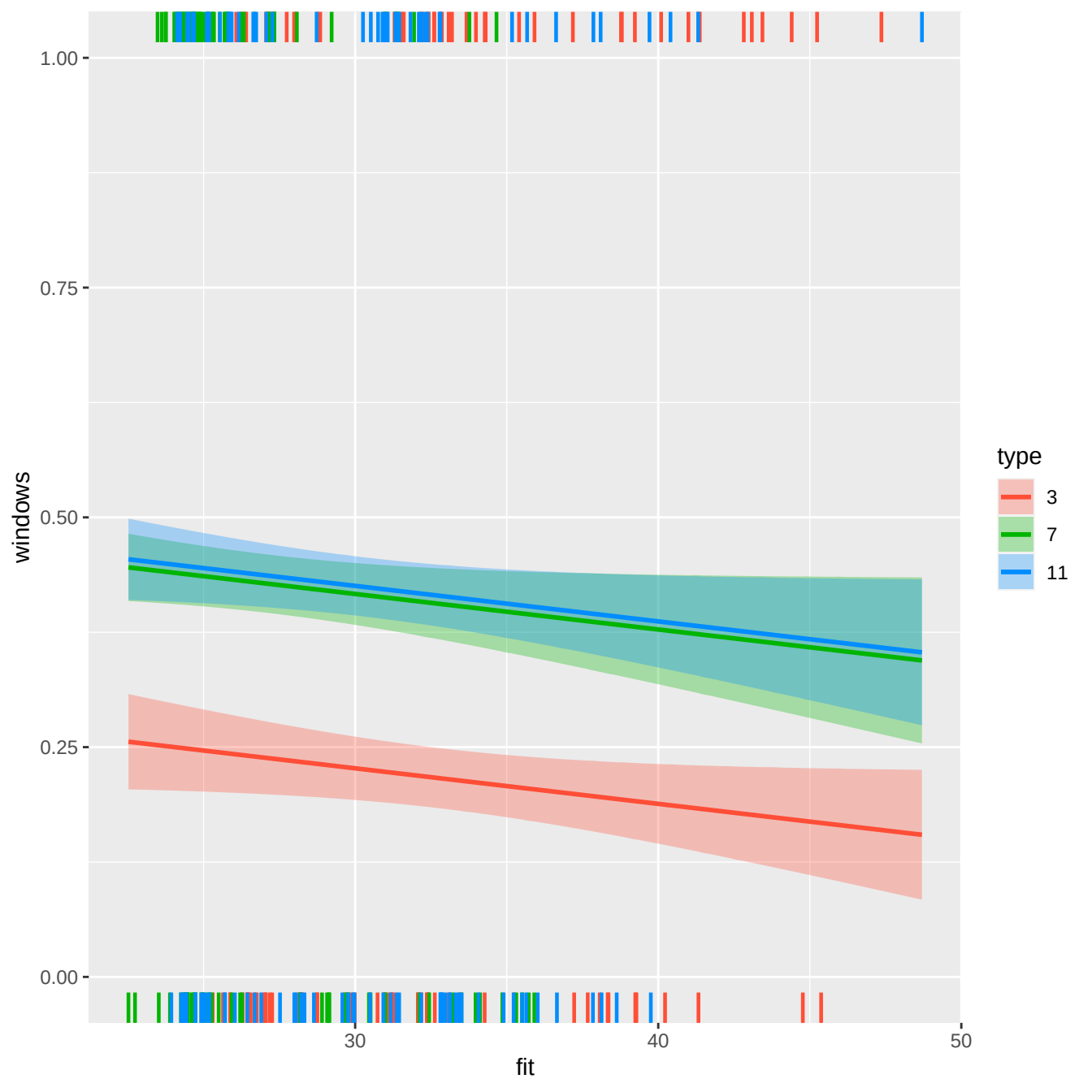

ACT_BMP4_auto.activation.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

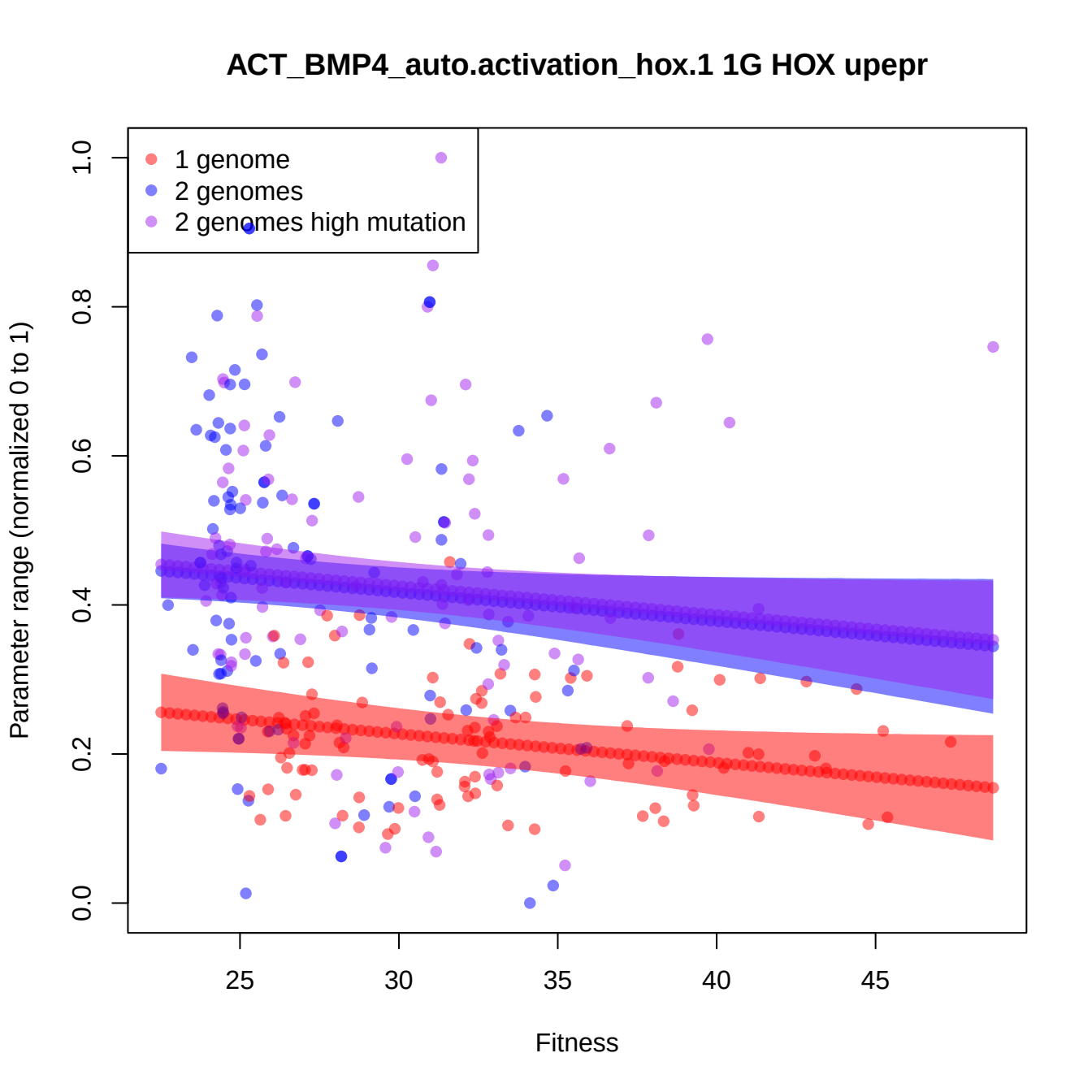

ACT_BMP4_auto.activation.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.22379894337947e−14
9.72912573719402e−35
2.87876541598975e−06
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

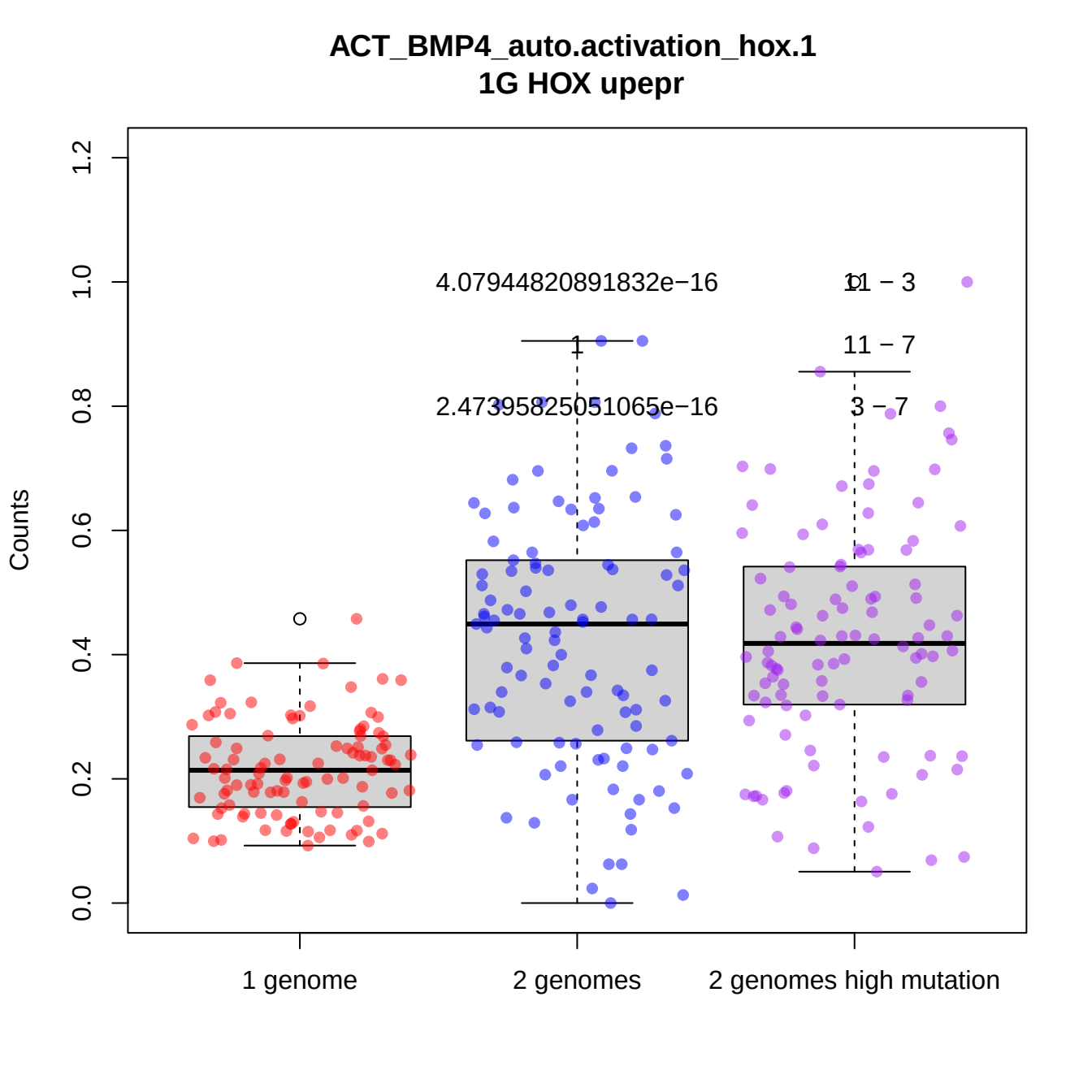

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

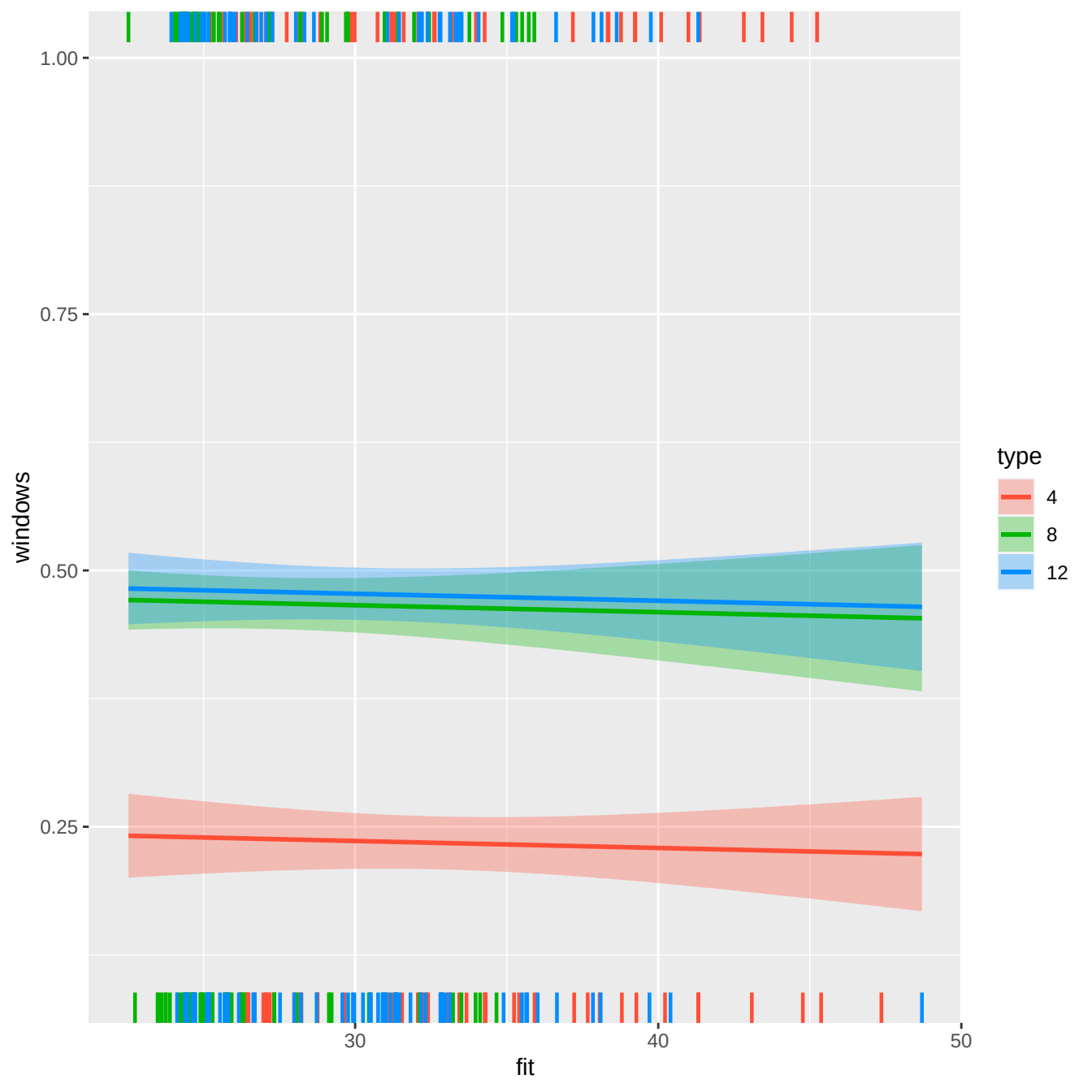

ACT_BMP4_auto.activation.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

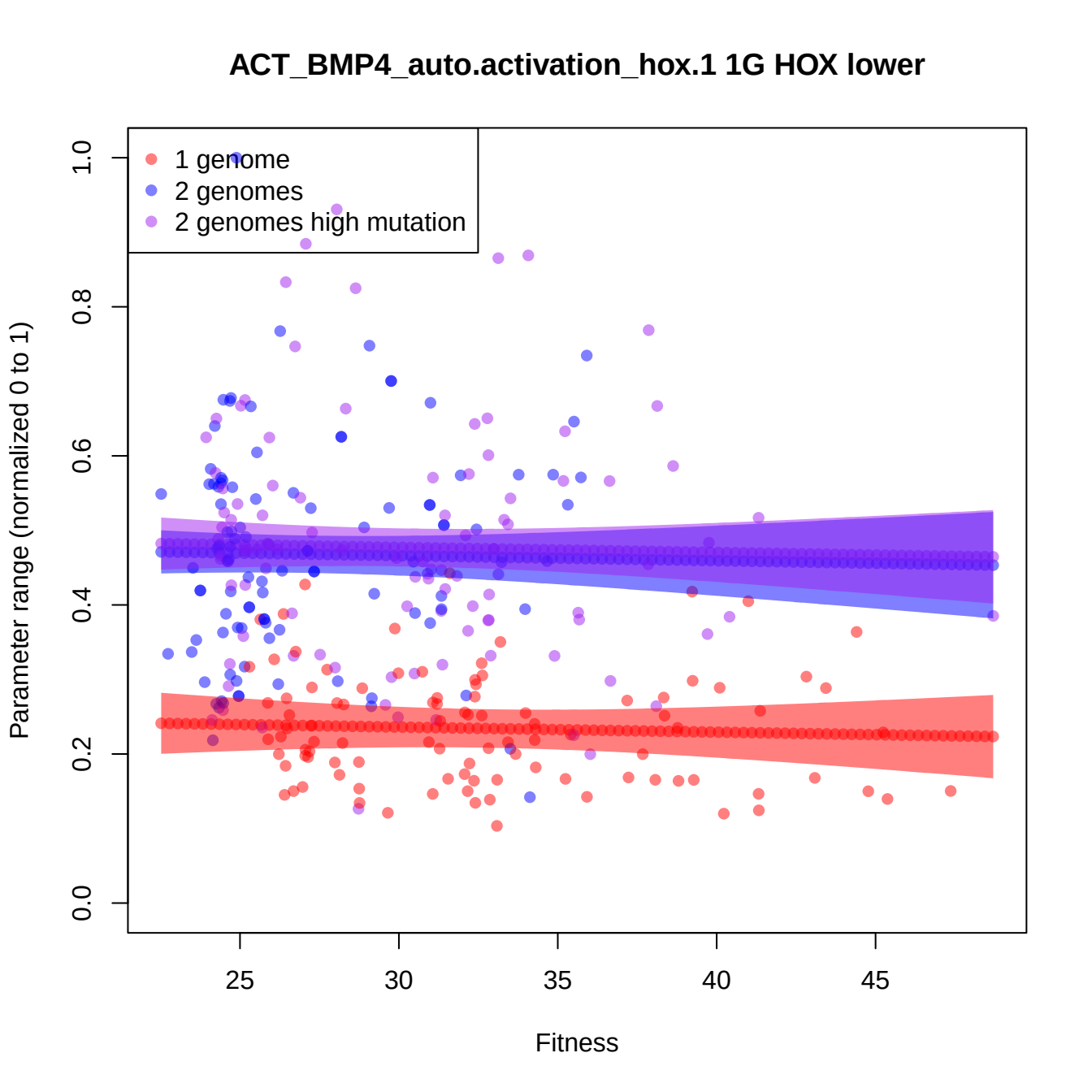

ACT_BMP4_auto.activation.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.3226032780833e−43
9.6403343188022e−10
9.16674427845333e−15
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

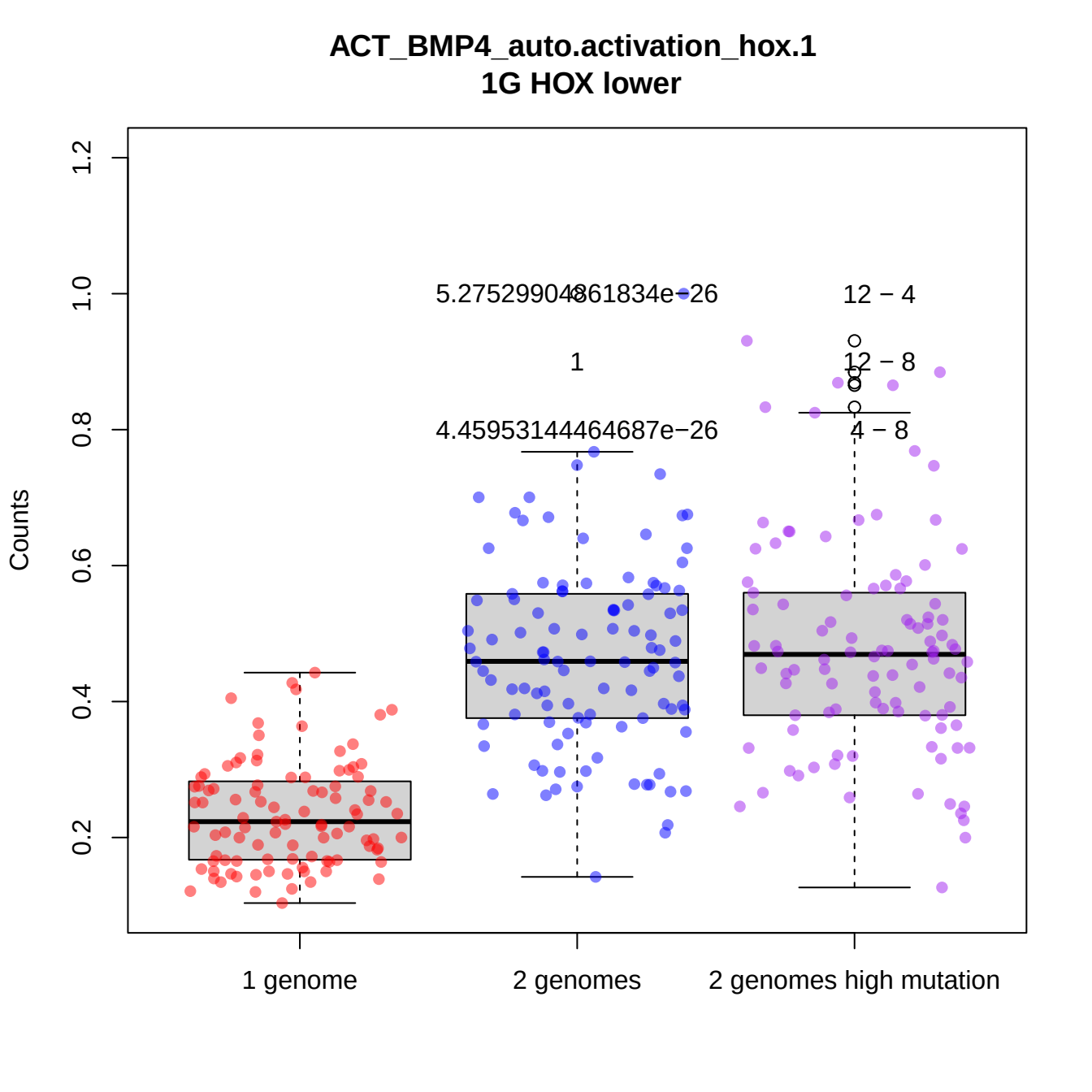

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

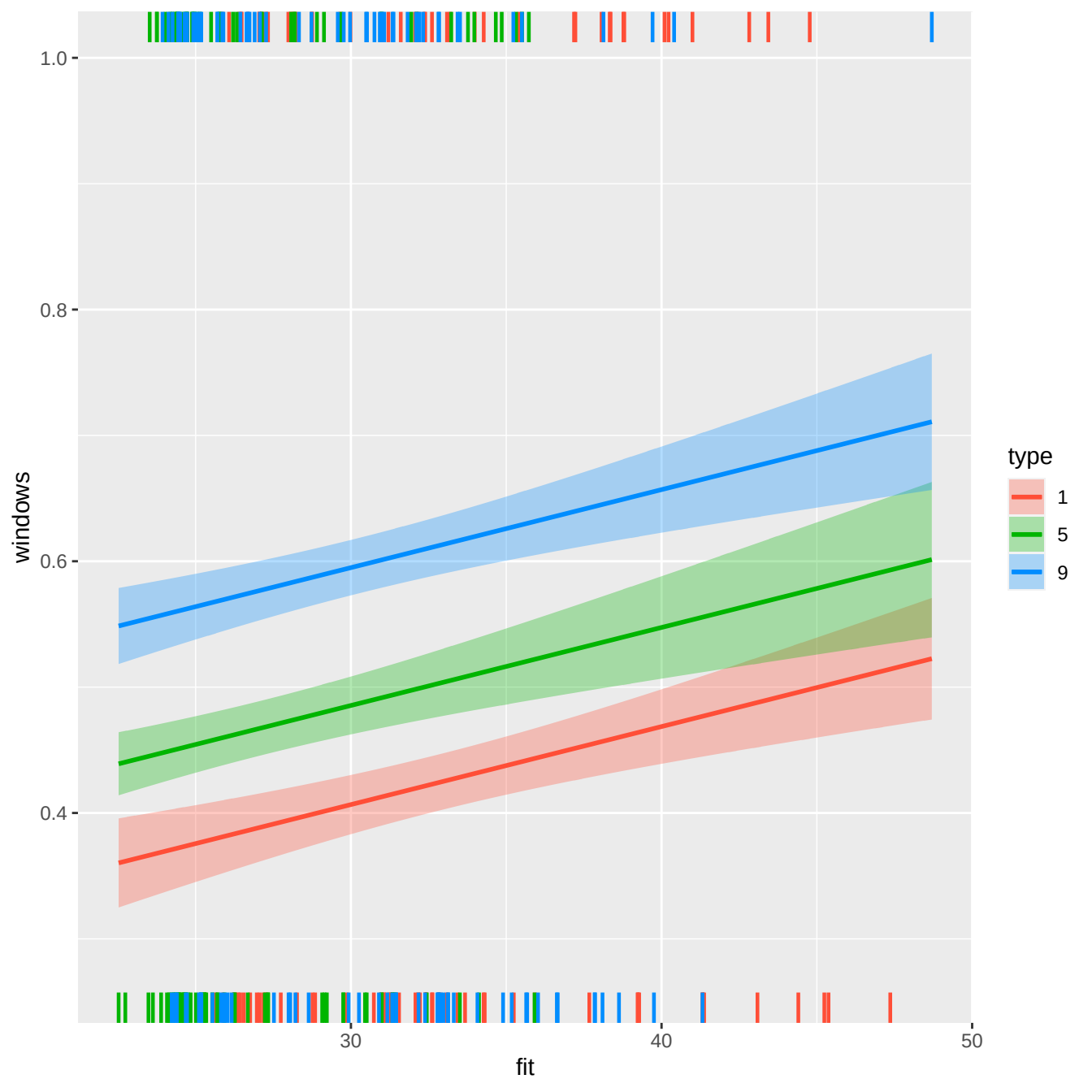

ACT_BMP4_auto.activation_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

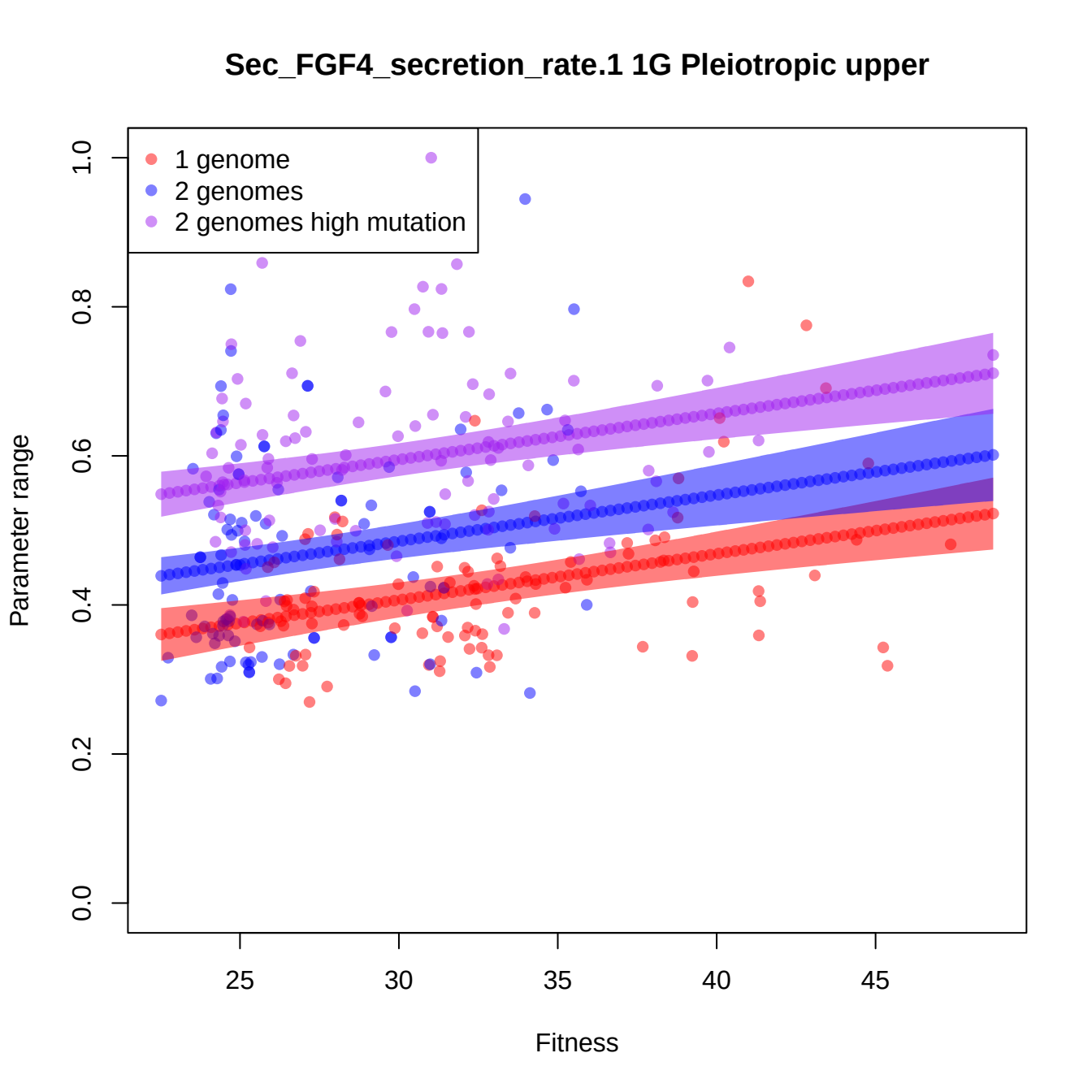

ACT_BMP4_auto.activation_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
4.07944820891832e−16
1
2.47395825051065e−16
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

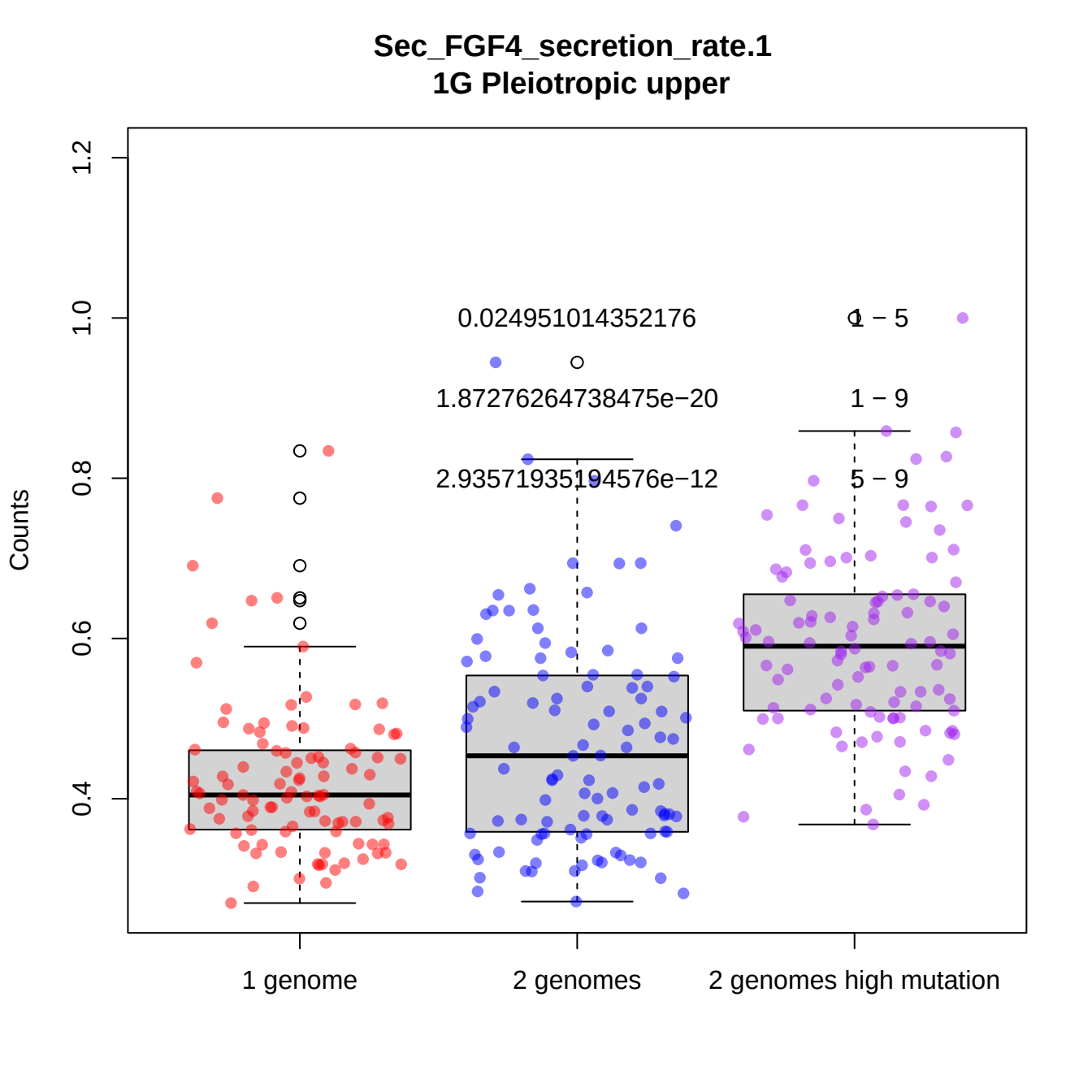

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
4
8
12

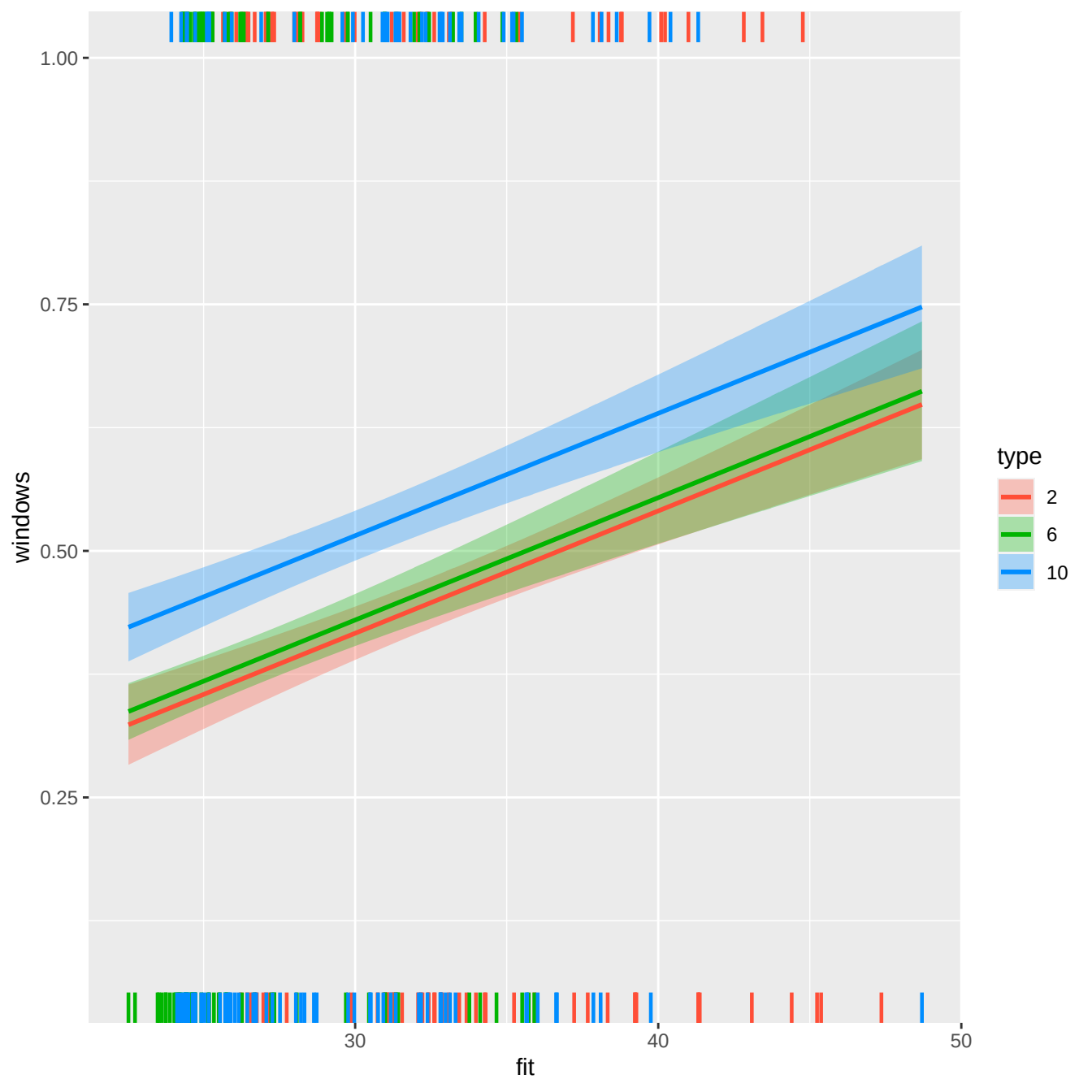

ACT_BMP4_auto.activation_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

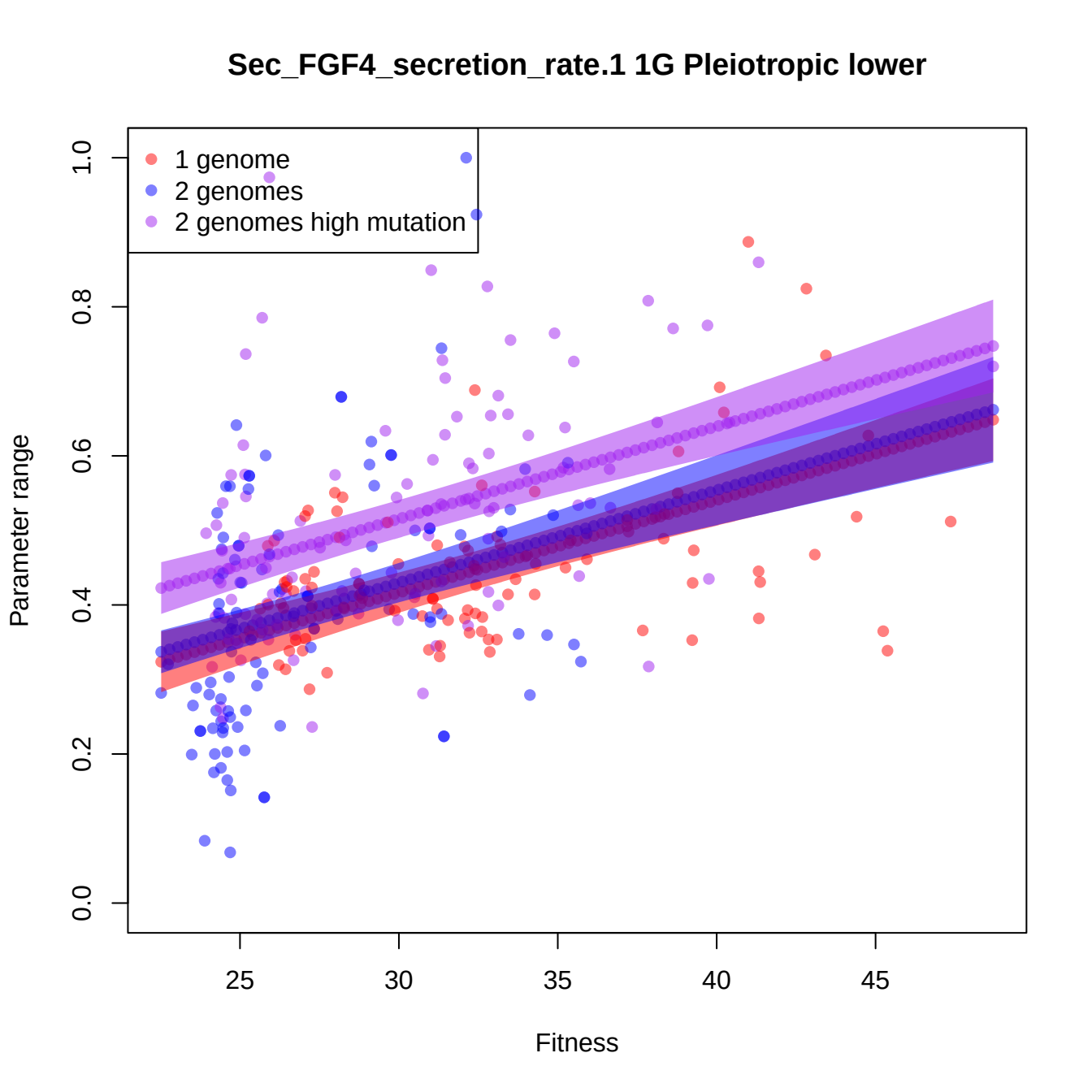

ACT_BMP4_auto.activation_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
5.27529904861834e−26
1
4.45953144464687e−26
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

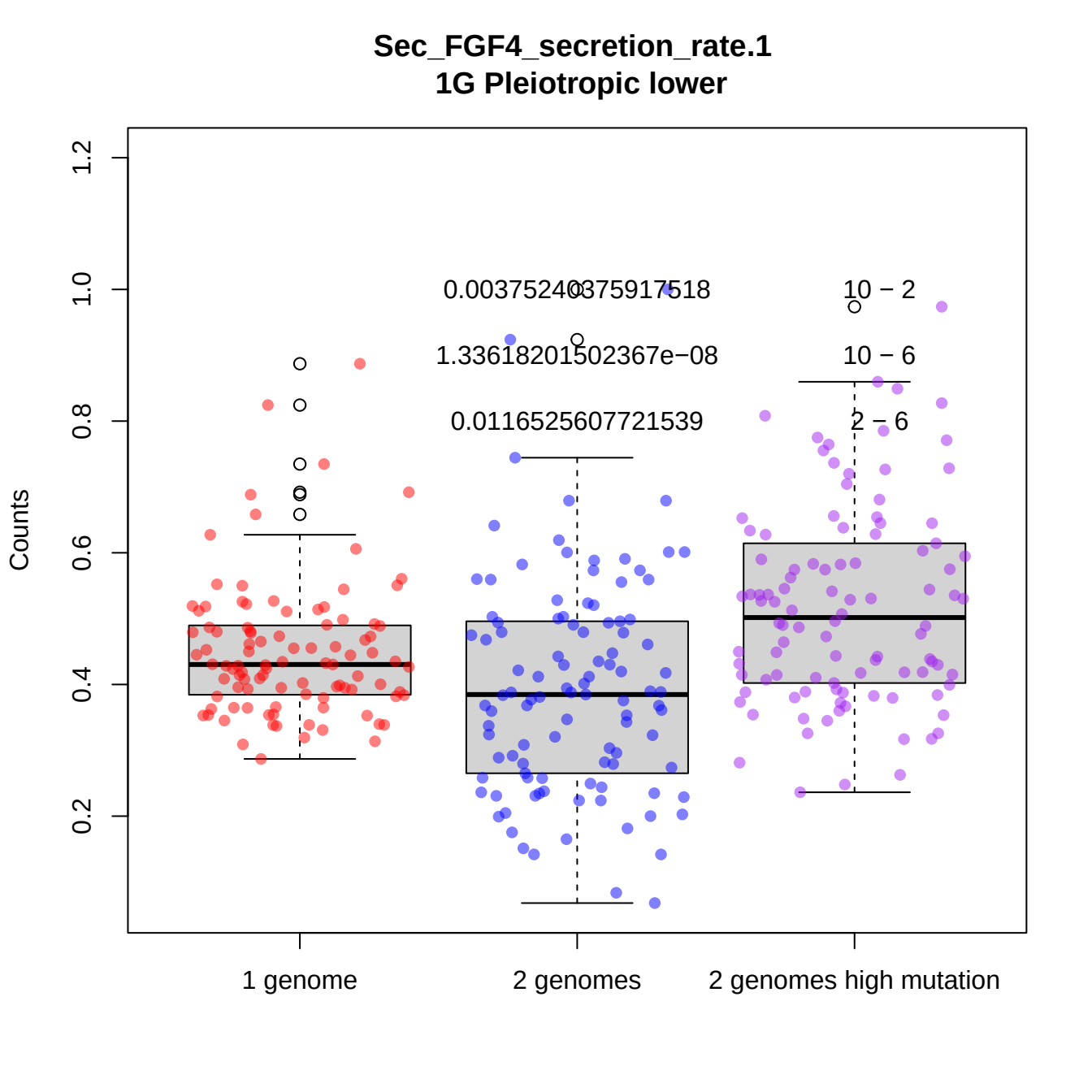

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
1
5
9

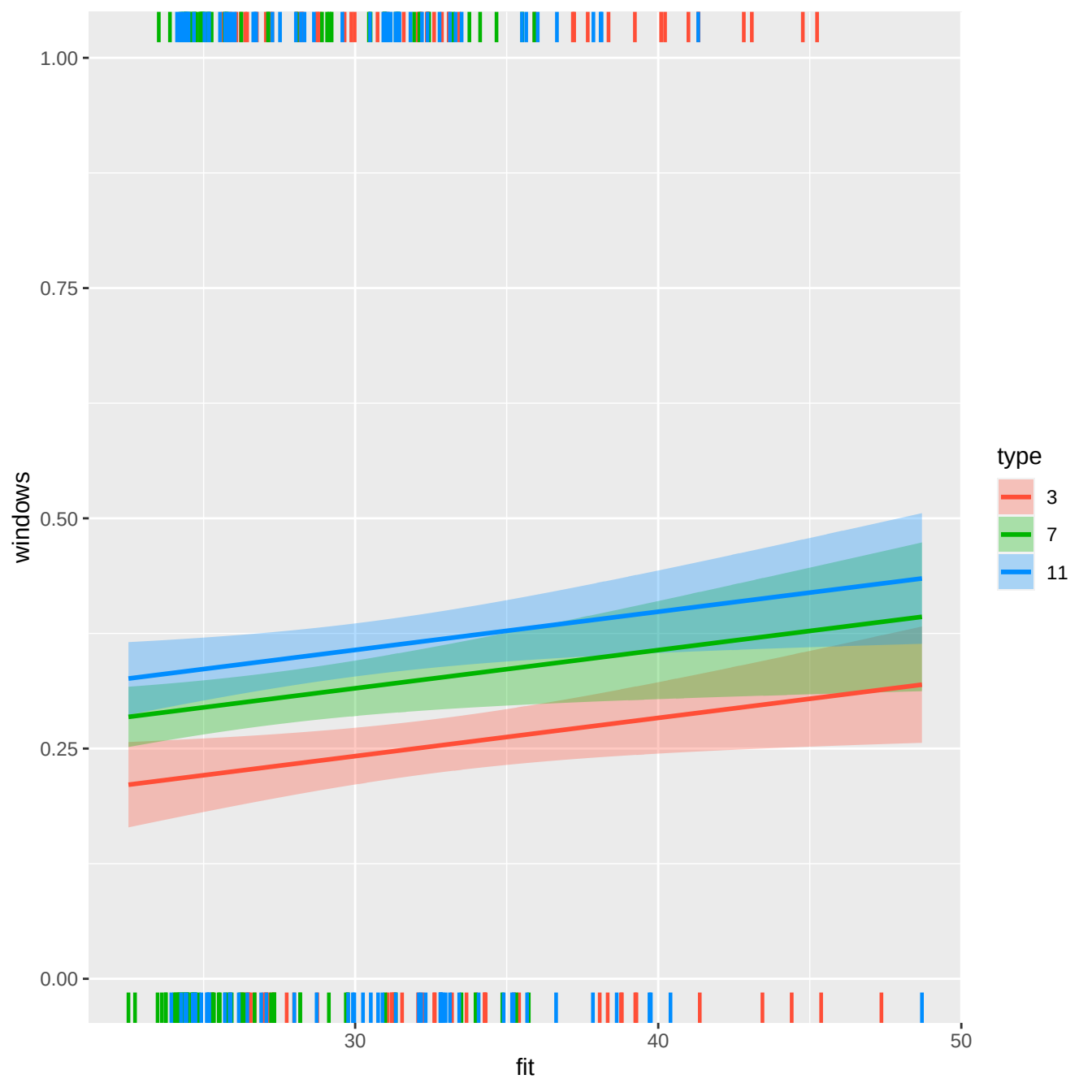

Sec_FGF4_secretion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

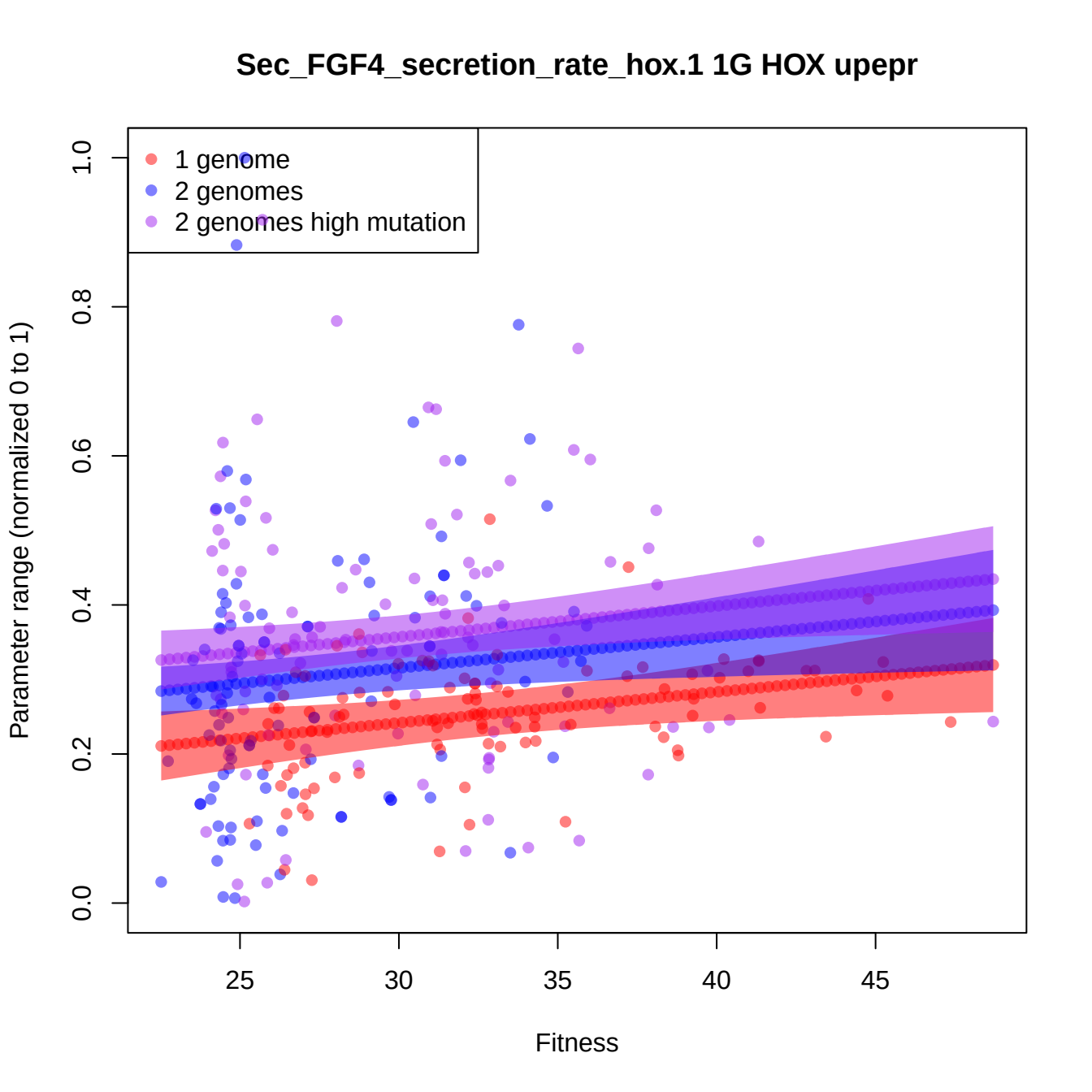

Sec_FGF4_secretion_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.024951014352176
1.87276264738475e−20
2.93571935194576e−12
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

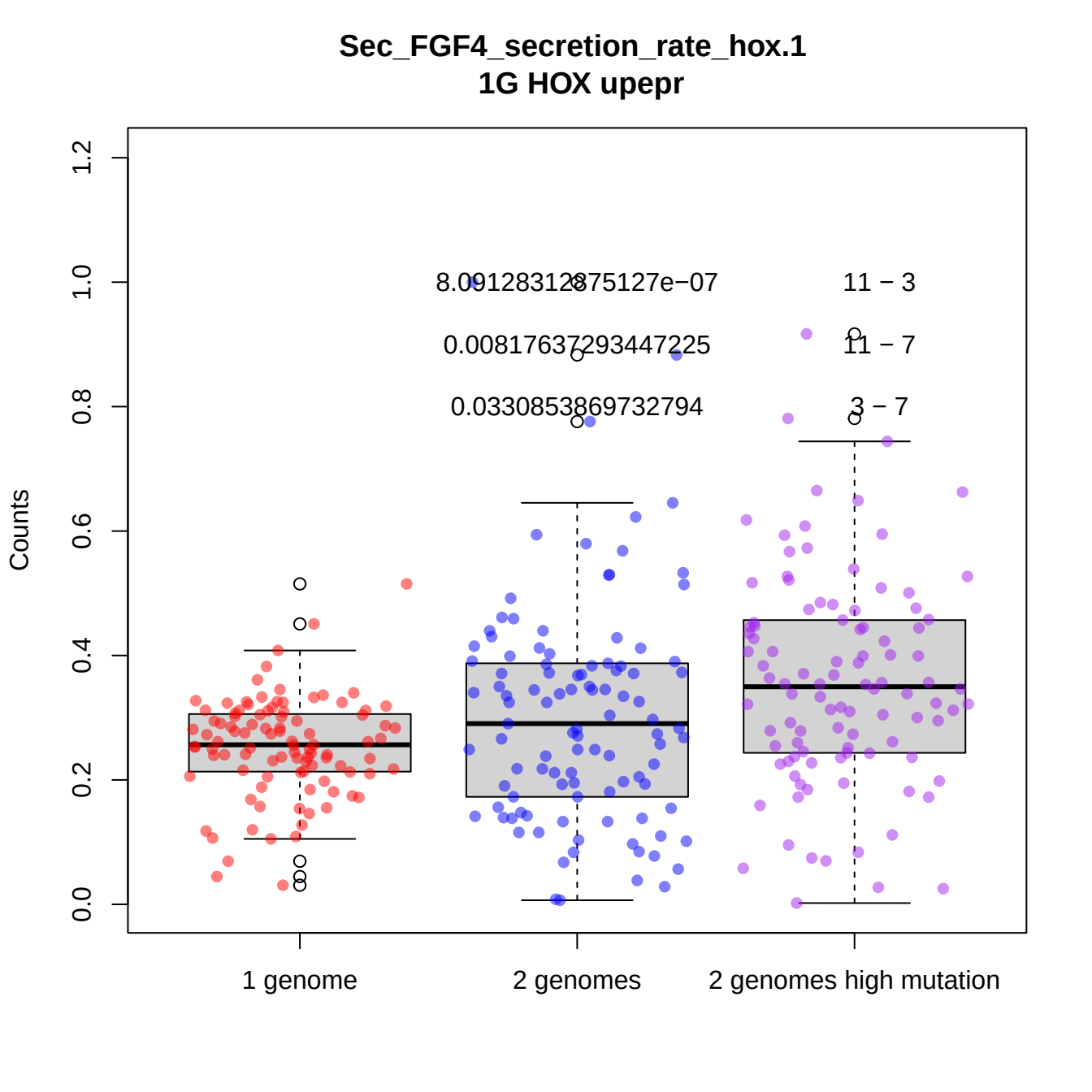

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

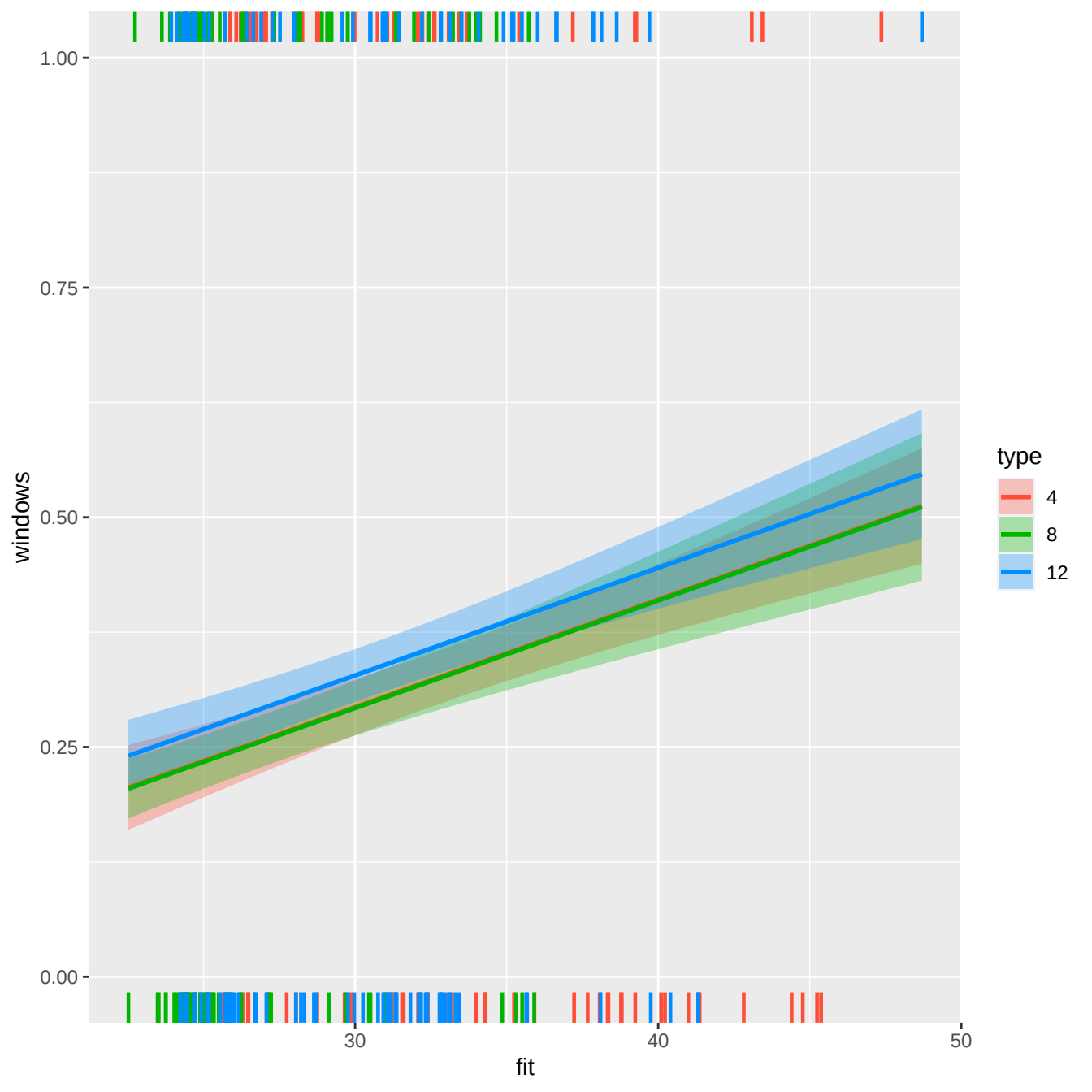

Sec_FGF4_secretion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

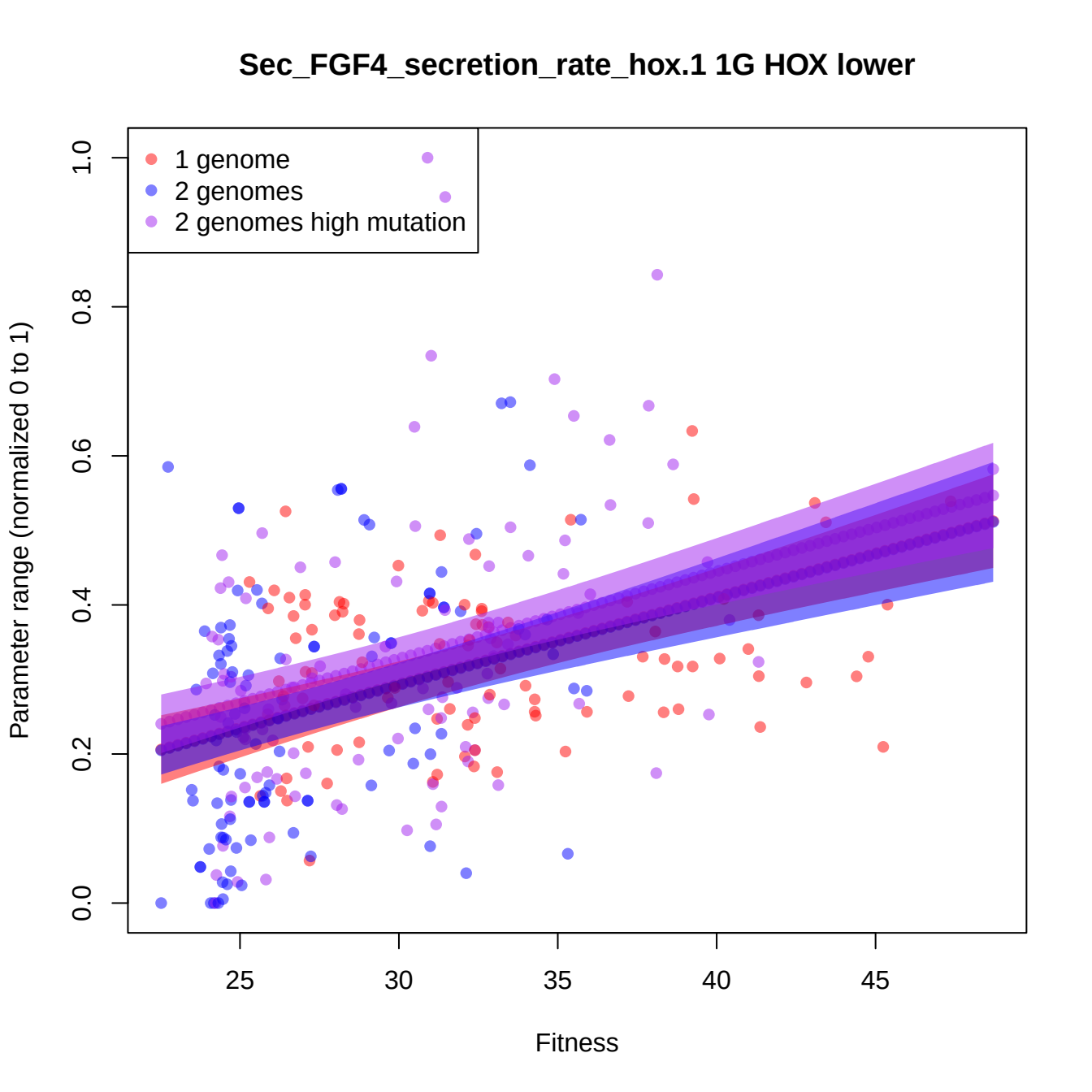

Sec_FGF4_secretion_rate.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0037524037591751 8
1.33618201502367e−08
0.0116525607721539
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

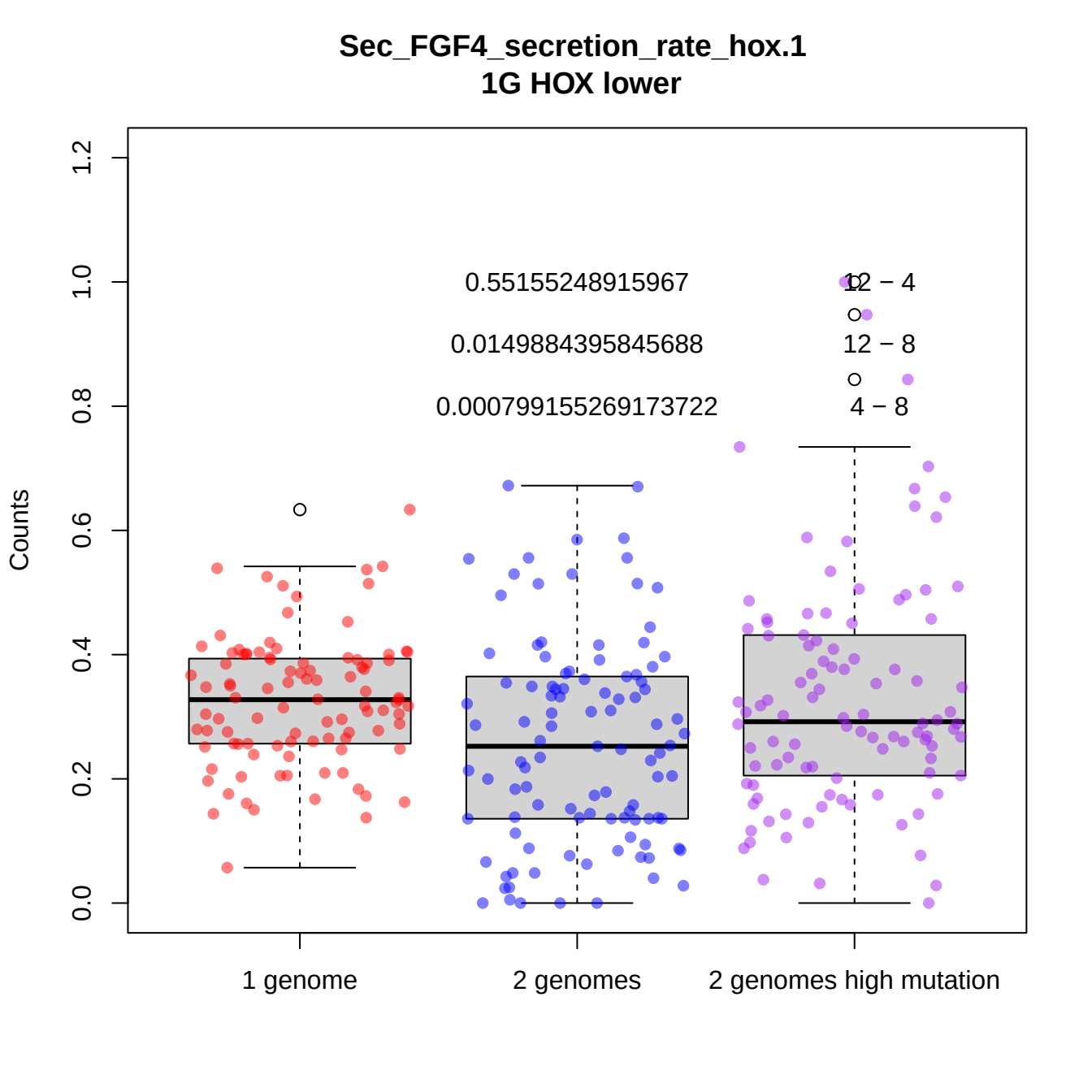

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

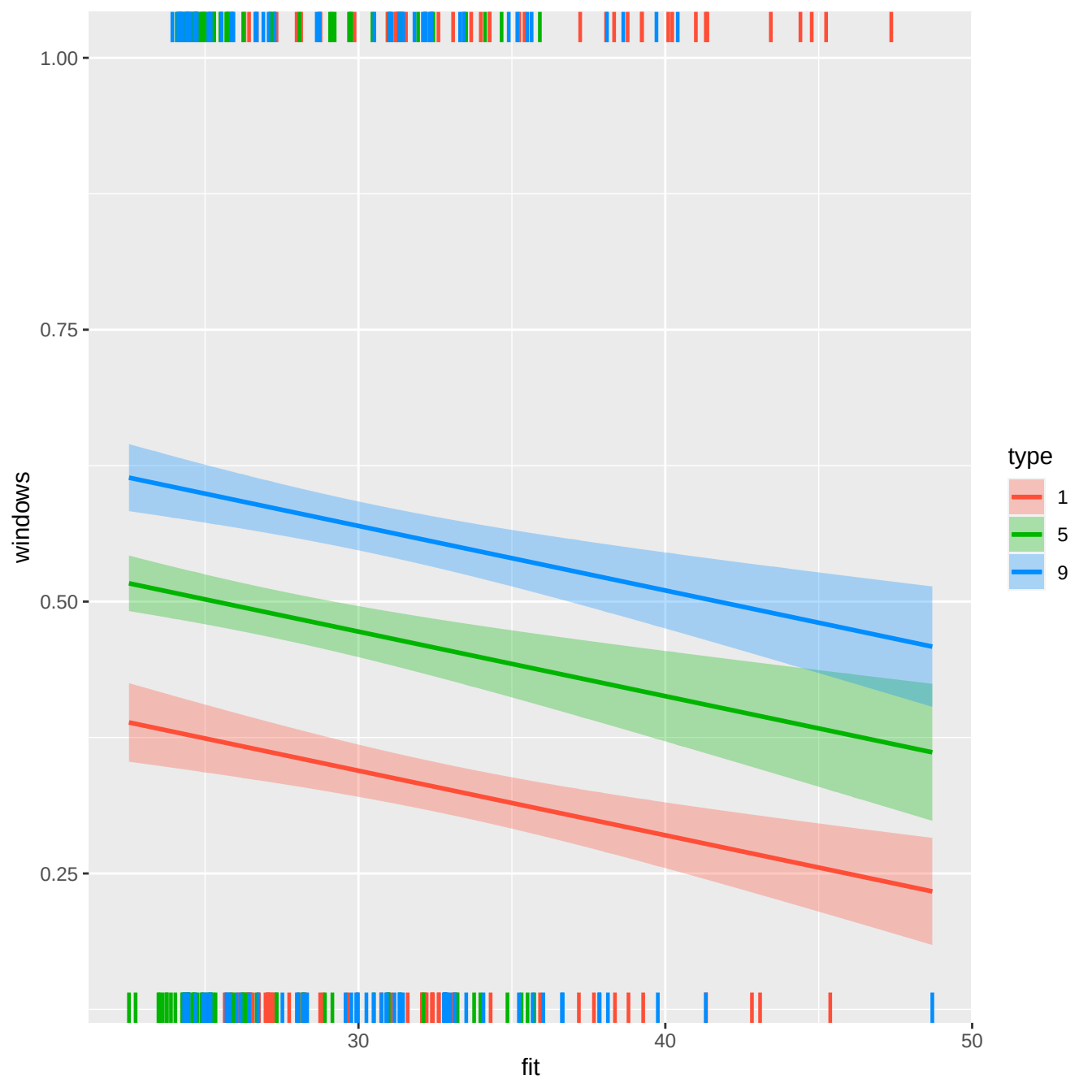

Sec_FGF4_secretion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

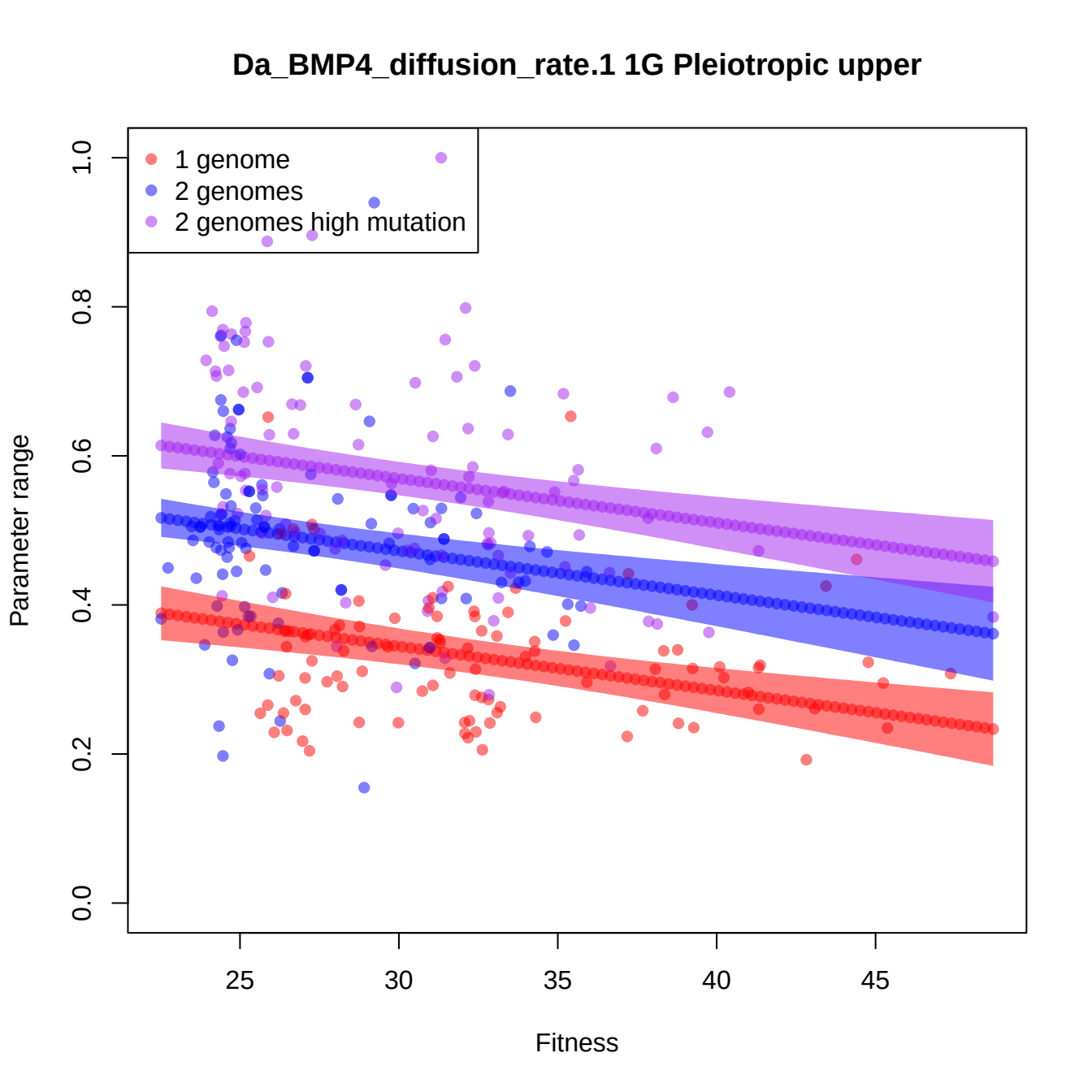

Sec_FGF4_secretion_rate_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
8.091283128751 27e−07
0.008176372934472 25
0.033085386973279 4
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

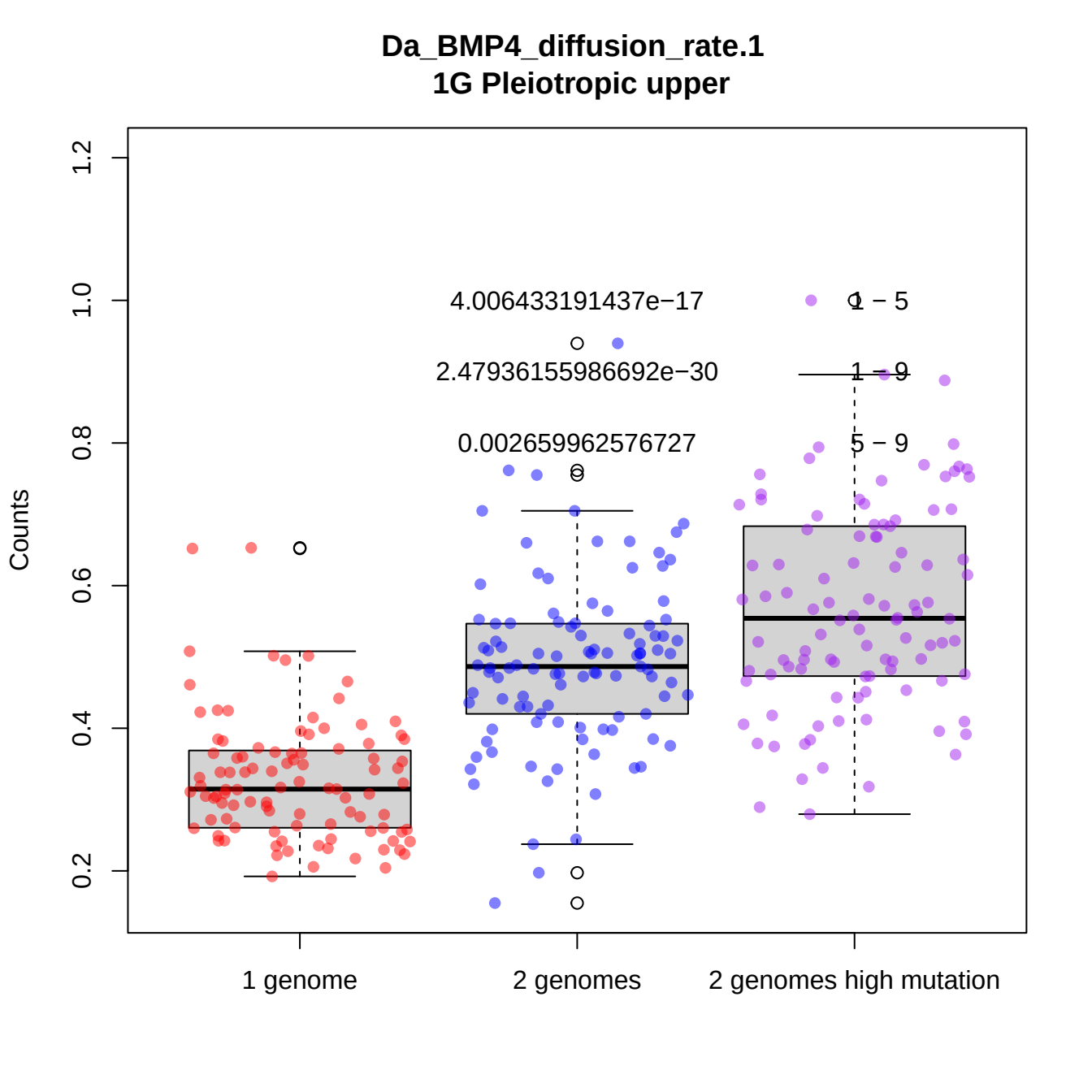

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

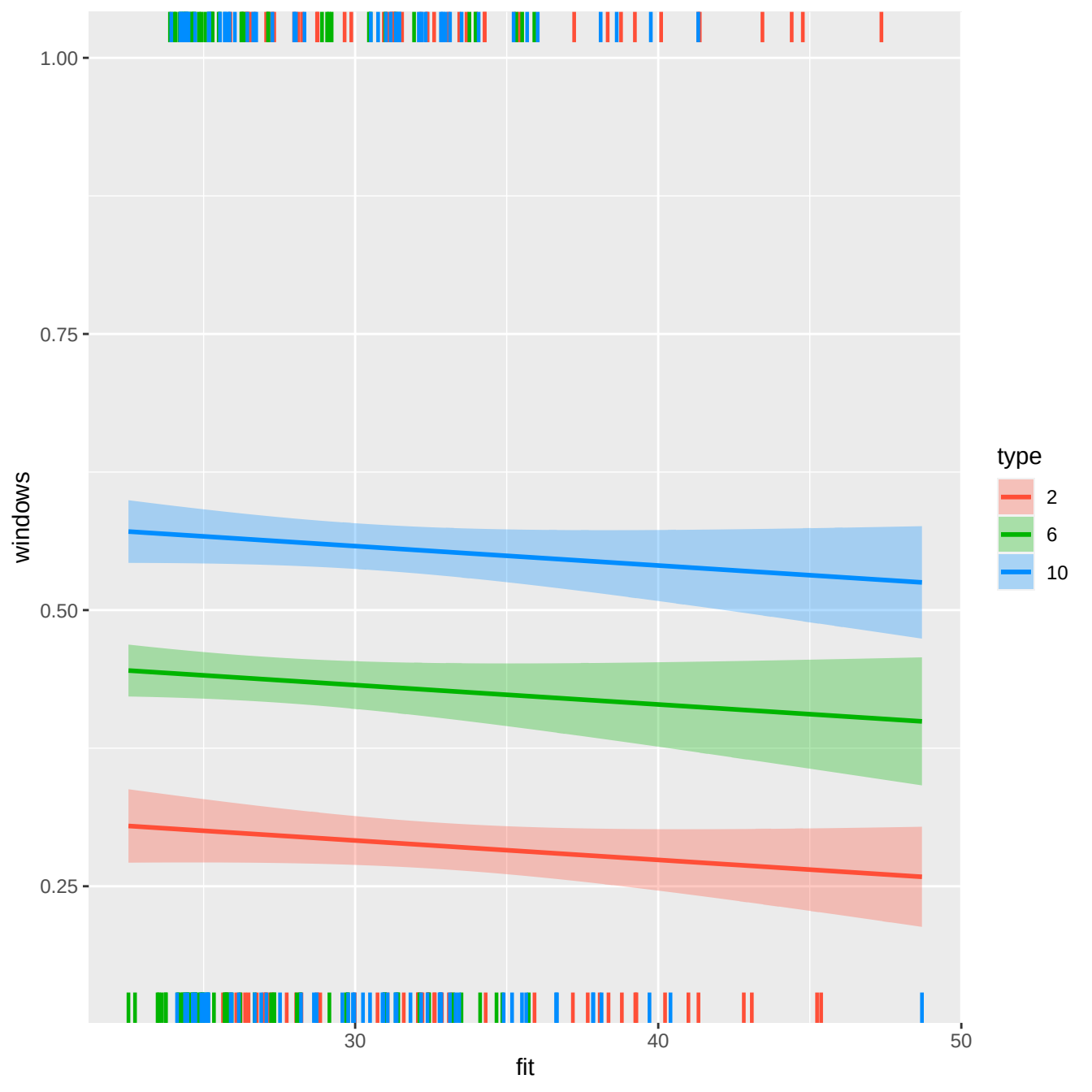

Sec_FGF4_secretion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

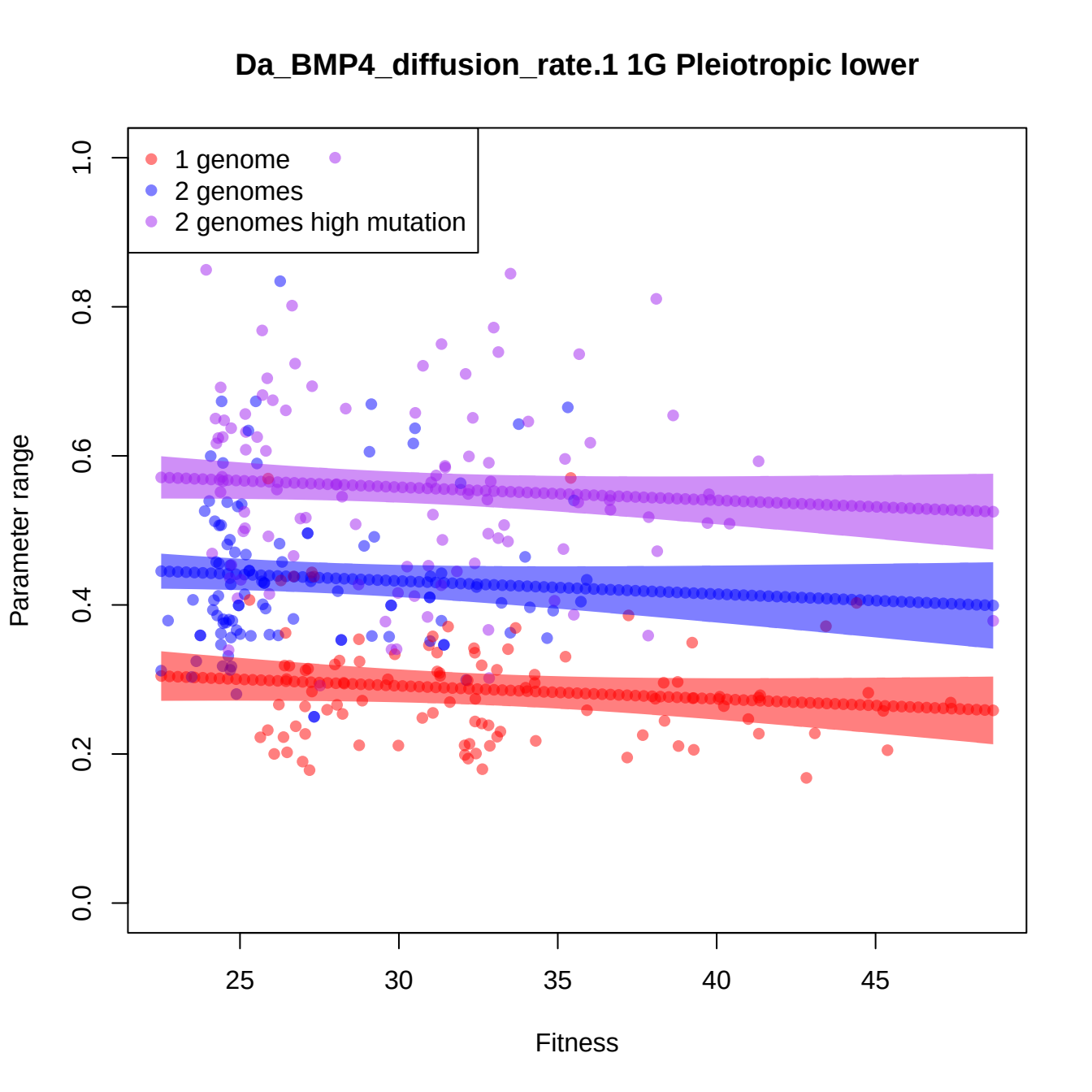

Sec_FGF4_secretion_rate_hox.1
1G HOX lower
0.55155248915967
0.0149884395845688
0.000799155269173722
12 − 4
12 − 8
4 − 8
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

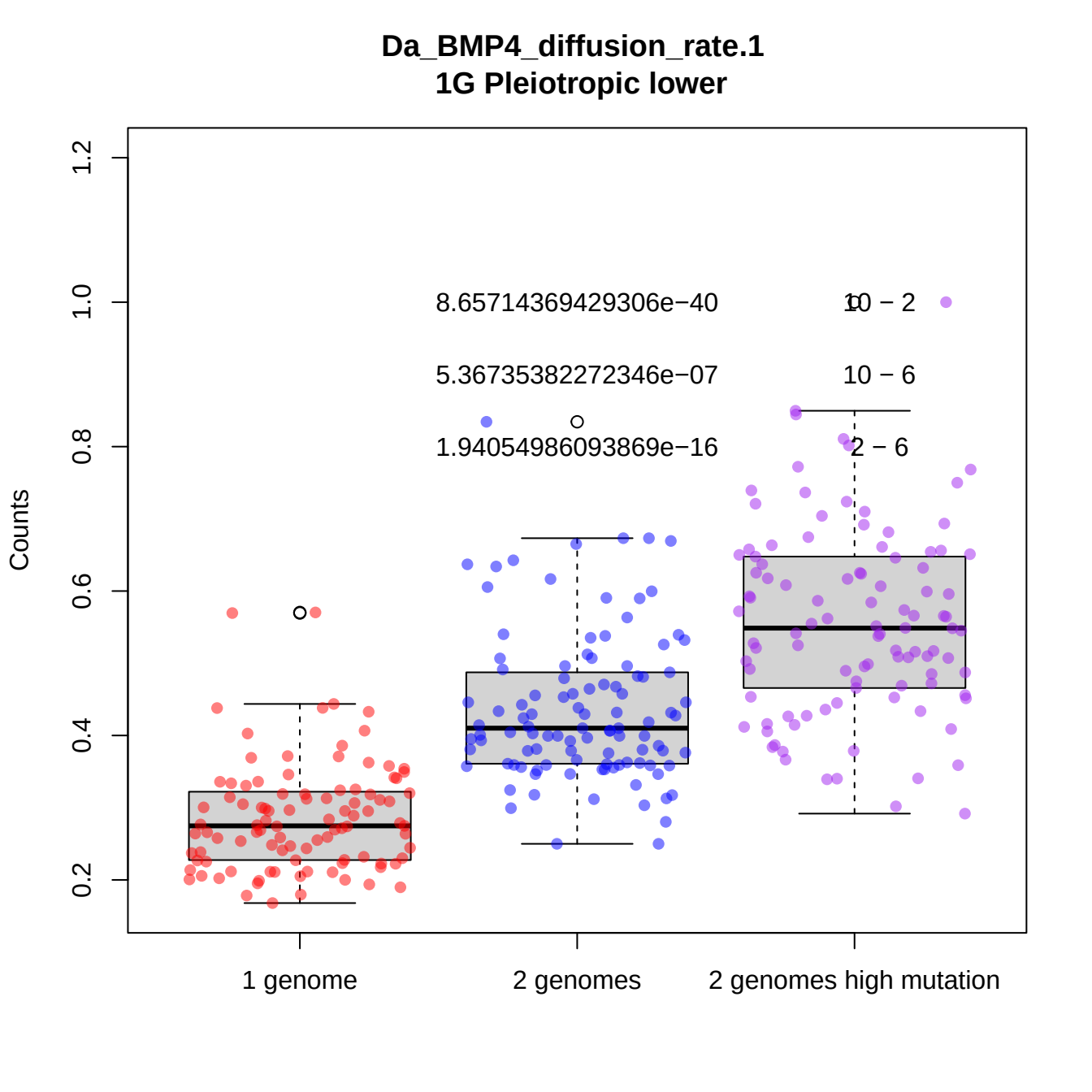

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

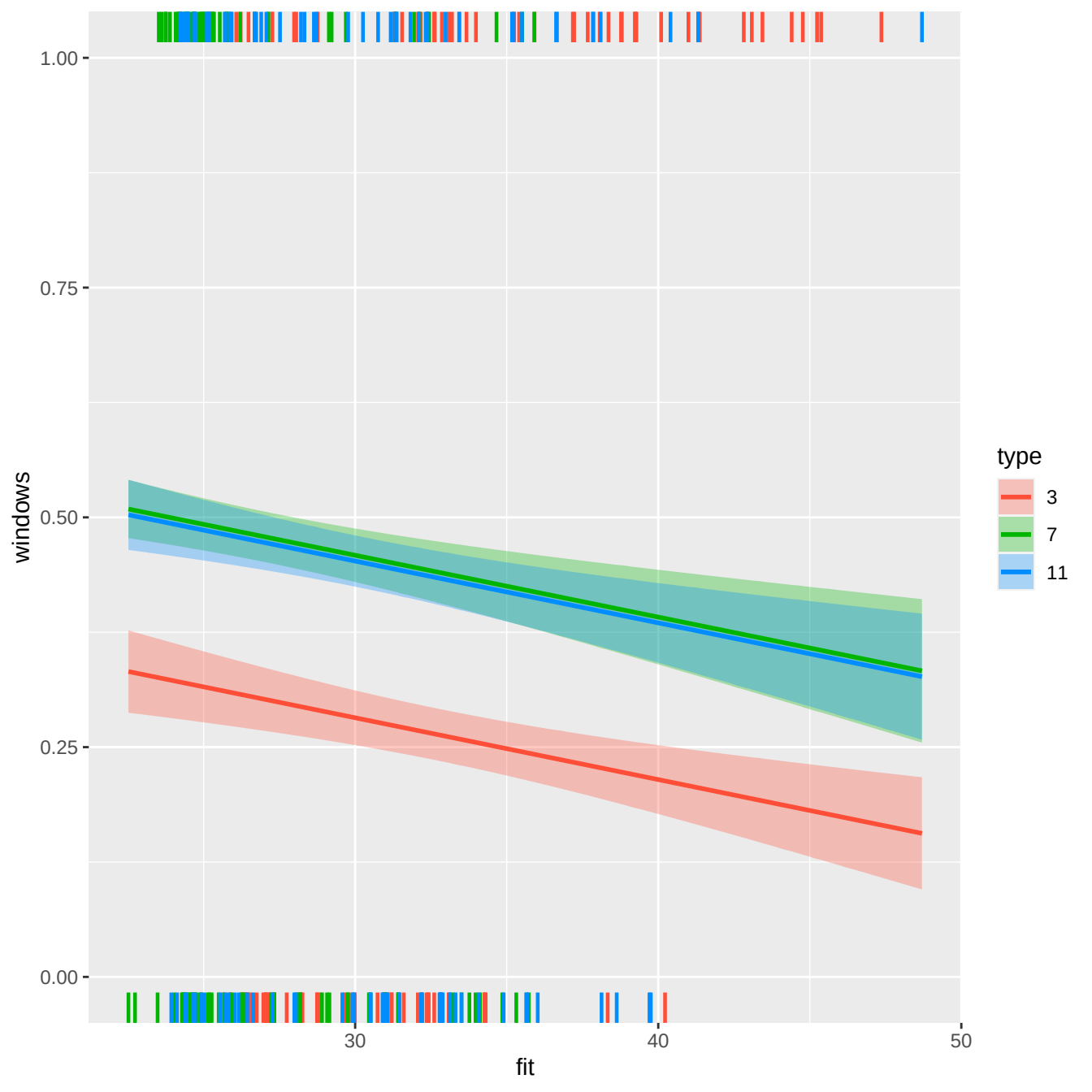

Da_BMP4_diffusion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

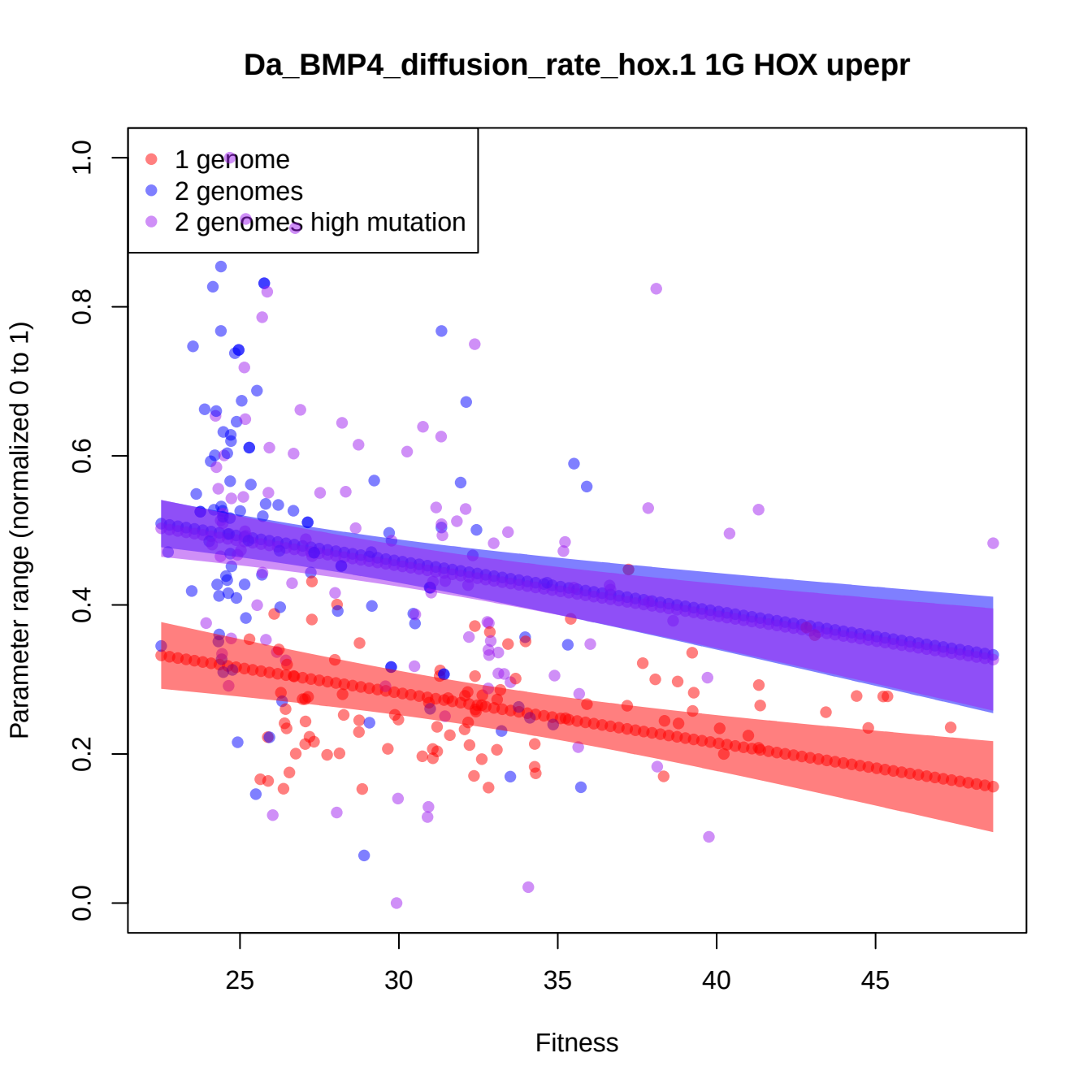

Da_BMP4_diffusion_rate.1
1G Pleiotropic upper
Counts
0.2
0.4
0.6
0.8
1.0
1.2
4.006433191437e−17
2.47936155986692e−30
0.002659962576727
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

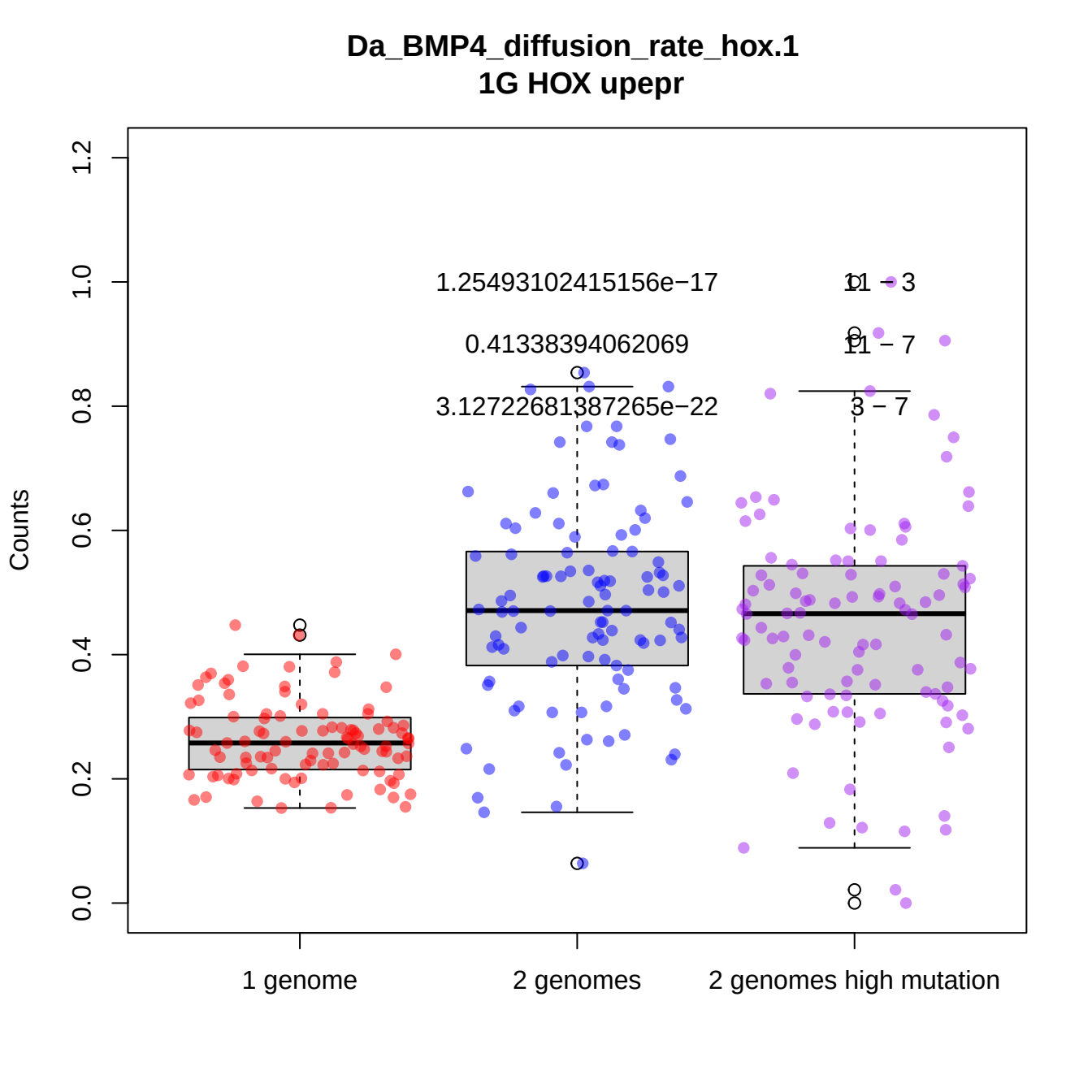

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

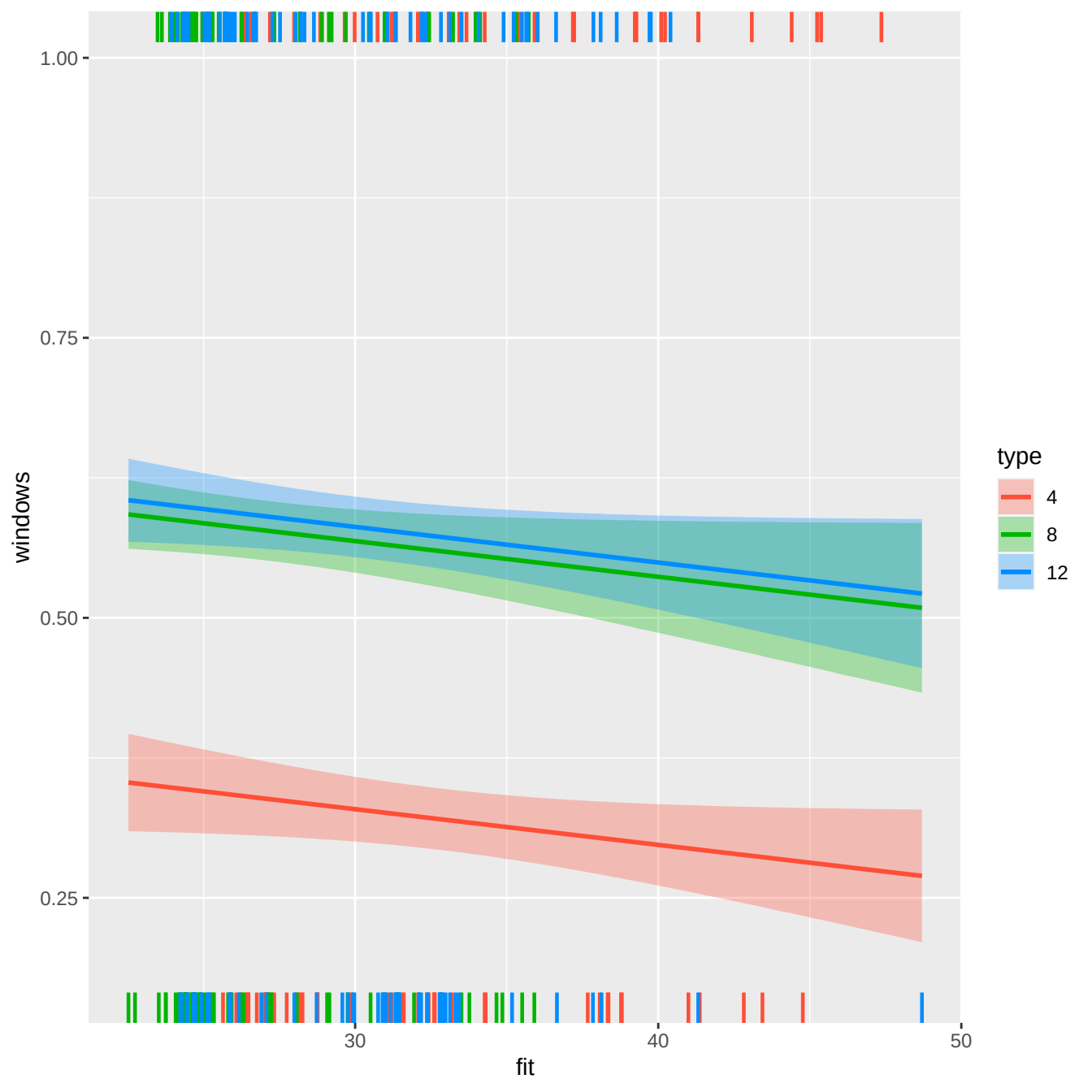

Da_BMP4_diffusion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

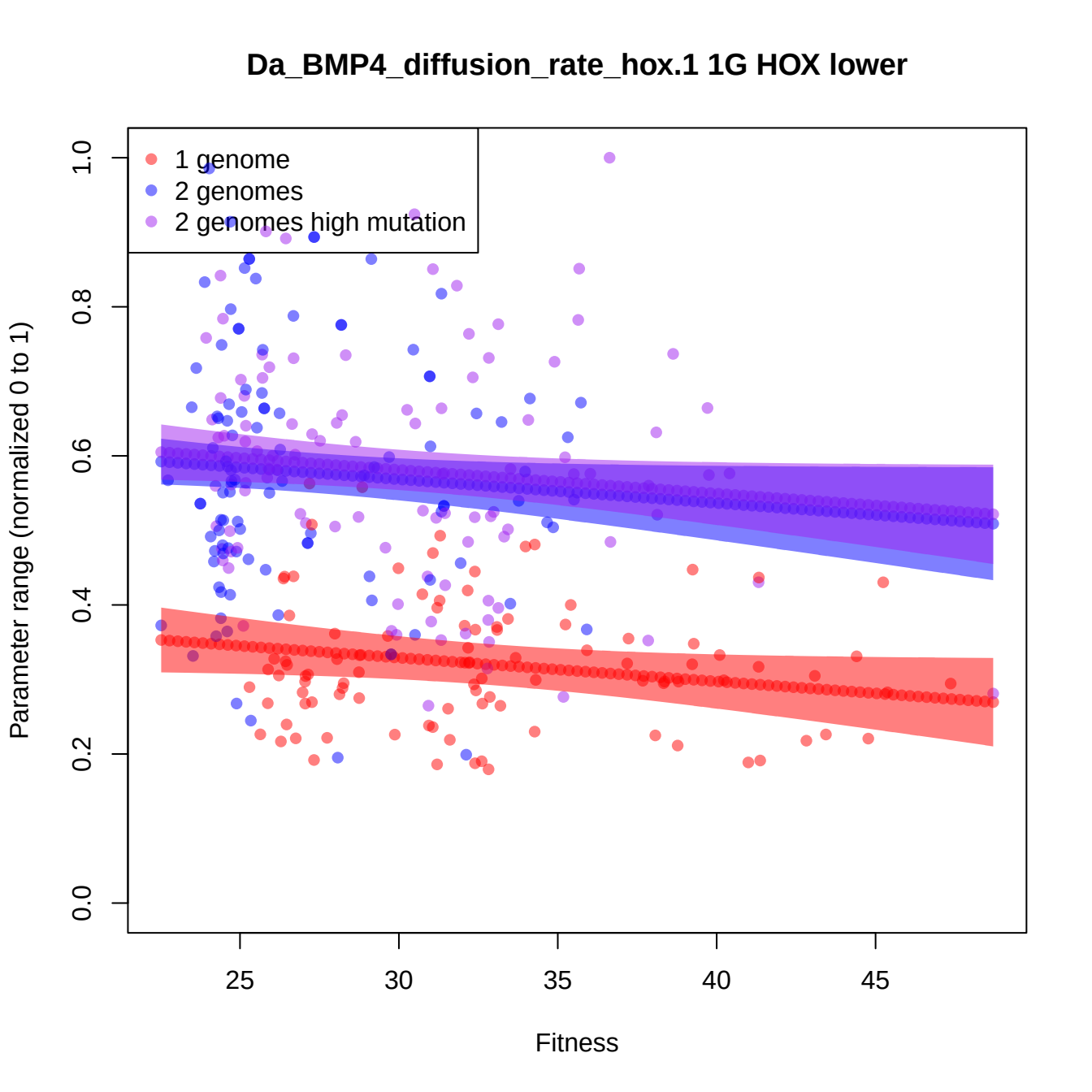

Da_BMP4_diffusion_rate.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
8.65714369429306e−40
5.36735382272346e−07
1.94054986093869e−16
1 genome
2 genomes
2 genomes high mutation

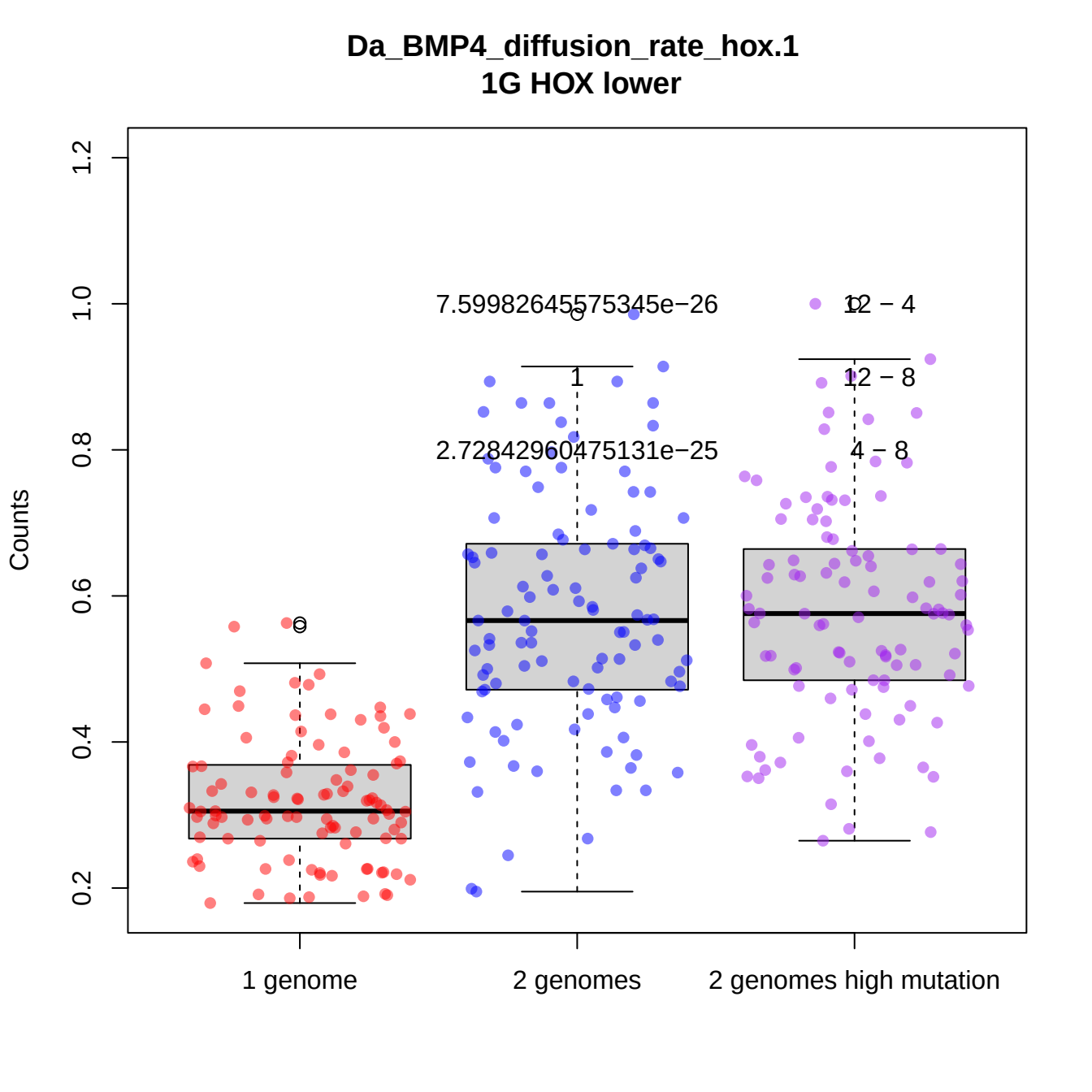

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

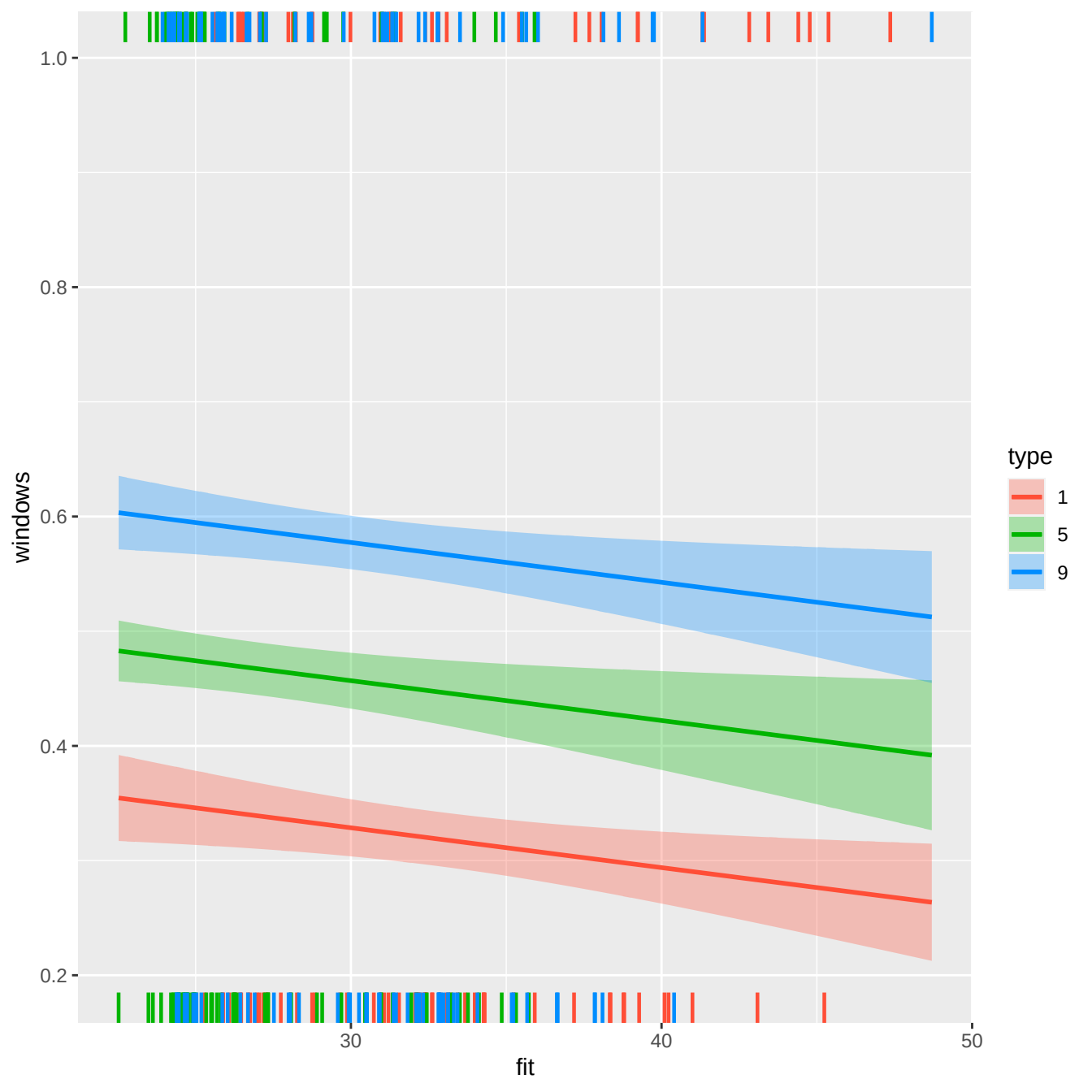

Da_BMP4_diffusion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

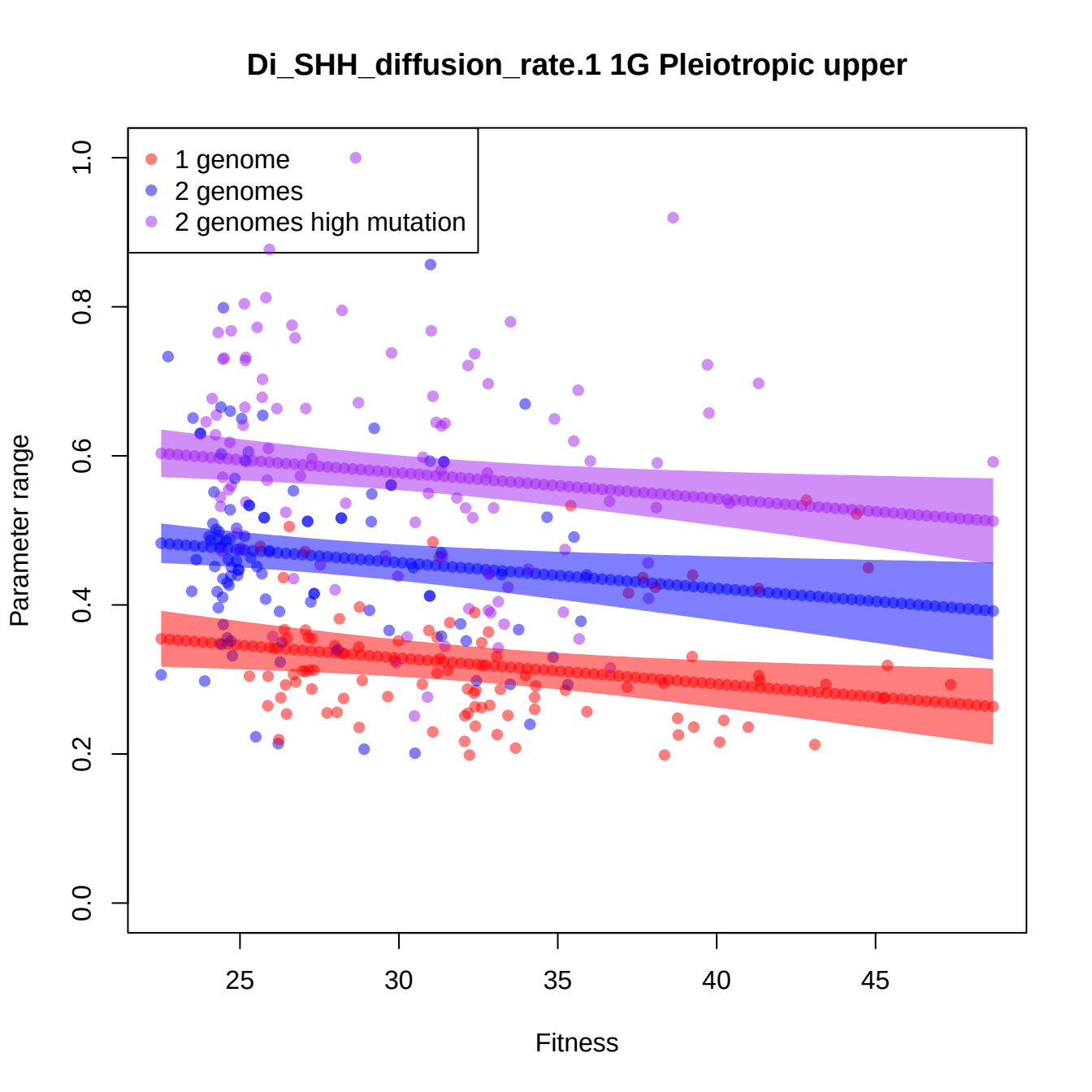

Da_BMP4_diffusion_rate_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.25493102415156e−17
0.41338394062069
3.12722681387265e−22
1 − 3
1 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

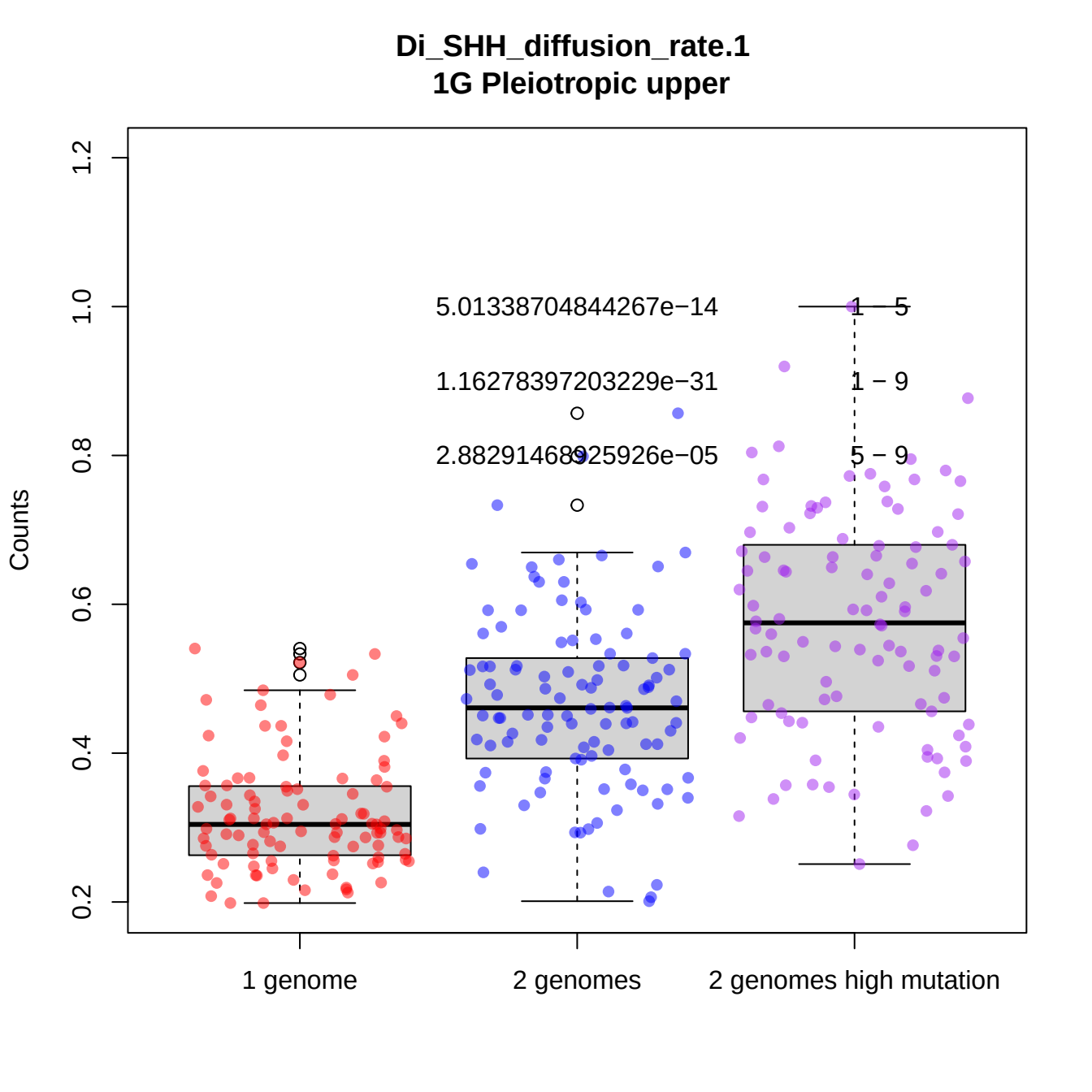

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
4
8
12

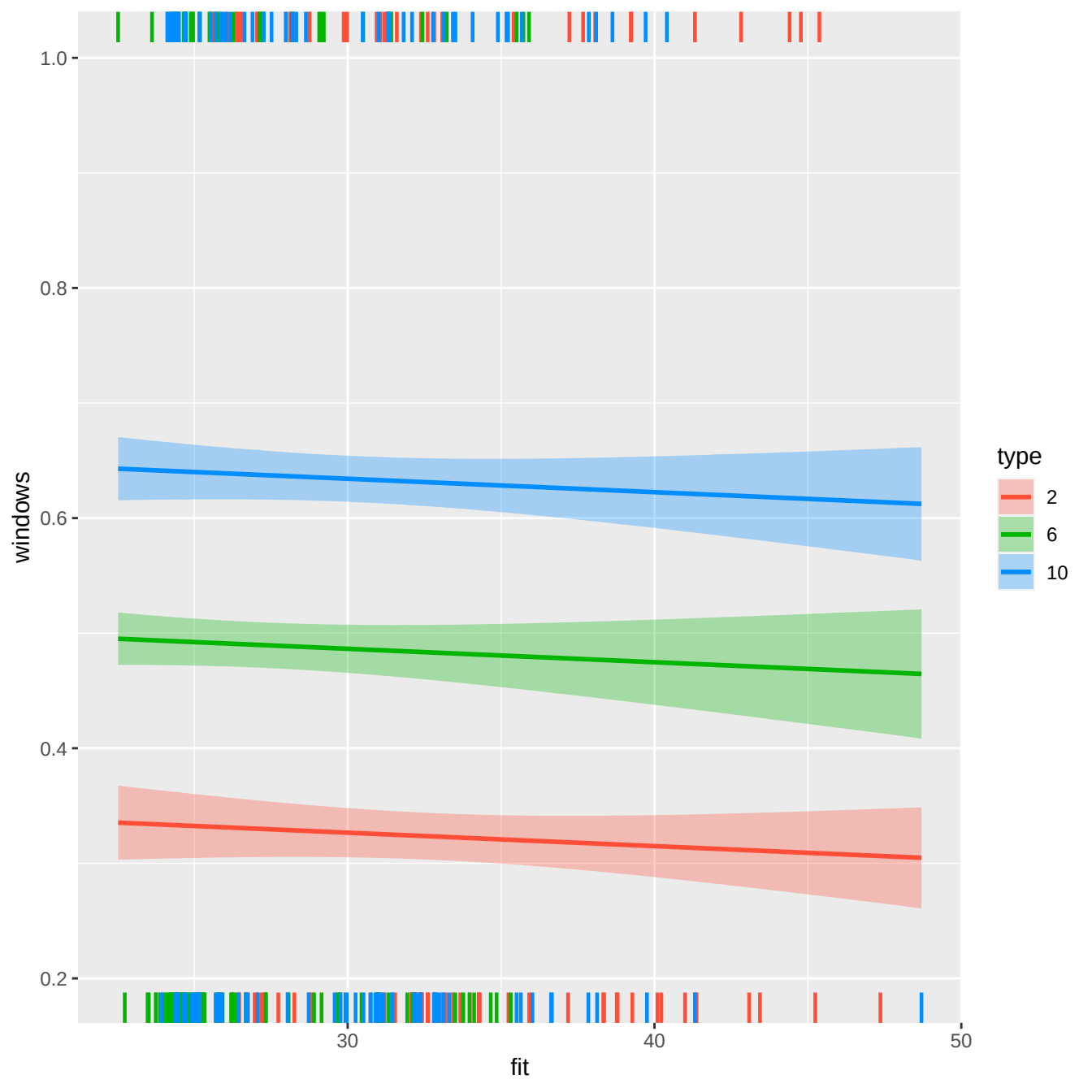

Da_BMP4_diffusion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

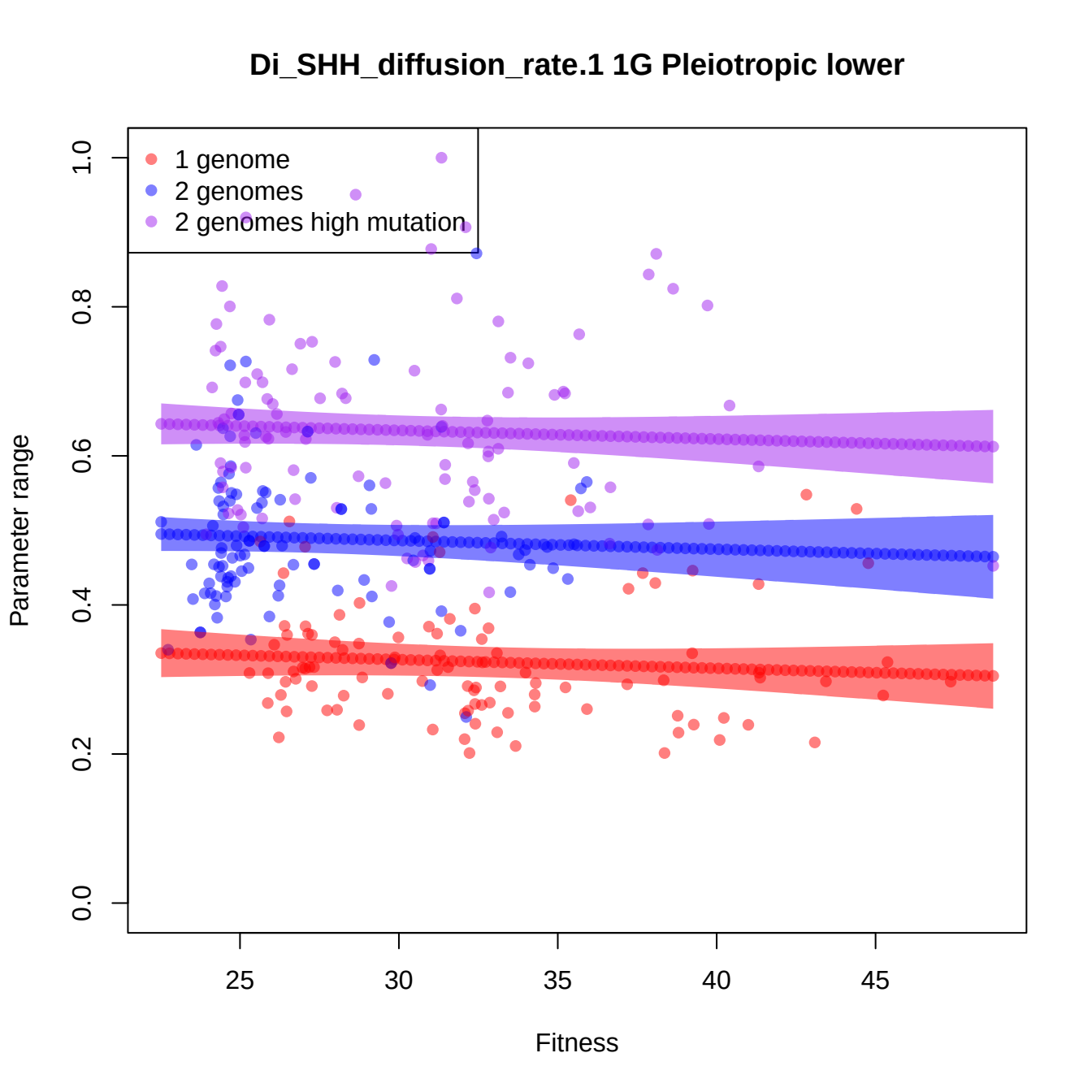

Da_BMP4_diffusion_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
7.599826455753345e−26
2.728429604751316e−25
0
1
2
1 genome
2 genomes
2 genomes high mutation

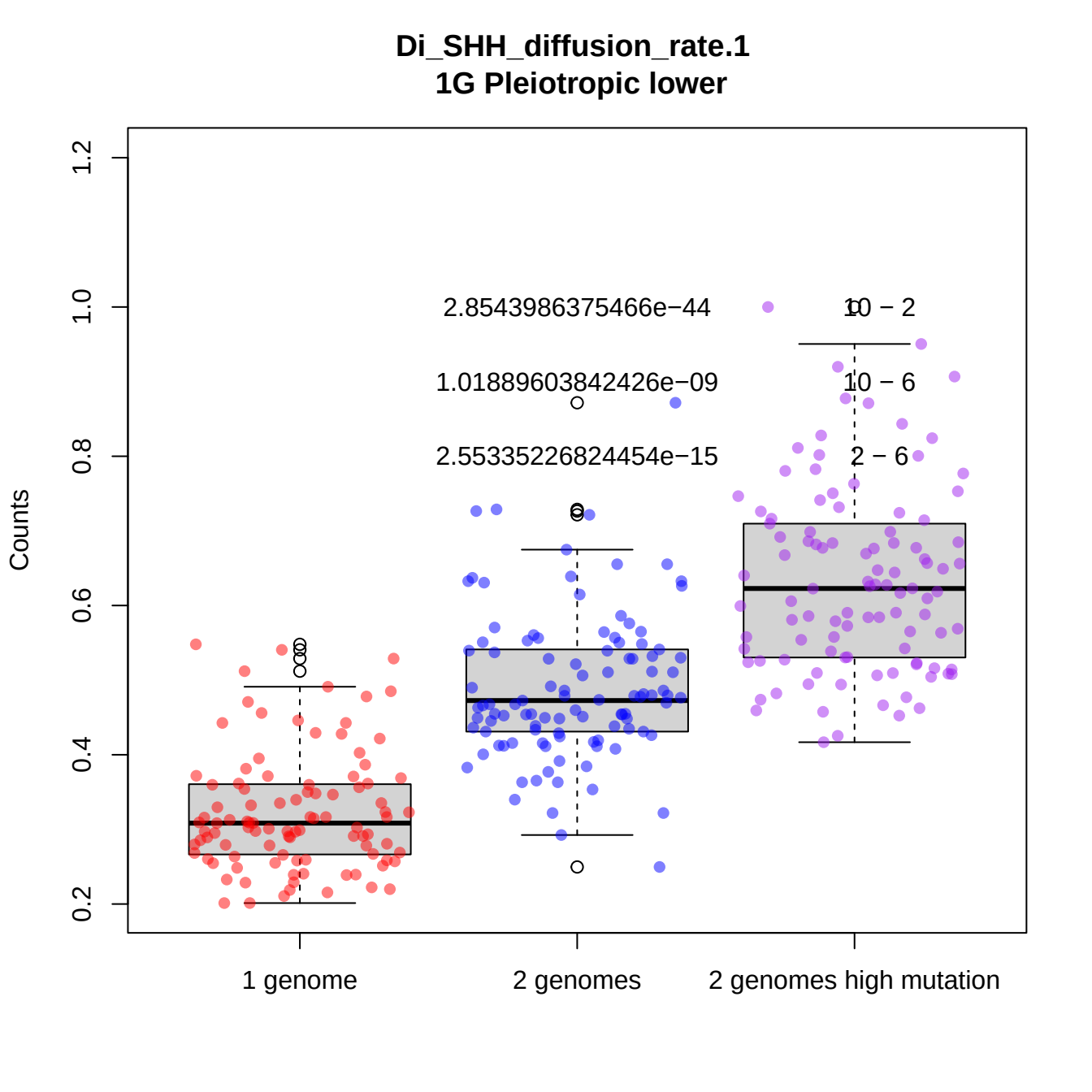

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

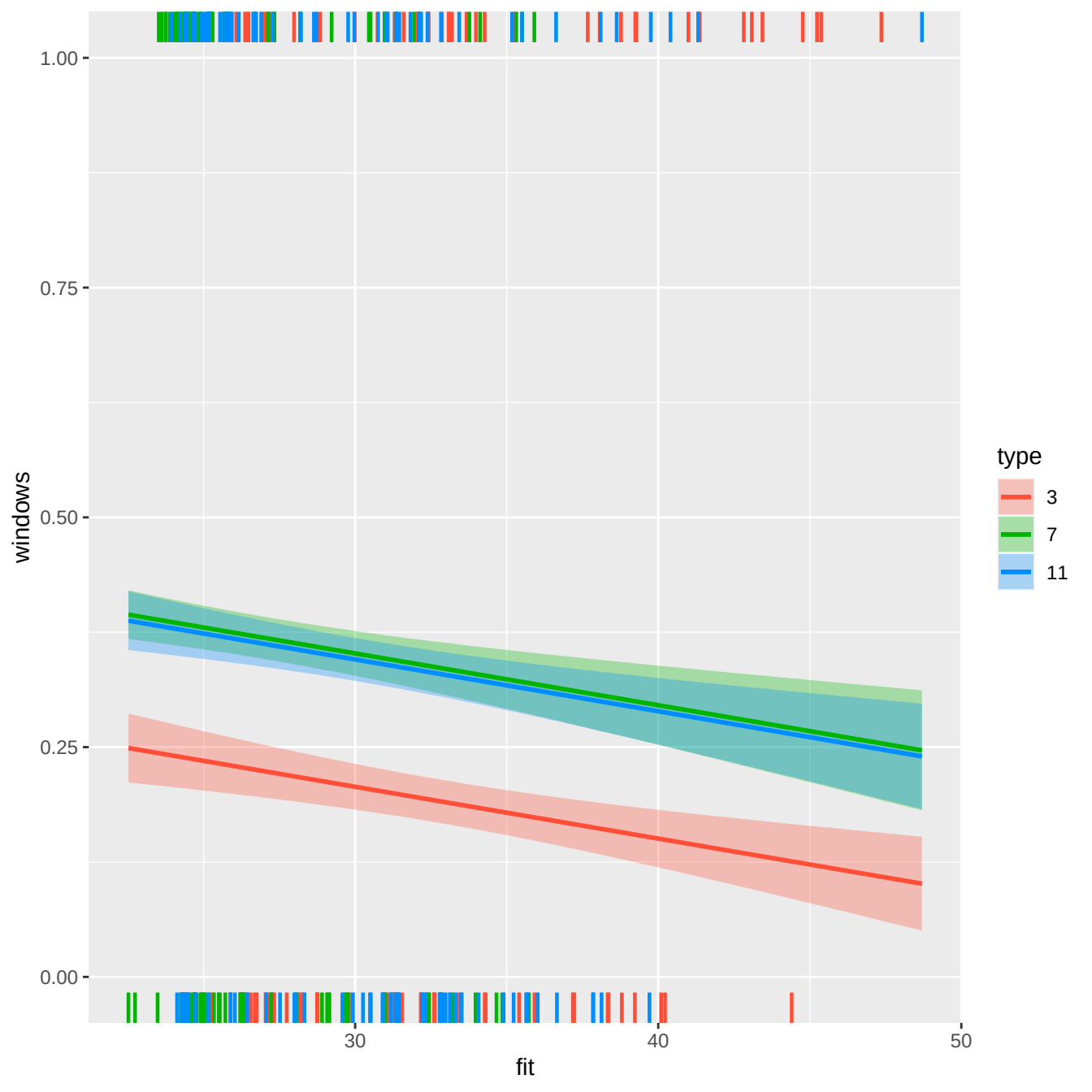

Di_SHH_diffusion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

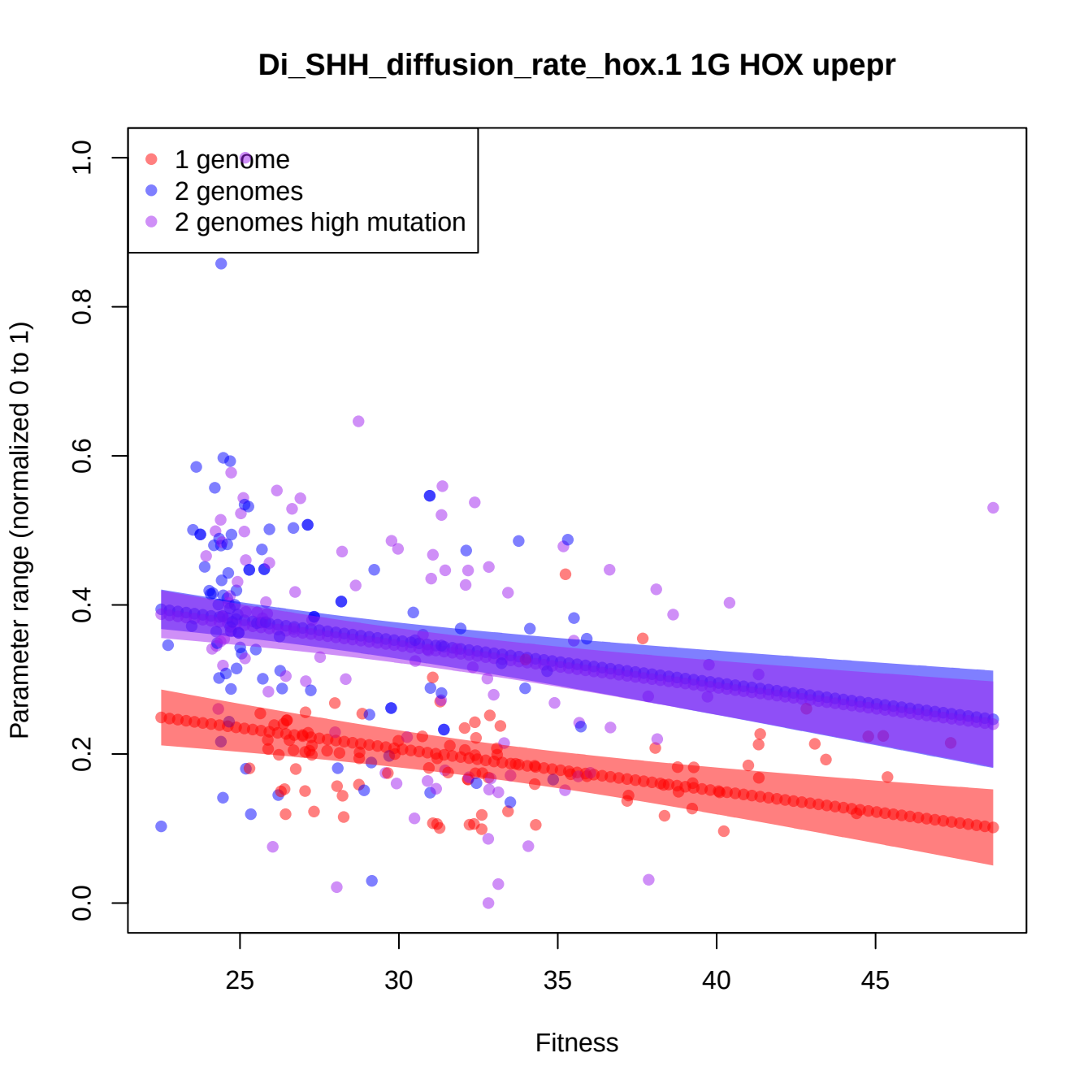

Di_SHH_diffusion_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
5.01338704844267e−14
1.16278397203229e−31
2.88291468925926e−05
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

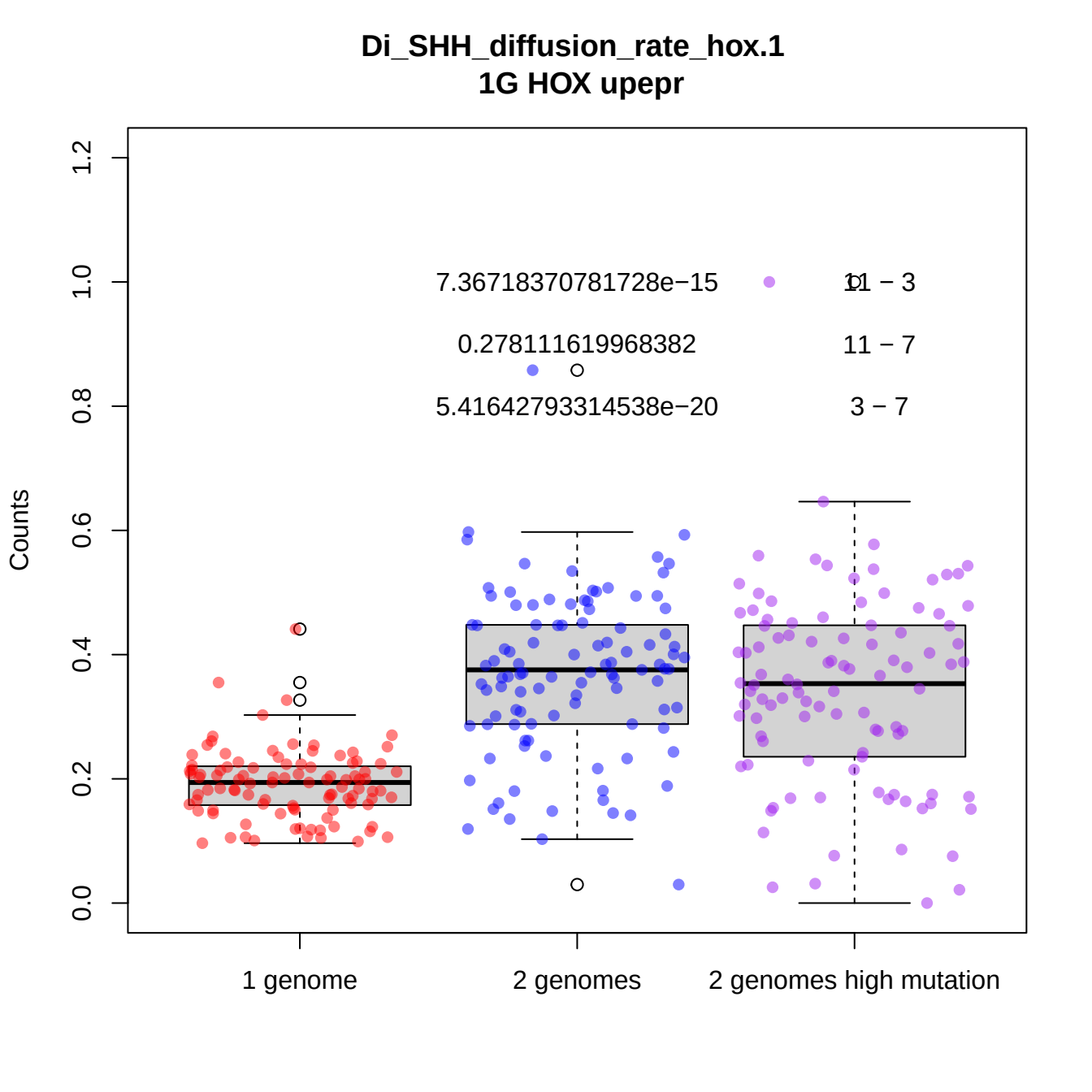

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
2
6
10

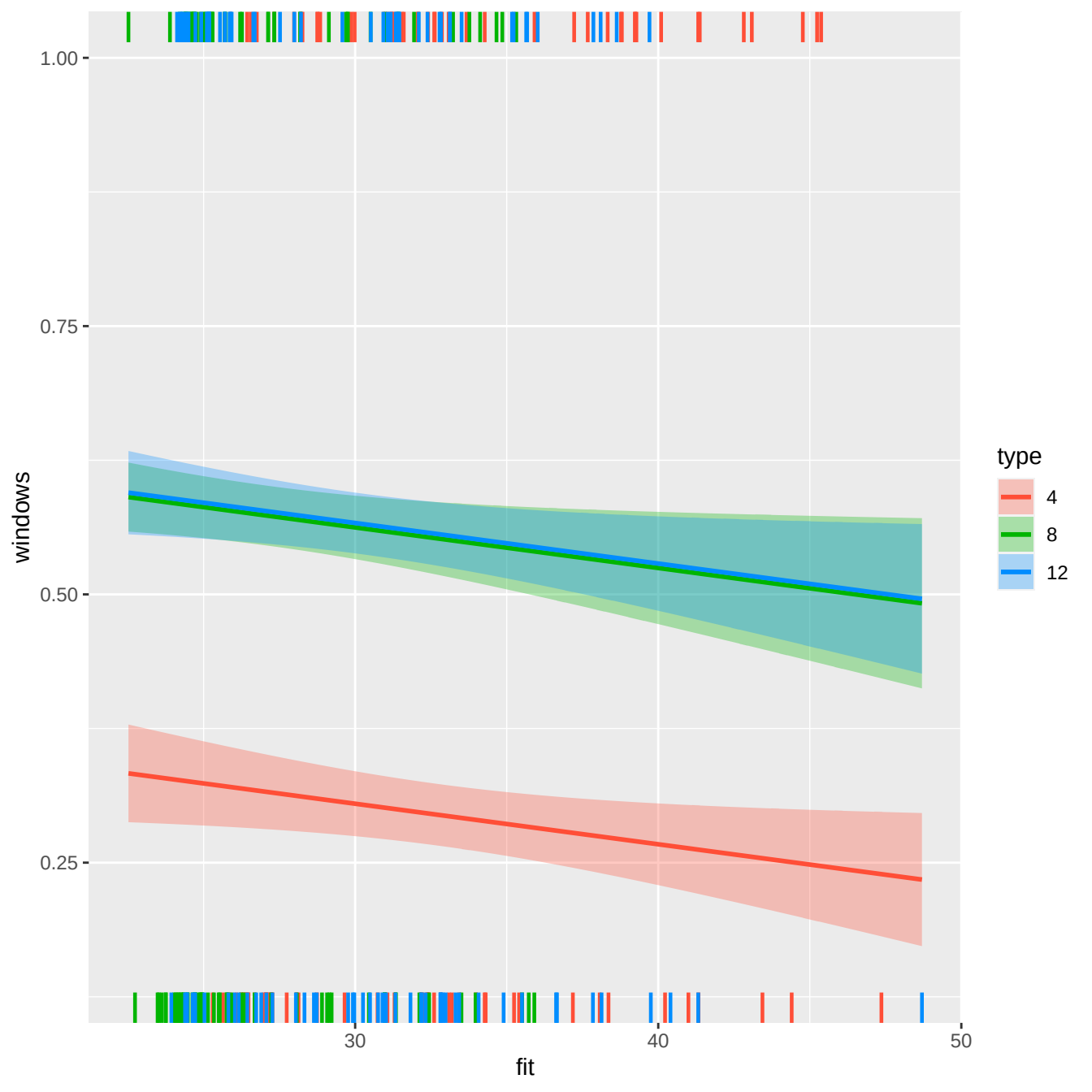

Di_SHH_diffusion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

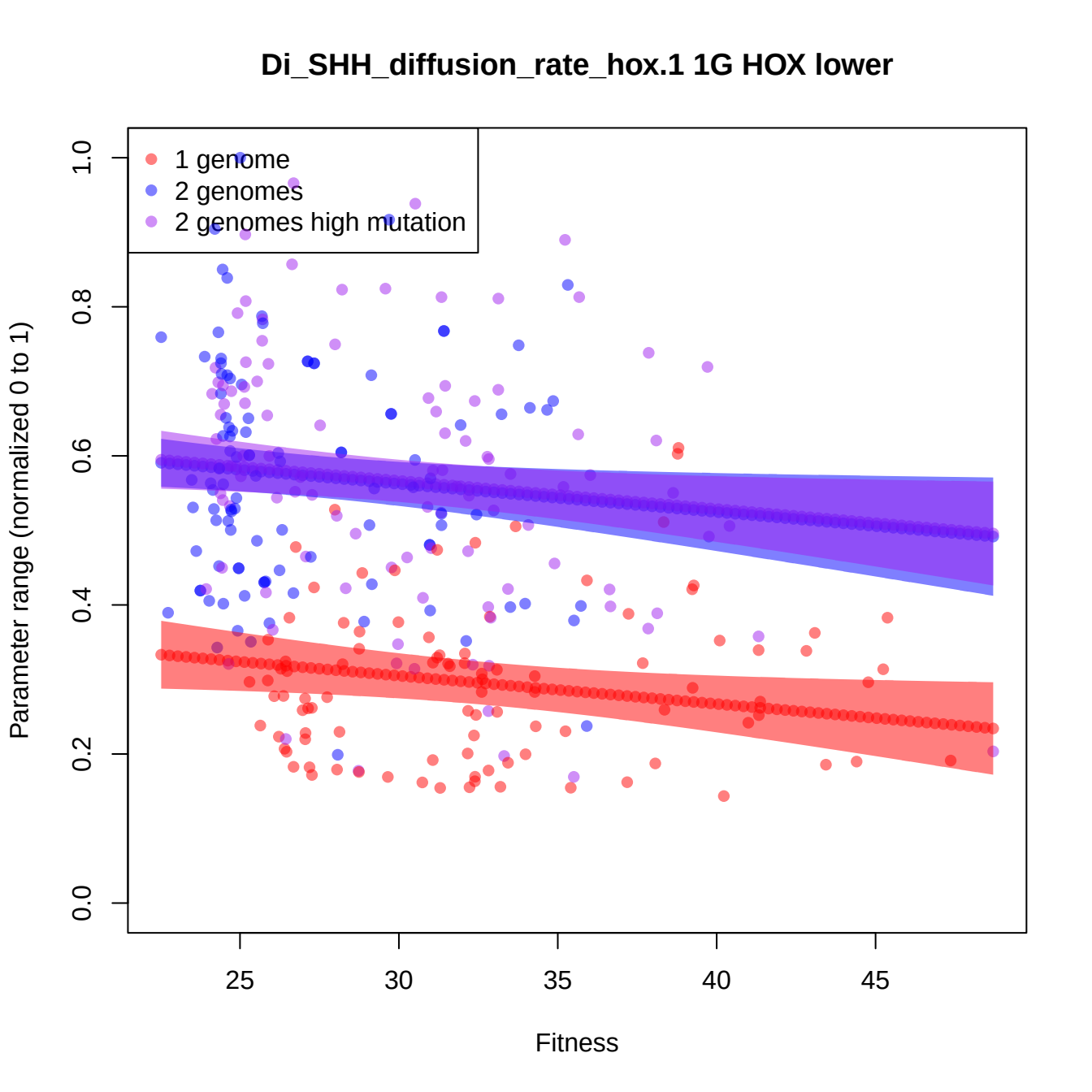

Di_SHH_diffusion_rate.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.8543986375466e−44
1.01889603842426e−09
2.55335226824454e−15
1 − 2
1 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

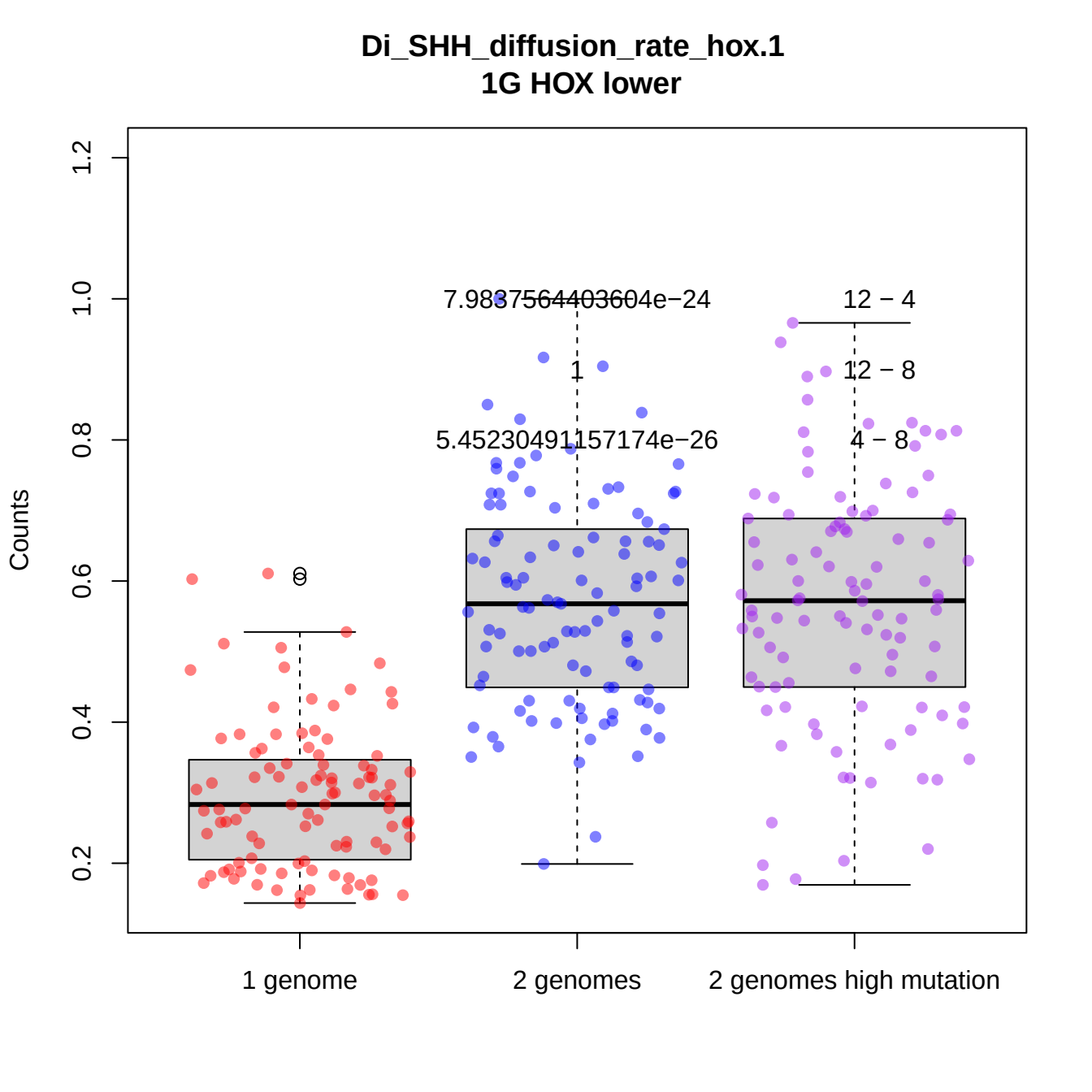

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

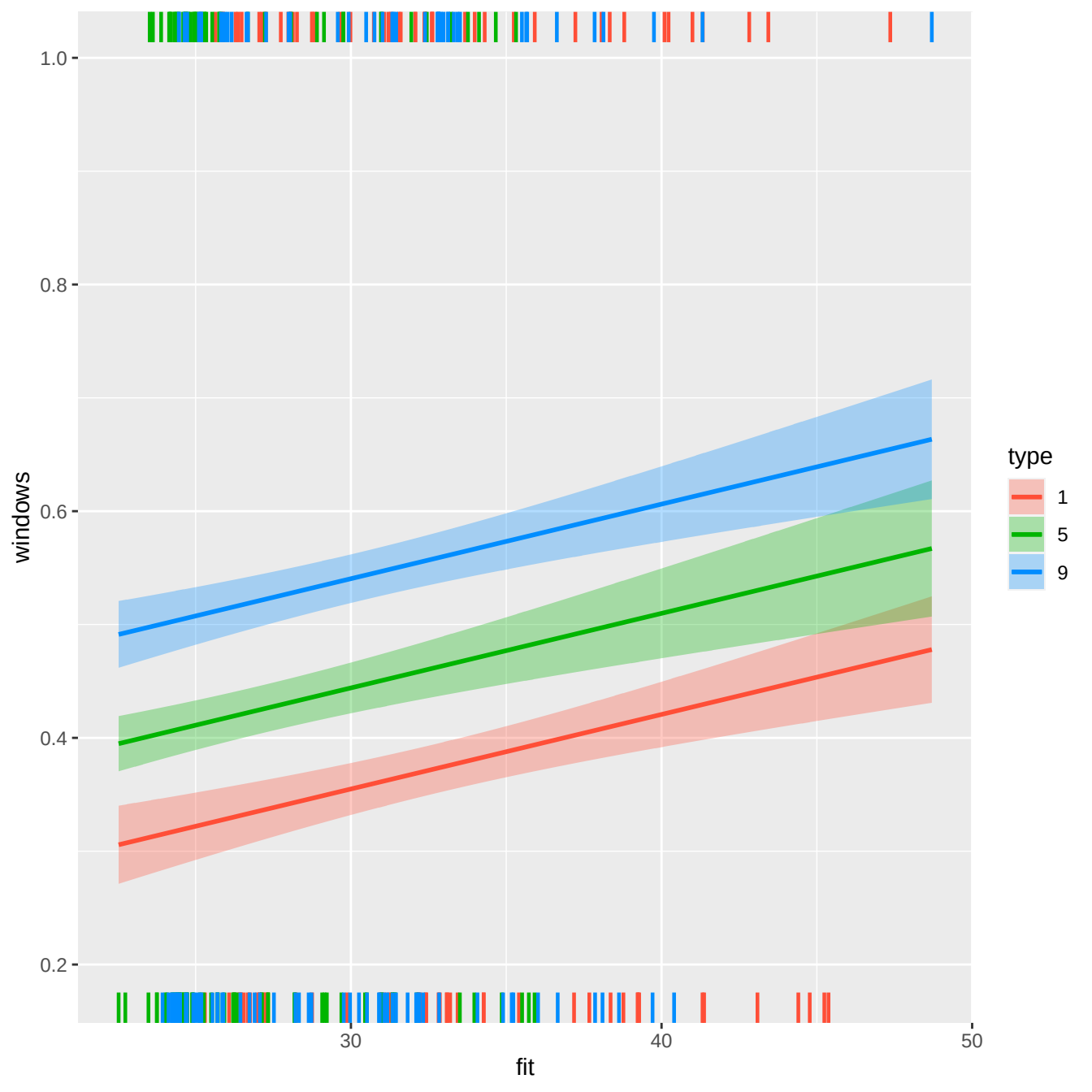

Di_SHH_diffusion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

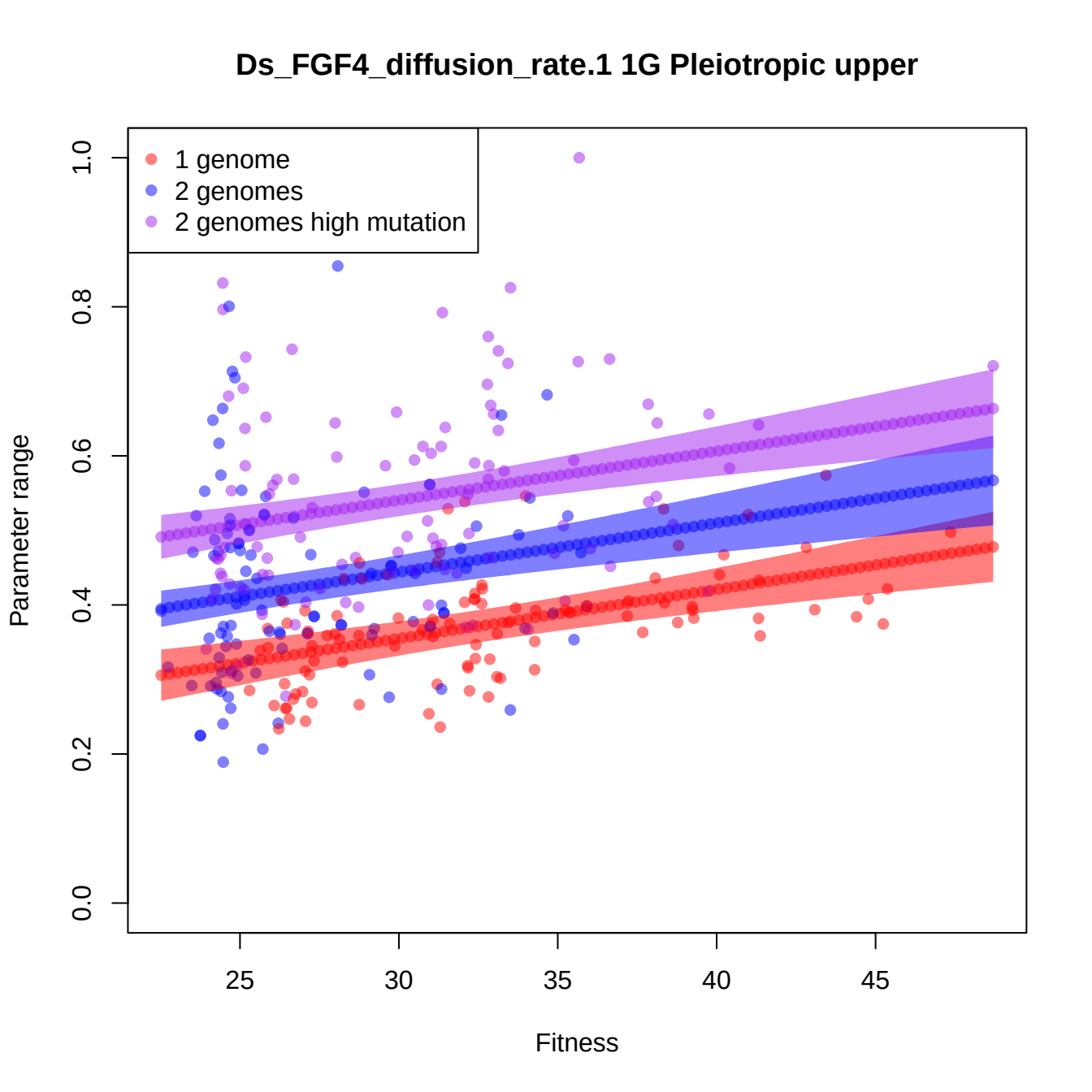

Di_SHH_diffusion_rate_hox.1
1G HOX upepr
7.36718370781728e−15
11 − 3
0.278111619968382
11 − 7
5.41642793314538e−20
3 − 7
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

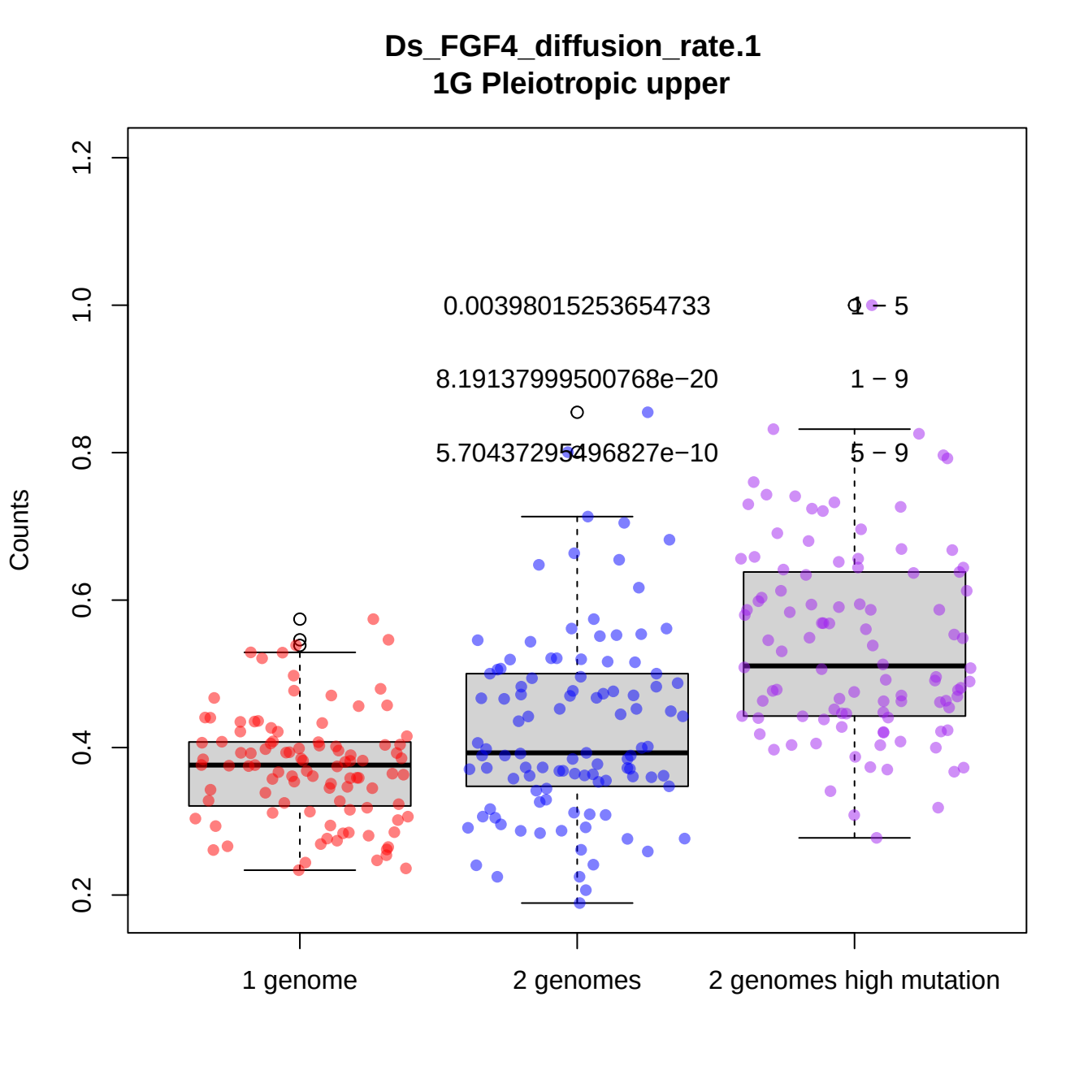

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
4
8
12

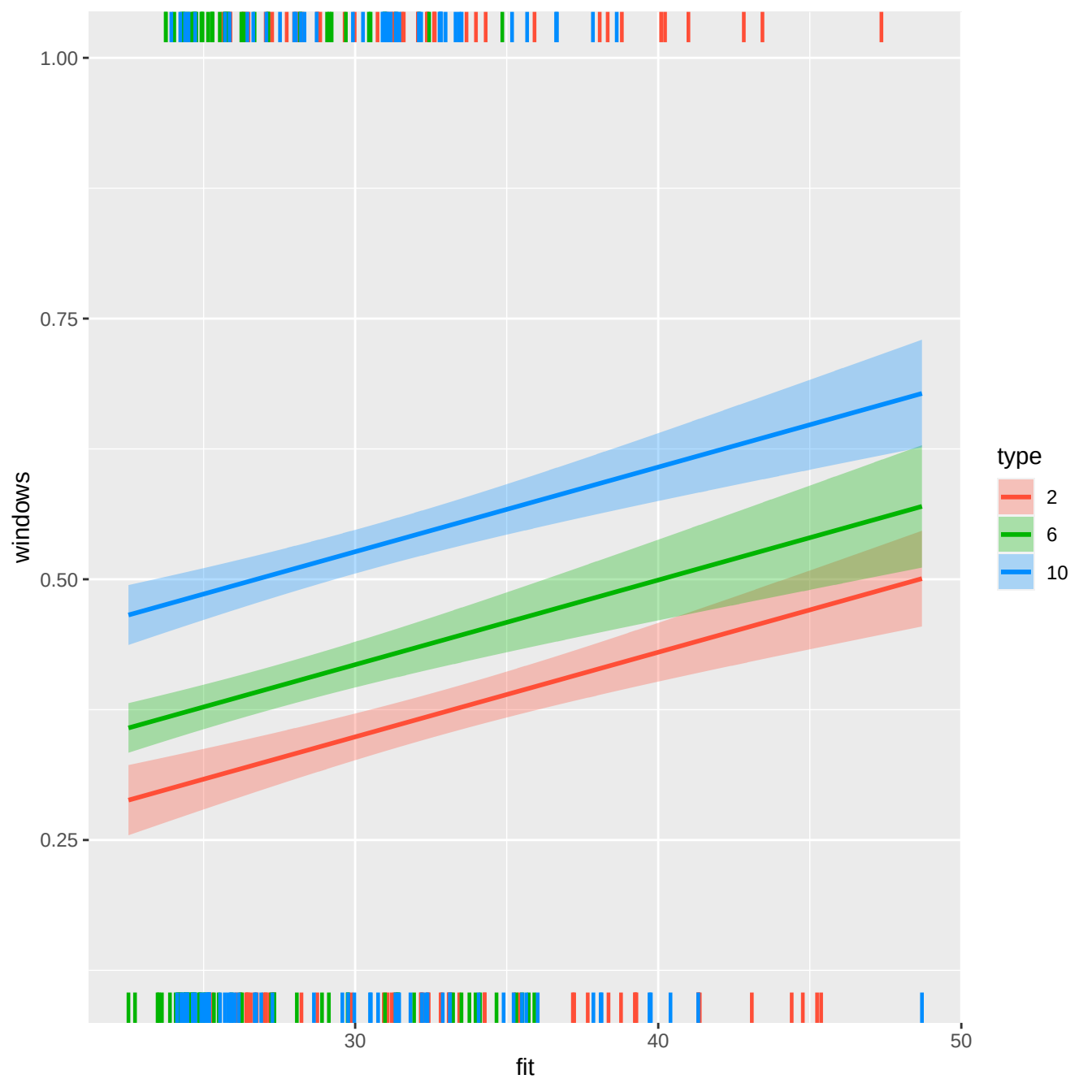

Di_SHH_diffusion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

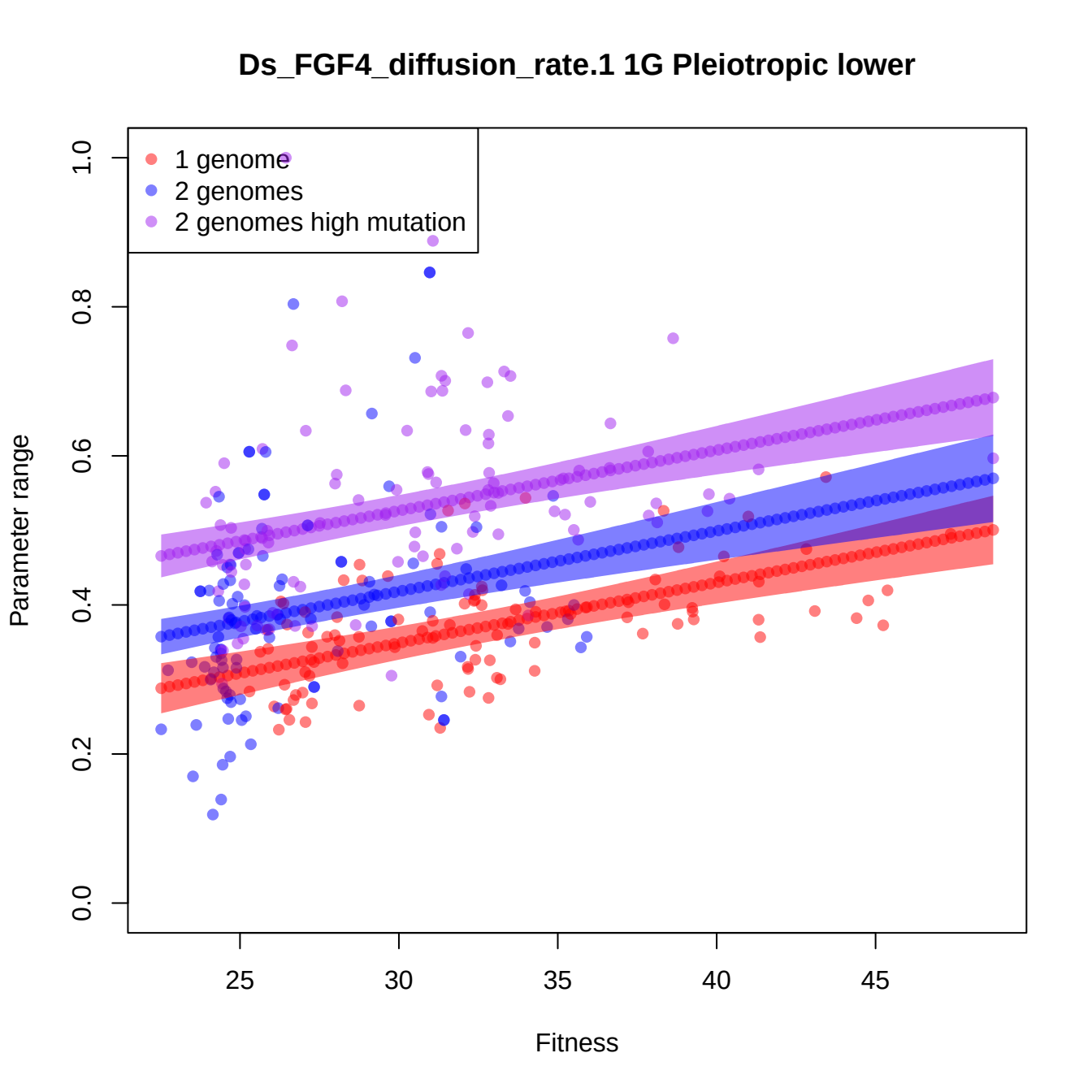

Di_SHH_diffusion_rate_hox.1
1G HOX lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0
7.98375644036e−24
1
5.45230491157174e−26
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

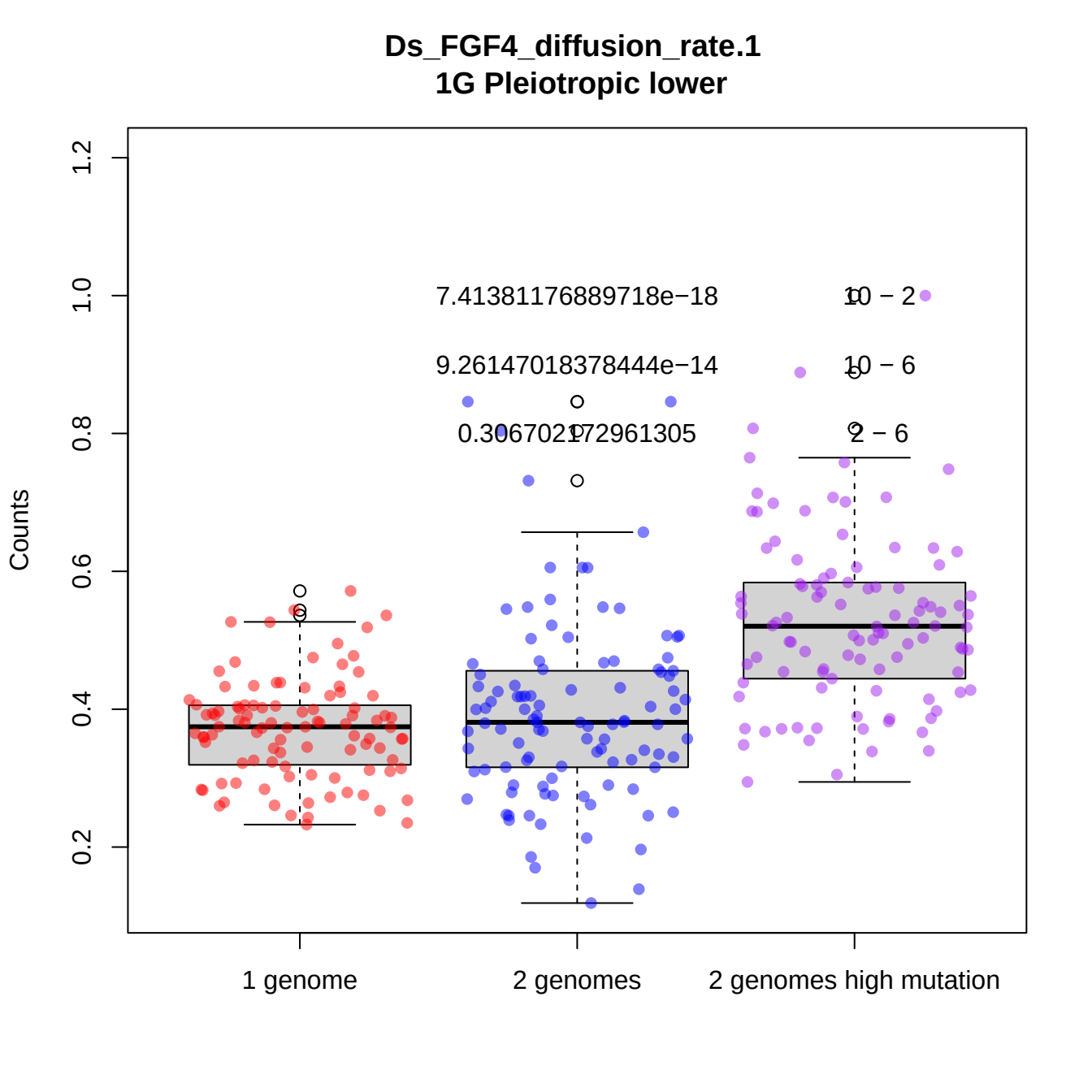

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

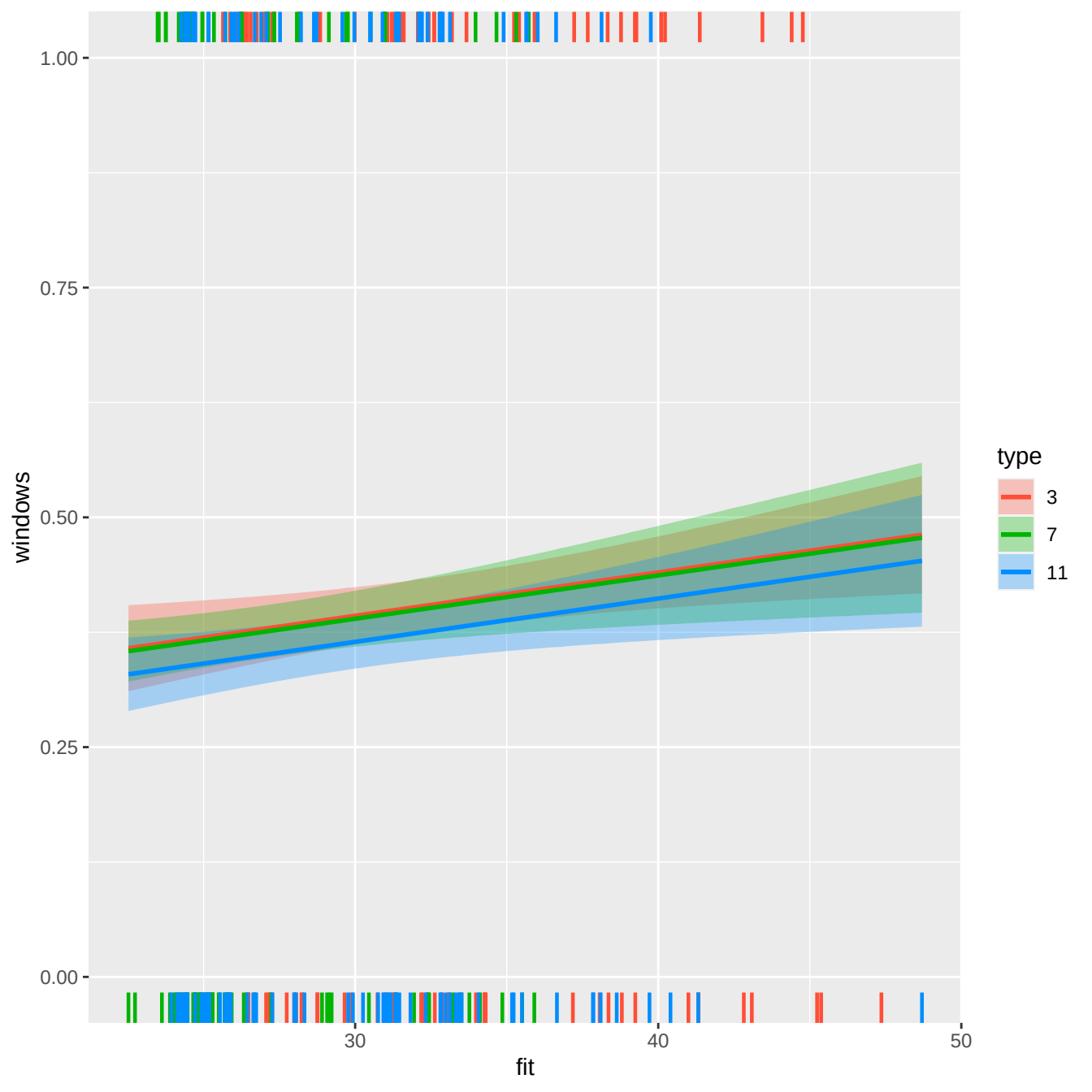

Ds_FGF4_diffusion_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

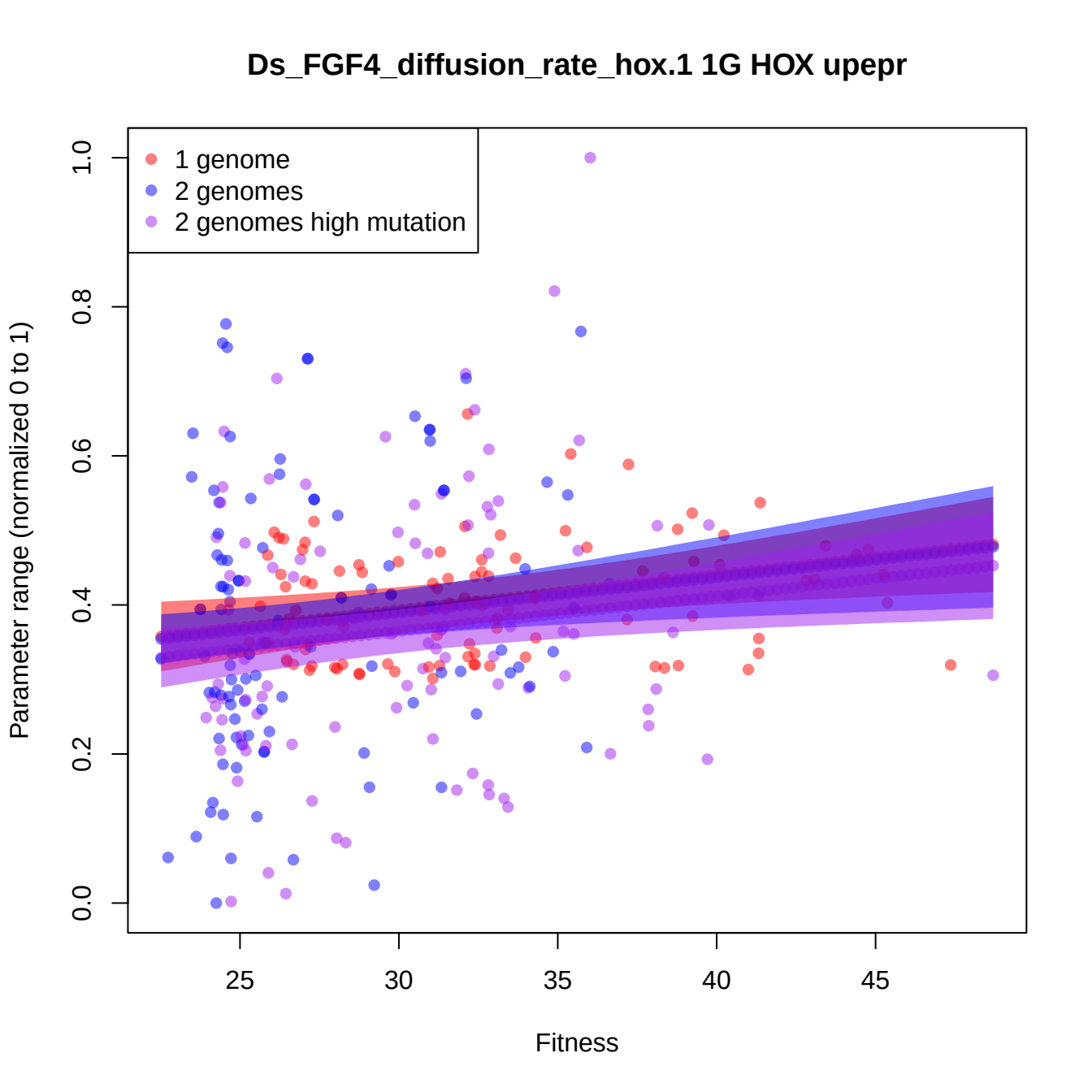

Ds_FGF4_diffusion_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.00398015253654733
8.19137999500768e−20
5.70437295496827e−10
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

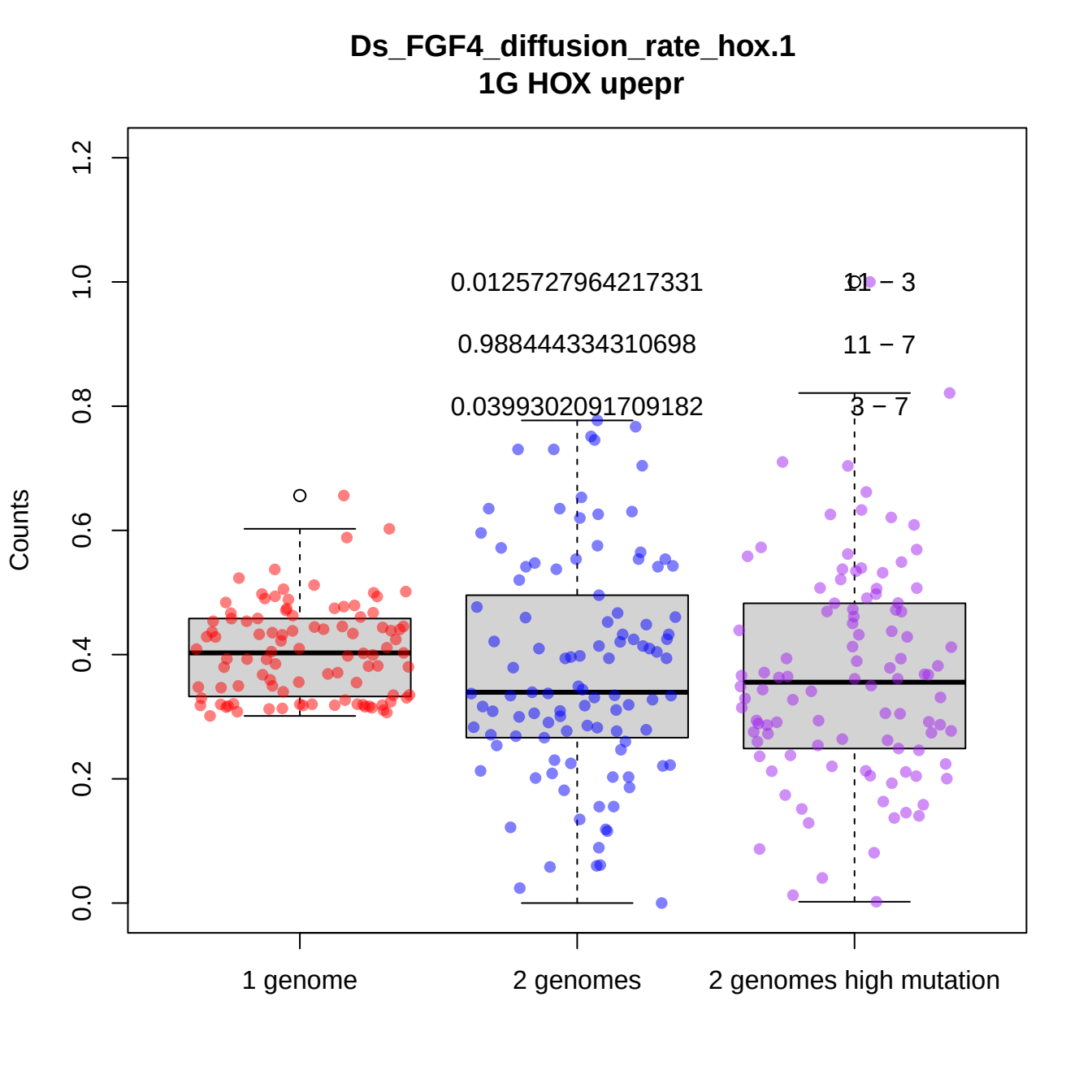

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

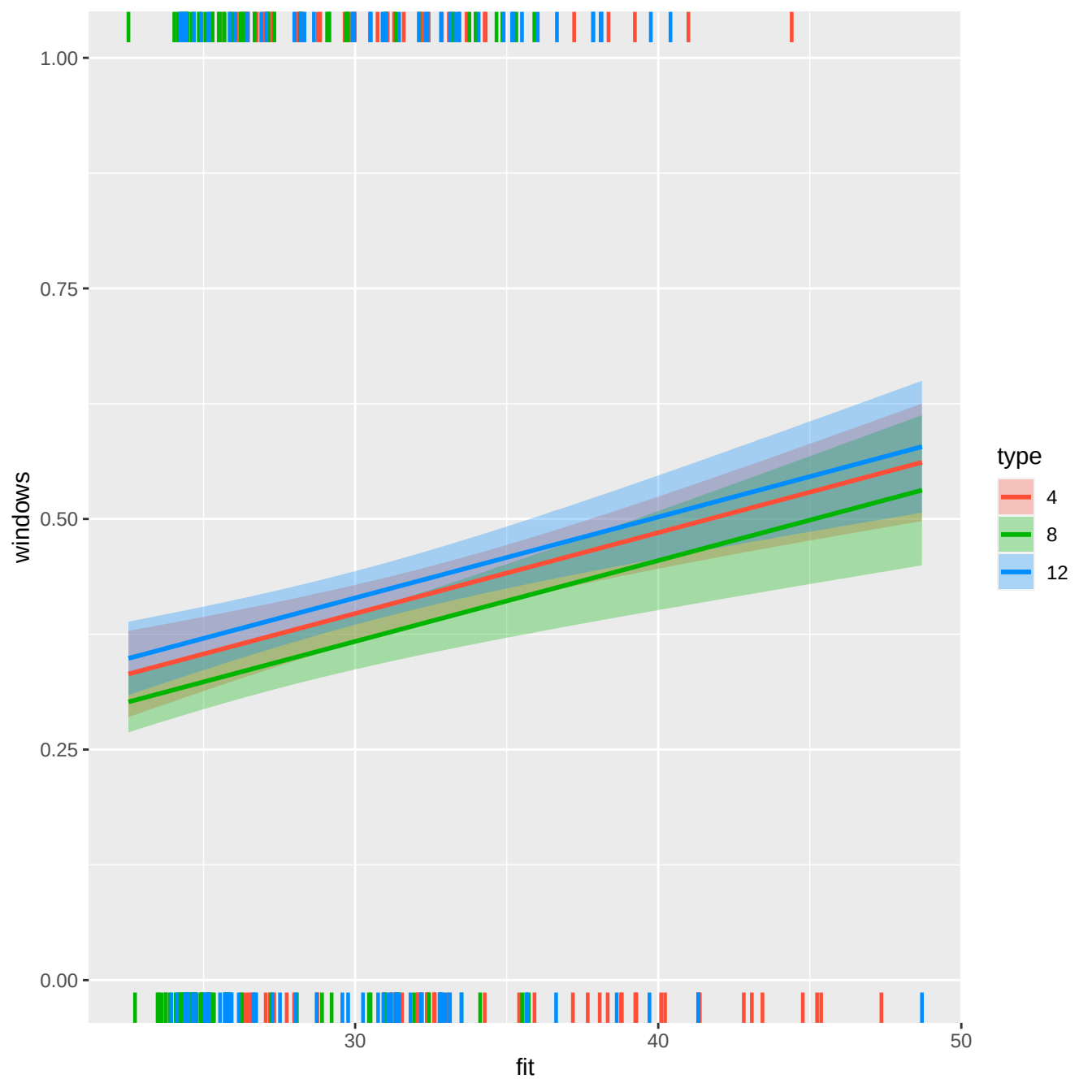

Ds_FGF4_diffusion_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

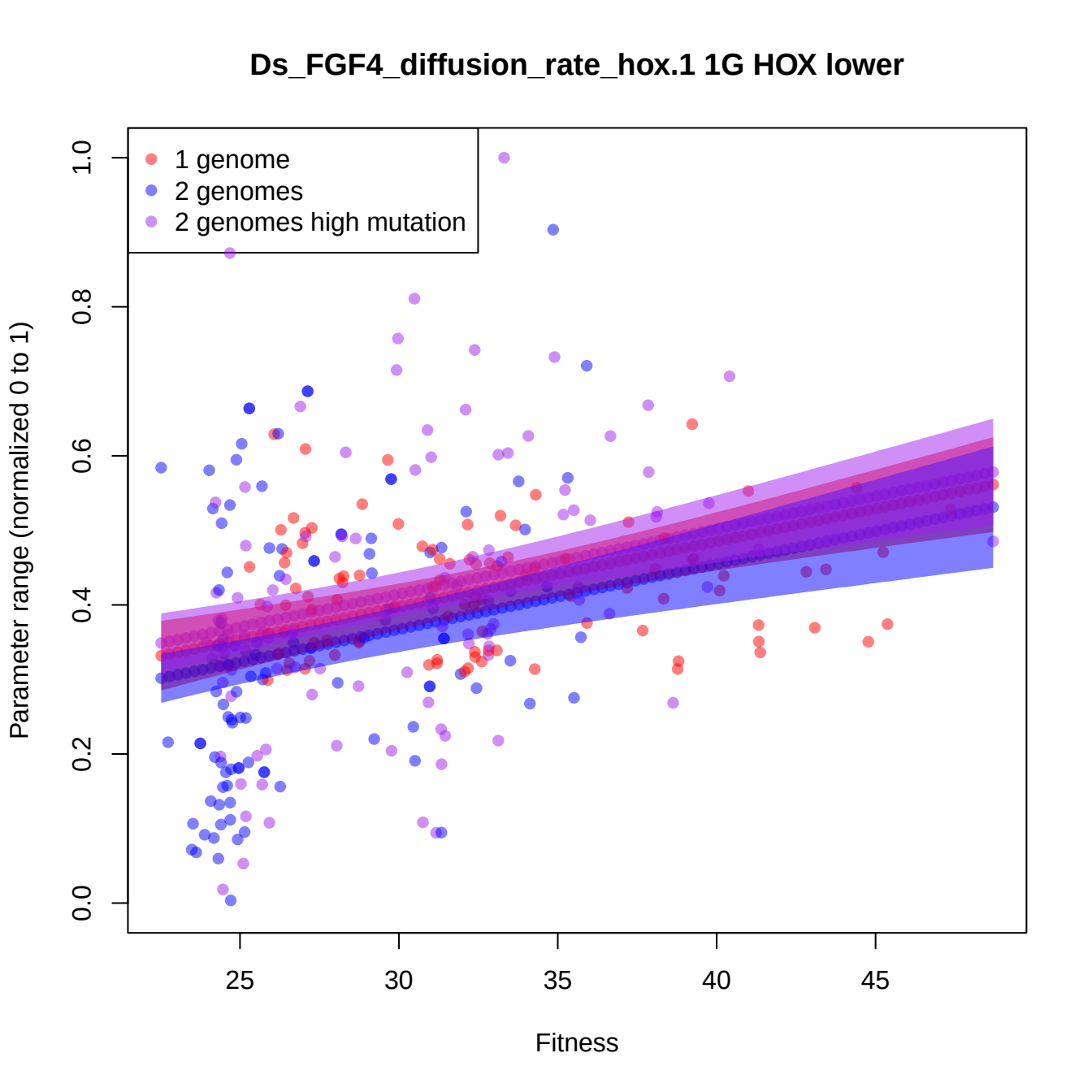

Ds_FGF4_diffusion_rate.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
7.41381176889718e−18
9.26147018378444e−14
0.30670217296130 5
1 0 − 2
1 0 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

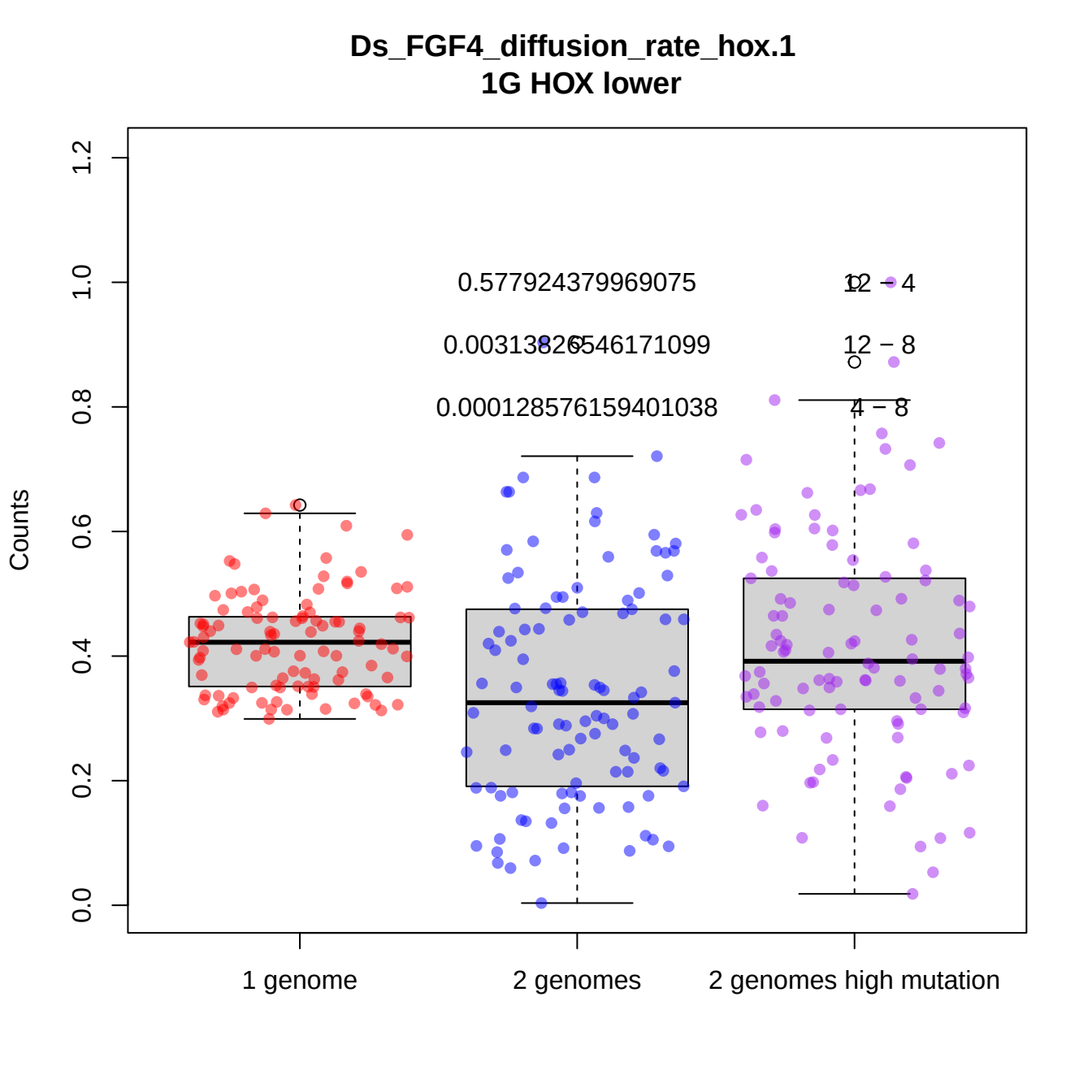

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

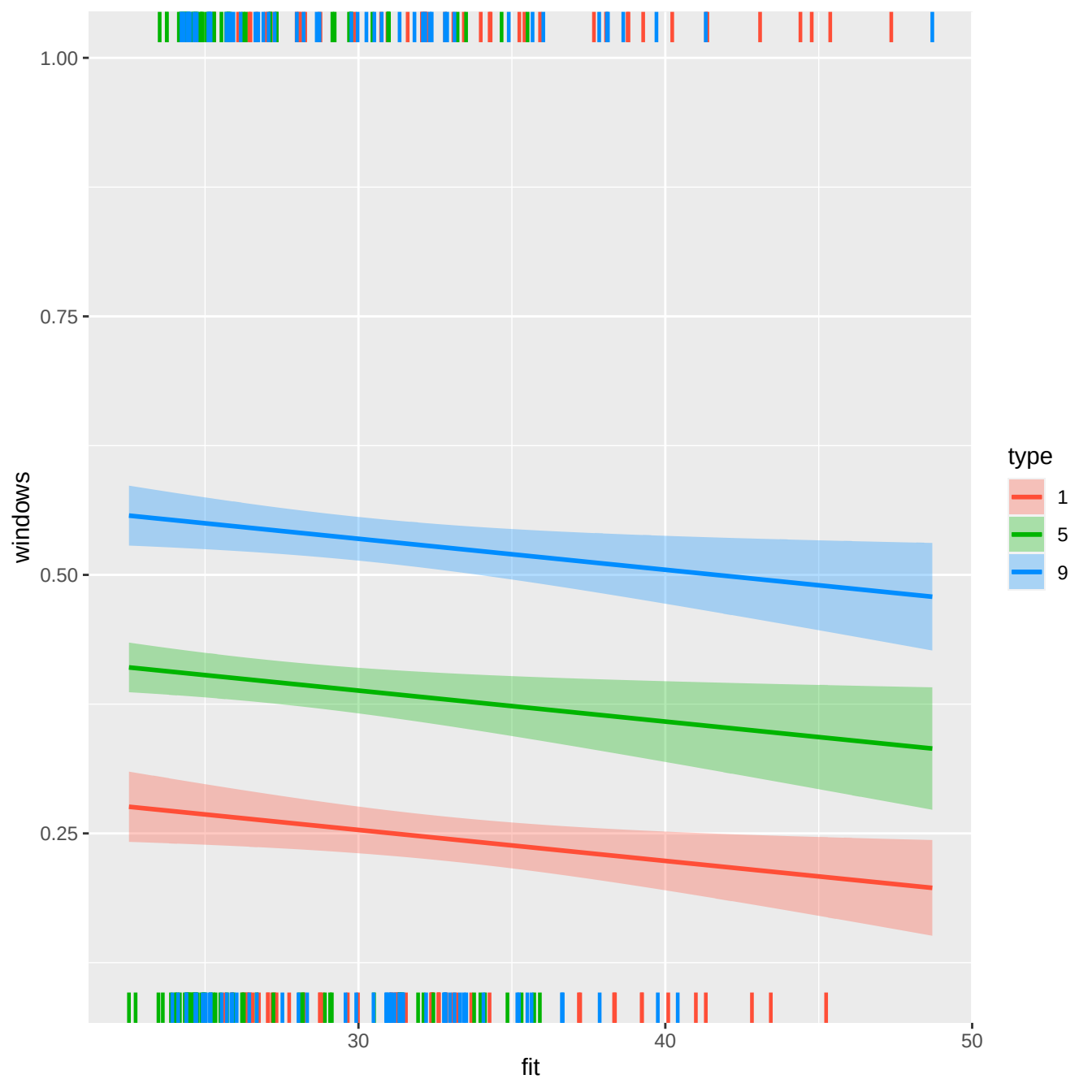

Ds_FGF4_diffusion_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

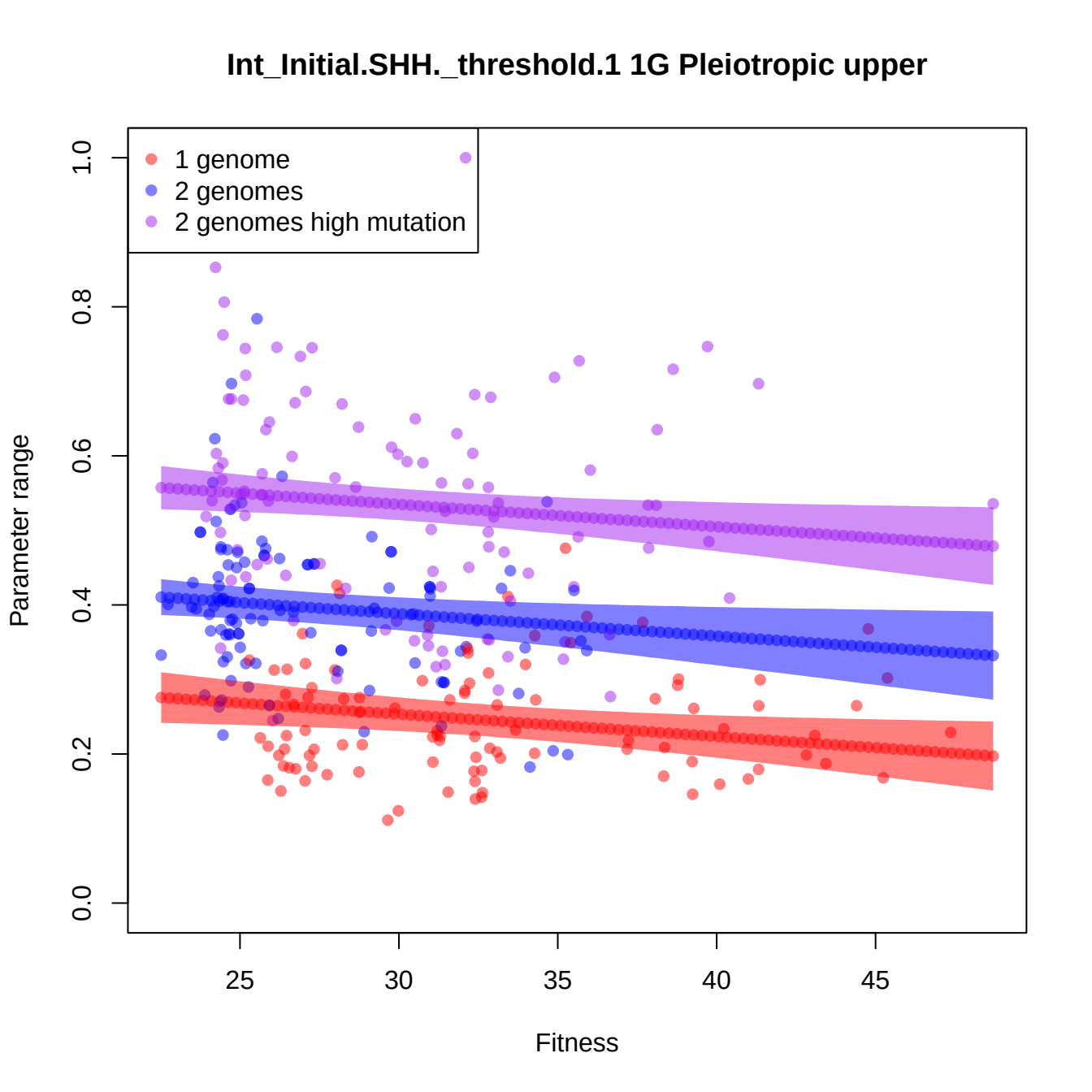

Ds_FGF4_diffusion_rate_hox.1
1G HOX upepr
0.0125727964217331
0.988444334310698
0.0399302091709182
1 − 3
1 − 7
3 − 7
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

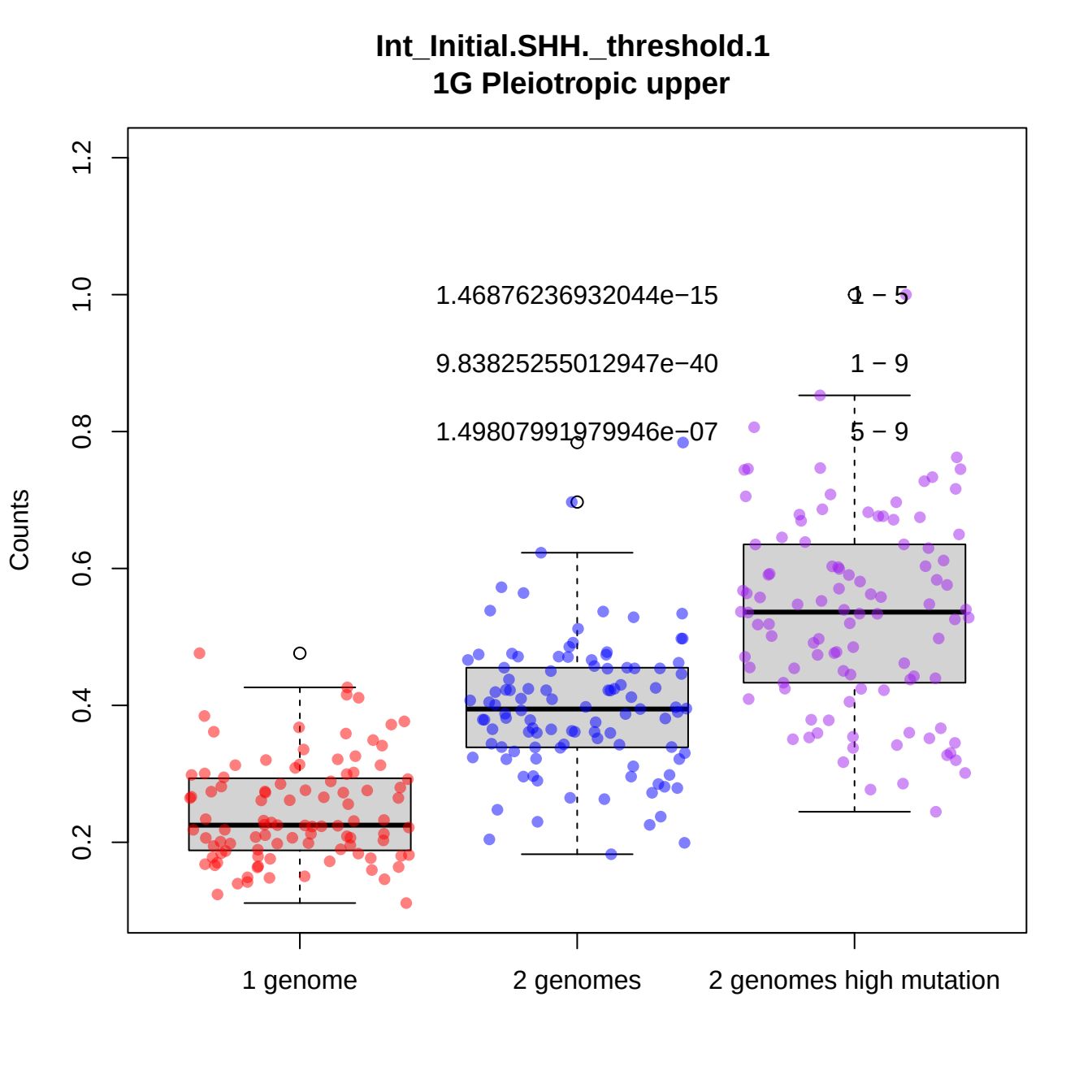

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

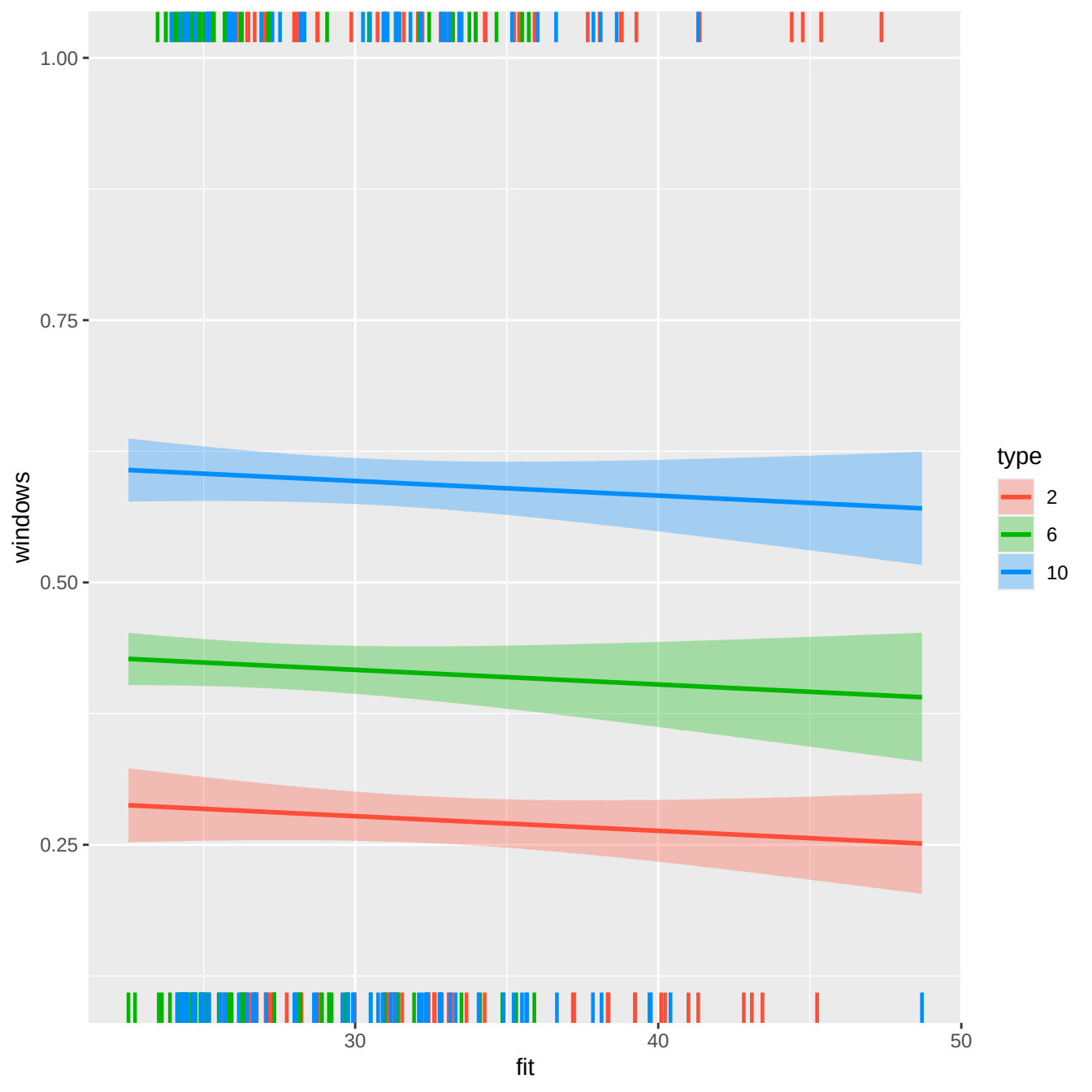

Ds_FGF4_diffusion_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

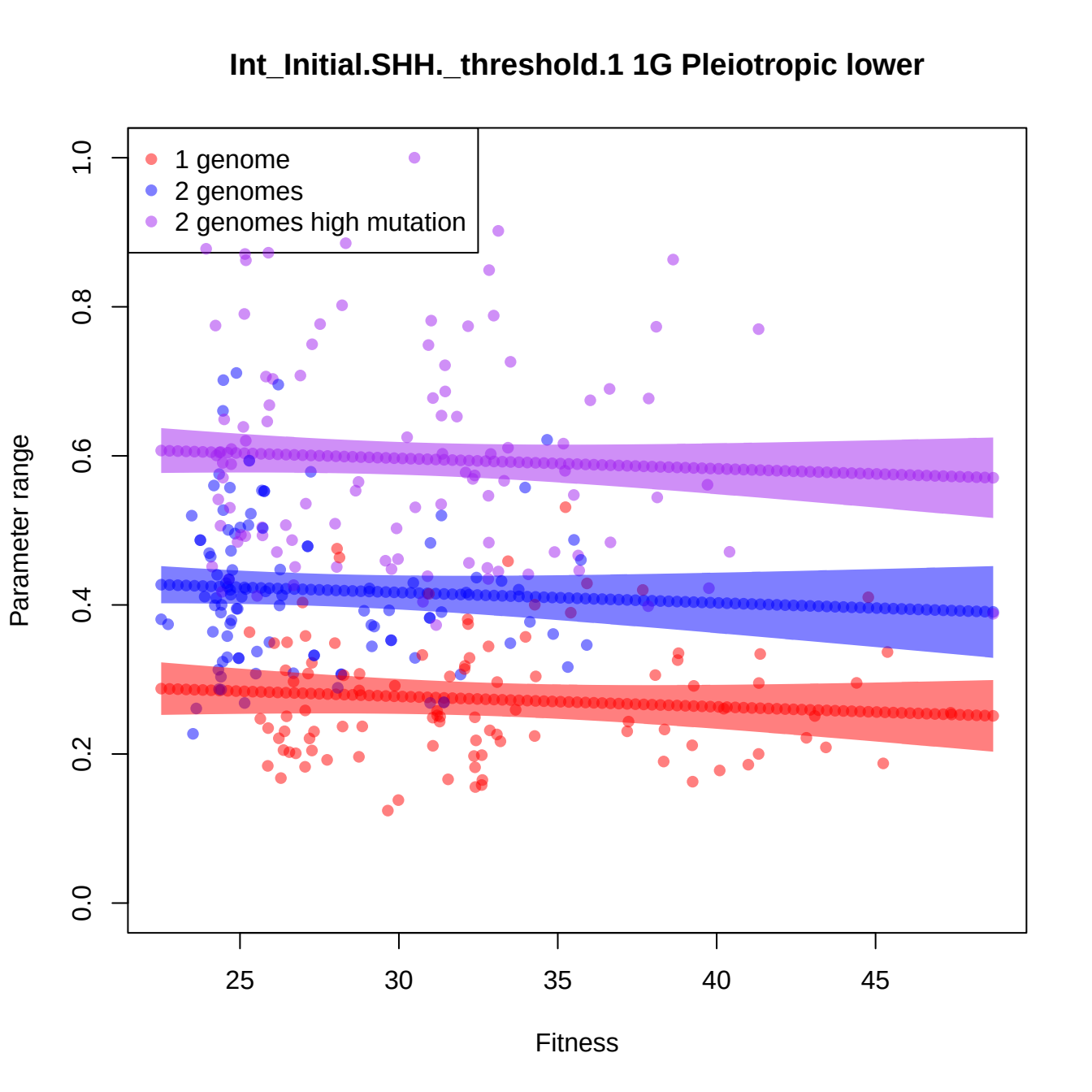

Ds_FGF4_diffusion_rate_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.577924379969075
0.0031382654617099
0.000128576159401038
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

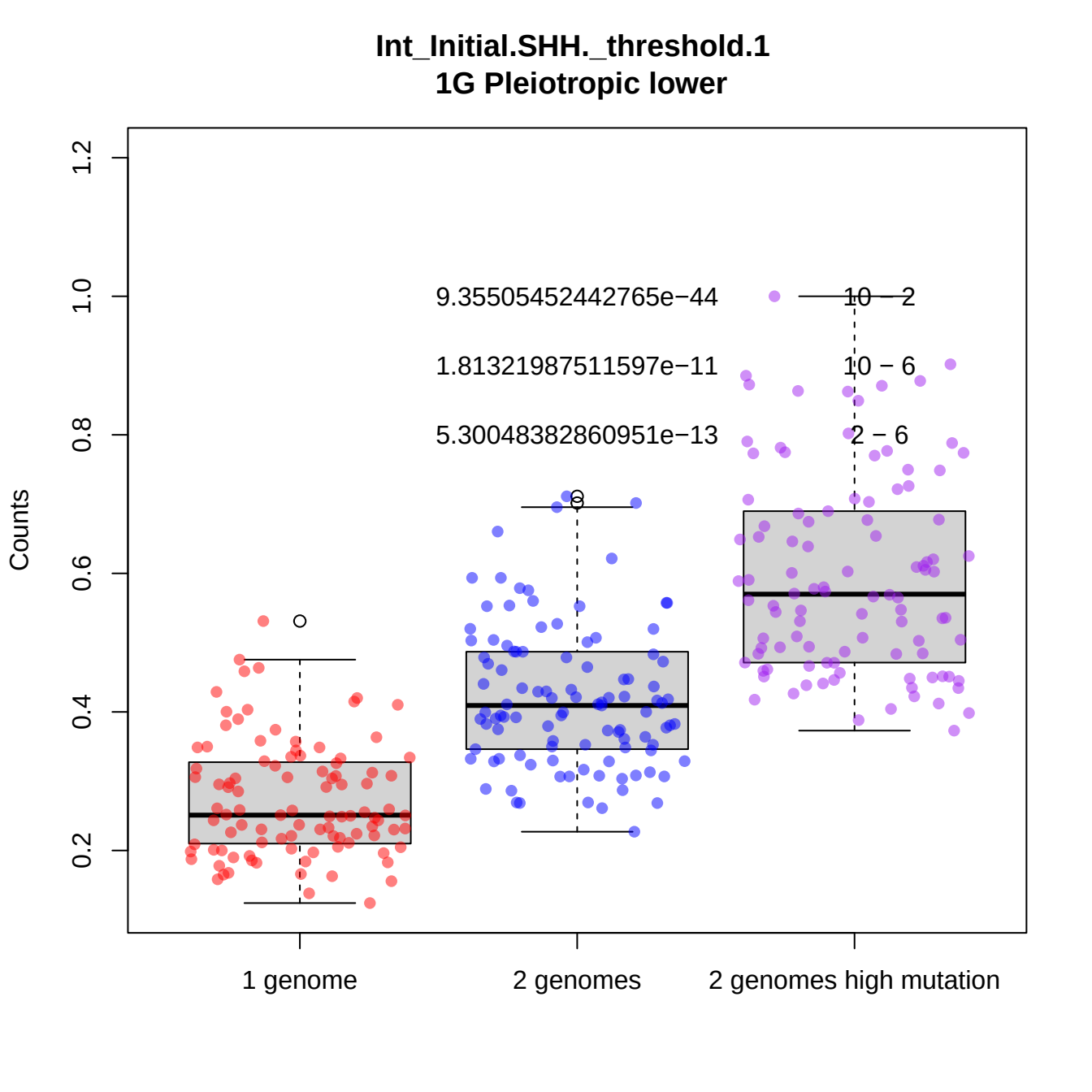

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

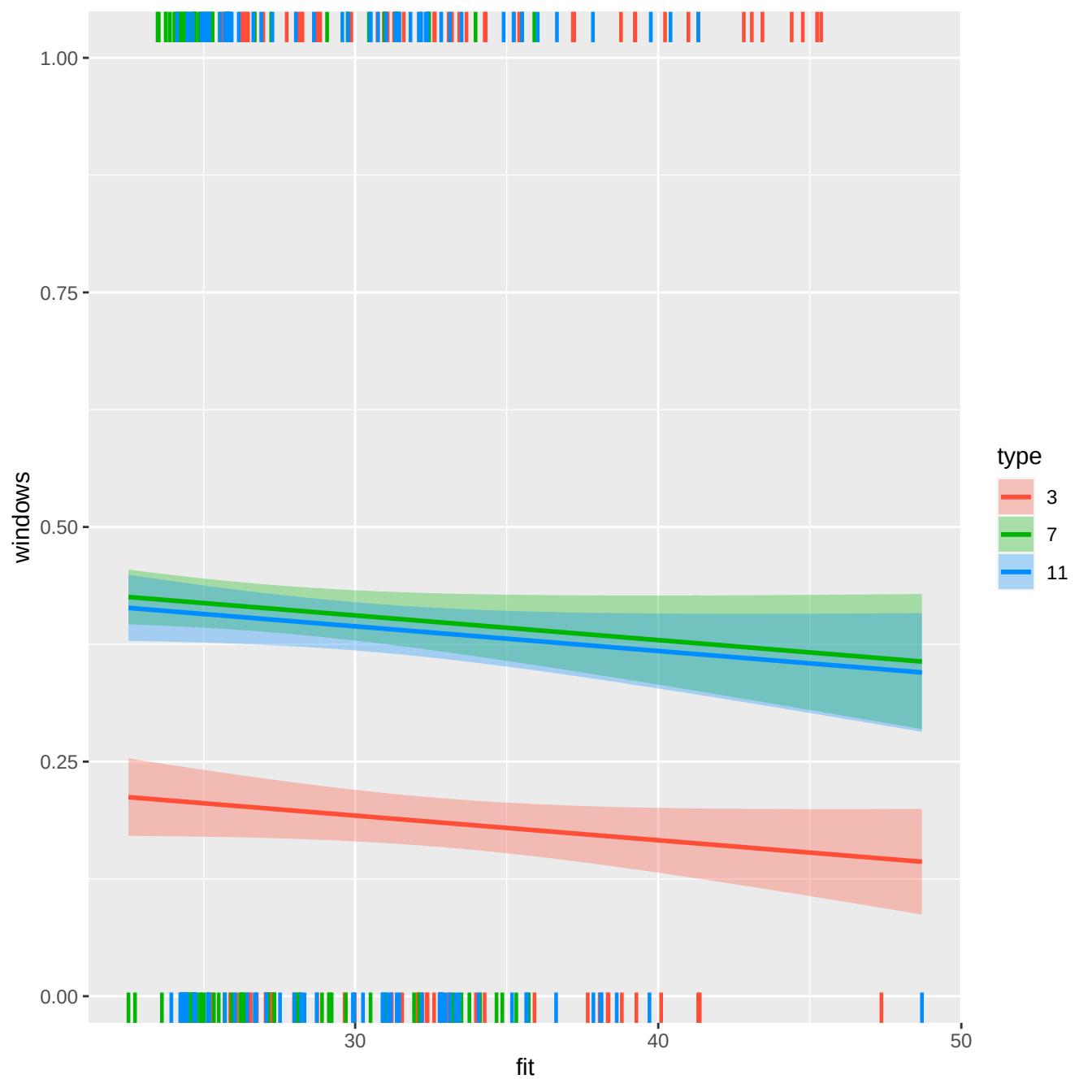

Int_Initial.SHH._threshold.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

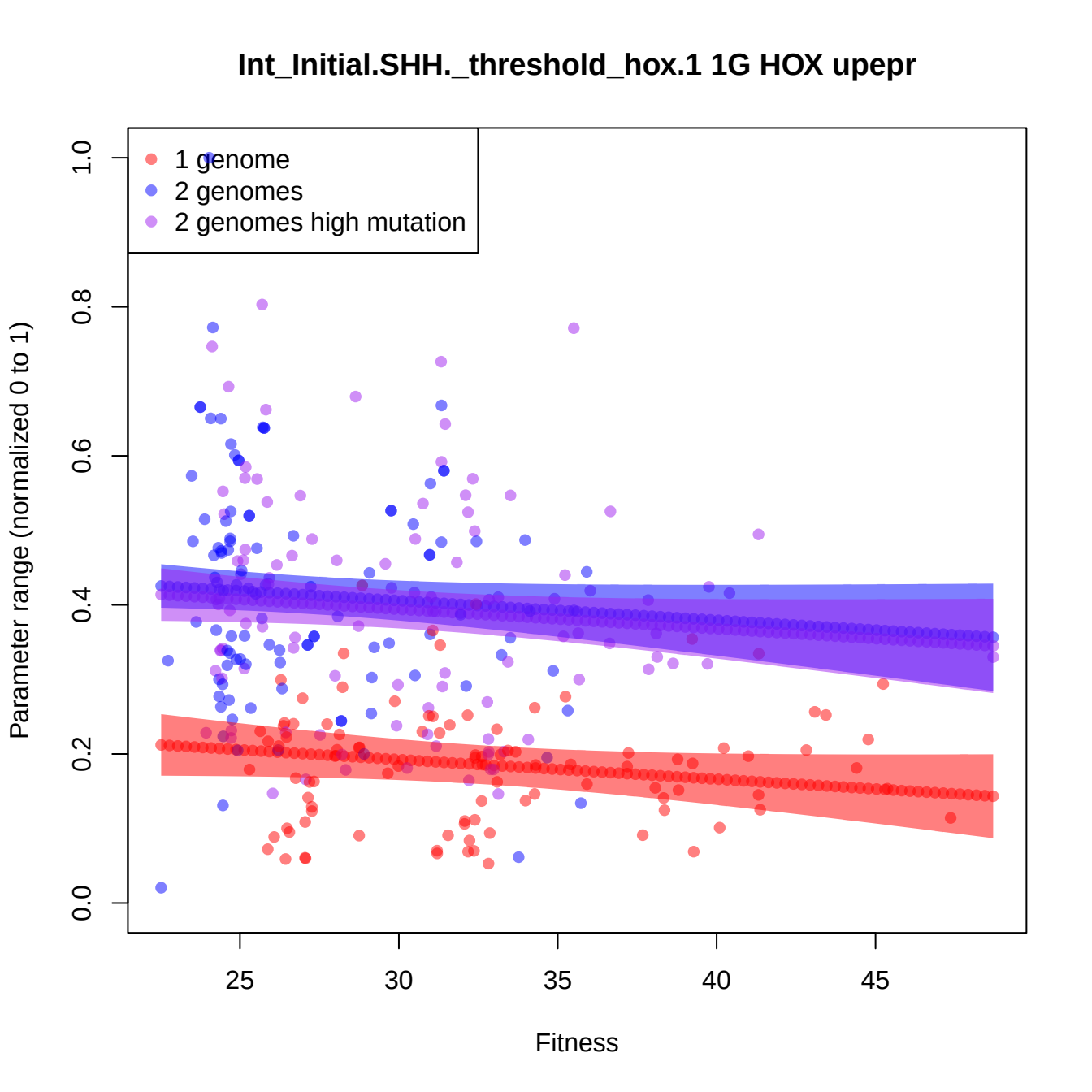

Int_Initial.SHH._threshold.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.46876236932044e−15
9.83825255012947e−40
1.49807991979946e−07
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

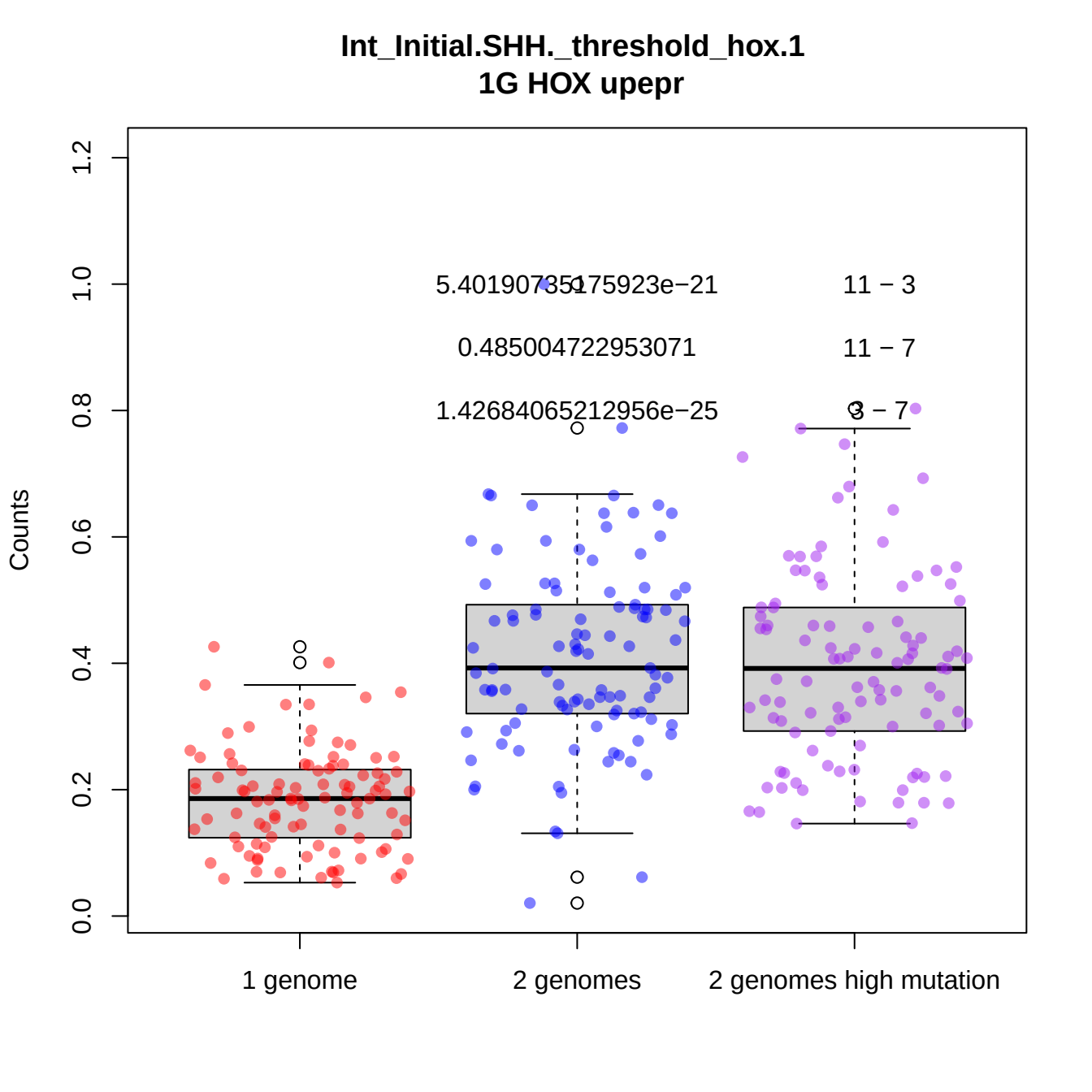

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

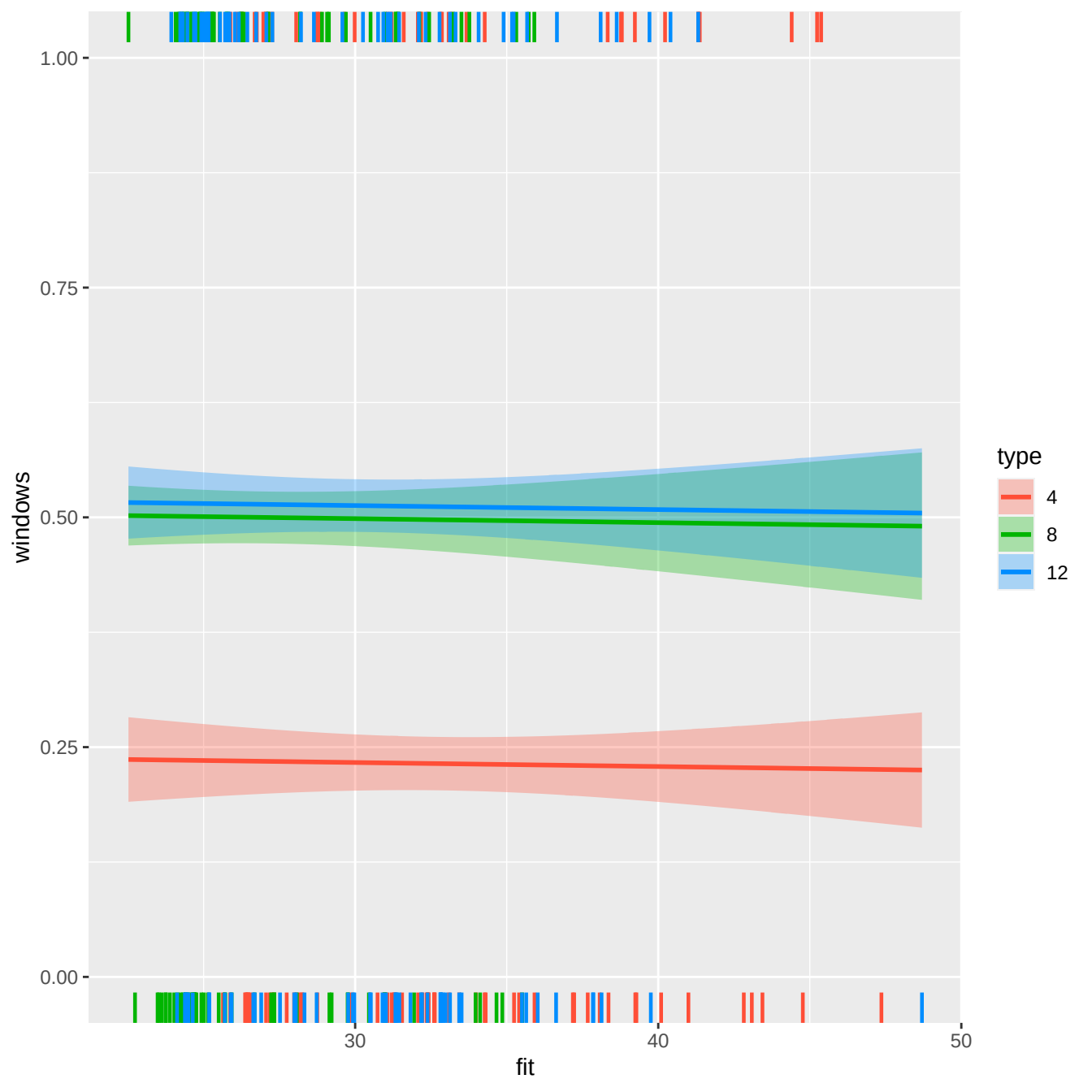

Int_Initial.SHH._threshold.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

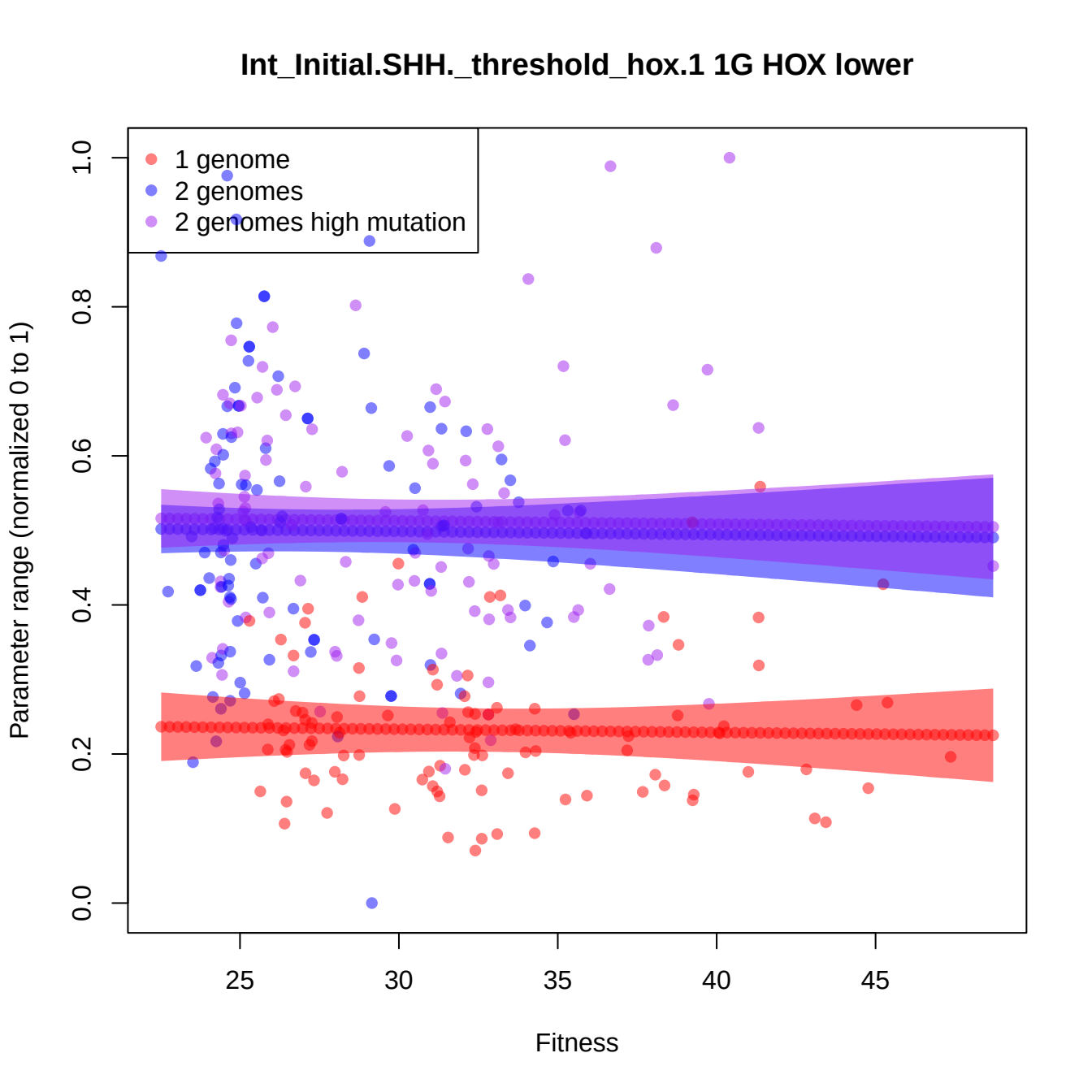

Int_Initial.SHH._threshold.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
9.35505452442765e−44
1.81321987511597e−11
5.30048382860951e−13
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

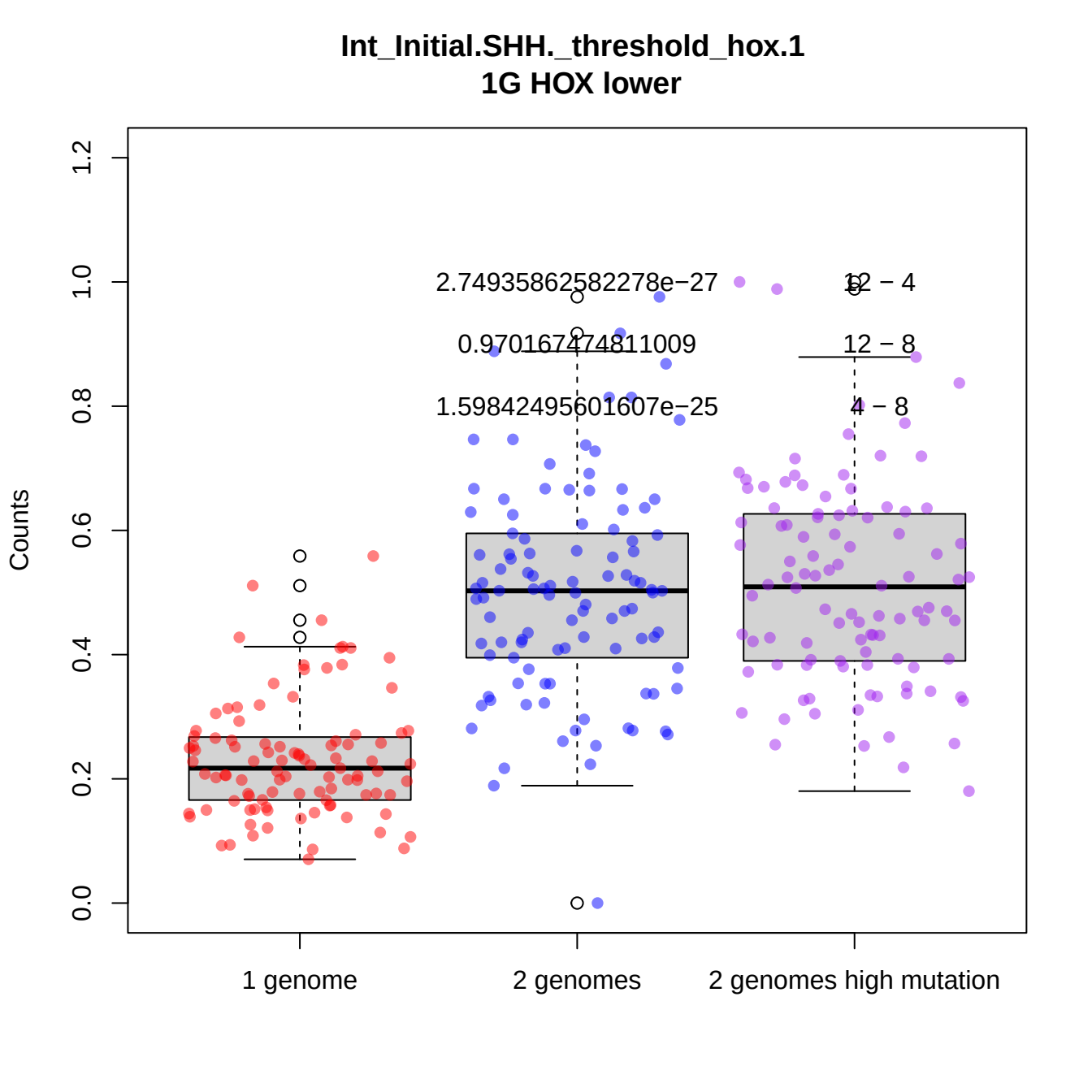

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

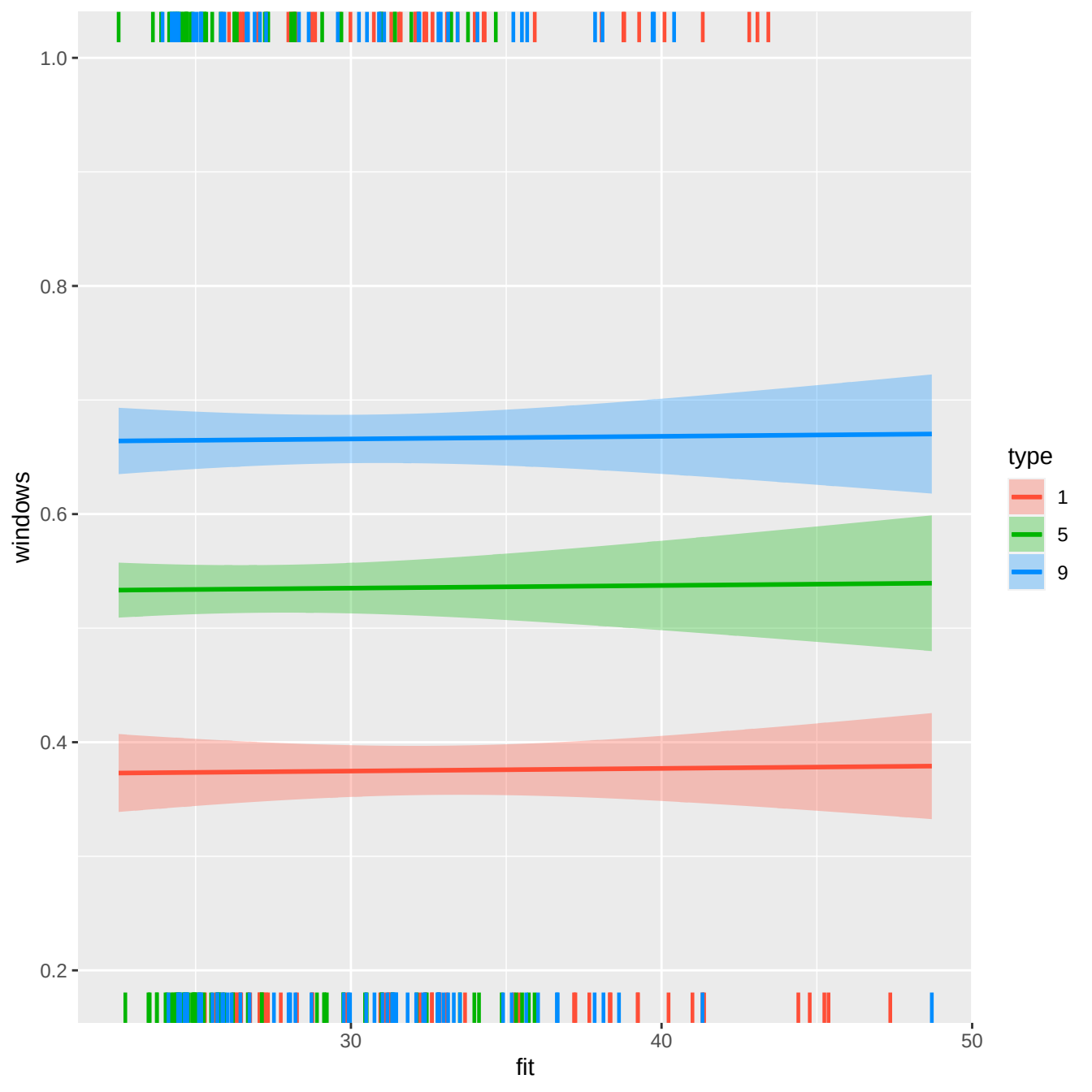

Int_Initial.SHH._threshold_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

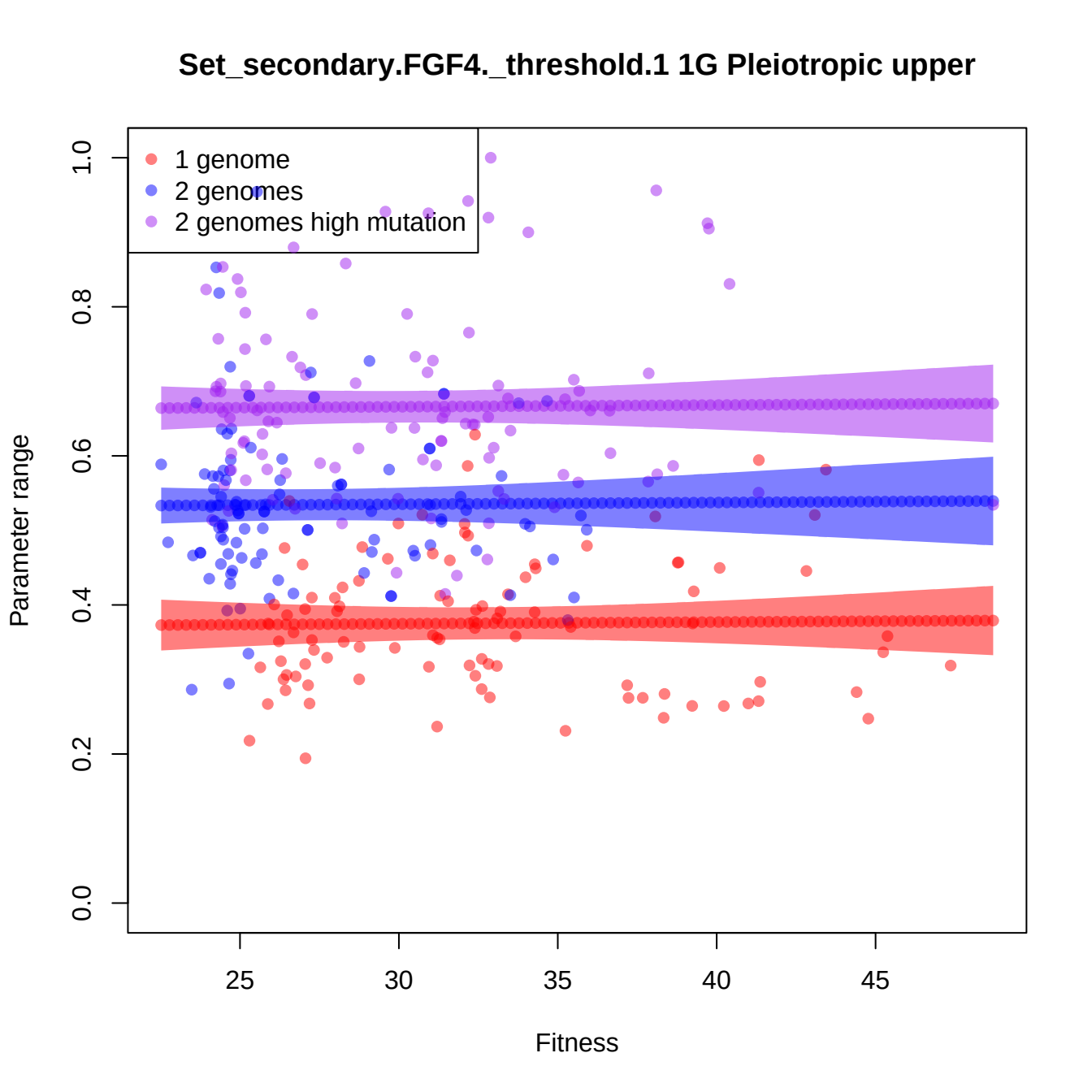

Int_Initial.SHH._threshold_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
5.401907351175923e−21
0.485004722953071
1.42684065212956e−25
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

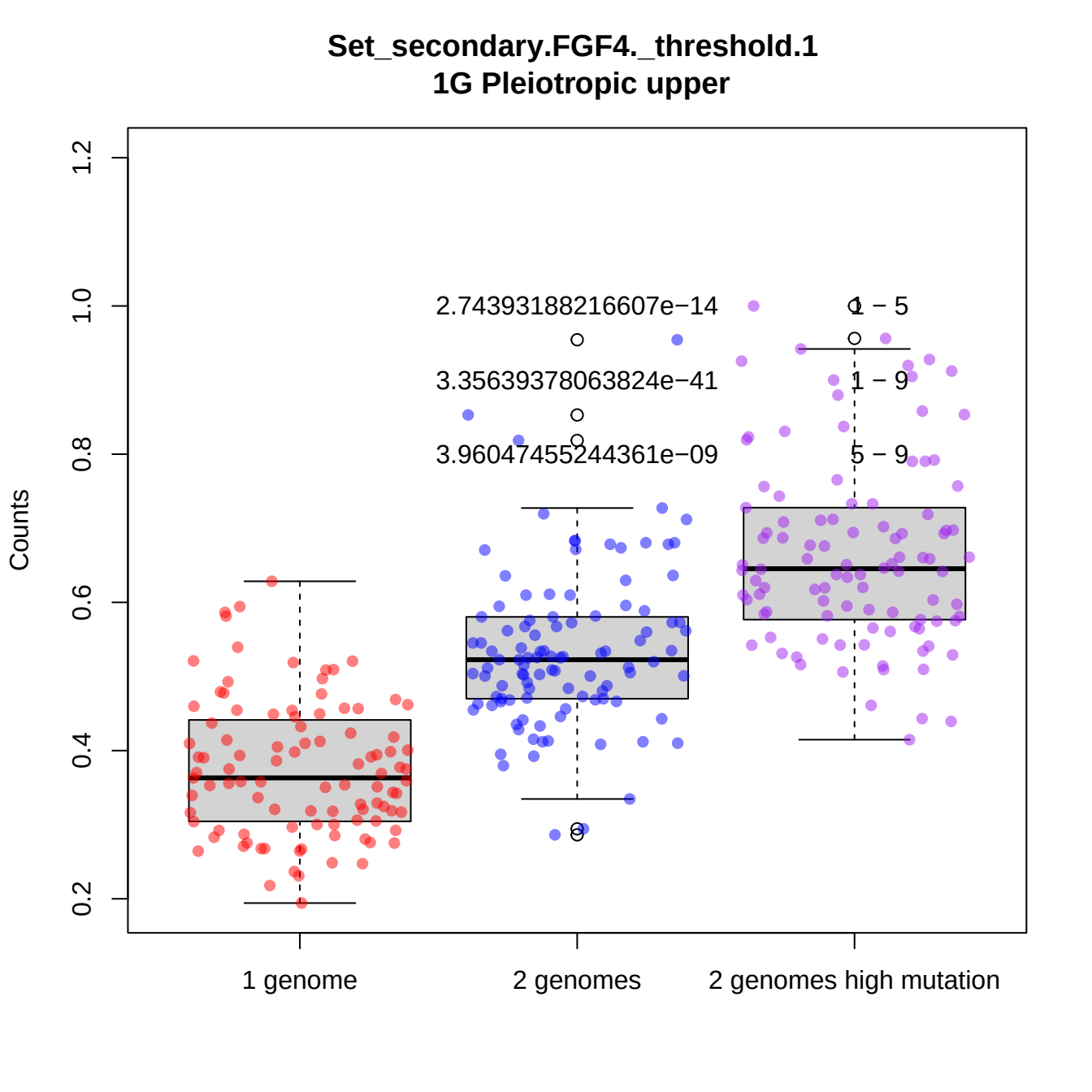

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

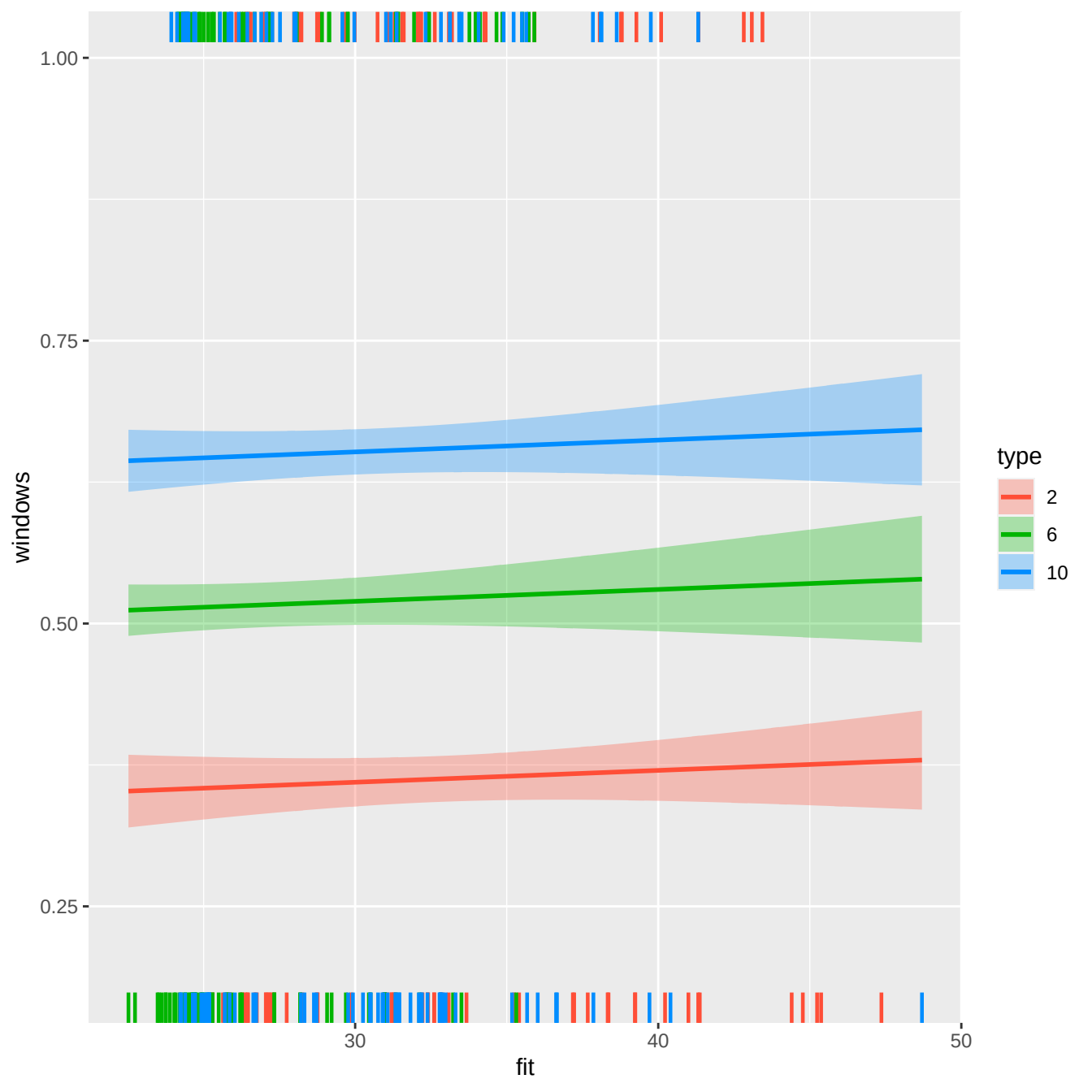

Int_Initial.SHH._threshold_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

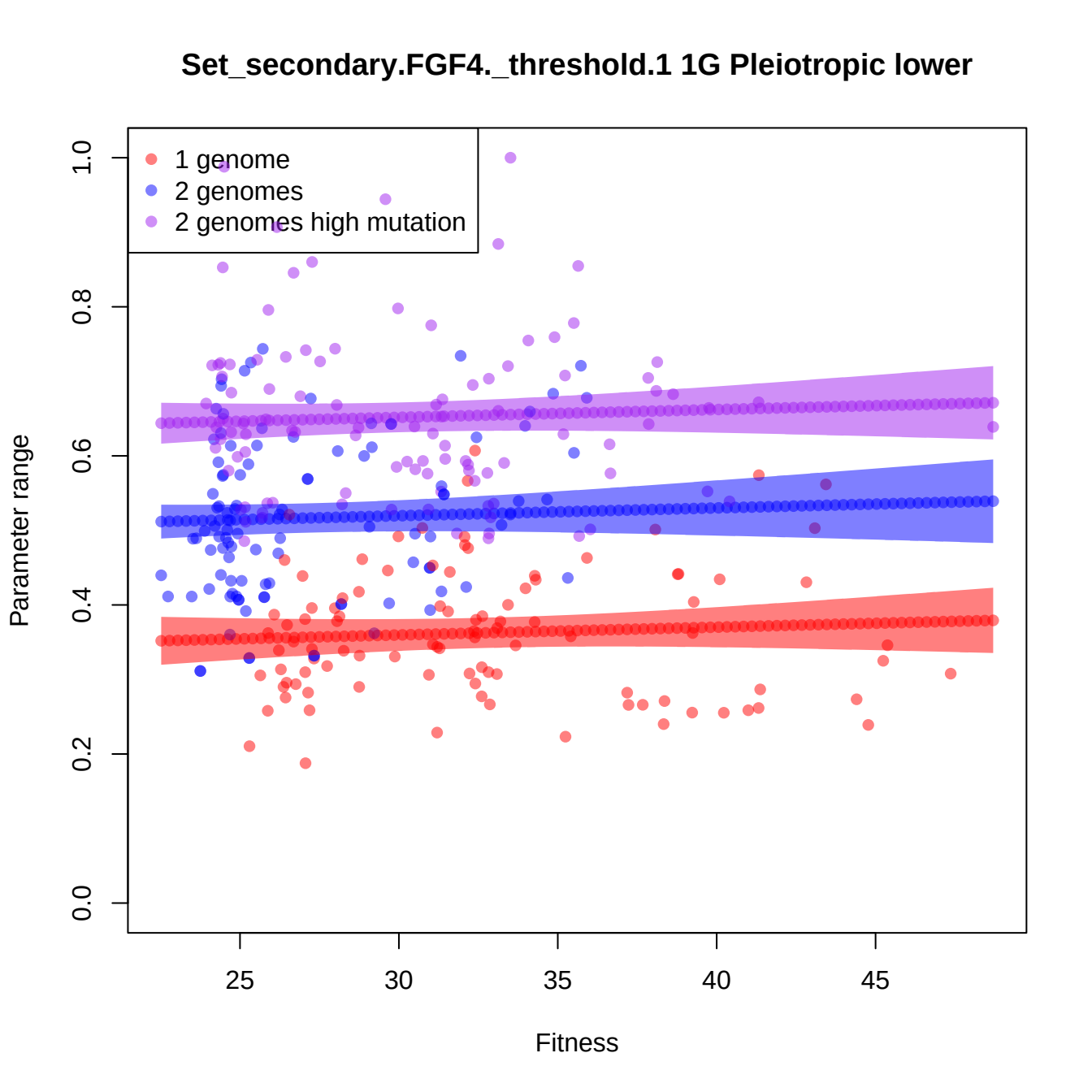

Int_Initial.SHH._threshold_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.74935862582278e−27
0.970167474811009
1.59842495601607e−25
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

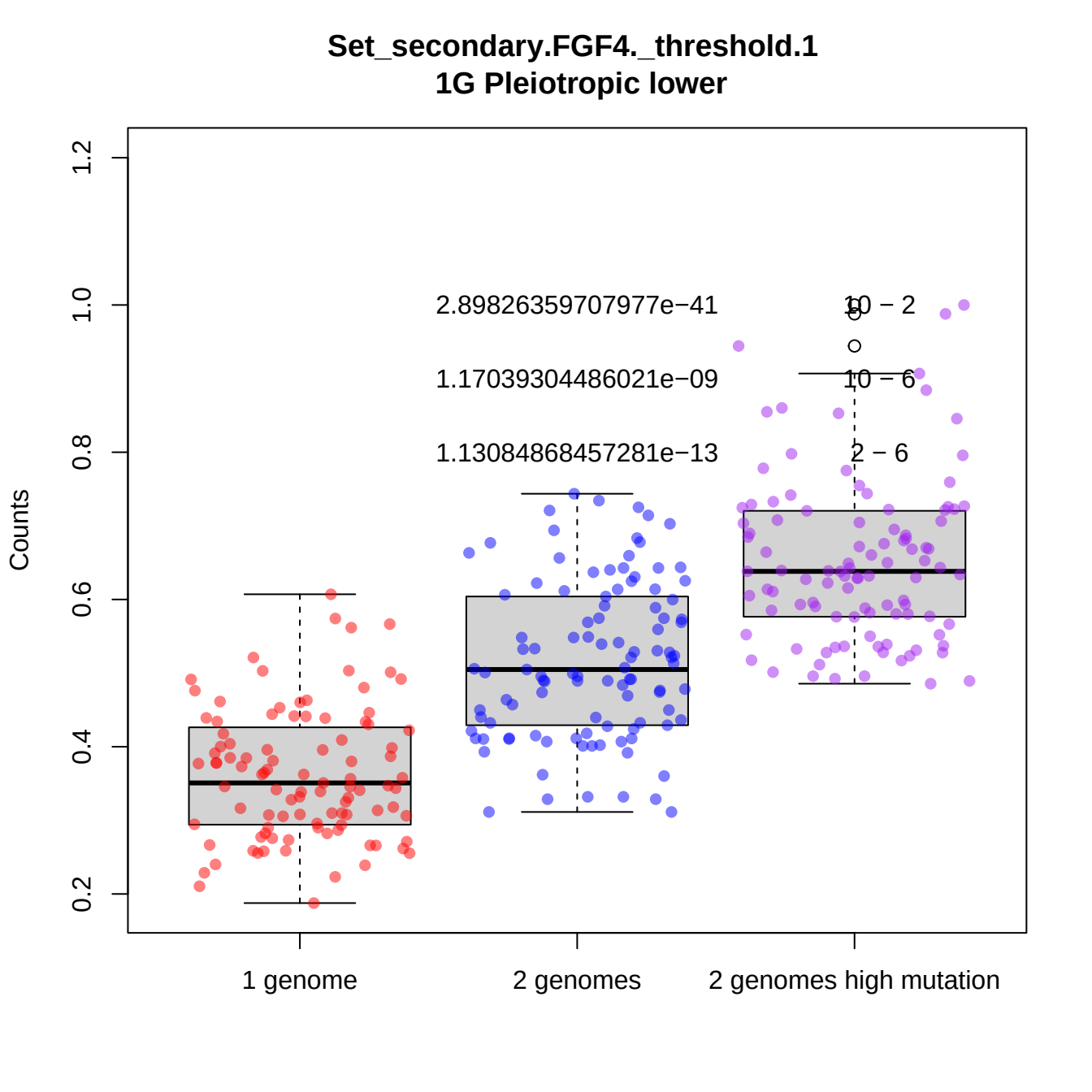

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

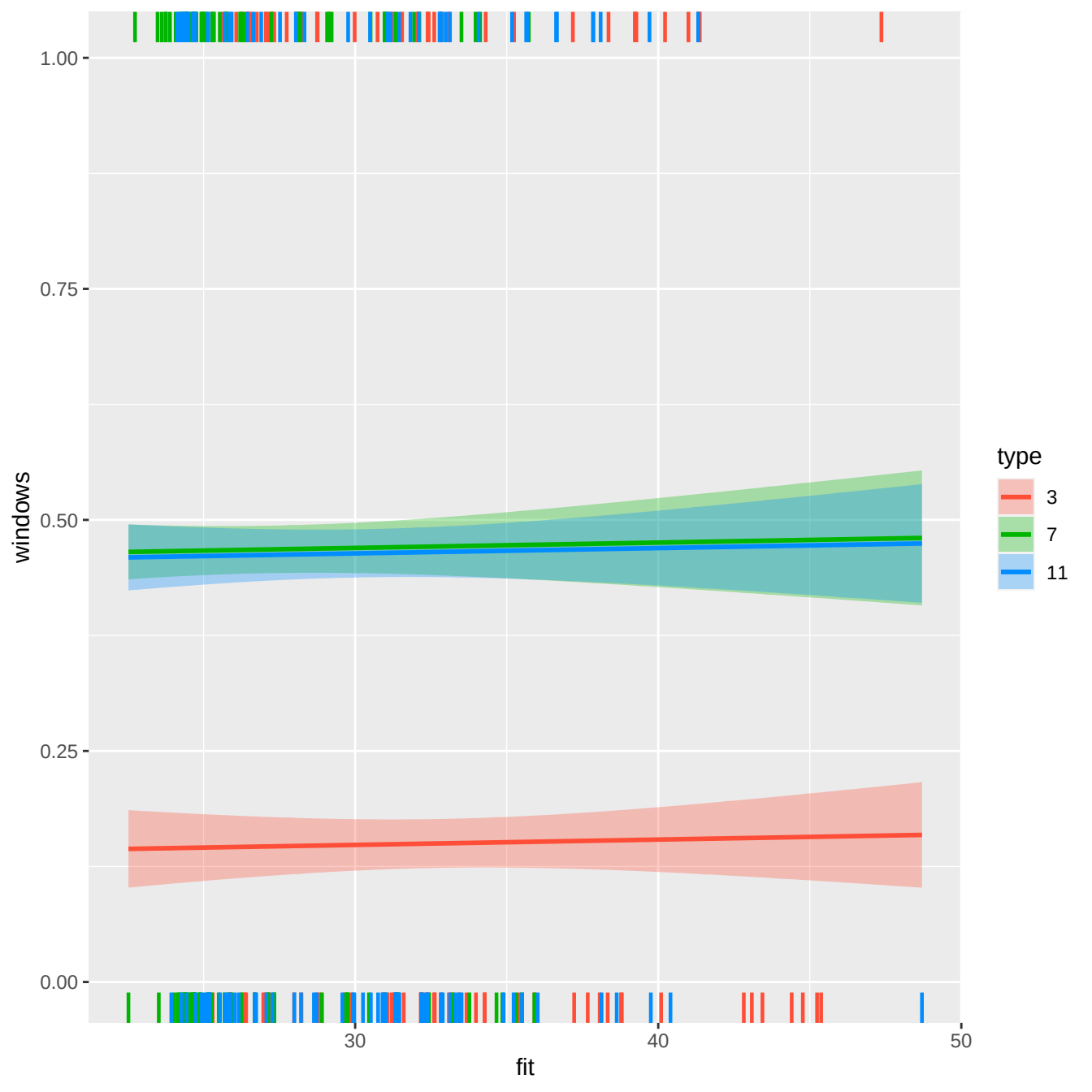

Set_secondary.FGF4._threshold.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

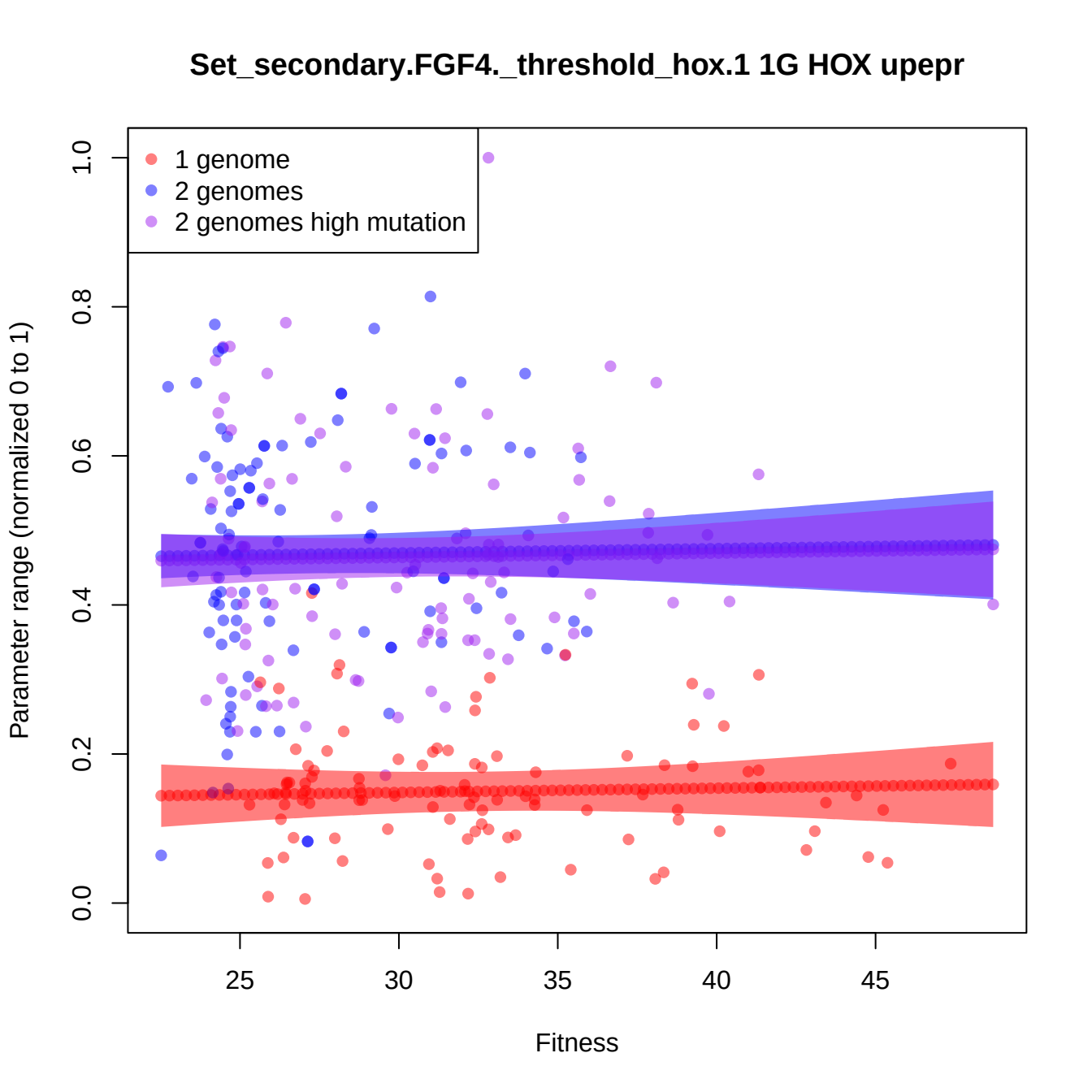

Set_secondary.FGF4._threshold.1
1G Pleiotropic upper
Counts
0.2
0.4
0.6
0.8
1.0
1.2
2.74393188216607e−14
3.35639378063824e−41
3.96047455244361e−09
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

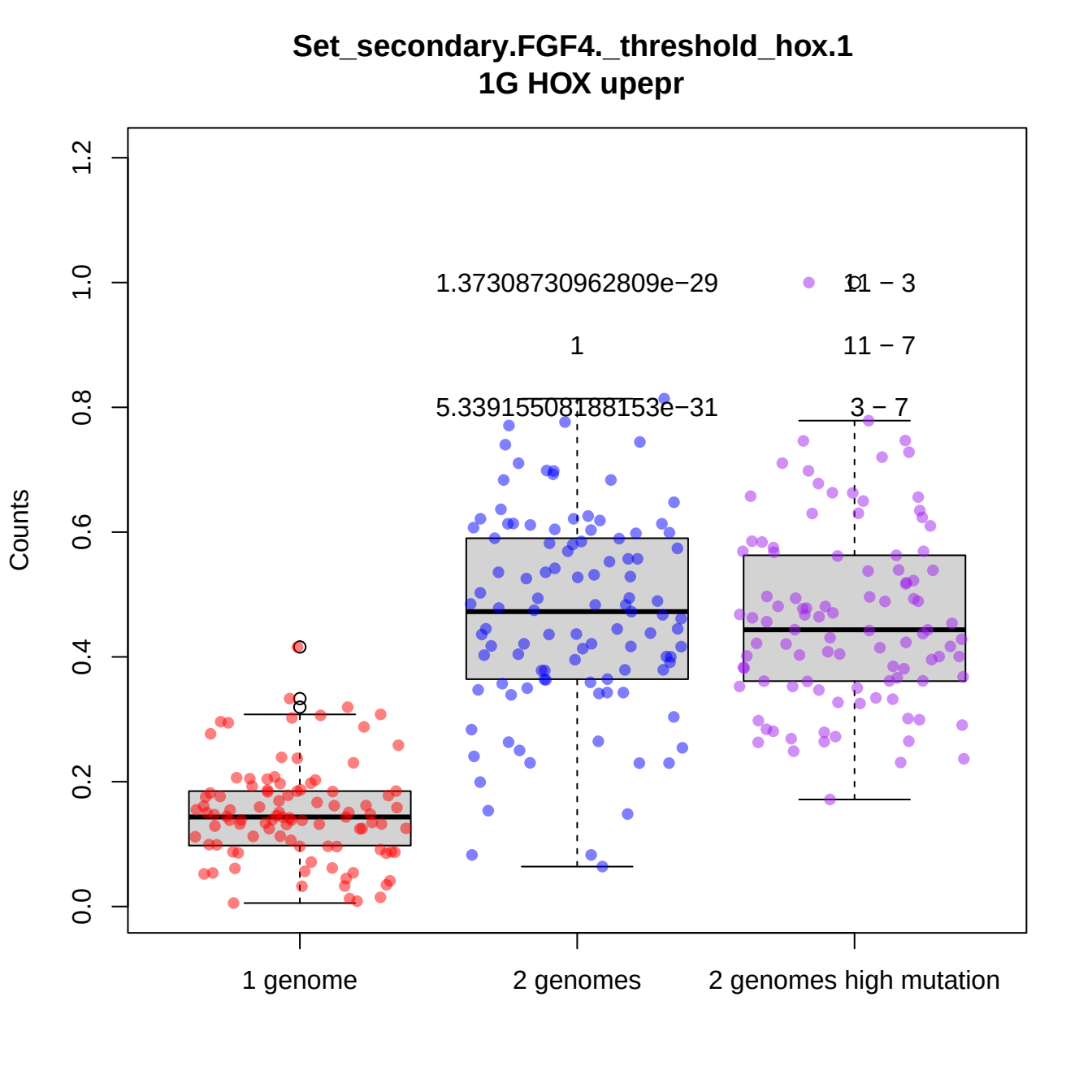

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

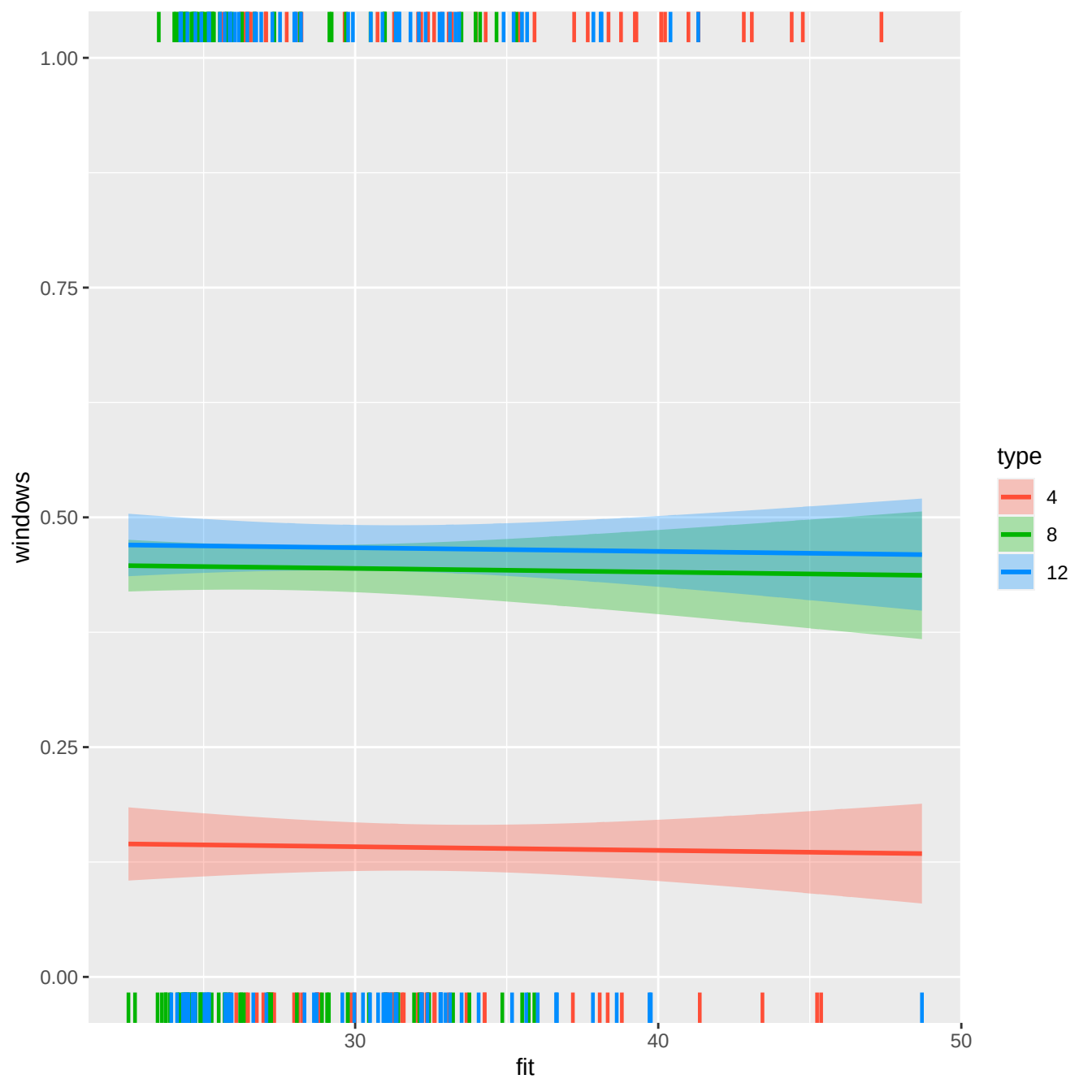

Set_secondary.FGF4._threshold.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

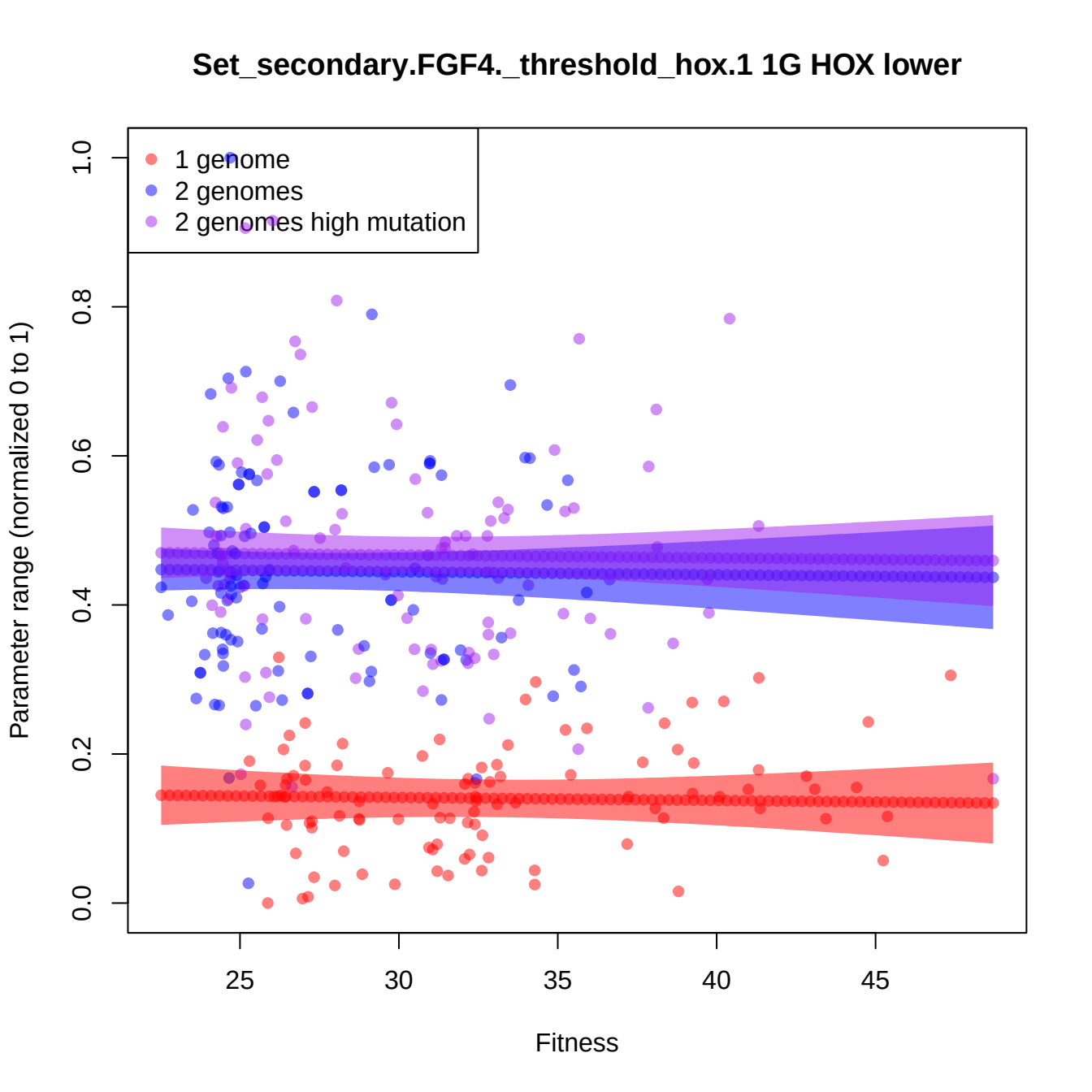

Set_secondary.FGF4._threshold.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.89826359707977e−41
1.17039304486021e−09
1.13084868457281e−13
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

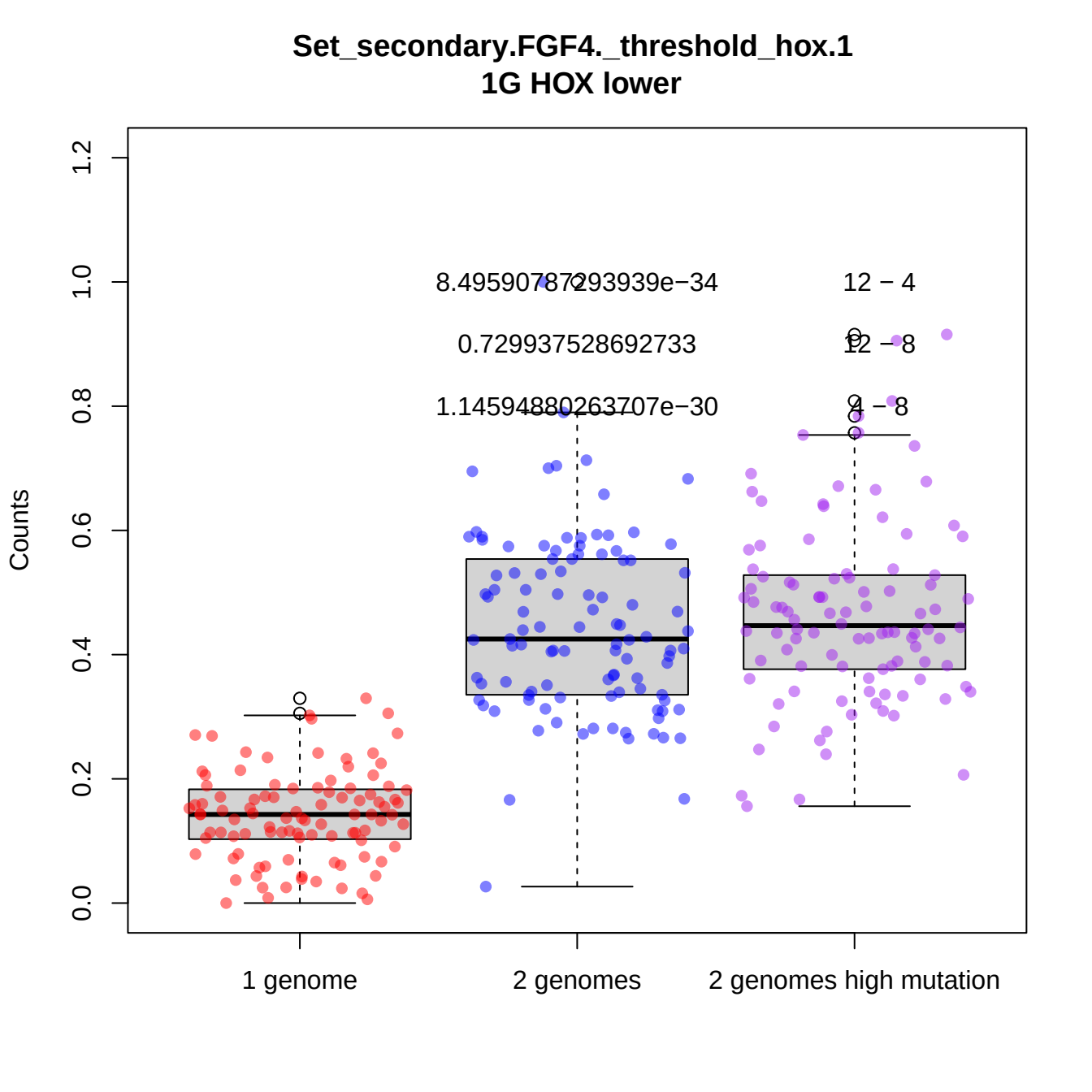

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

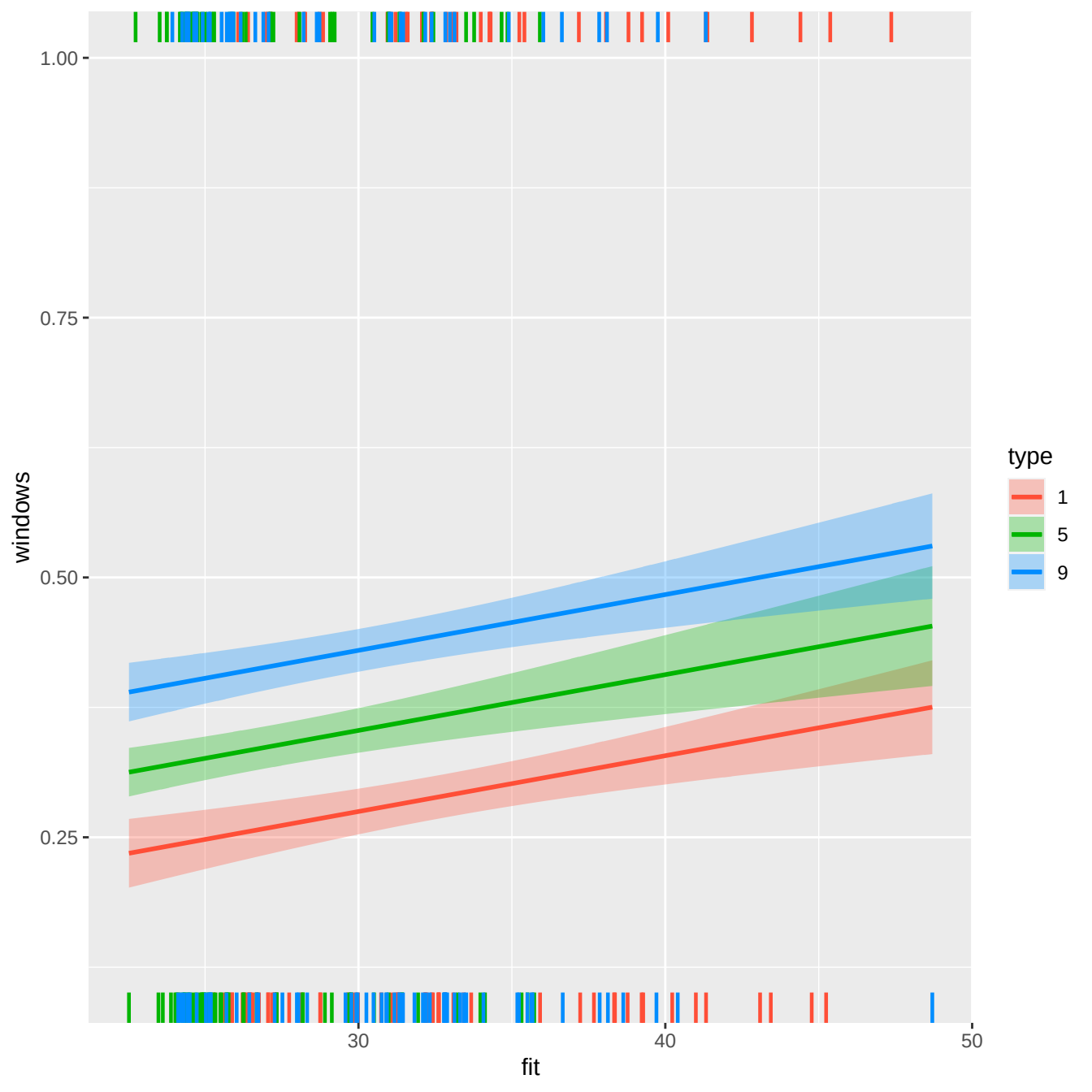

Set_secondary.FGF4._threshold_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

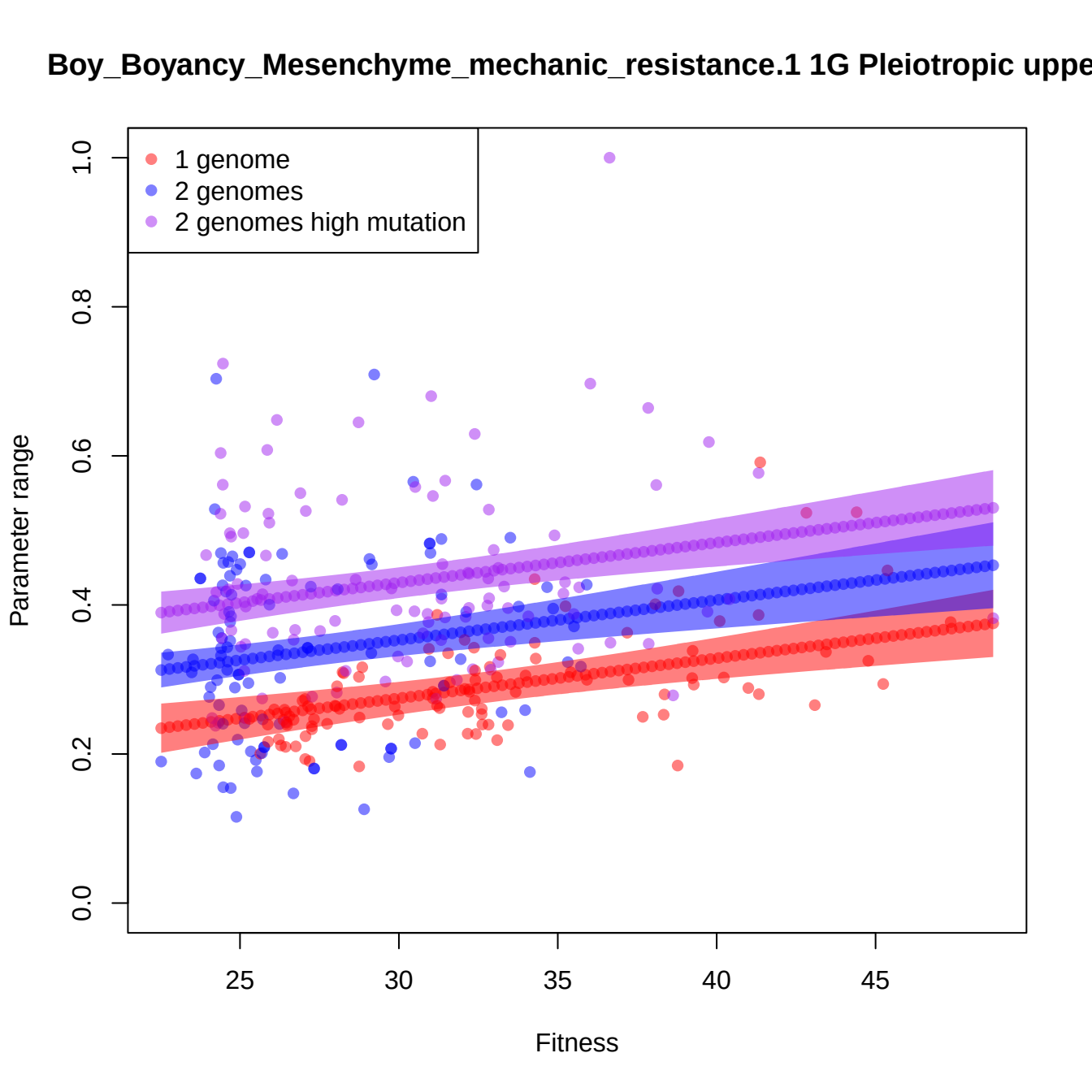

Set_secondary.FGF4._threshold_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.37308730962809e−29
1
5.33915508188153e−31
1 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

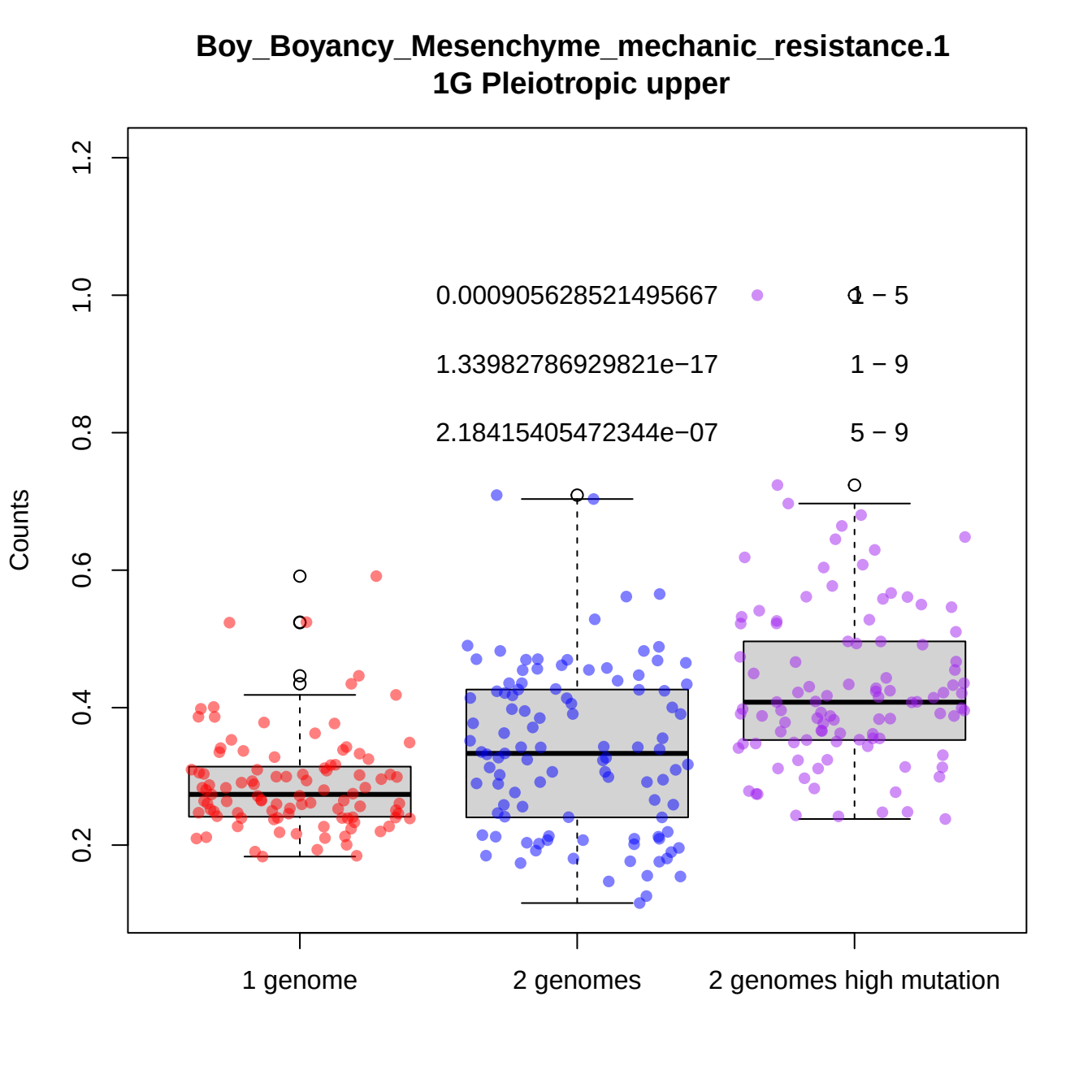

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

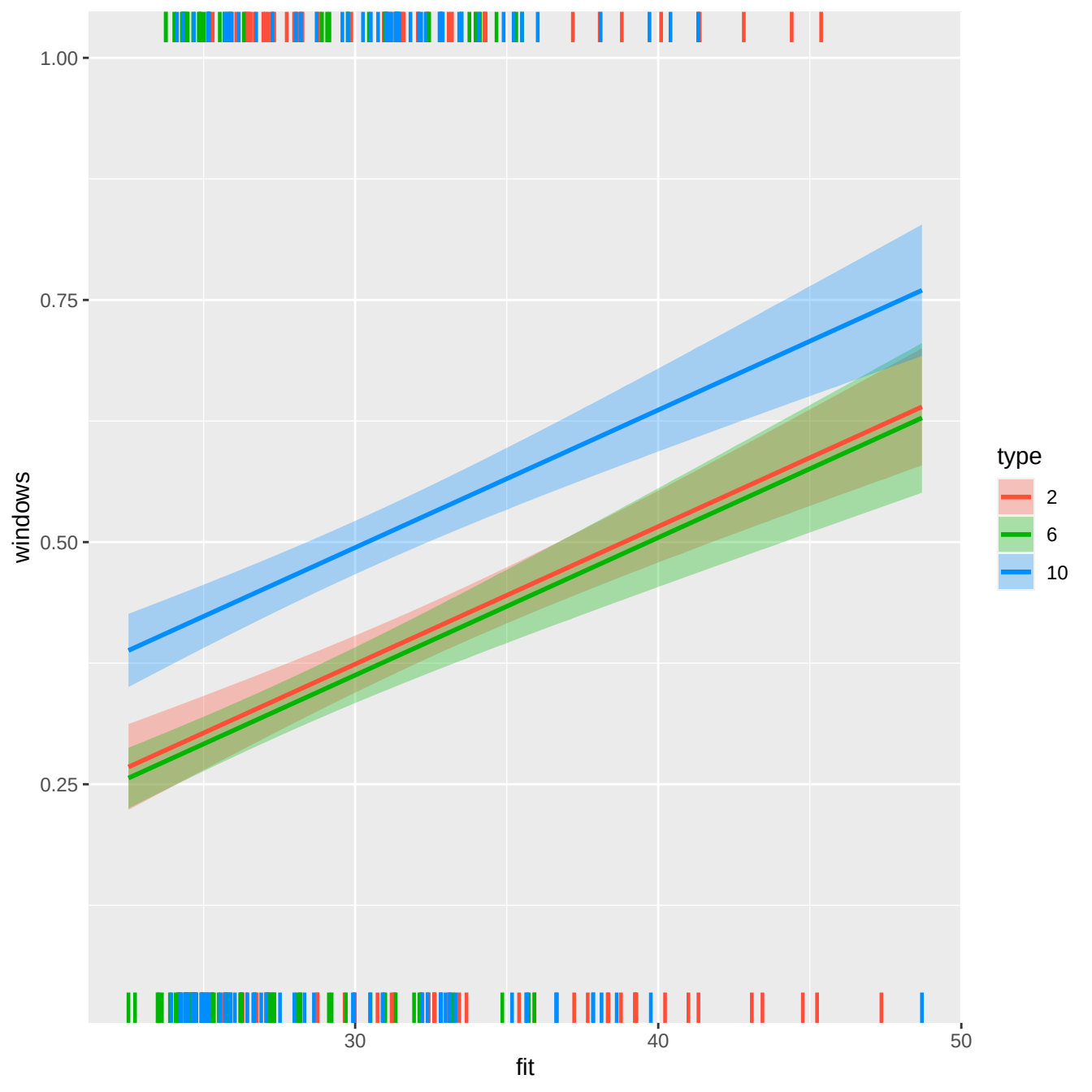

Set_secondary.FGF4._threshold_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

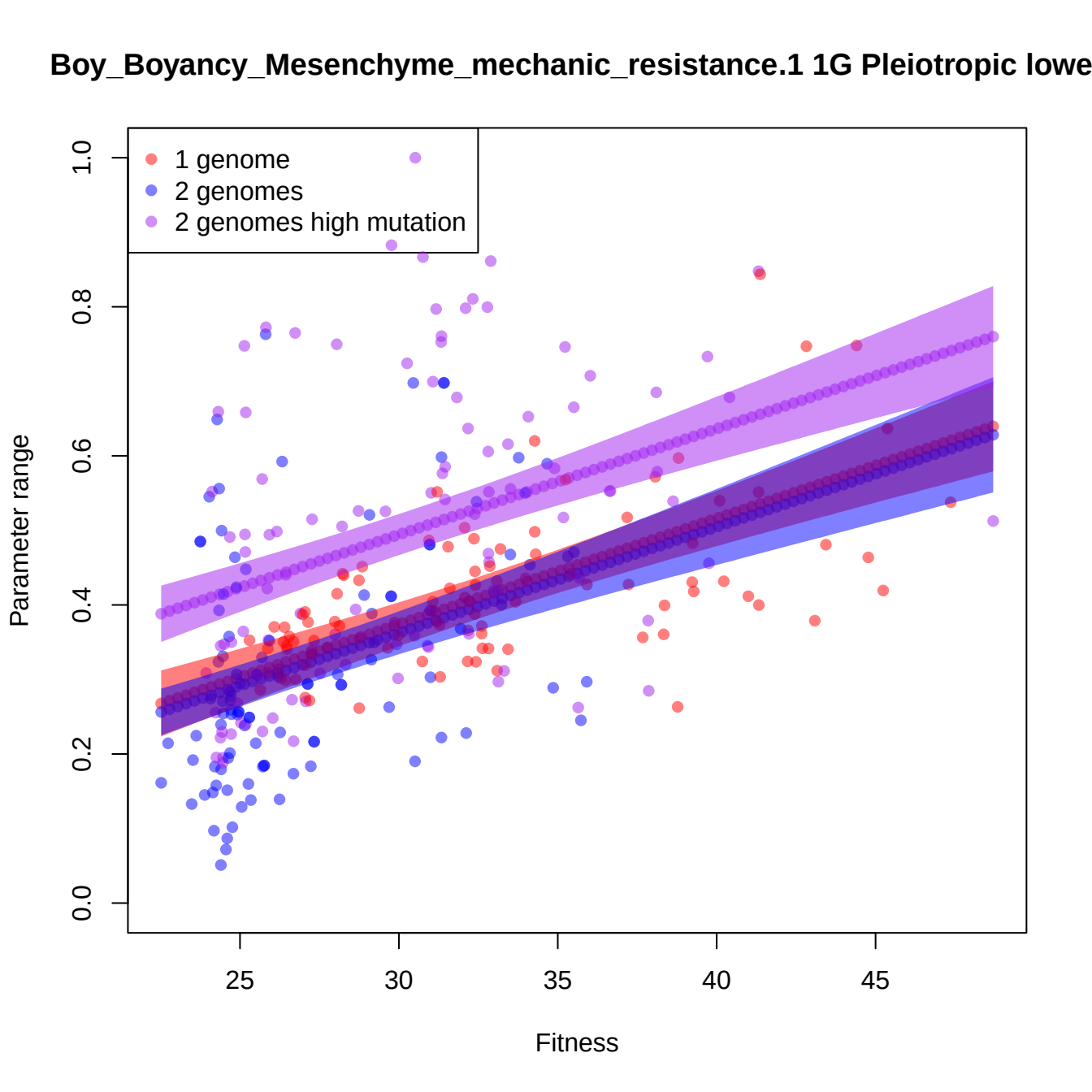

Set_secondary.FGF4._threshold_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
8.49590787293939e−34
0.729937528692733
1.14594880263707e−30
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

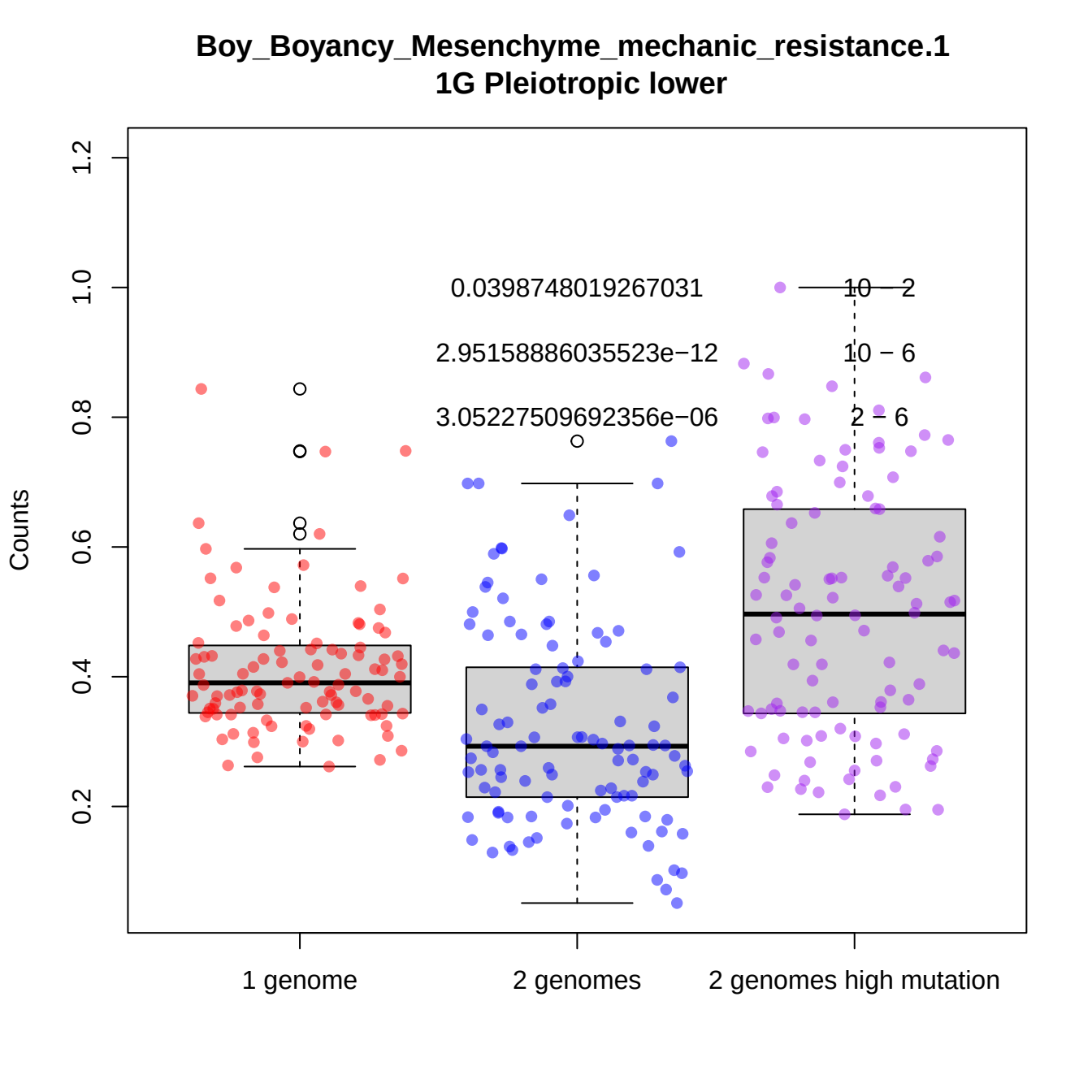

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

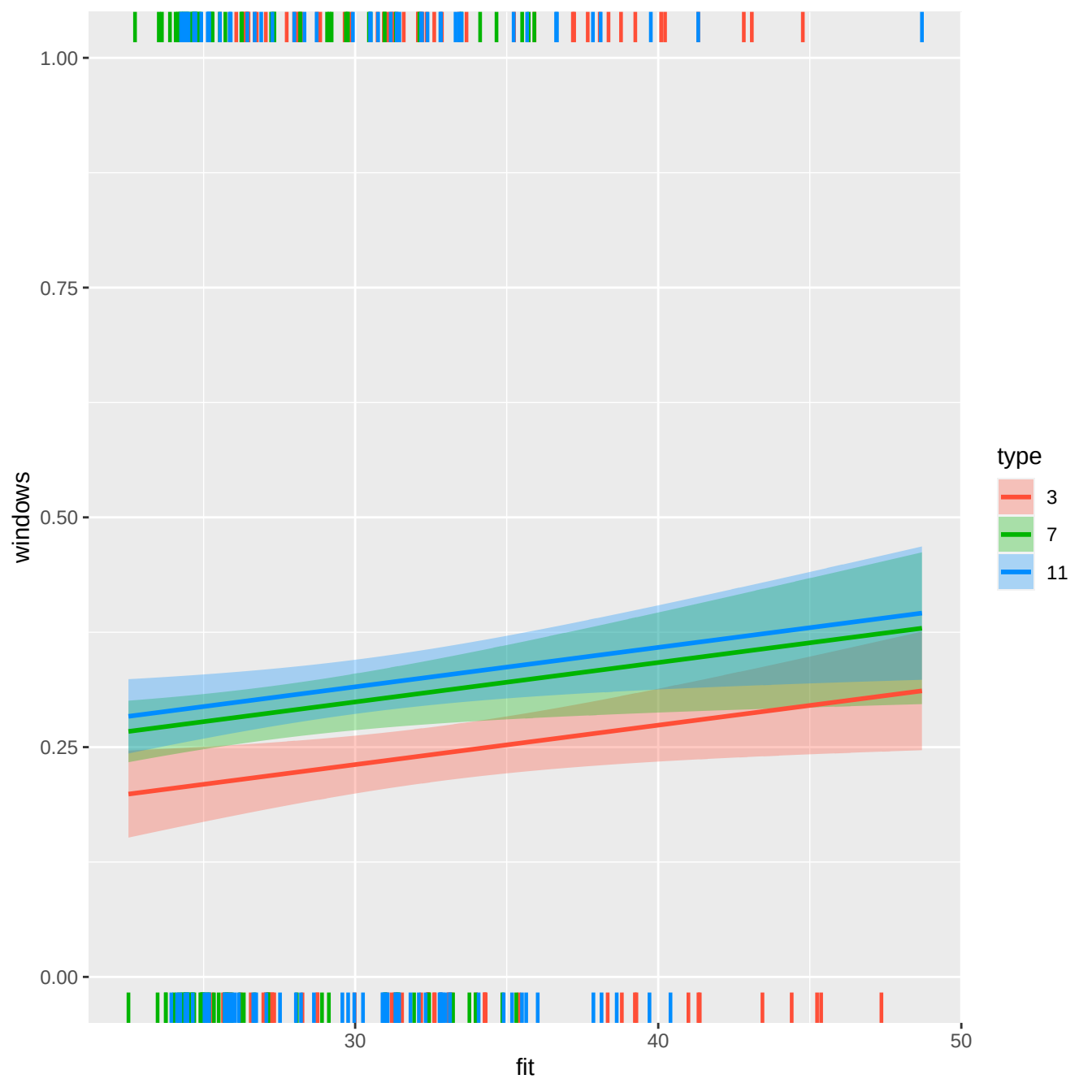

Boy_Boyancy_Mesenchyme_mechanic_resistance.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

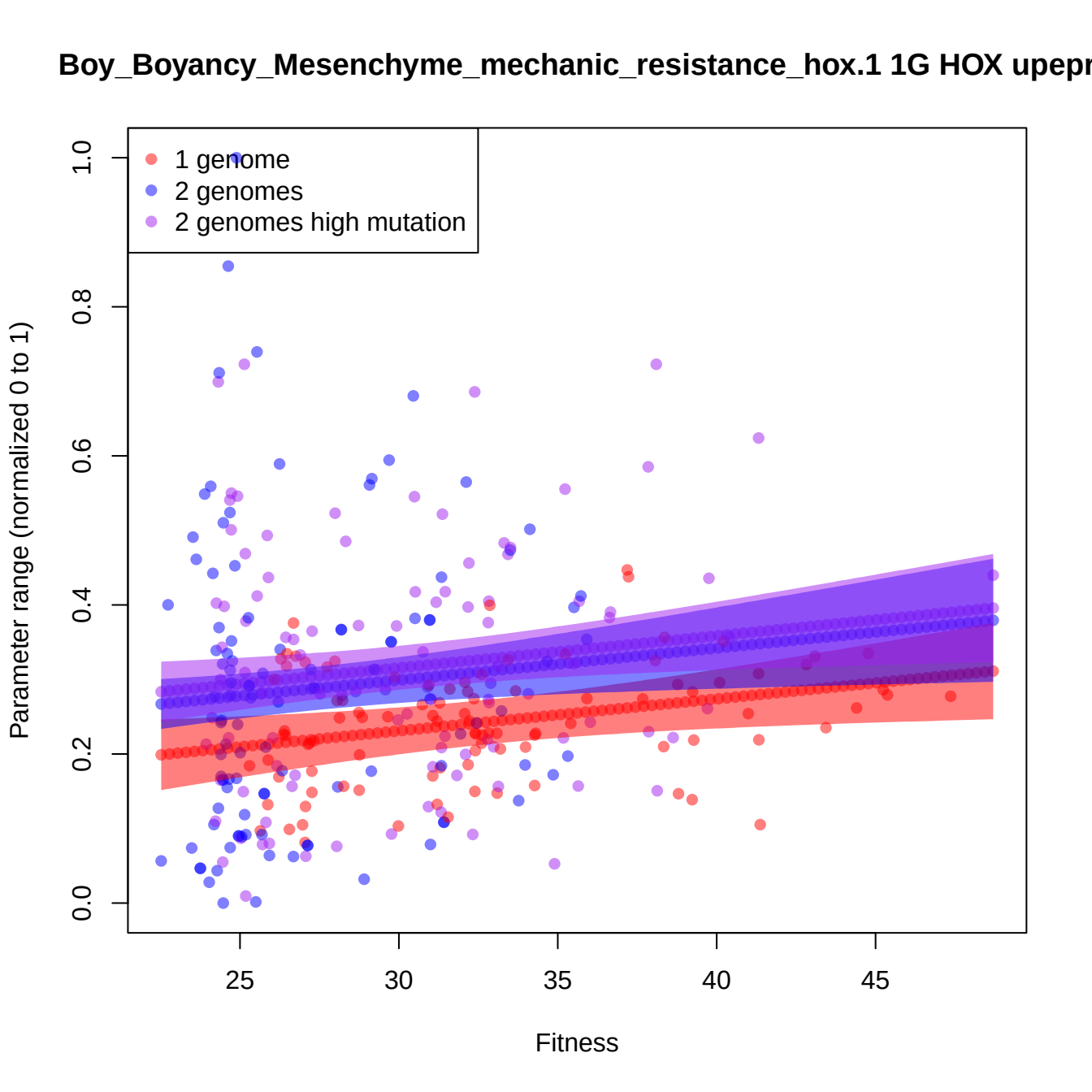

Boy_Boyancy_Mesenchyme_mechanic_resistance.1
1G Pleiotropic upper
Counts
0.2
0.4
0.6
0.8
1.0
1.2
0.000905628521495667
1 − 5
1.33982786929821e−17
1 − 9
2.18415405472344e−07
5 − 9
1 genome
2 genomes
2 genomes high mutation

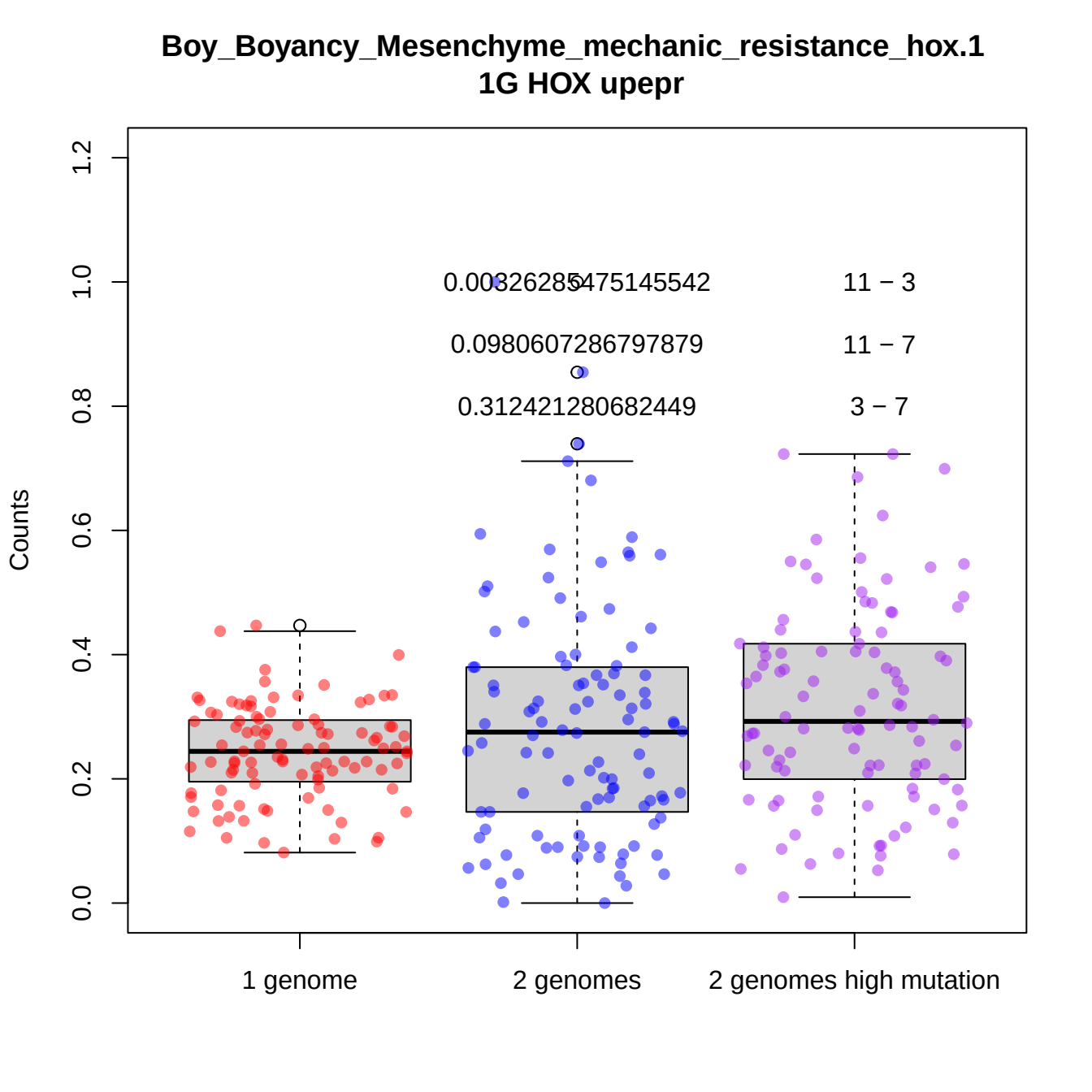

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

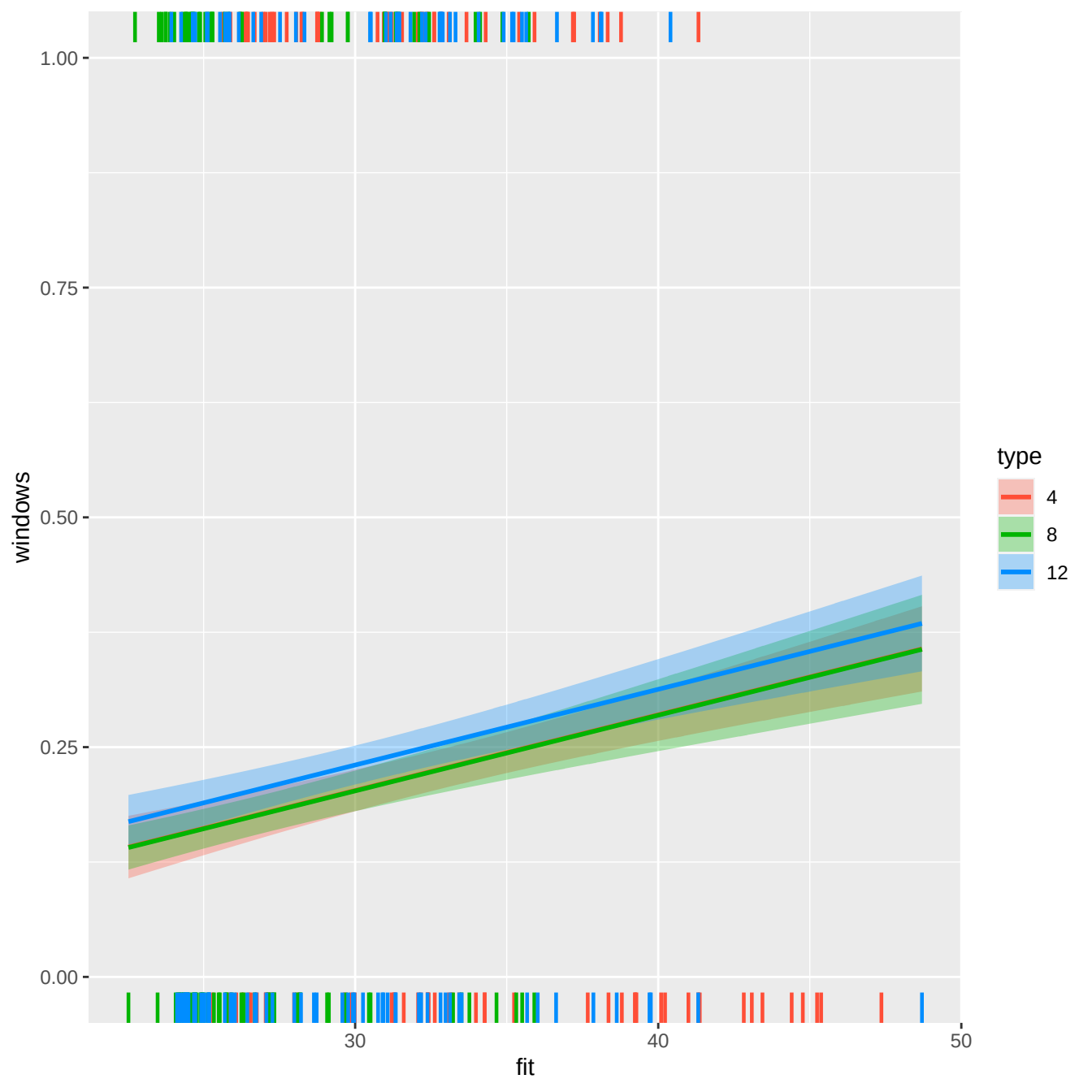

Boy_Boyancy_Mesenchyme_mechanic_resistance.1 1G Pleiotropic lowe
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

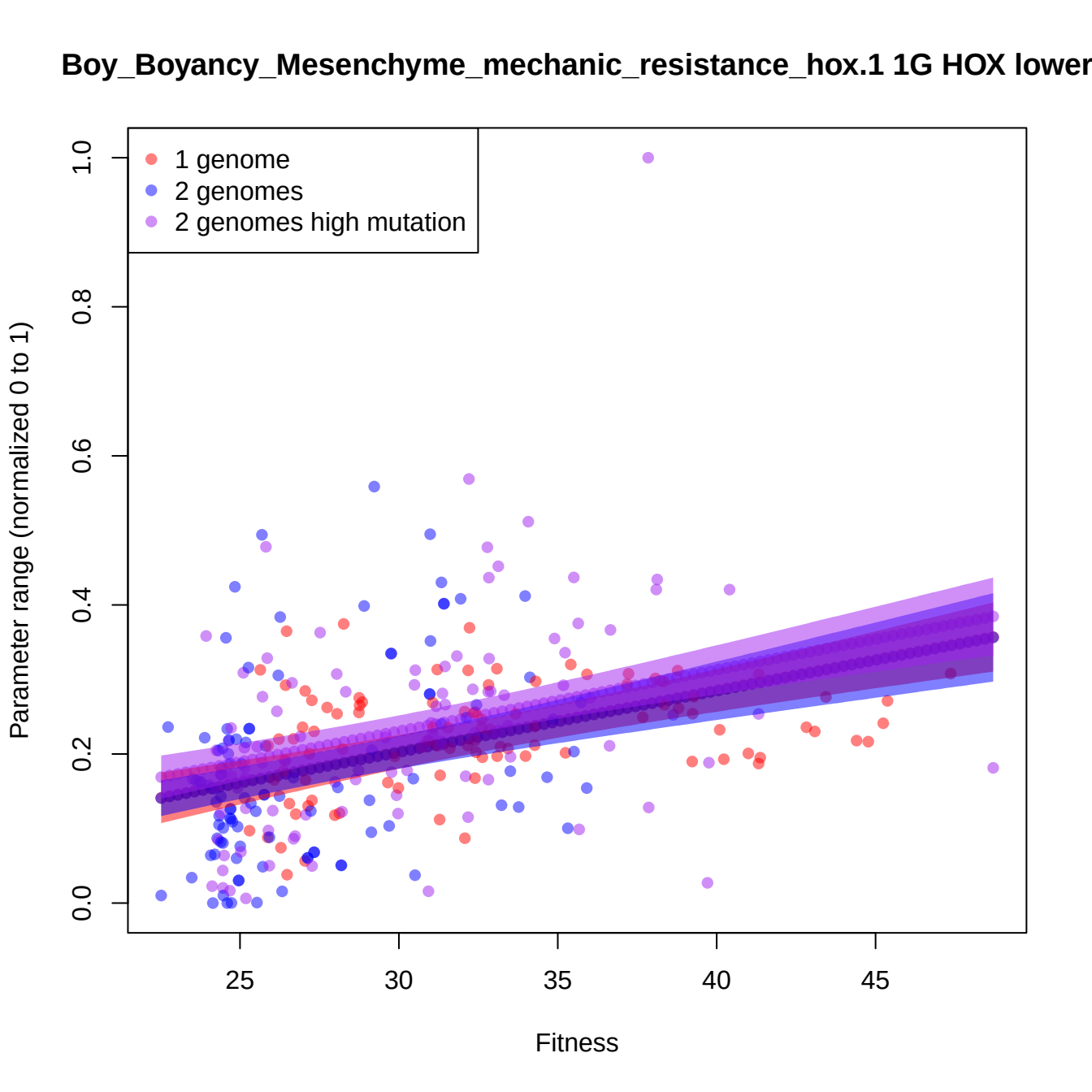

Boy_Boyancy_Mesenchyme_mechanic_resistance.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0398748019267031
2.95158886035523e−12
3.05227509692356e−06
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

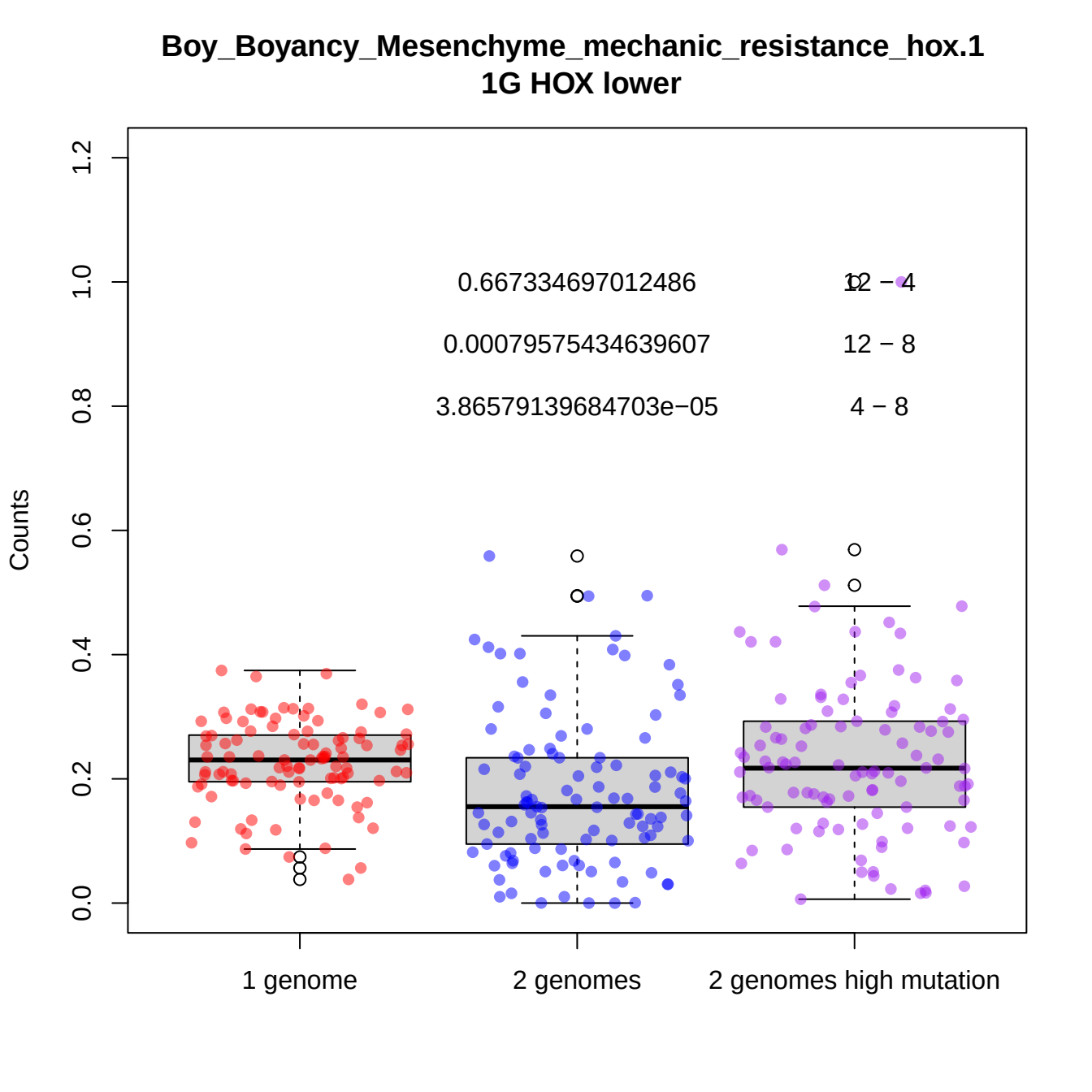

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

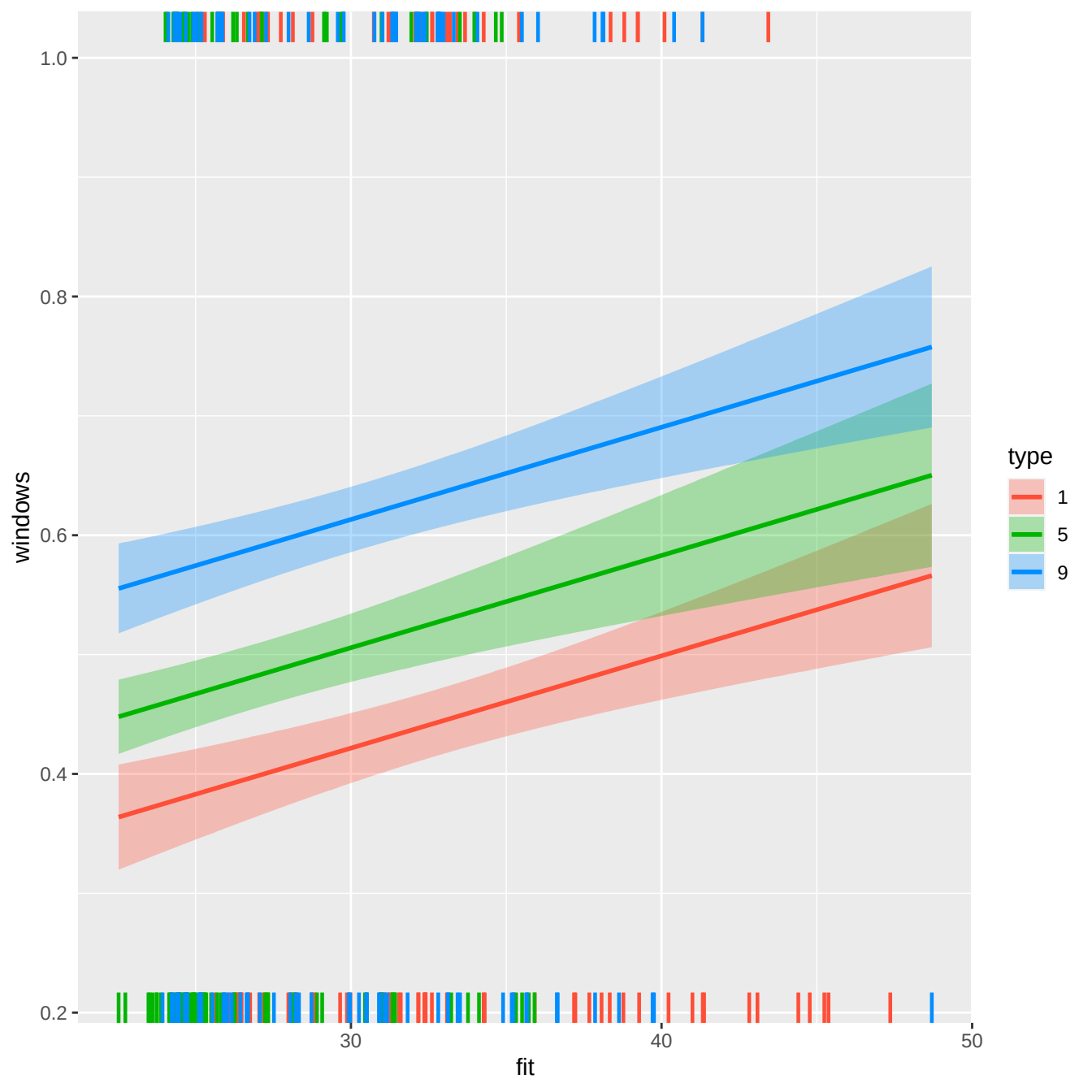

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

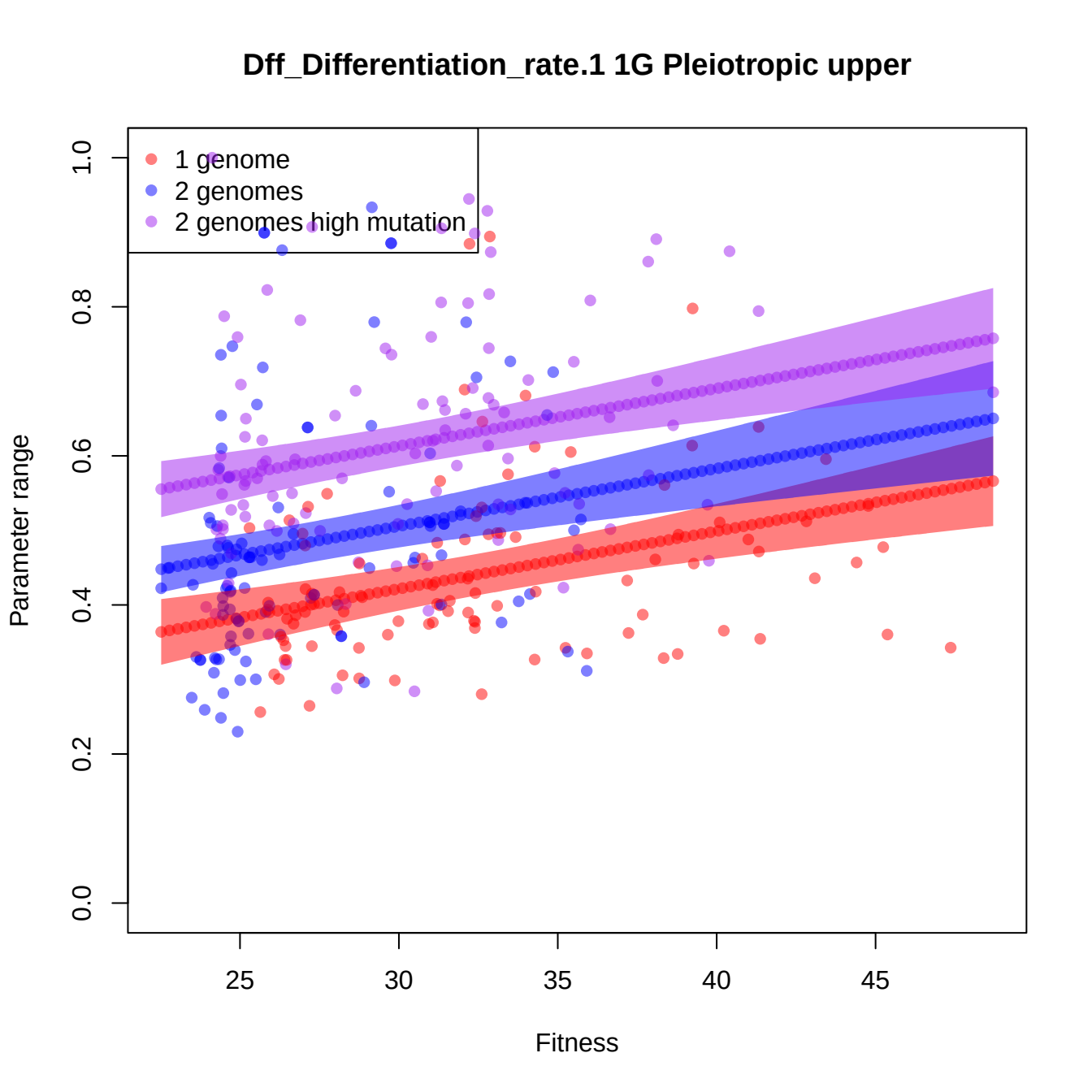

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.0032628547514554 2
0.0980607286797879
0.312421280682449
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

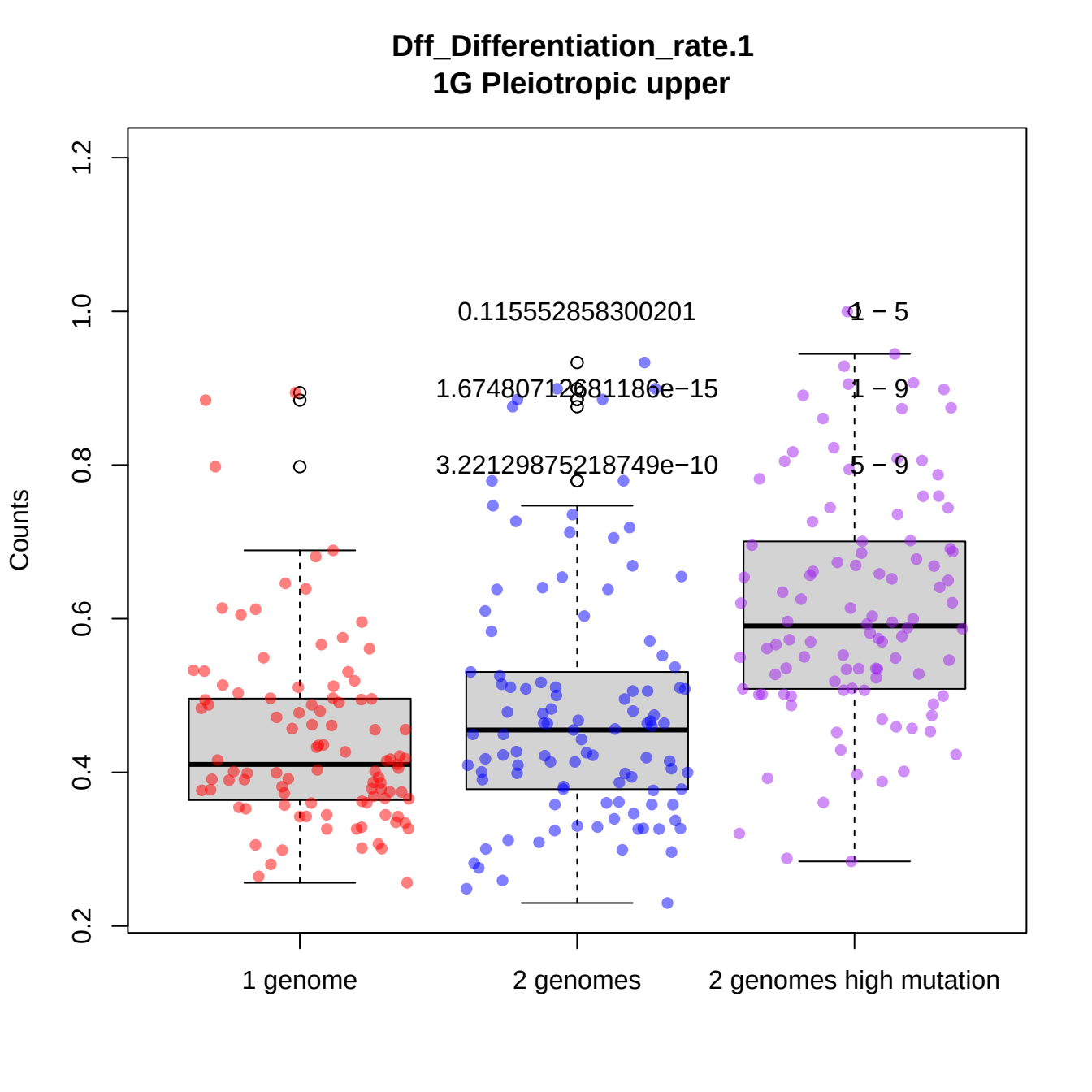

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

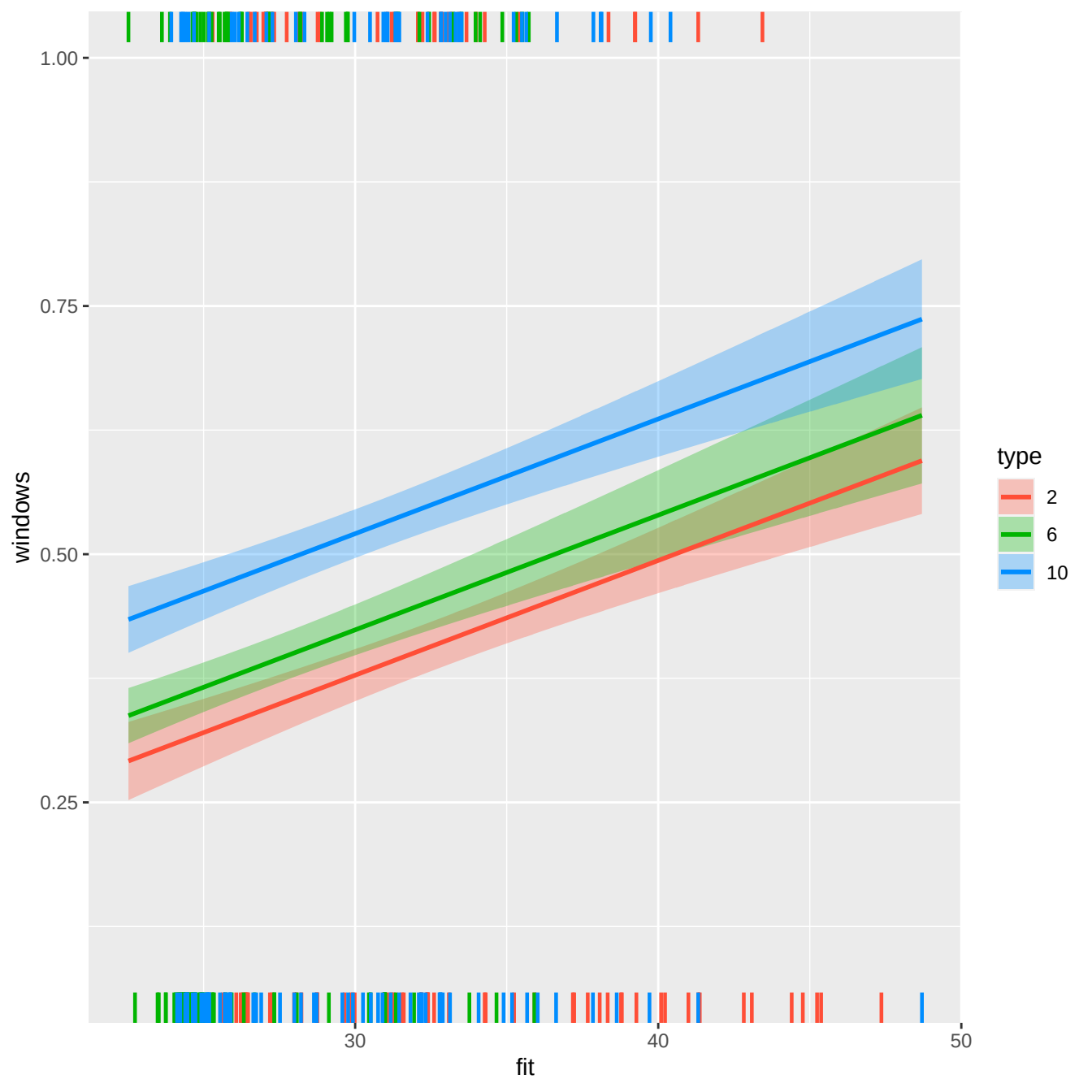

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

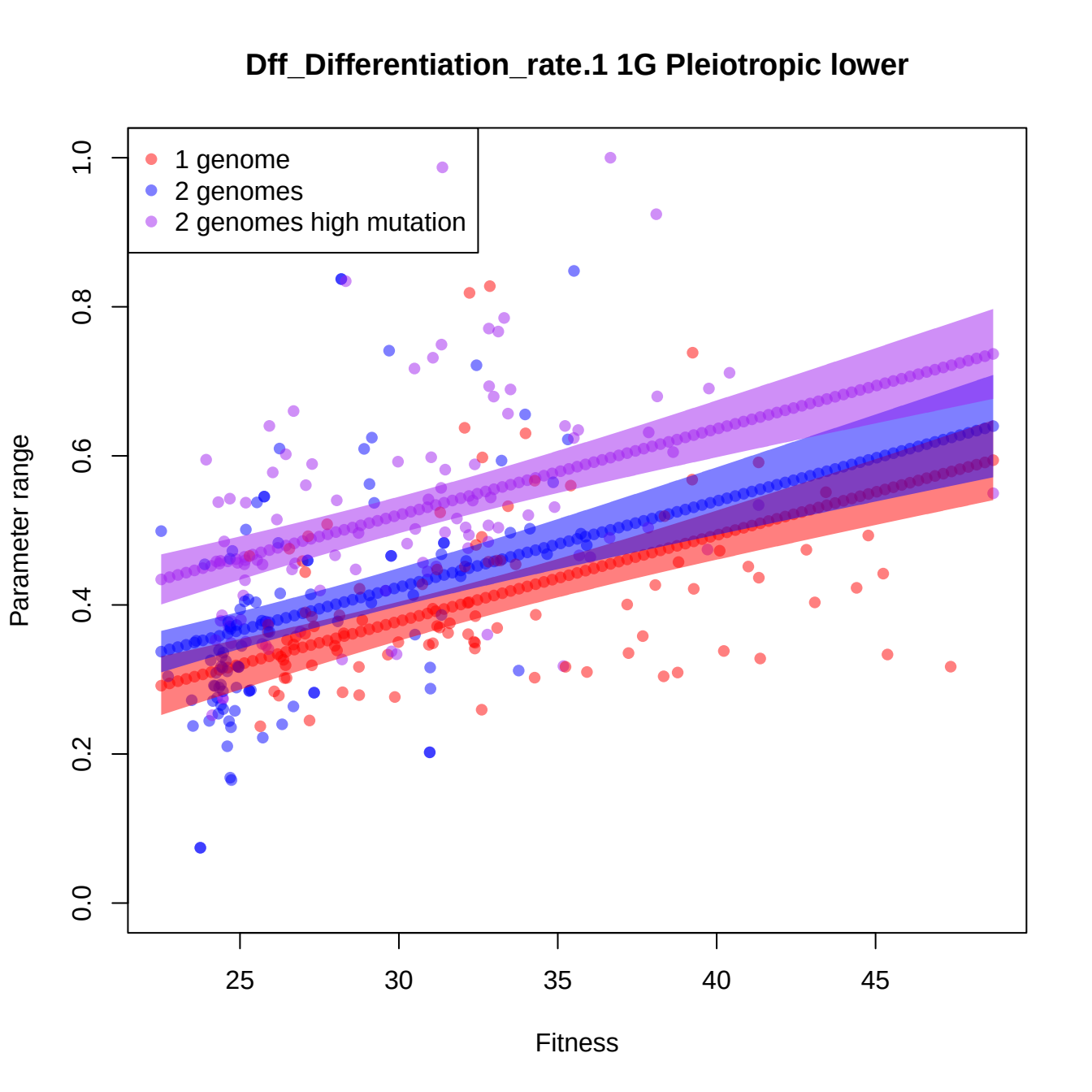

Boy_Boyancy_Mesenchyme_mechanic_resistance_hox.1
1G HOX lower
0.667334697012486
12 − 4
0.00079575434639607
12 − 8
3.86579139684703e−05
4 − 8
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1 genome
2 genomes
2 genomes high mutation

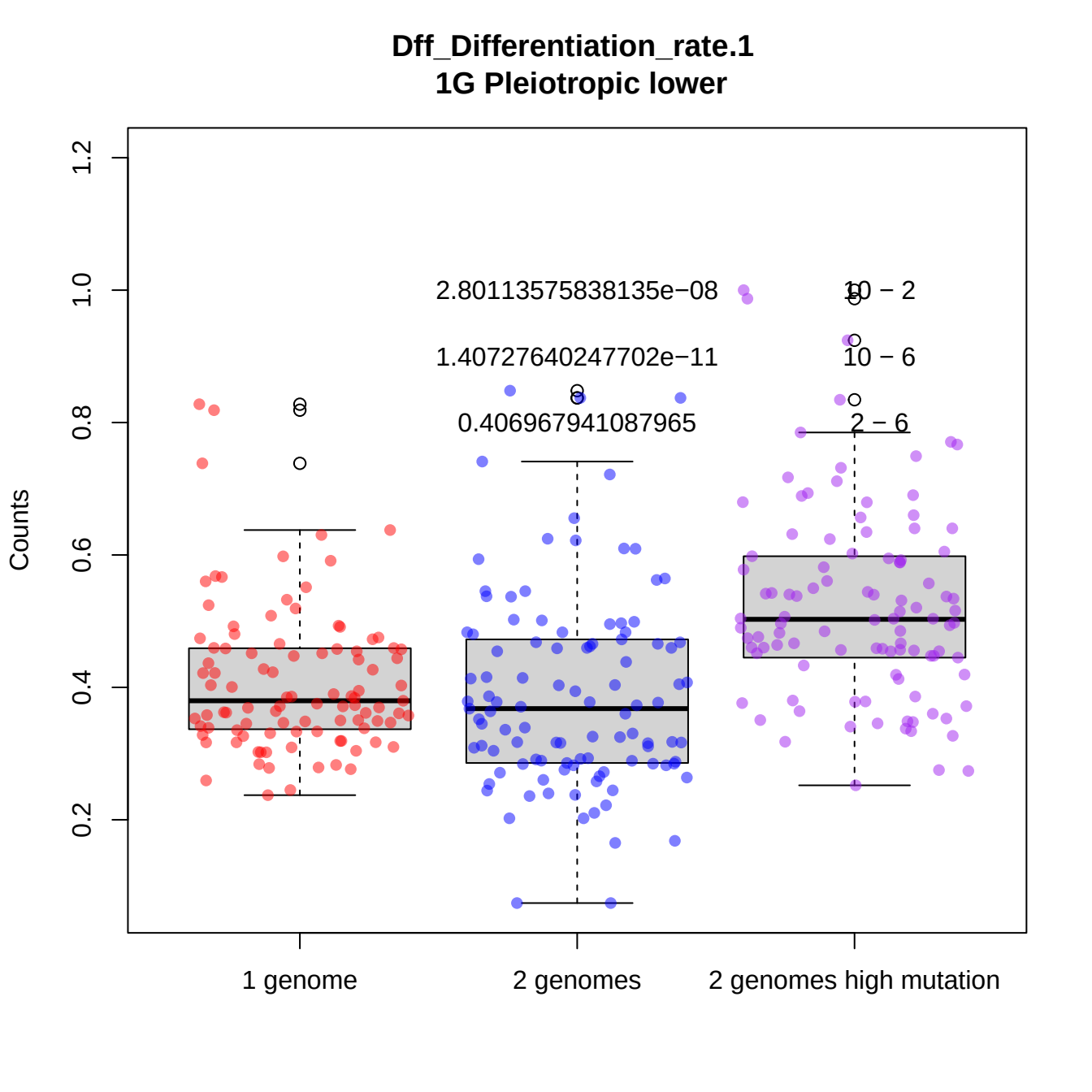

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

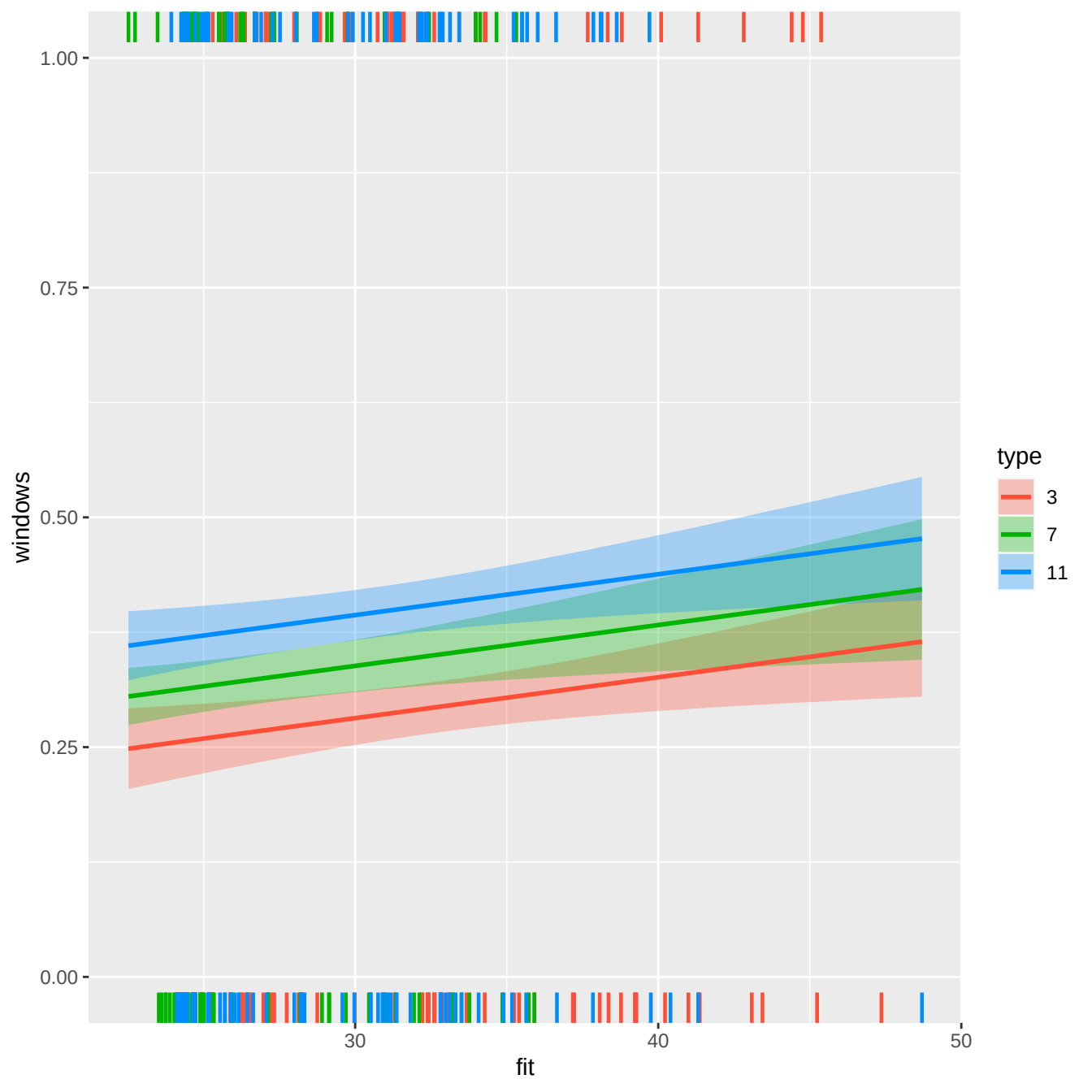

Dff_Differentiation_rate.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

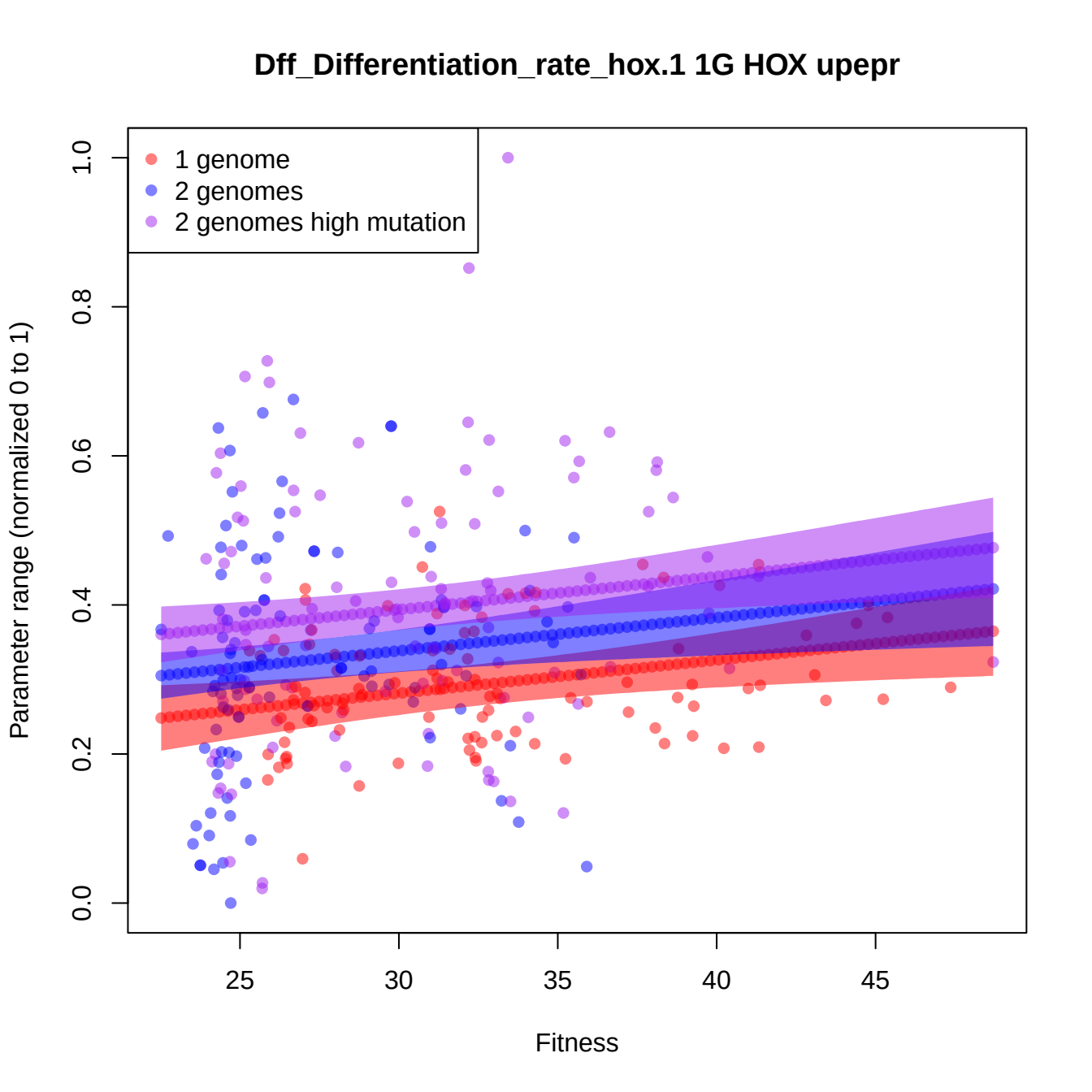

Dff_Differentiation_rate.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.115552858300201
1.6748071268118 6e−15
3.22129875218749e−10
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

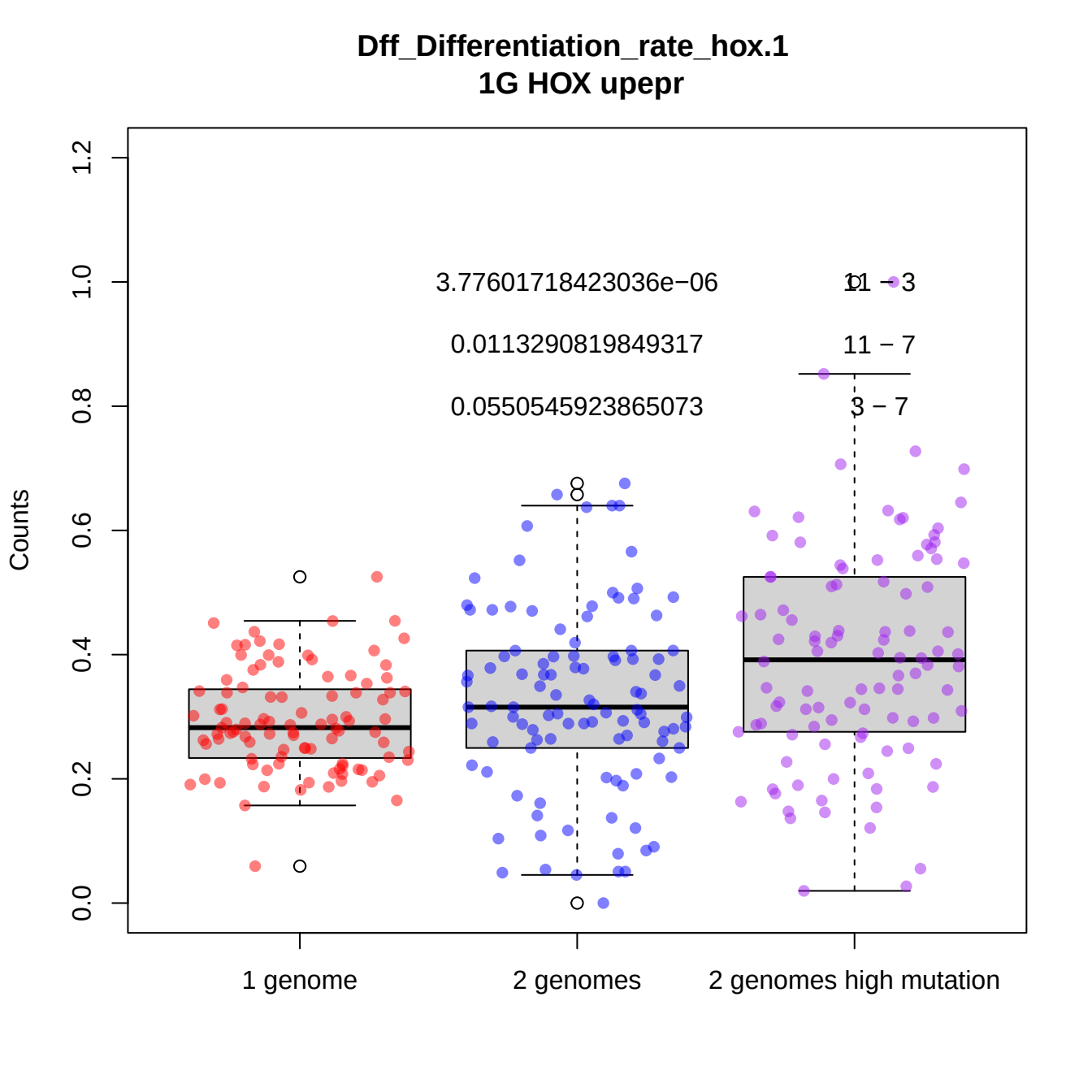

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

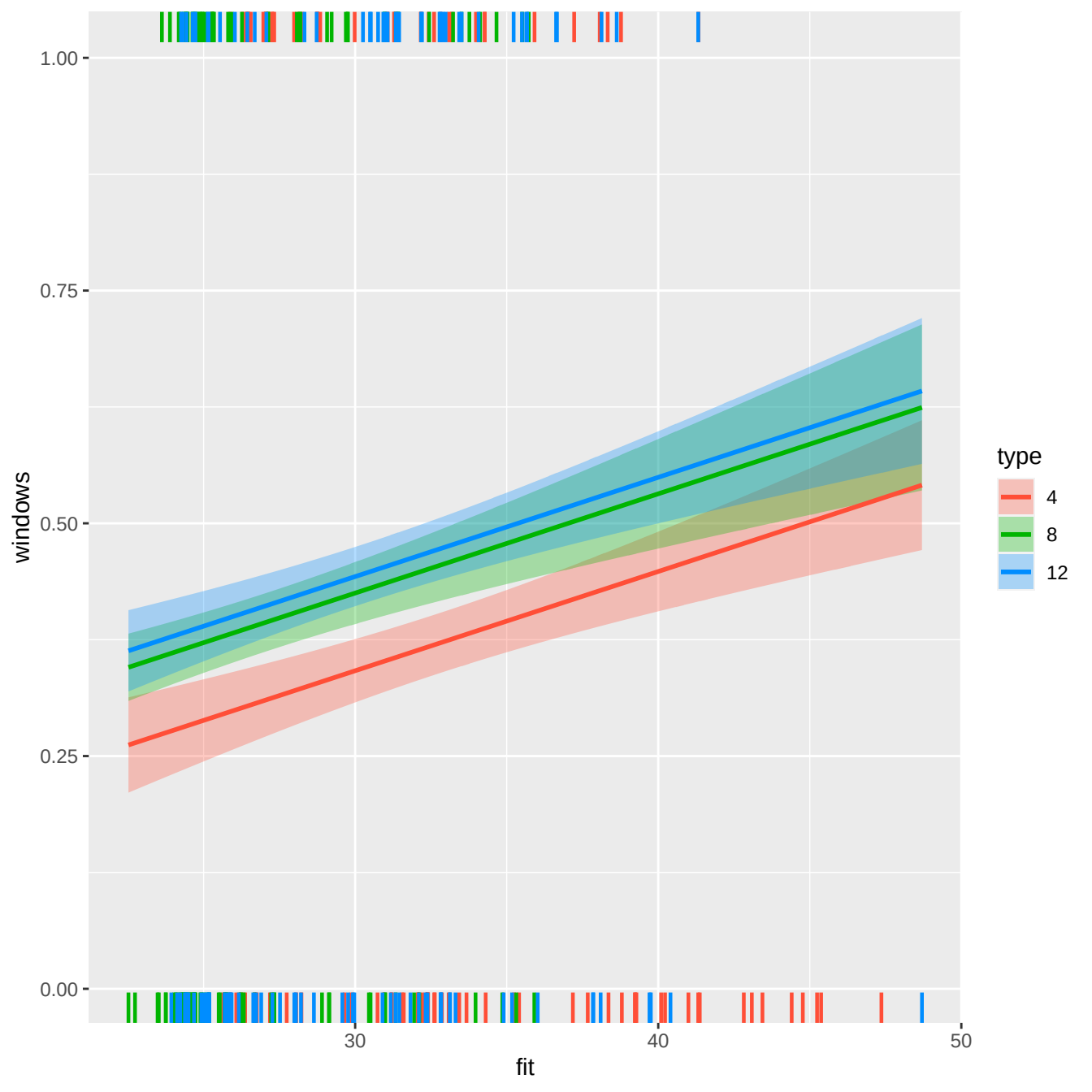

Dff_Differentiation_rate.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

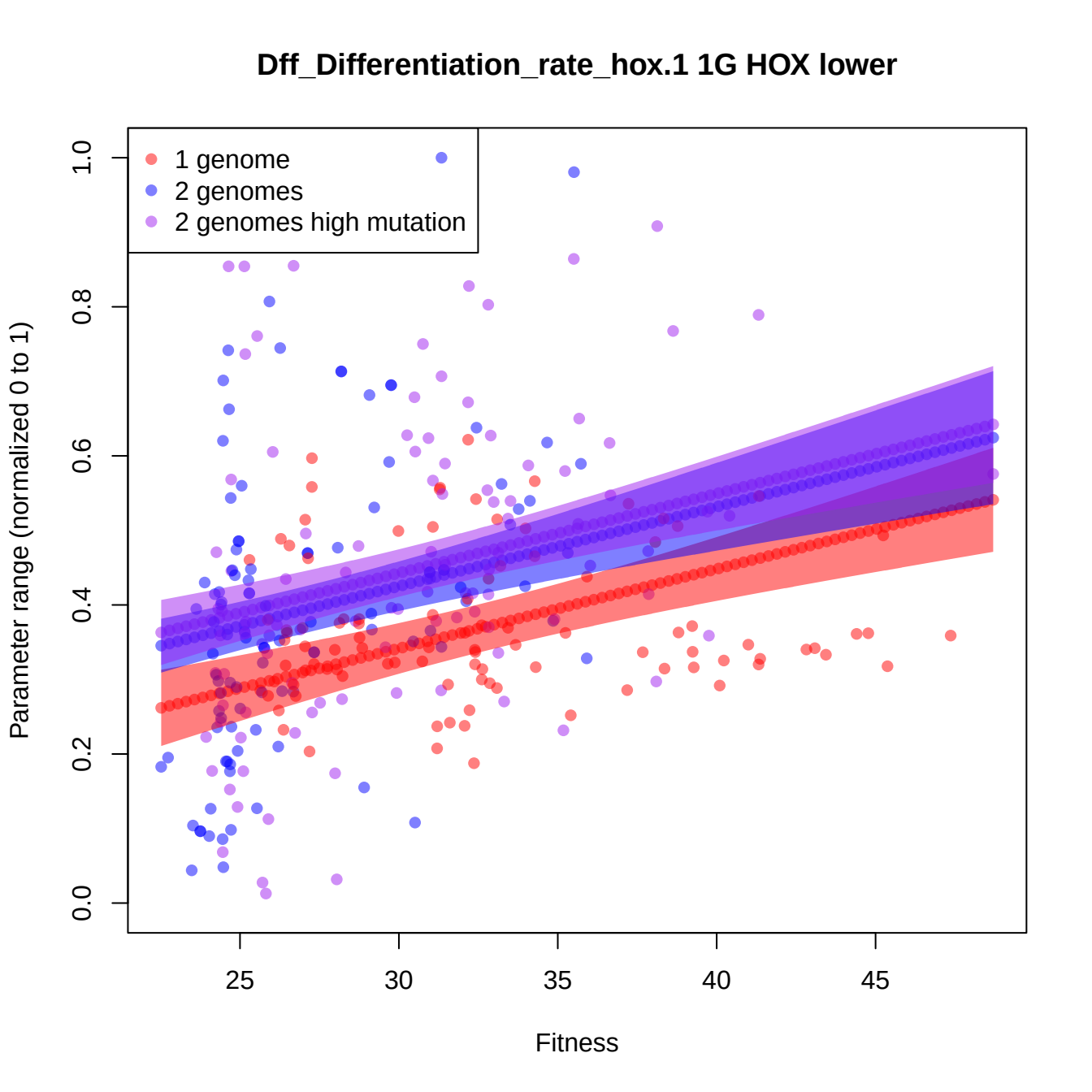

Dff_Differentiation_rate.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.80113575838135e−08
1.40727640247702e−11
0.406967941087965
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

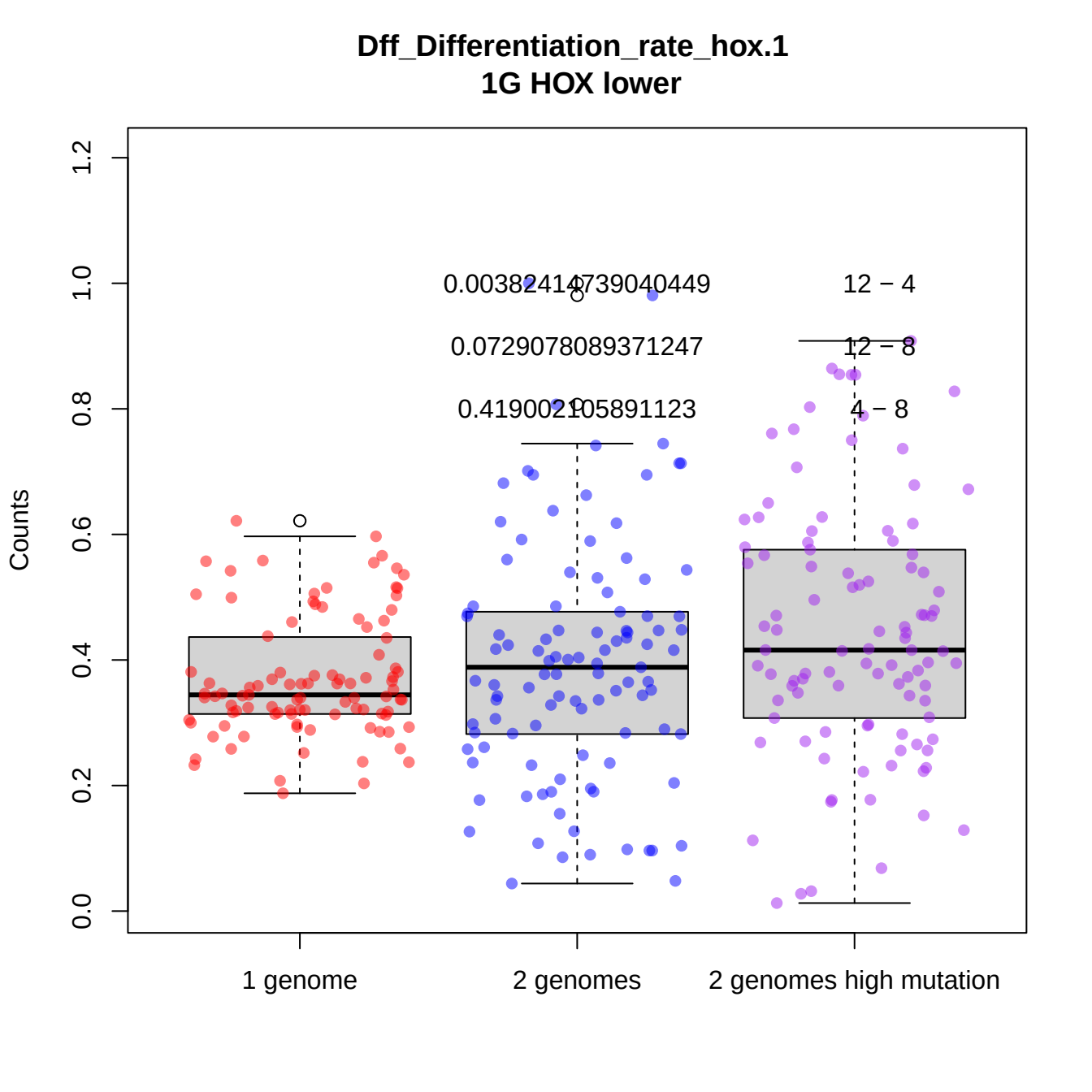

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

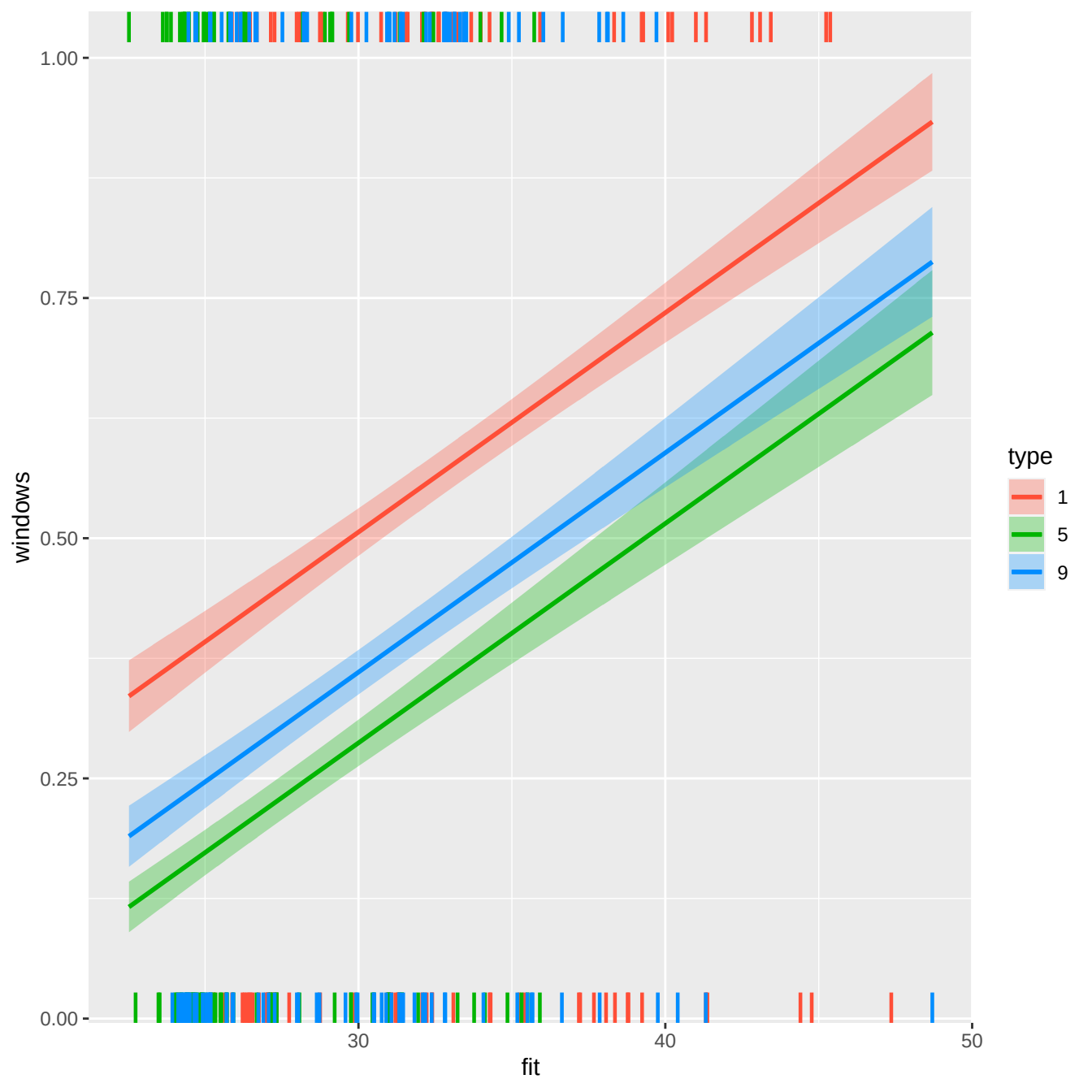

Dff_Differentiation_rate_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

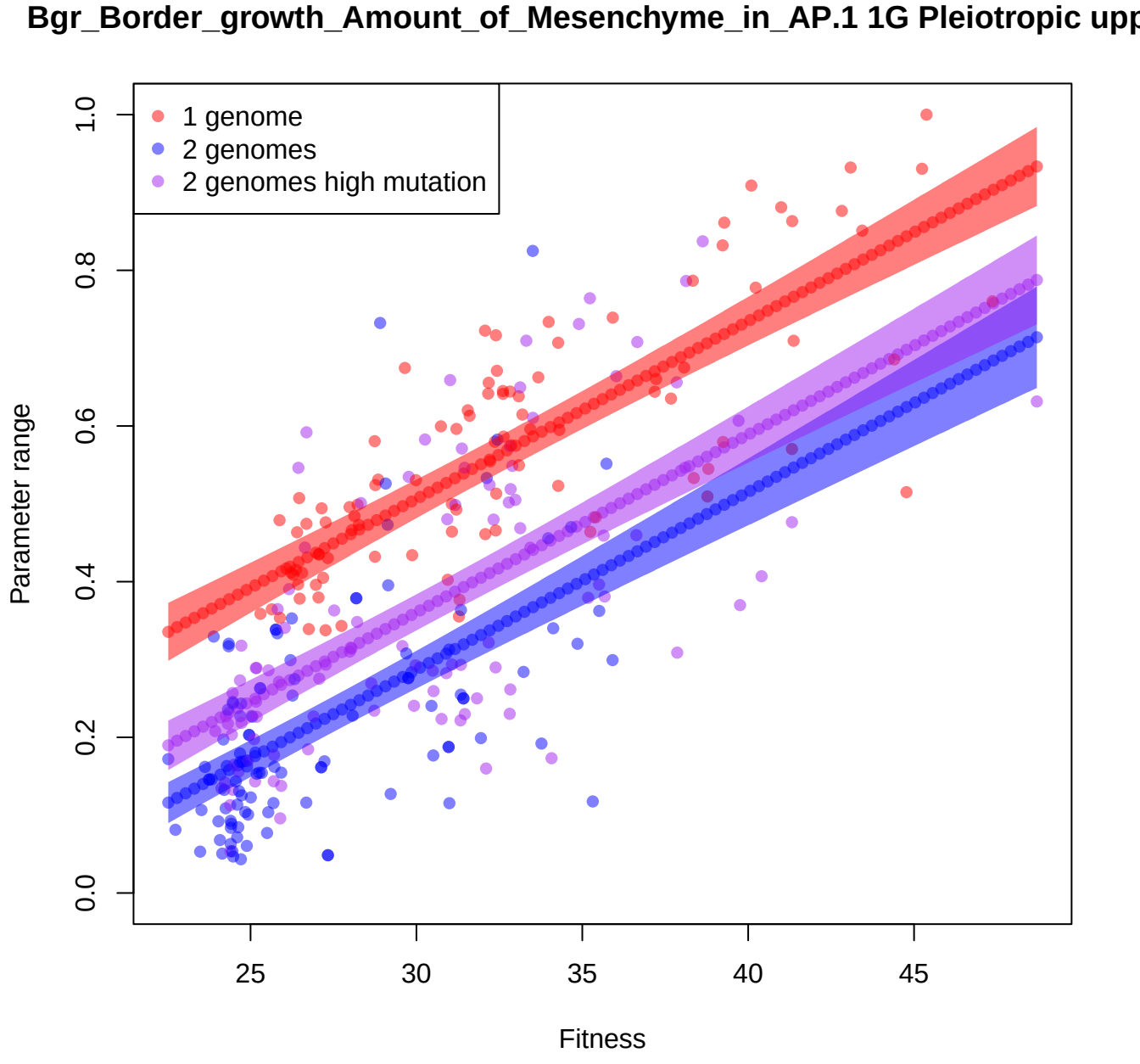

Dff_Differentiation_rate_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
3.77601718423036e−06
0.0113290819849317
0.0550545923865073
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

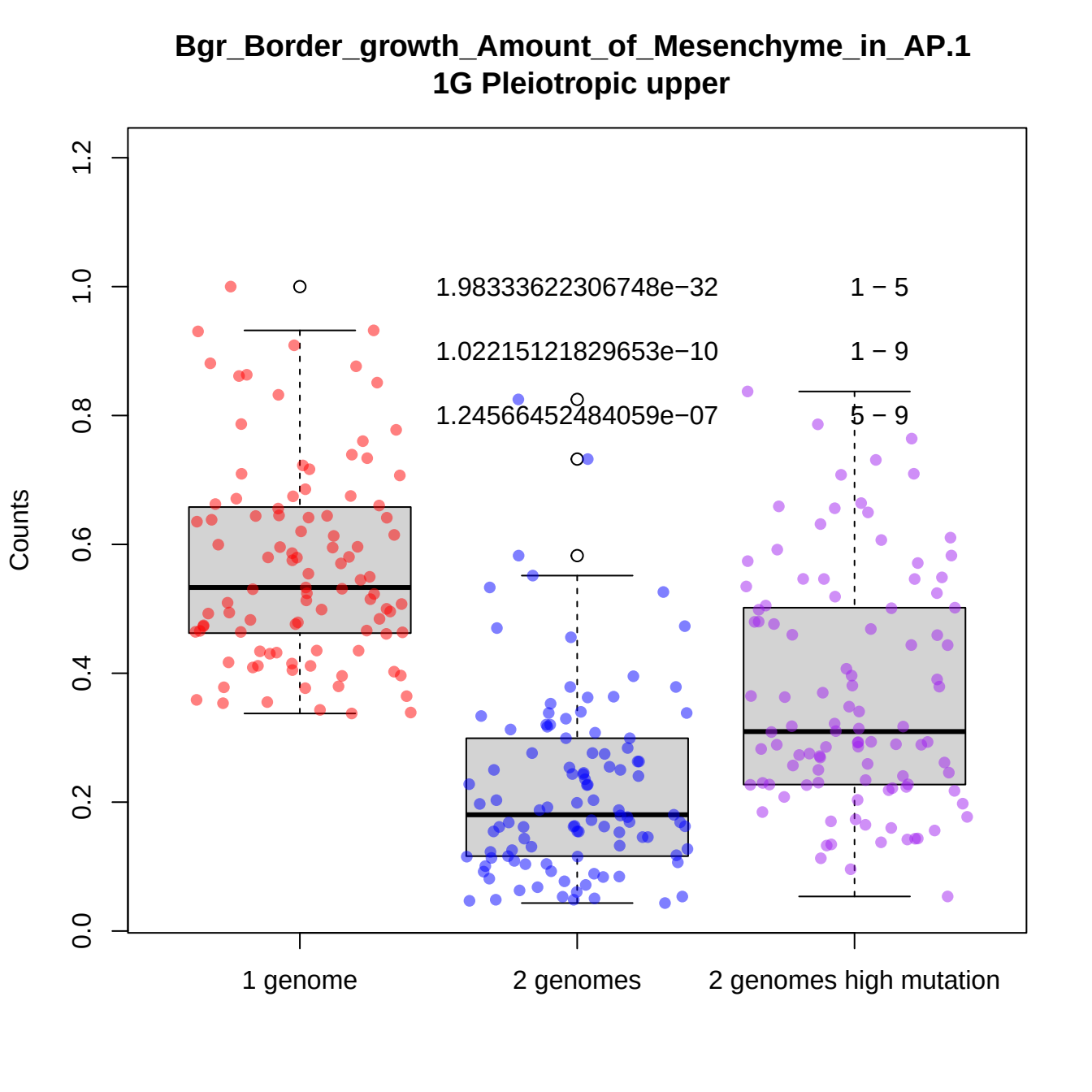

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

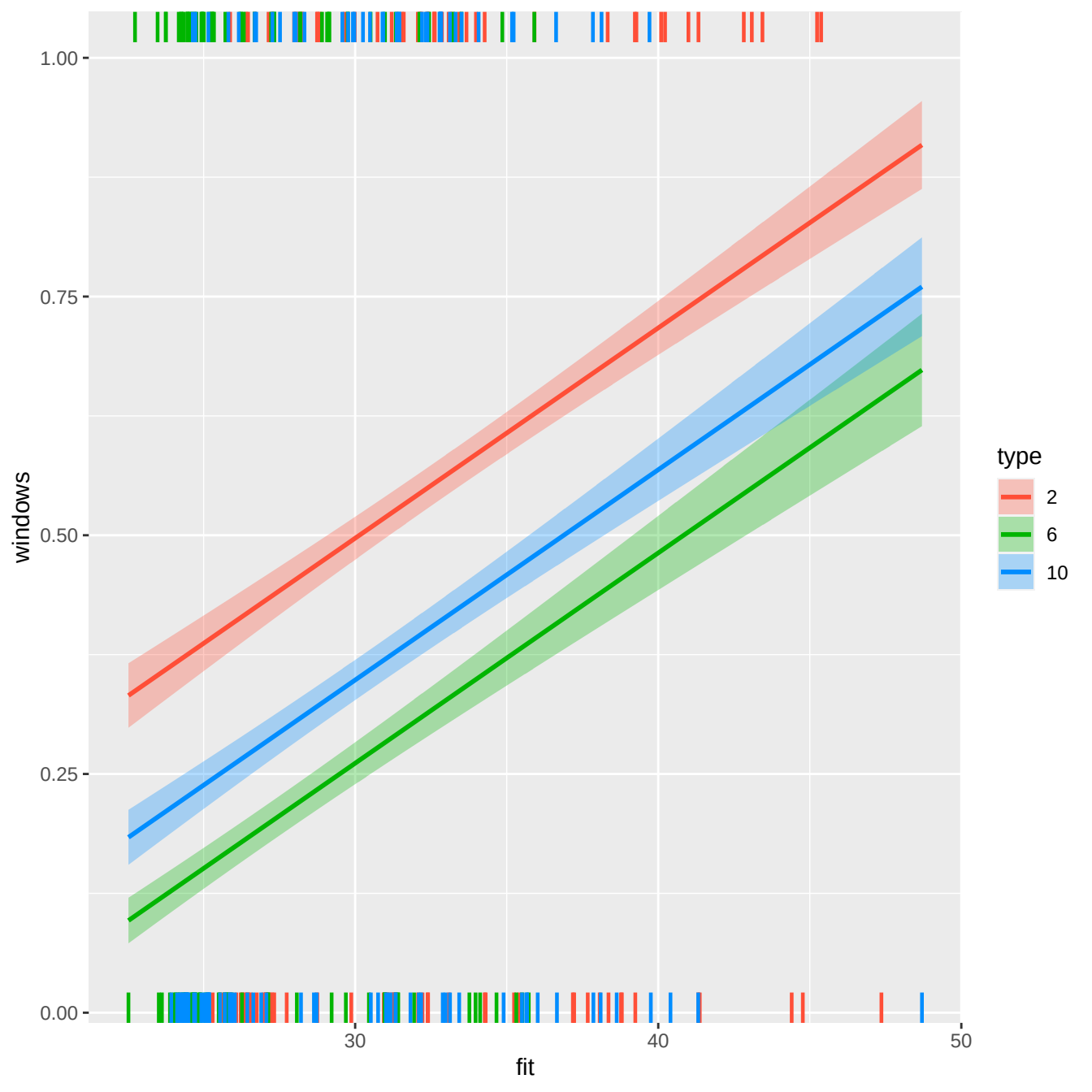

Dff_Differentiation_rate_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

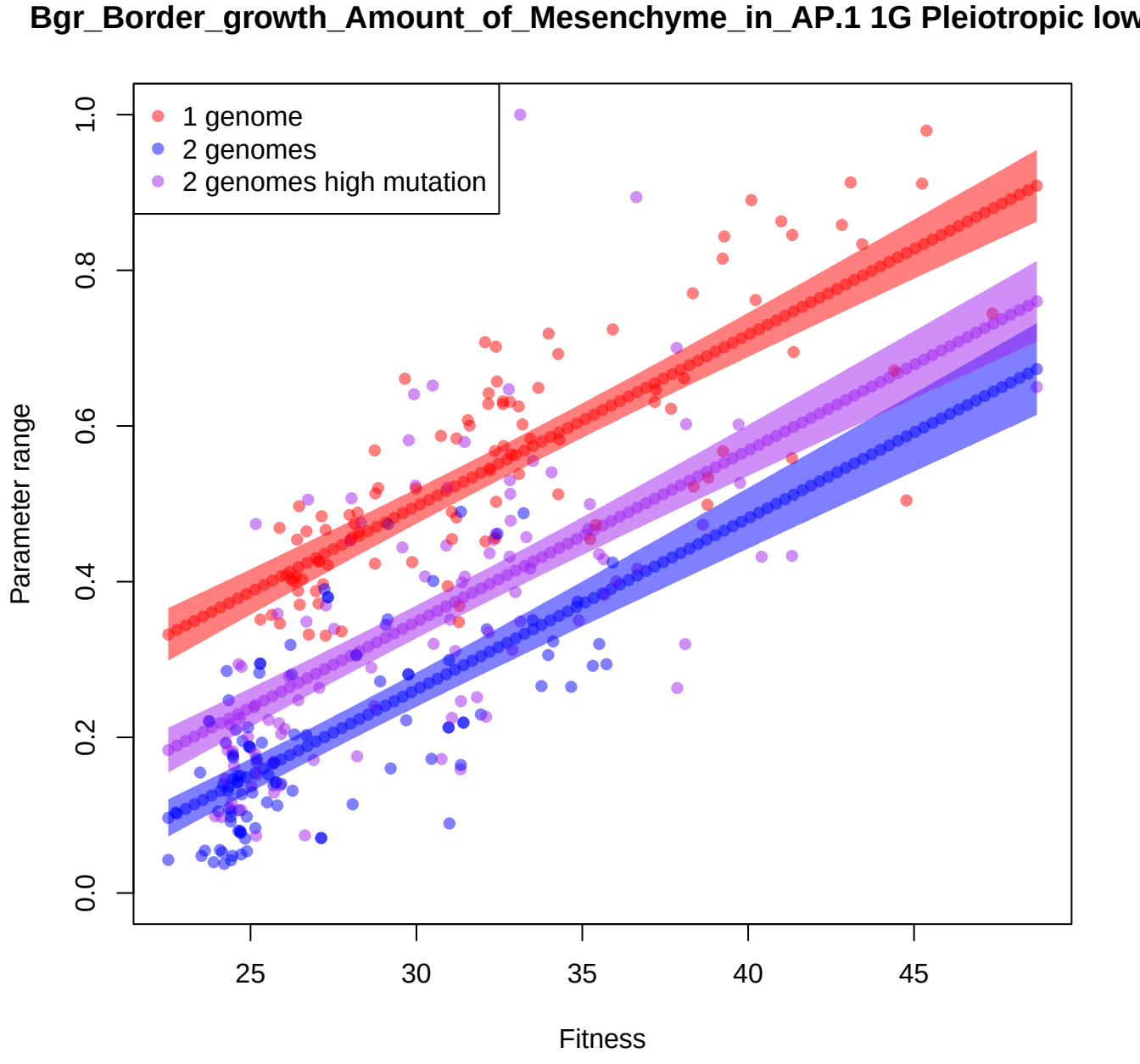

Dff_Differentiation_rate_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.0038241473904049
0.0729078089371247
0.419002105891123
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

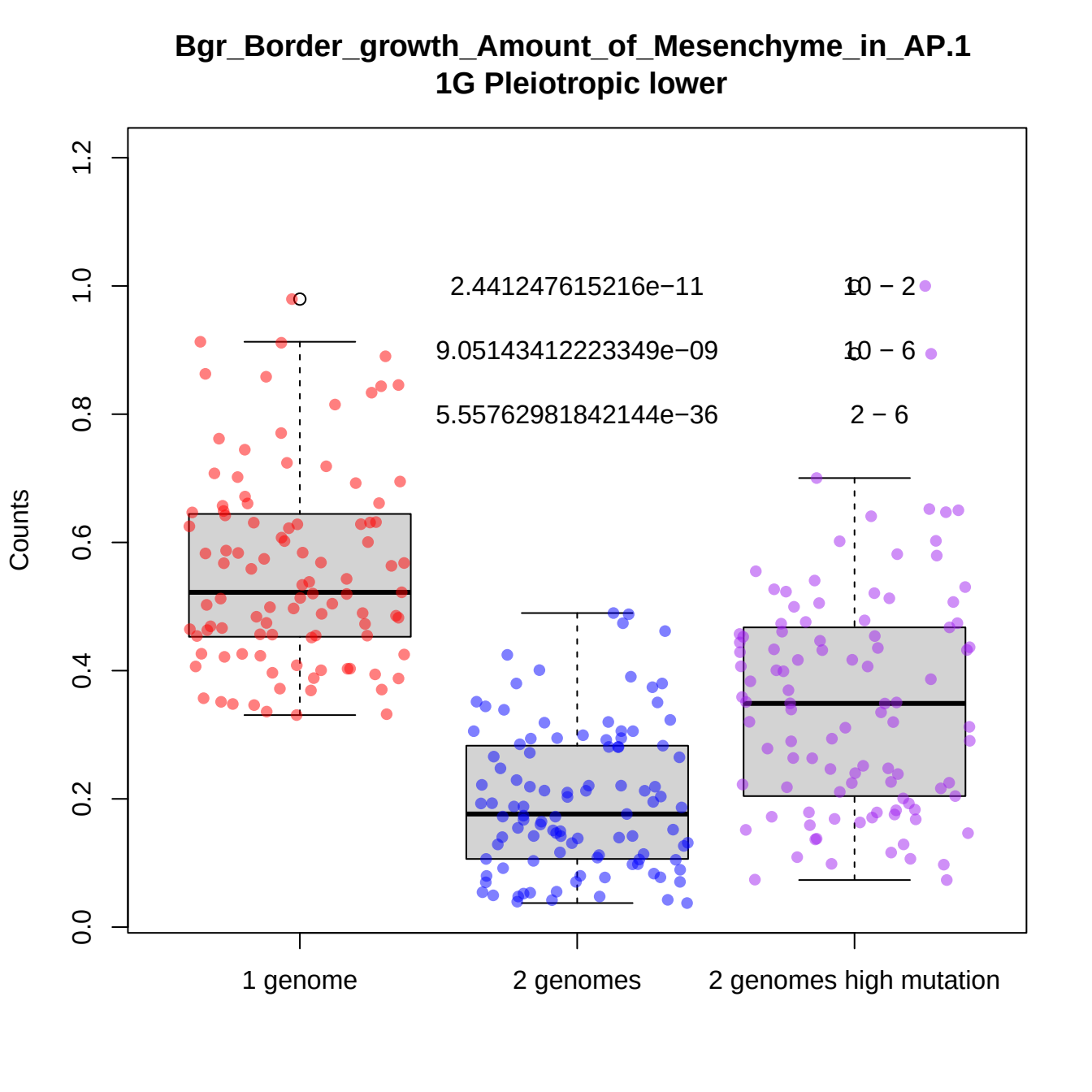

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
1
5
9

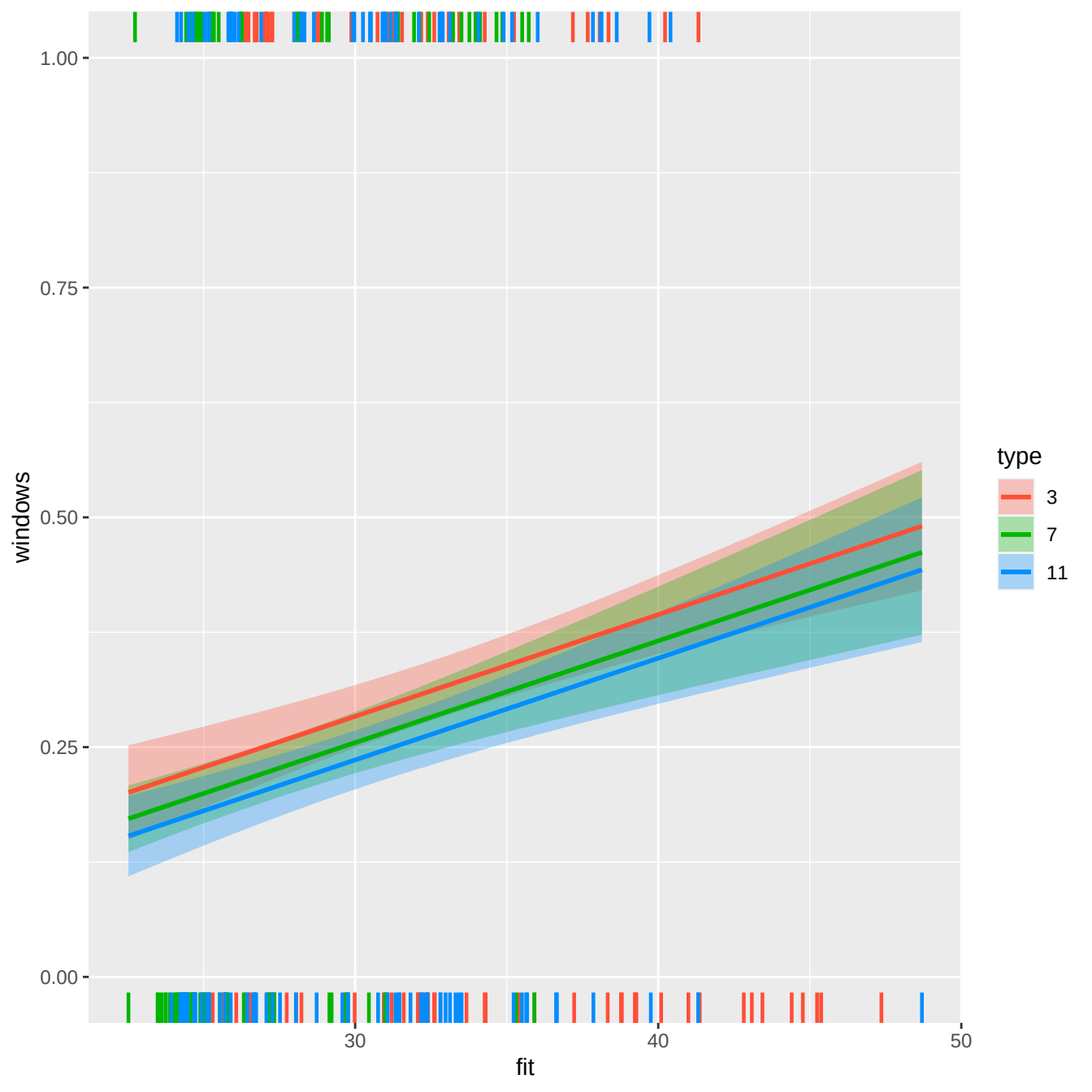

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1 1G Pleiotropic upp
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

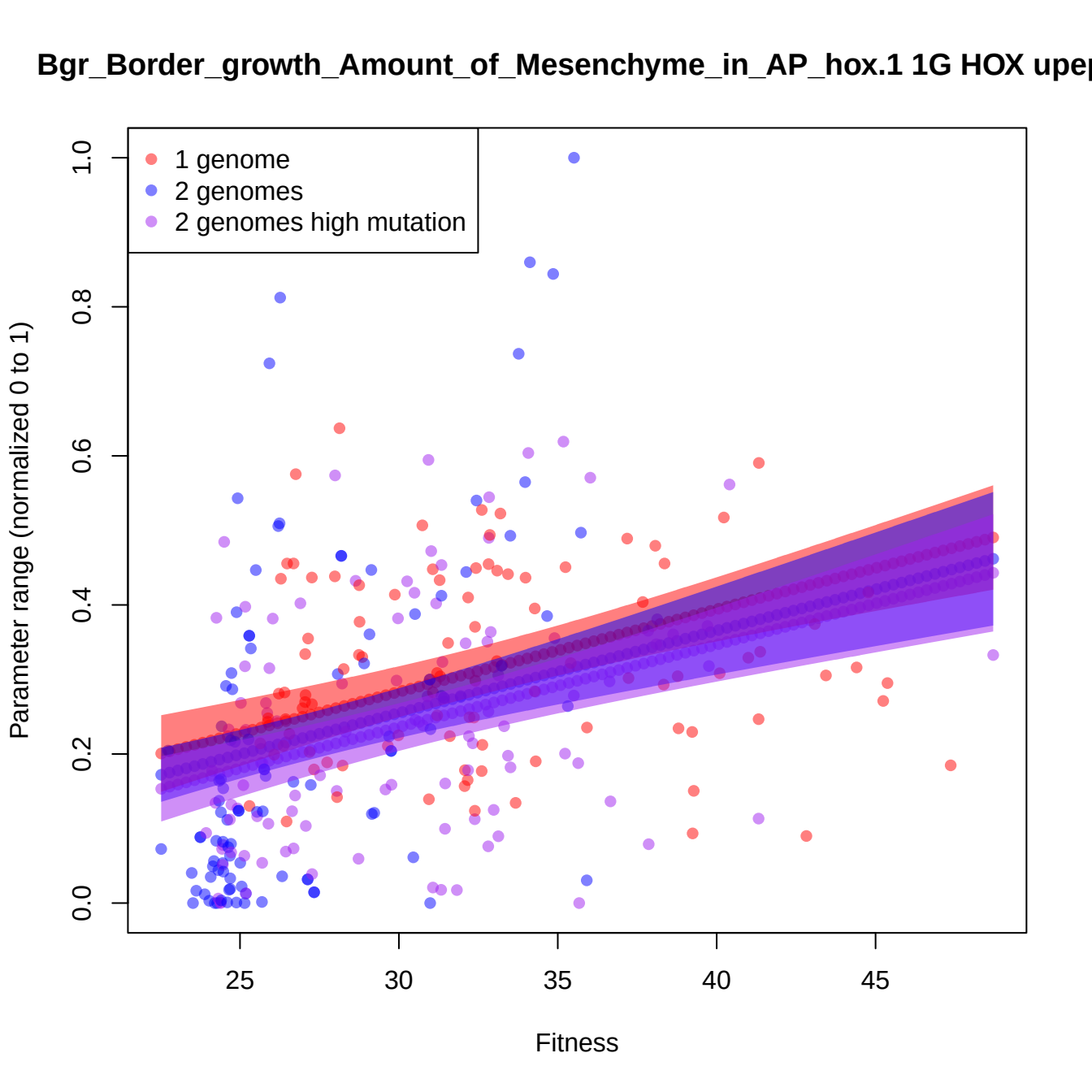

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1
1G Pleiotropic upper
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.98333622306748e−32
1.02215121829653e−10
1.24566452484059e−07
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

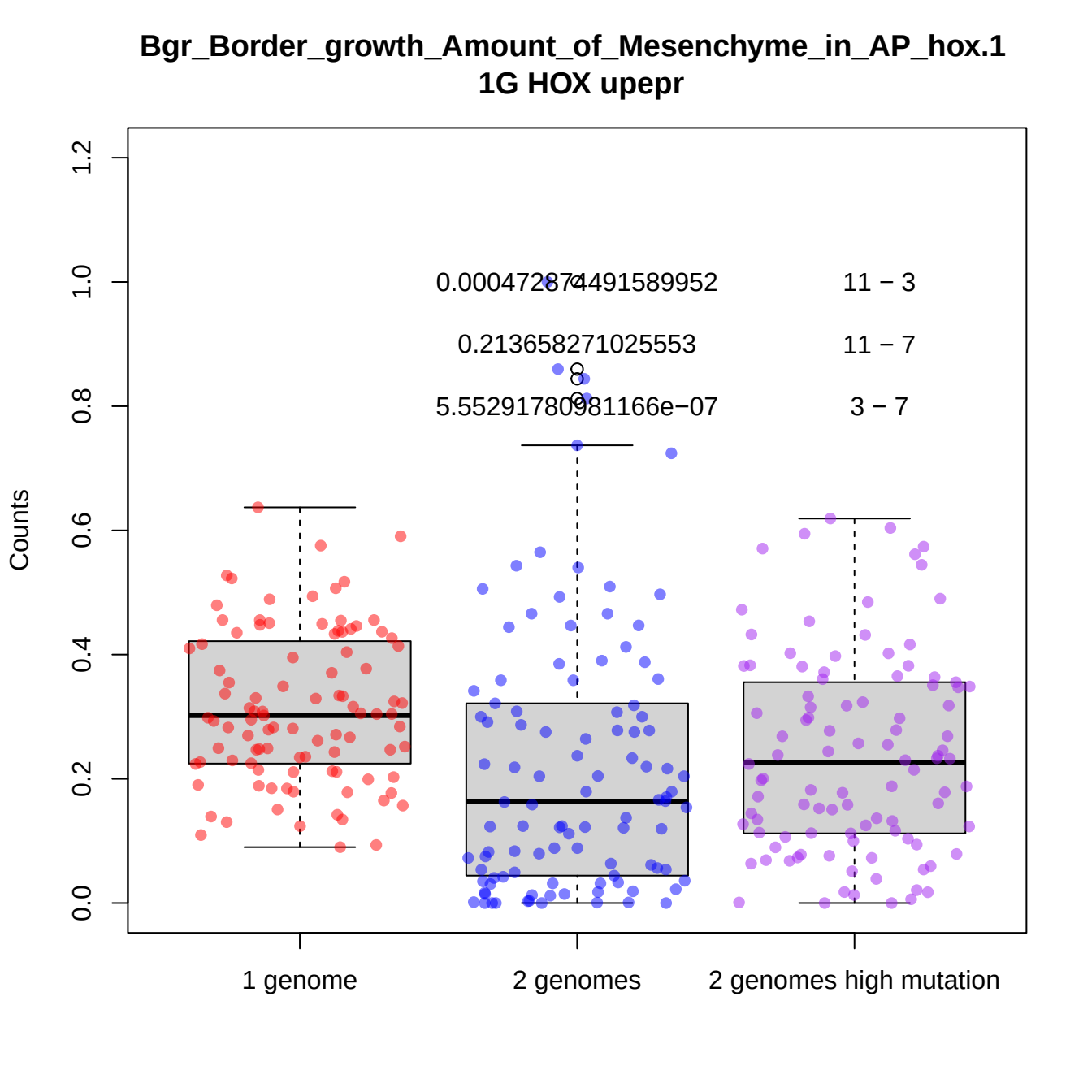

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
2
6
10

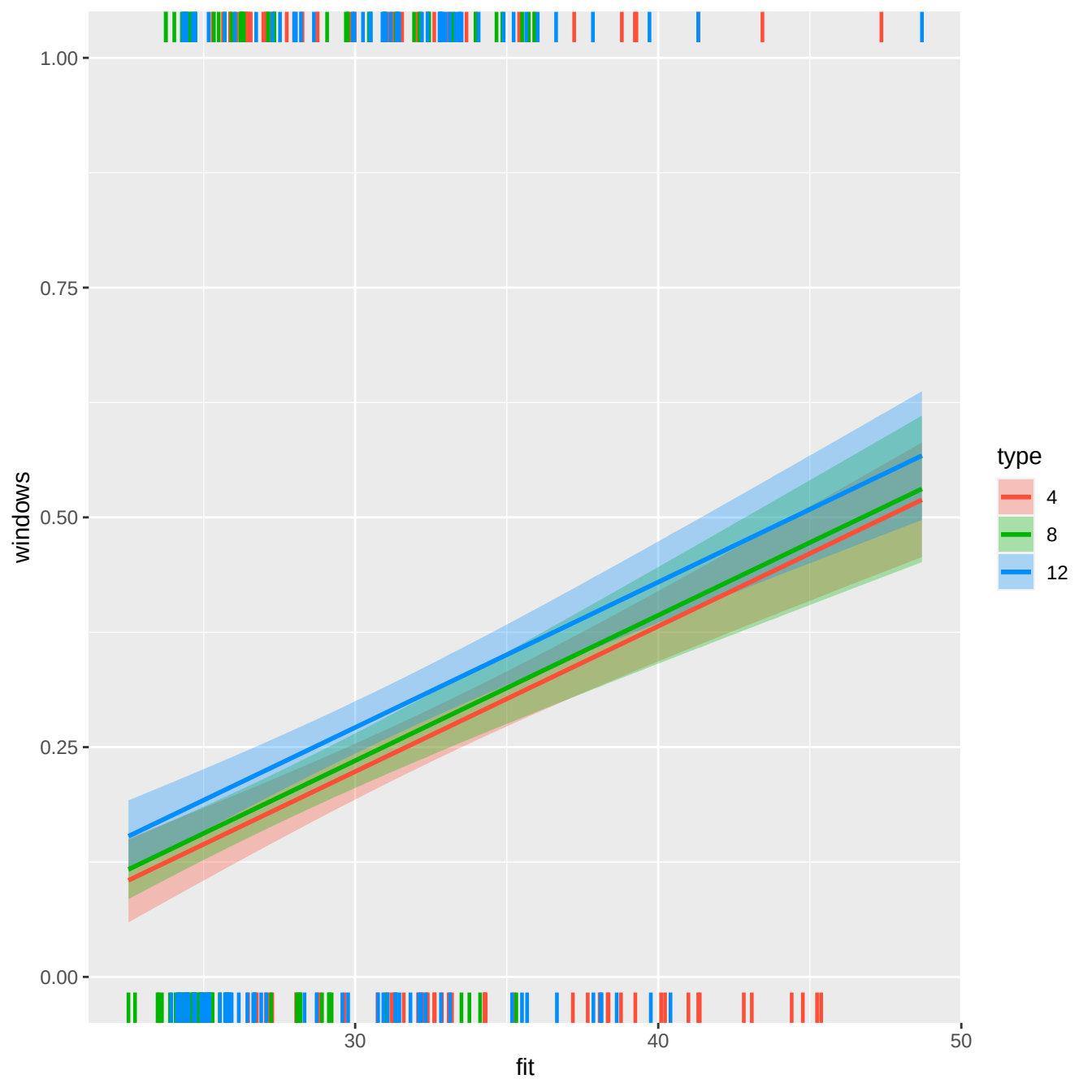

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1 1G Pleiotropic low
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

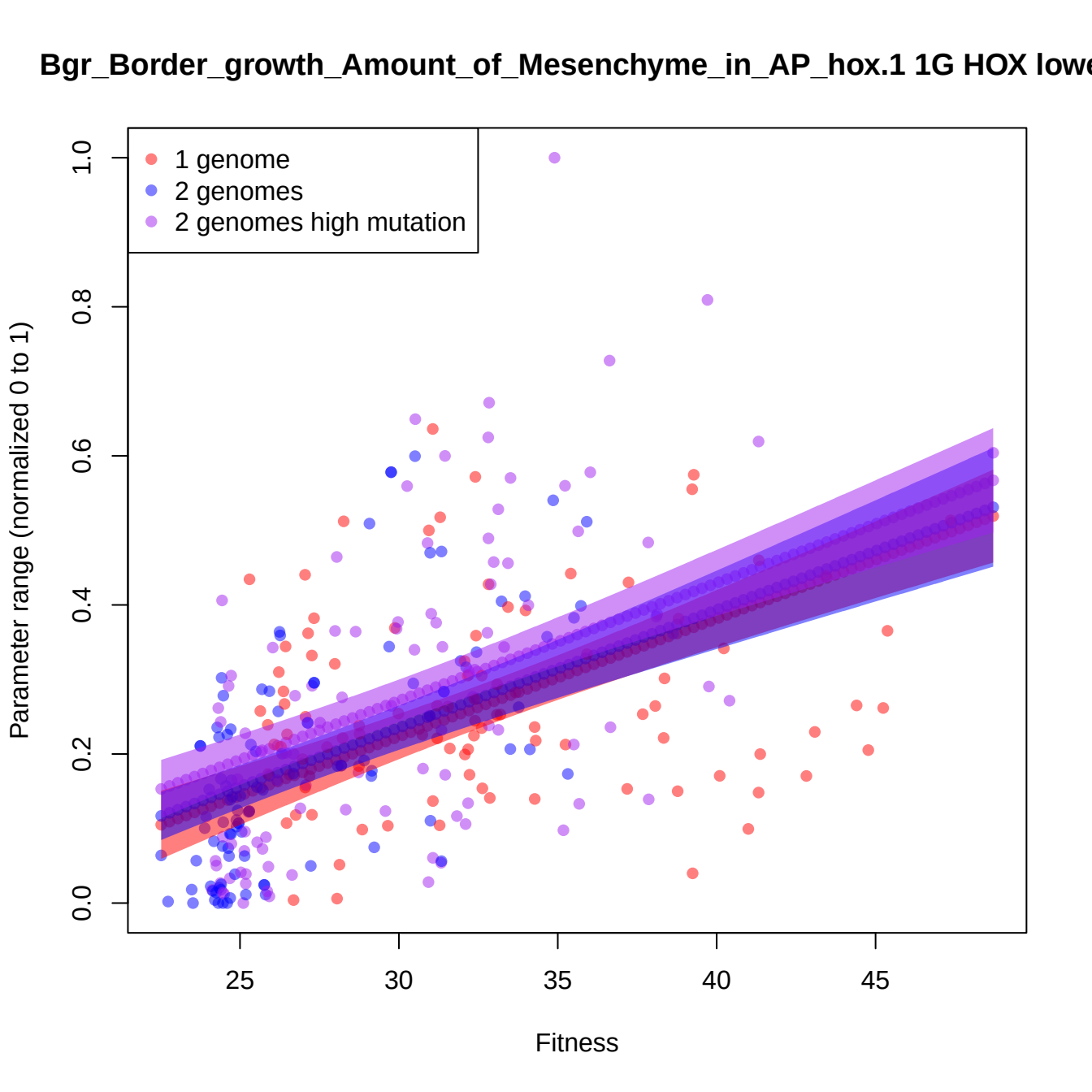

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP.1
1G Pleiotropic lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.441247615216e−11
9.05143412223349e−09
5.55762981842144e−36
1 − 2
1 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

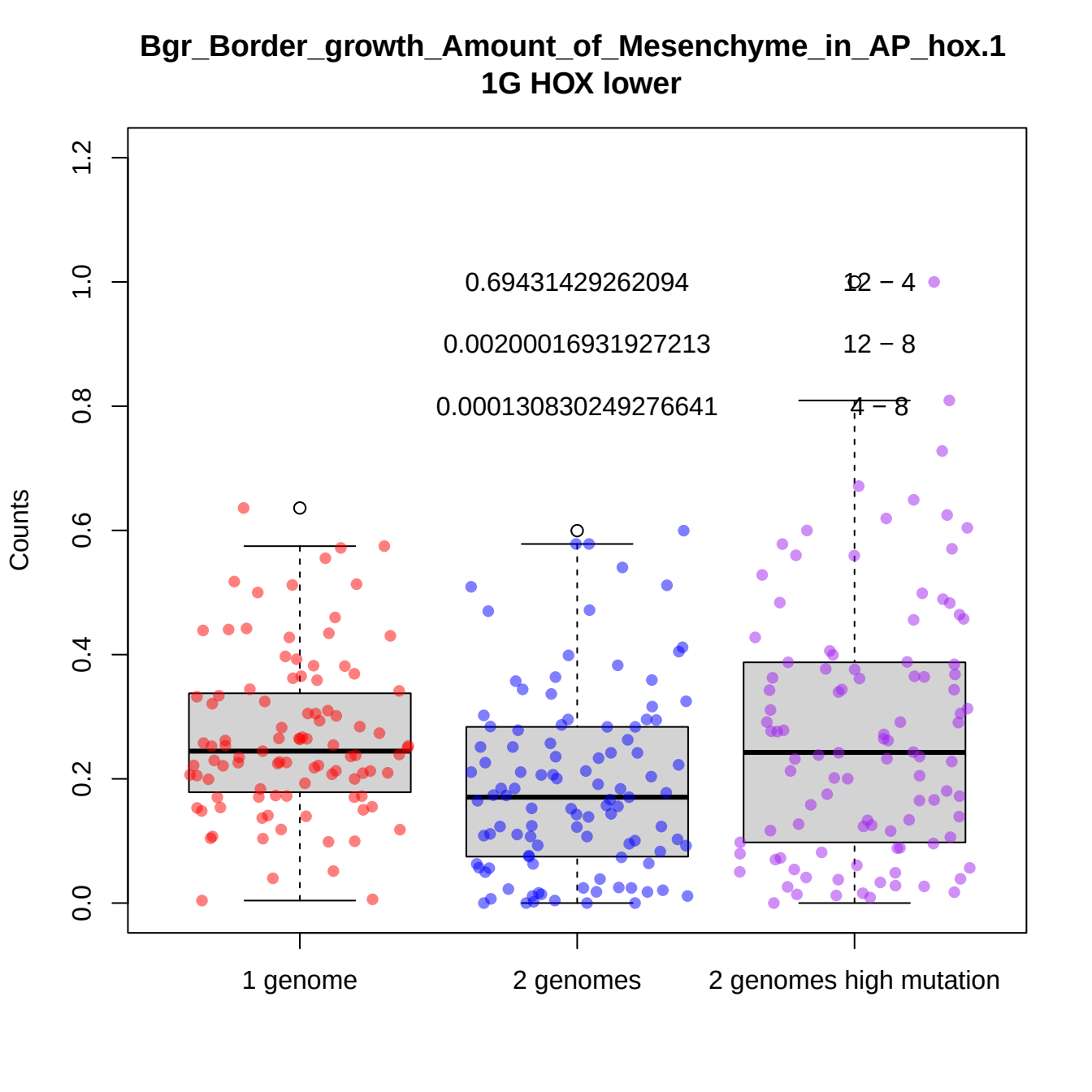

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

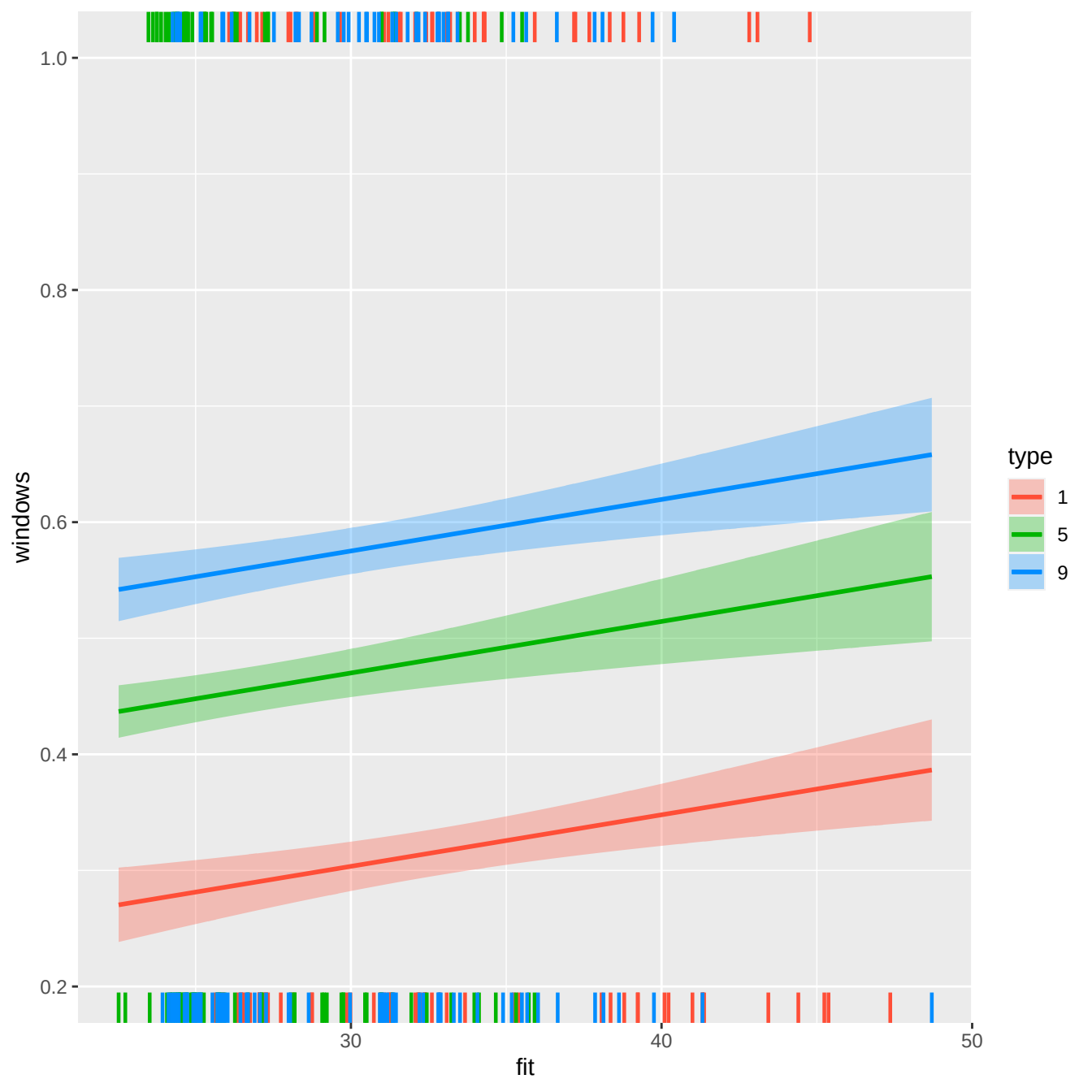

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1 1G HOX upep
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

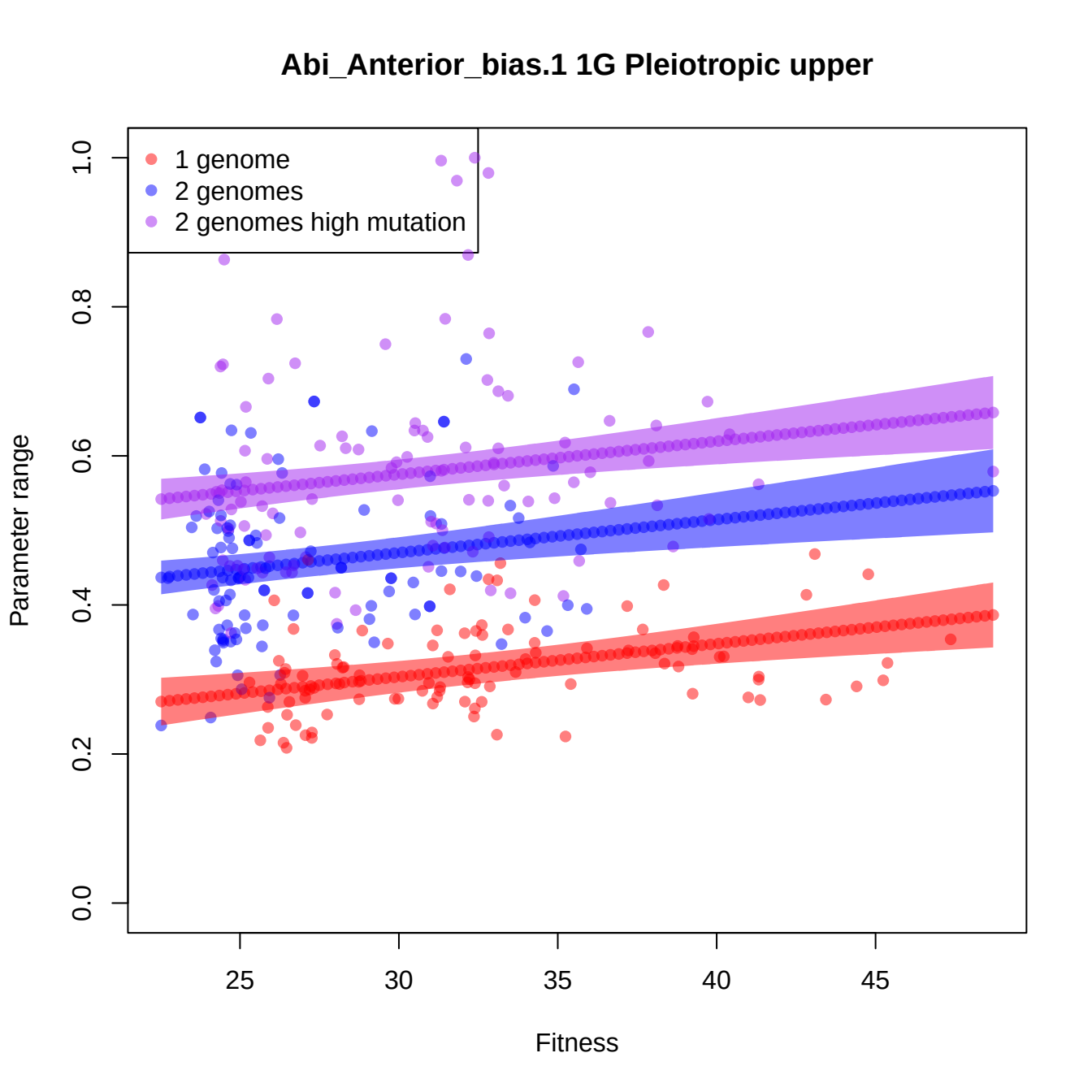

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.00047287
0.213658271025553
5.552917809811166e−07
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

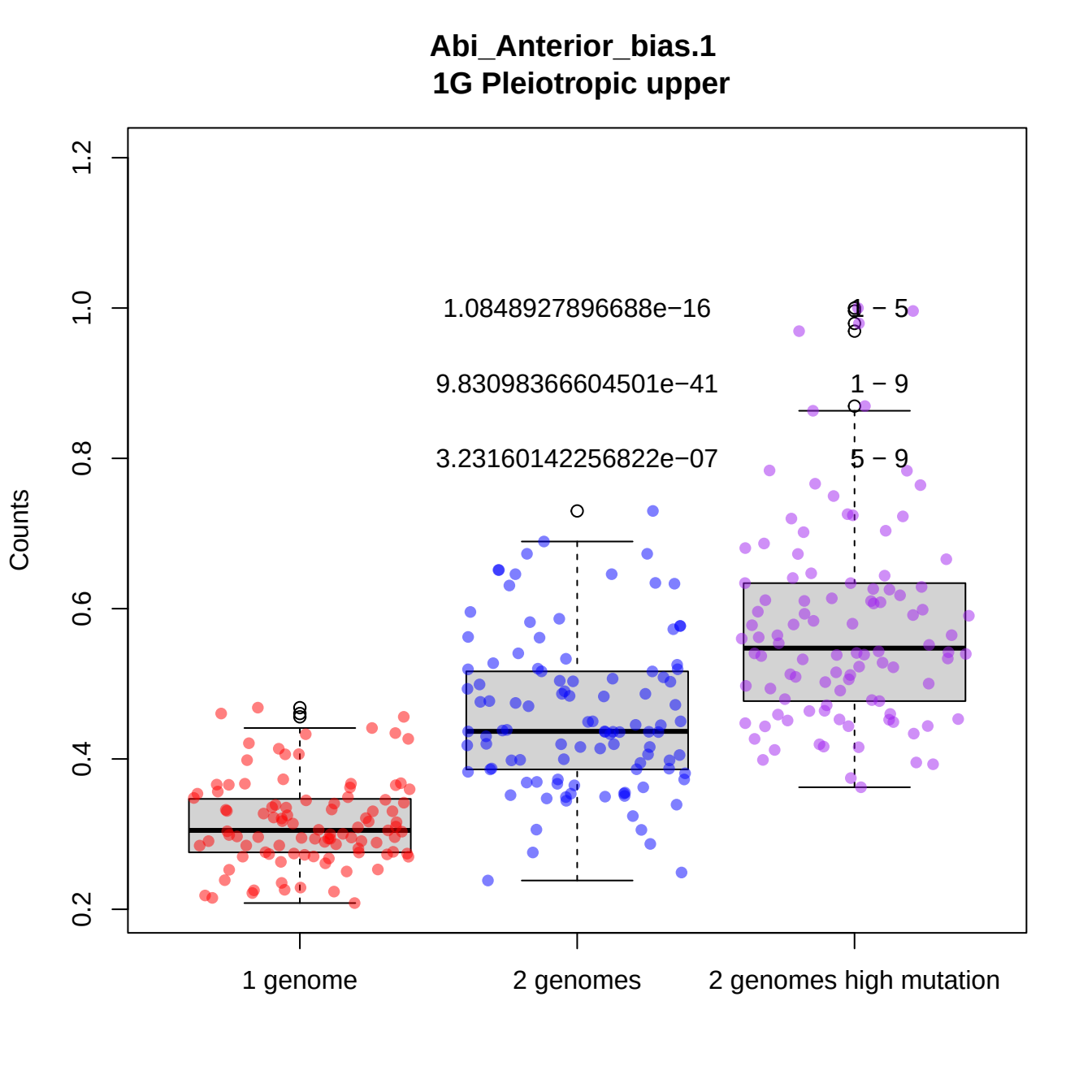

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

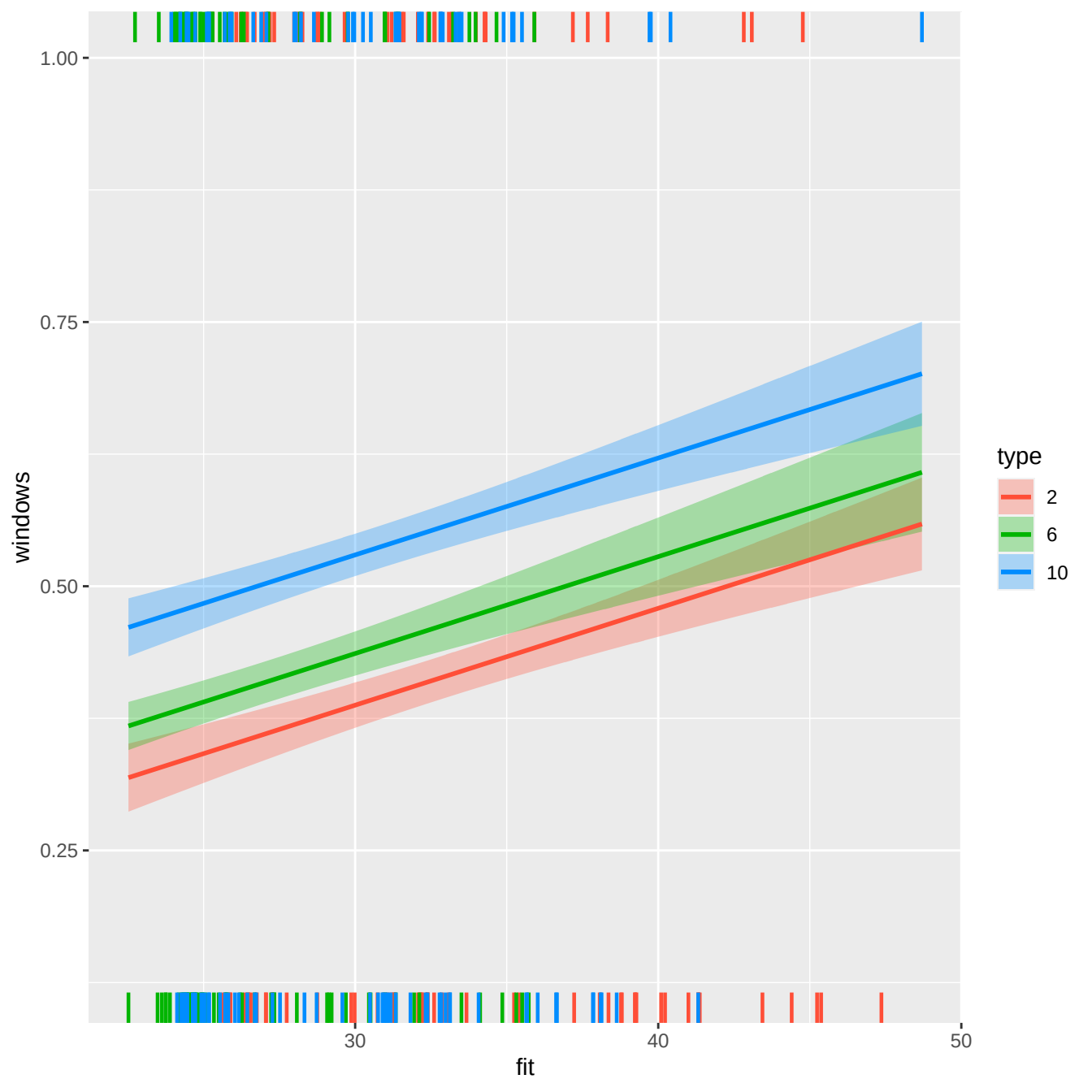

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1 1G HOX lowe
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

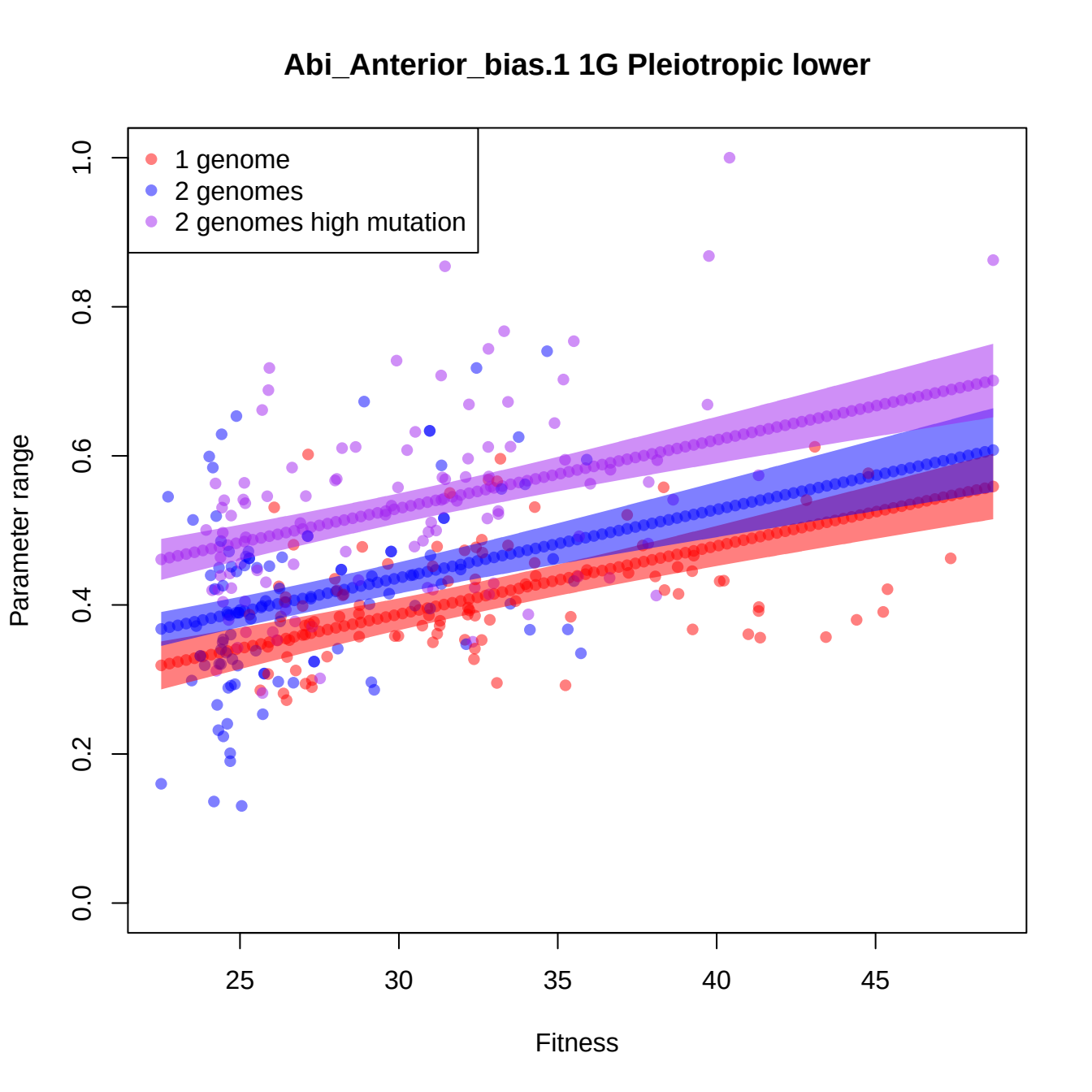

Bgr_Border_growth_Amount_of_Mesenchyme_in_AP_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.69431429262094
0.00200016931927213
0.000130830249276641
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

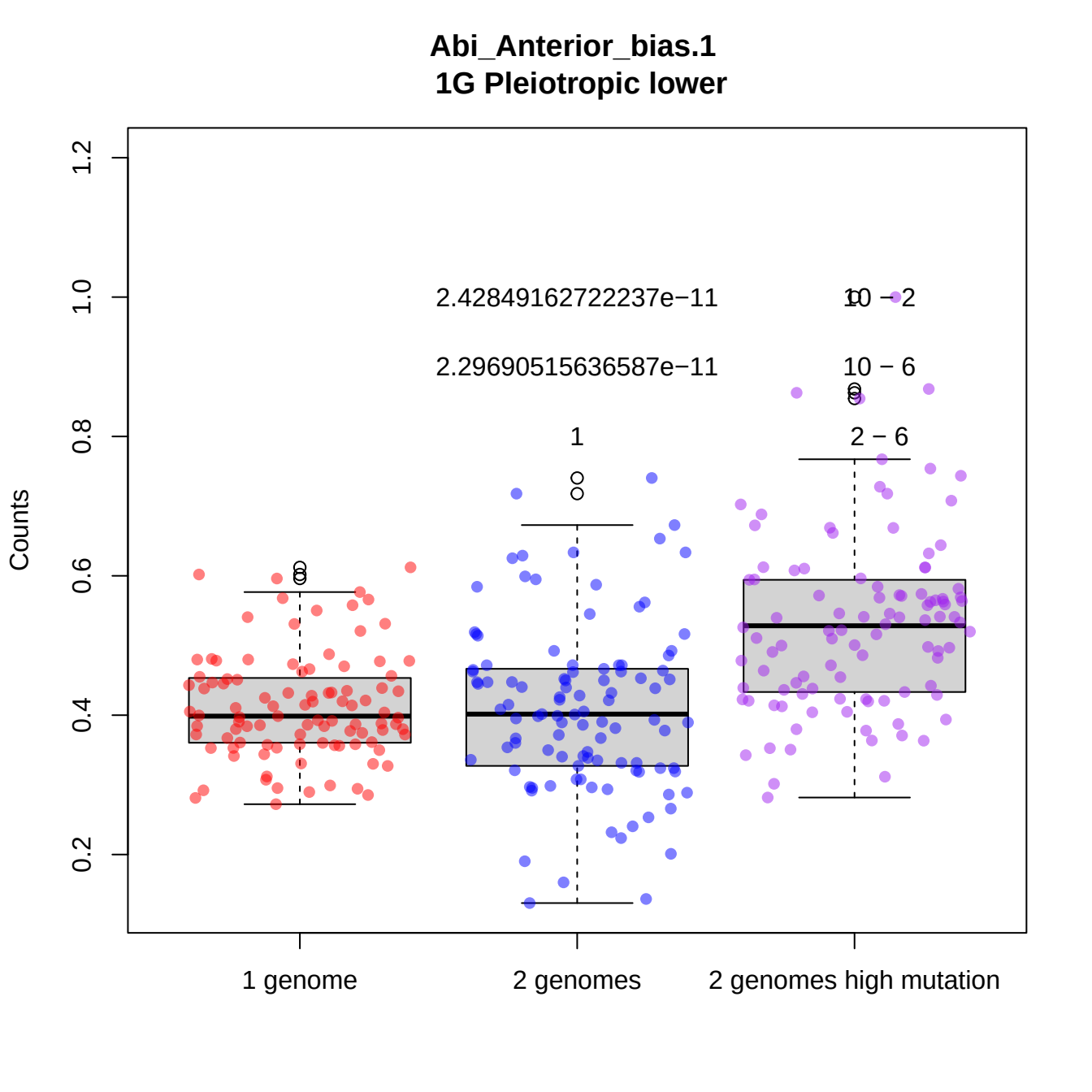

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

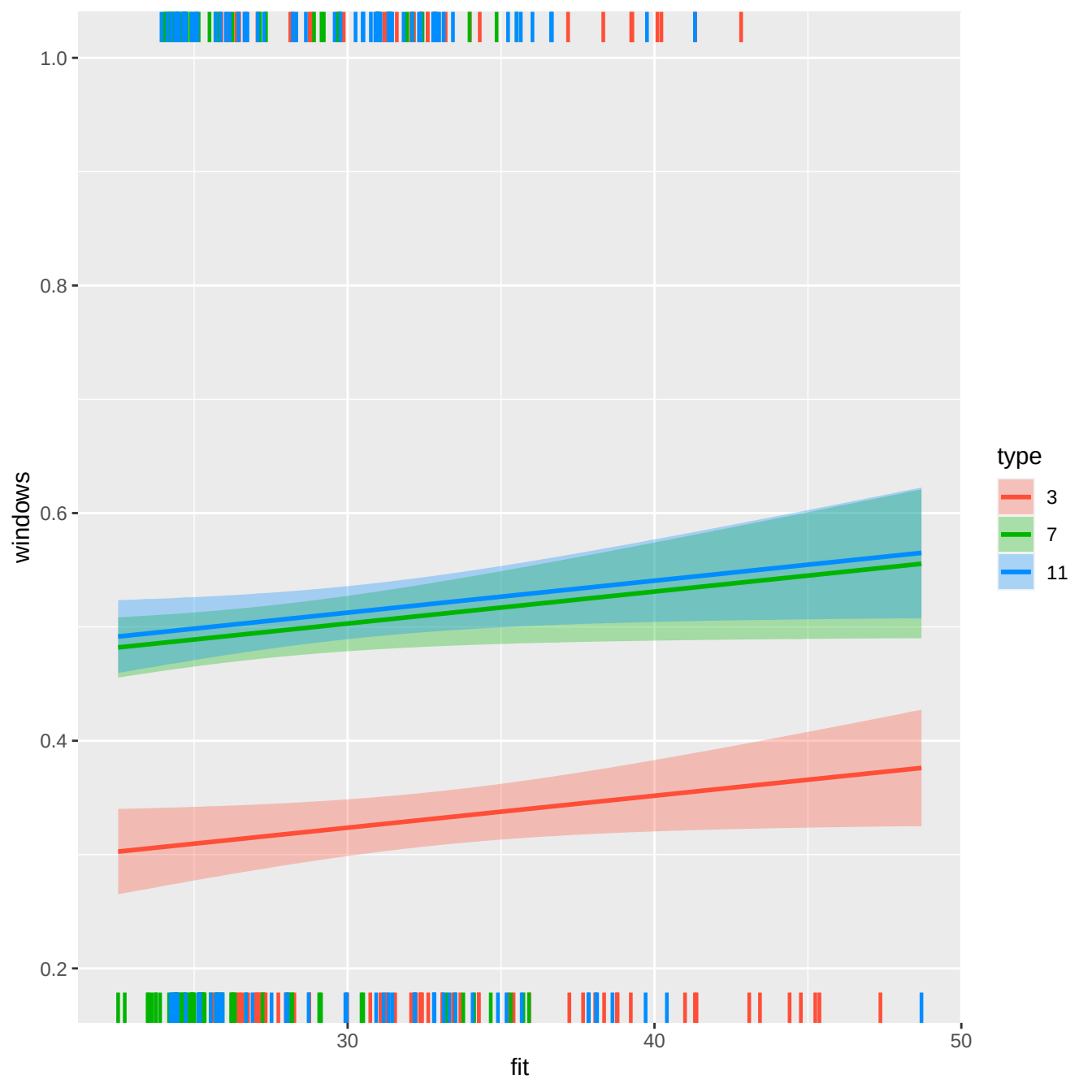

Abi_Anterior_bias.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

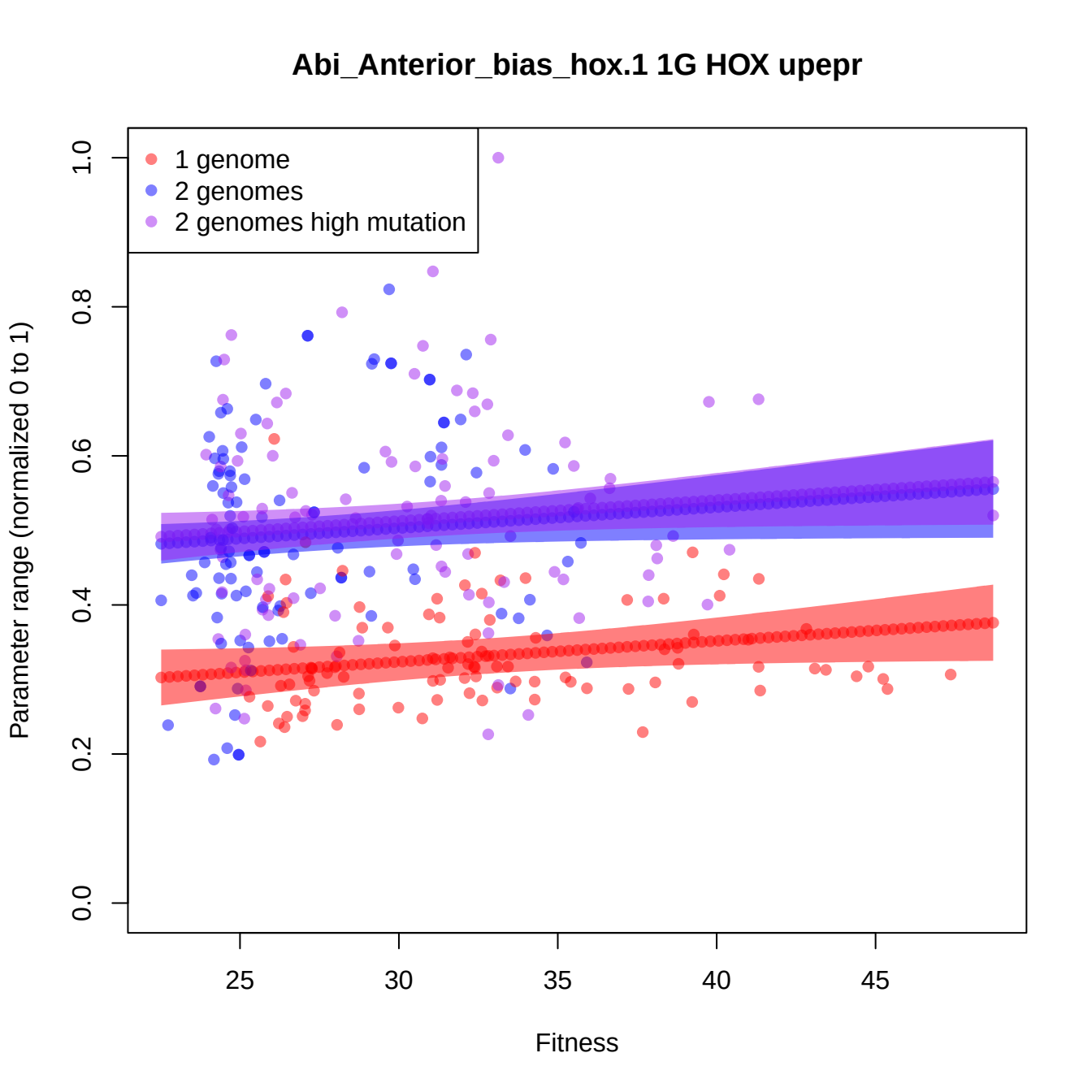

Abi_Anterior_bias.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.0848927896688e−16
9.83098366604501e−41
3.23160142256822e−07
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

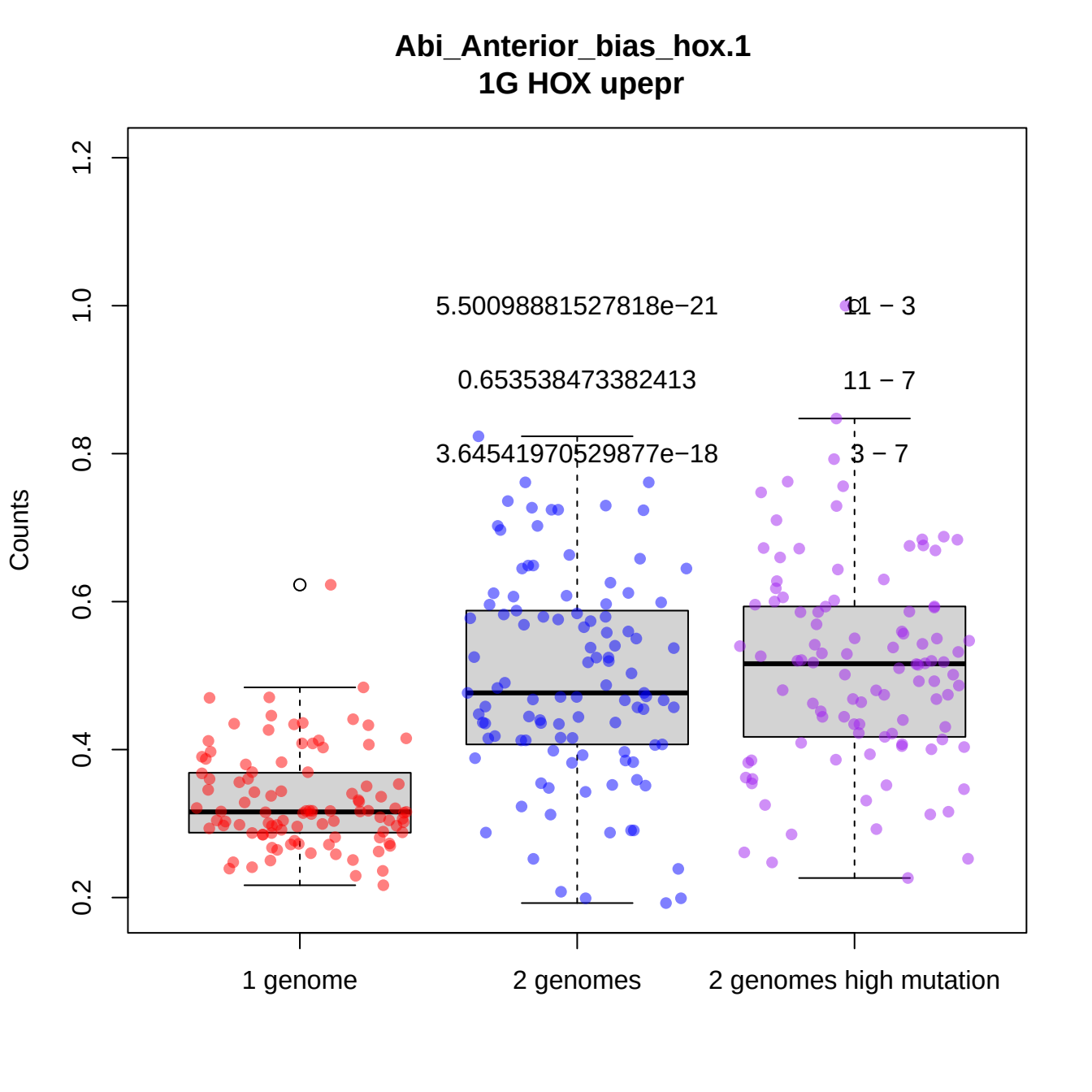

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

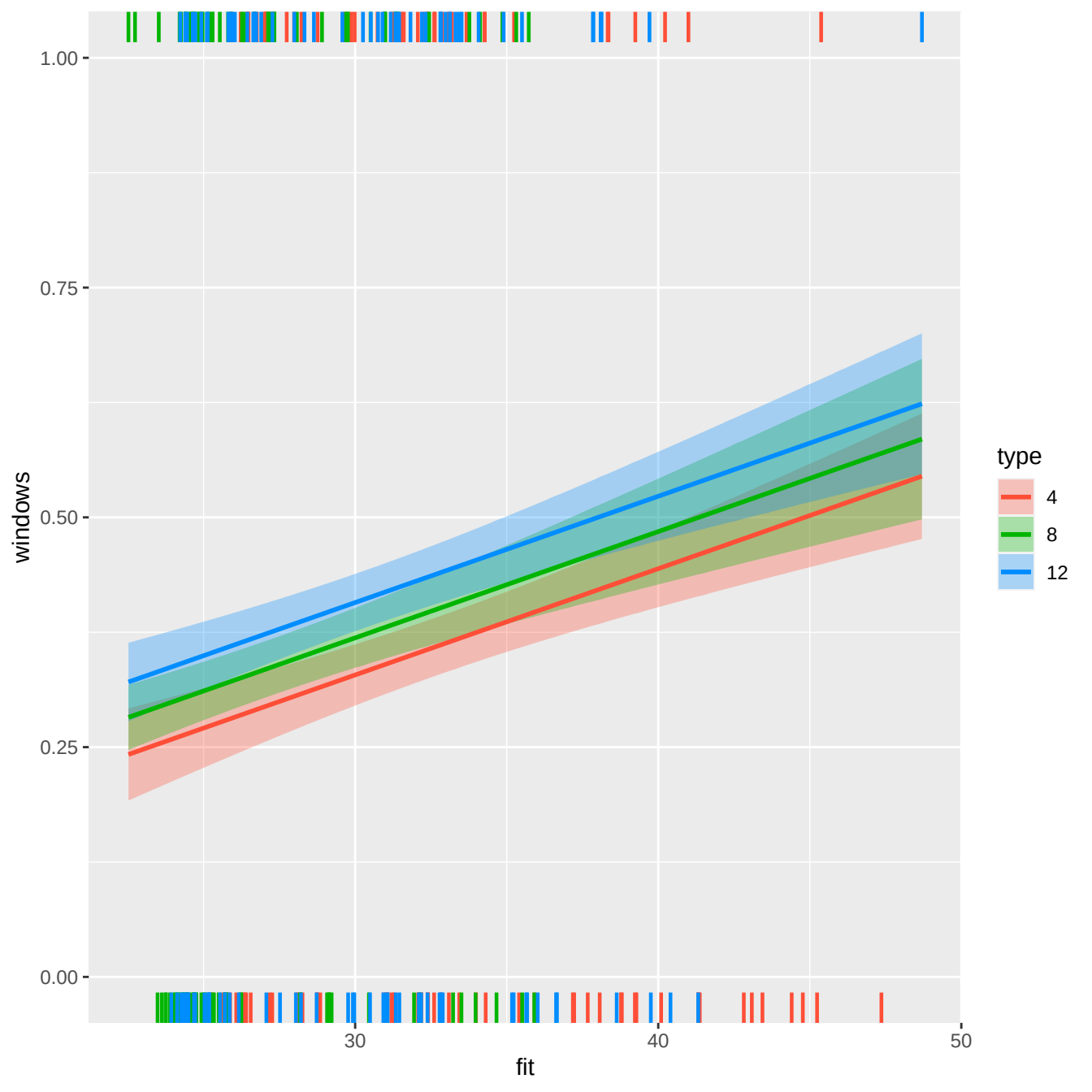

Abi_Anterior_bias.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

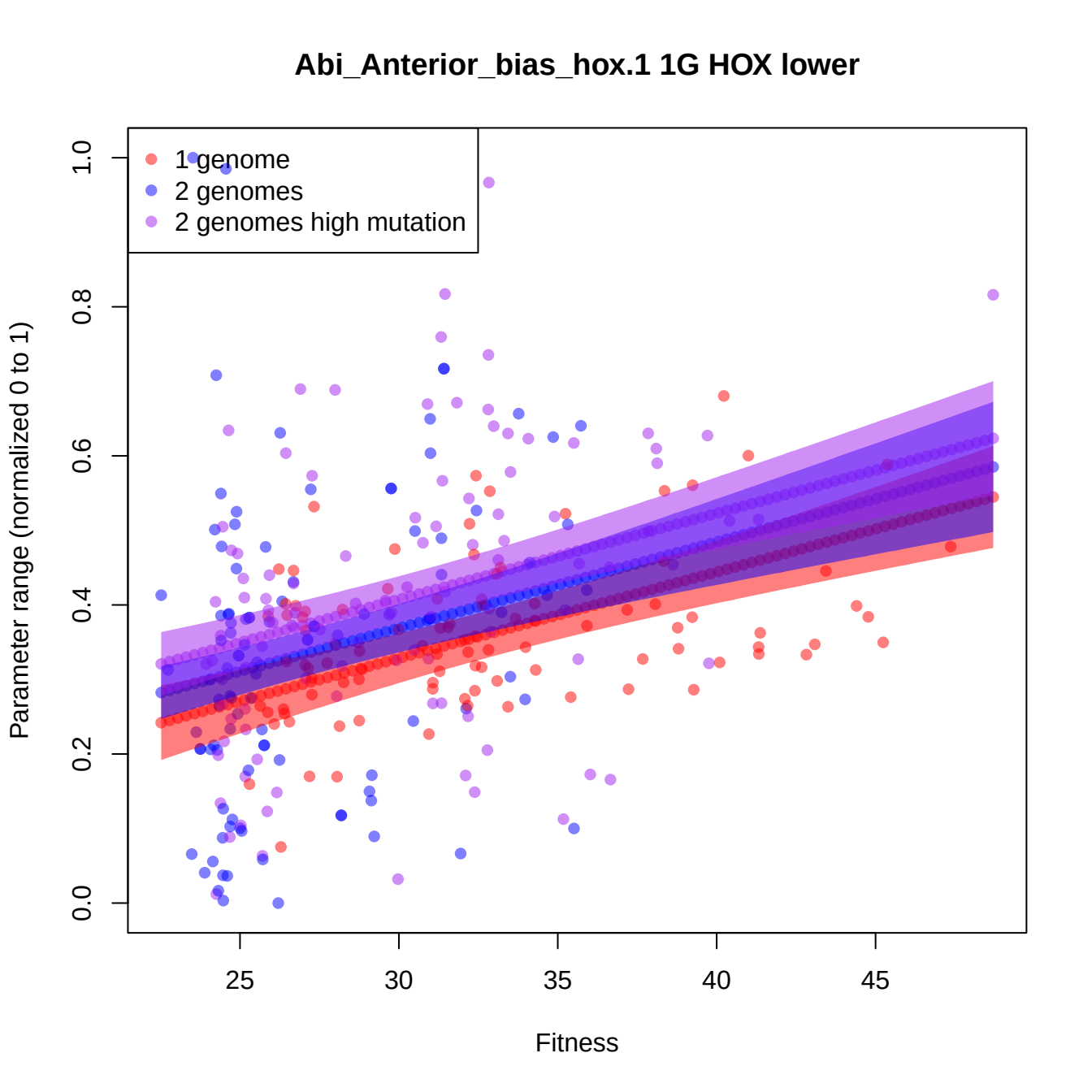

Abi_Anterior_bias.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.42849162722237e−11
2.29690515636587e−11
1
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

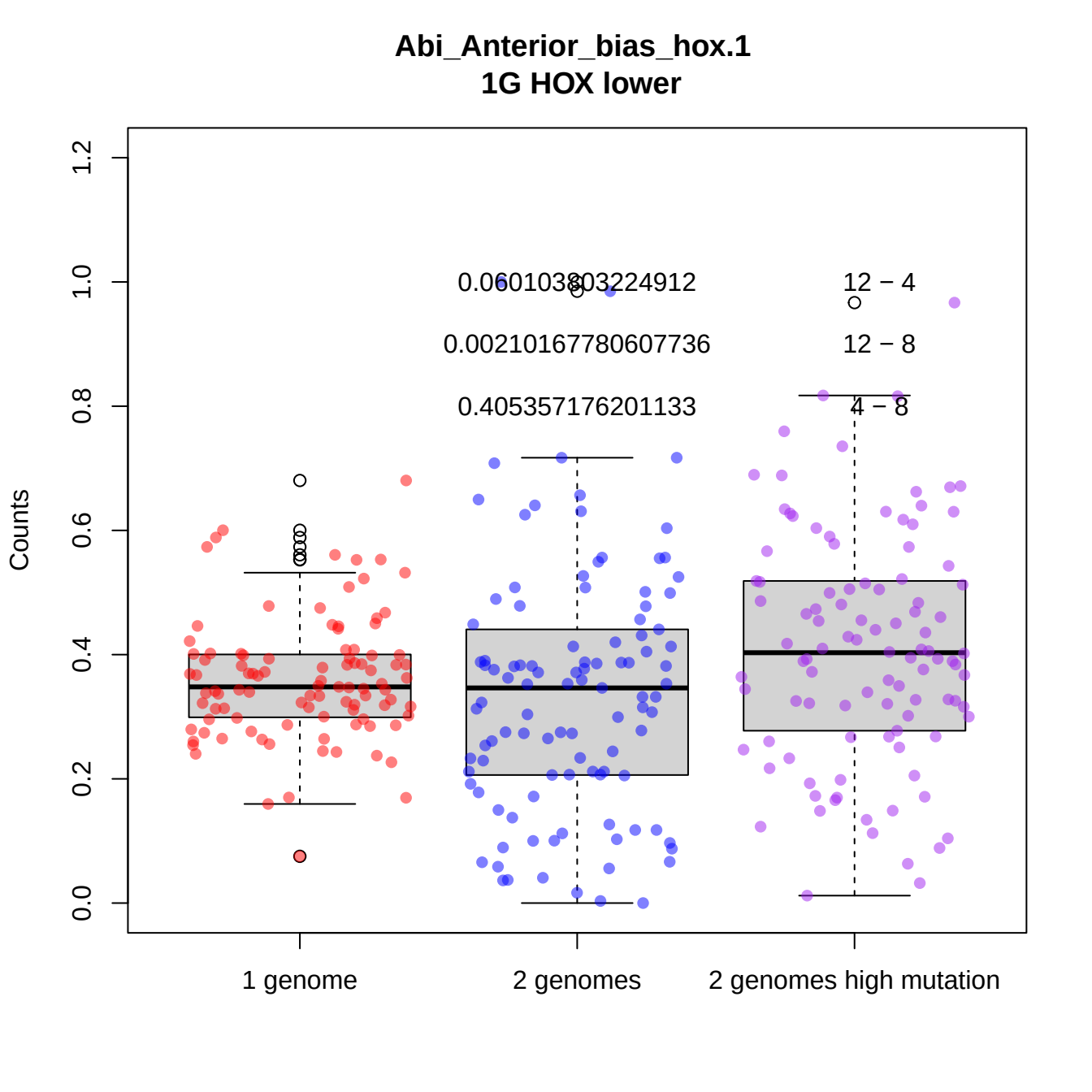

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
3
7
11

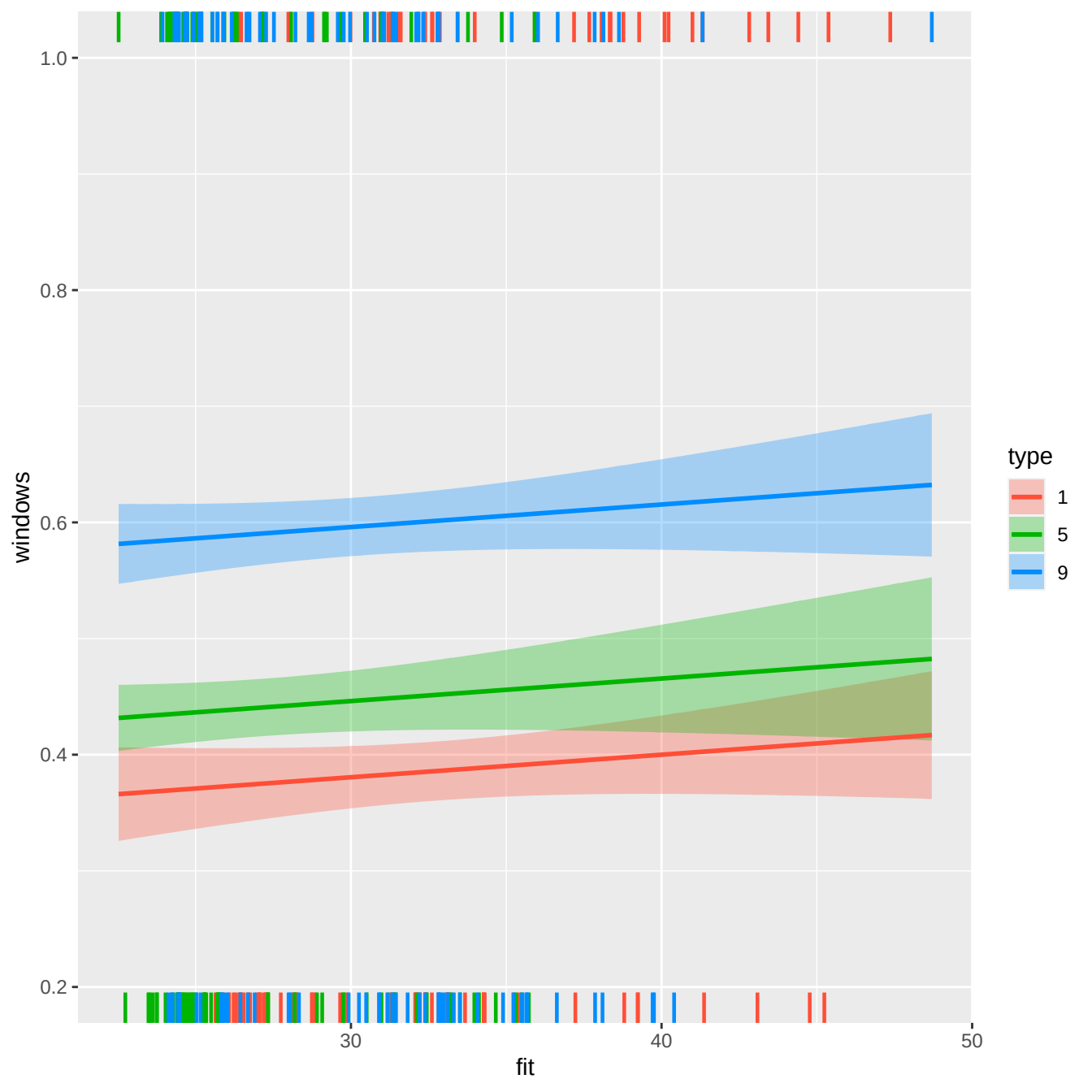

Abi_Anterior_bias_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

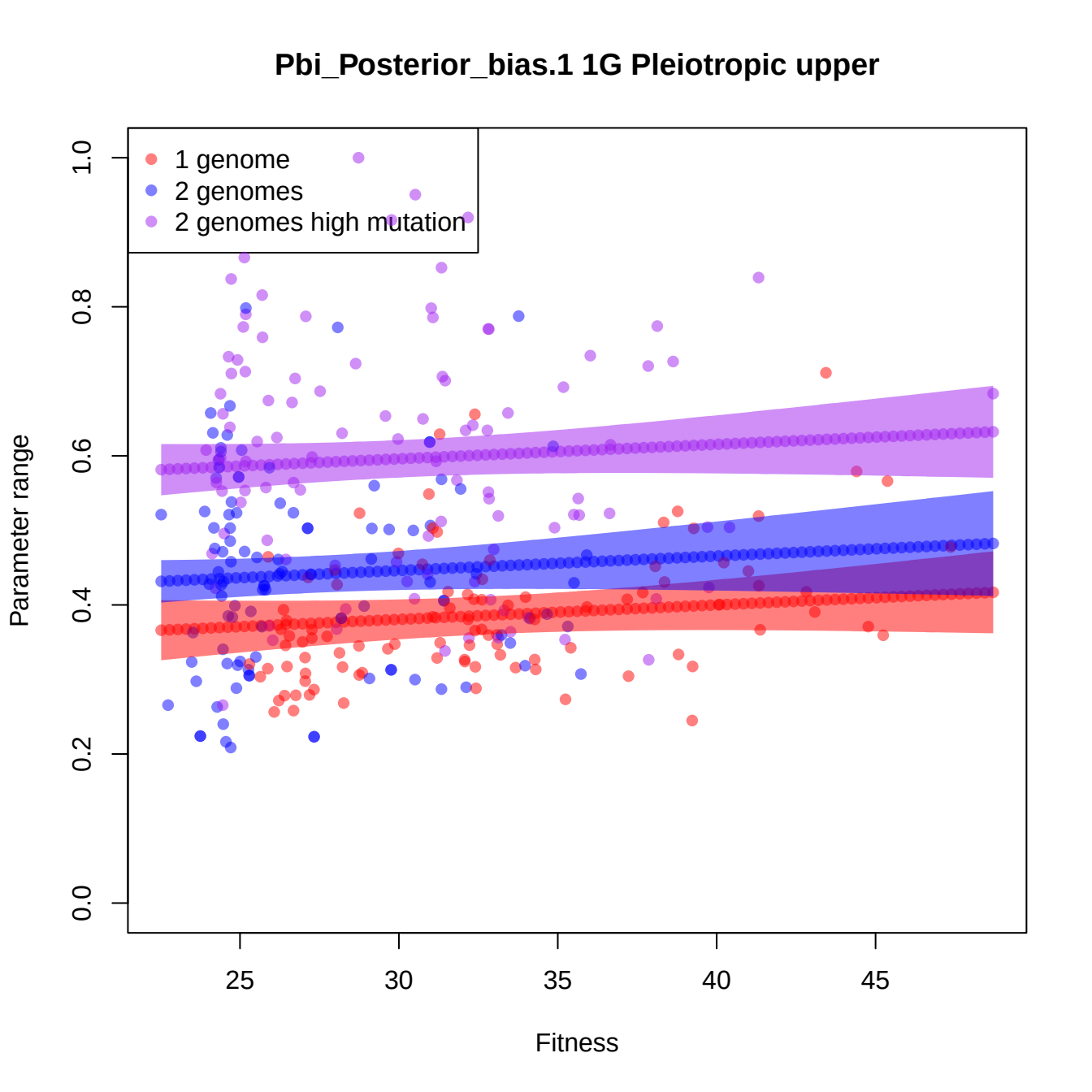

Abi_Anterior_bias_hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
5.50098881527818e−21
0.653538473382413
3.64541970529877e−18
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

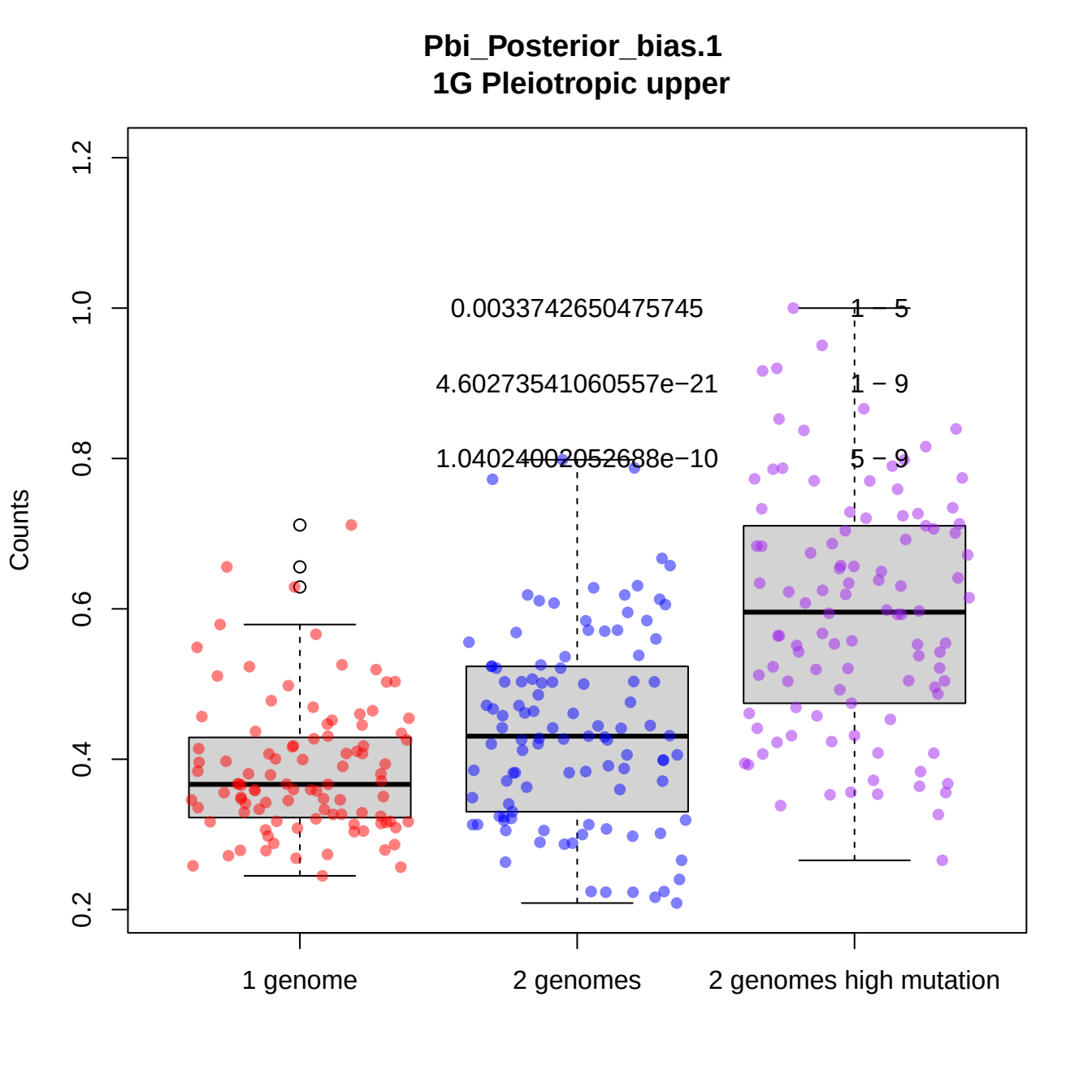

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

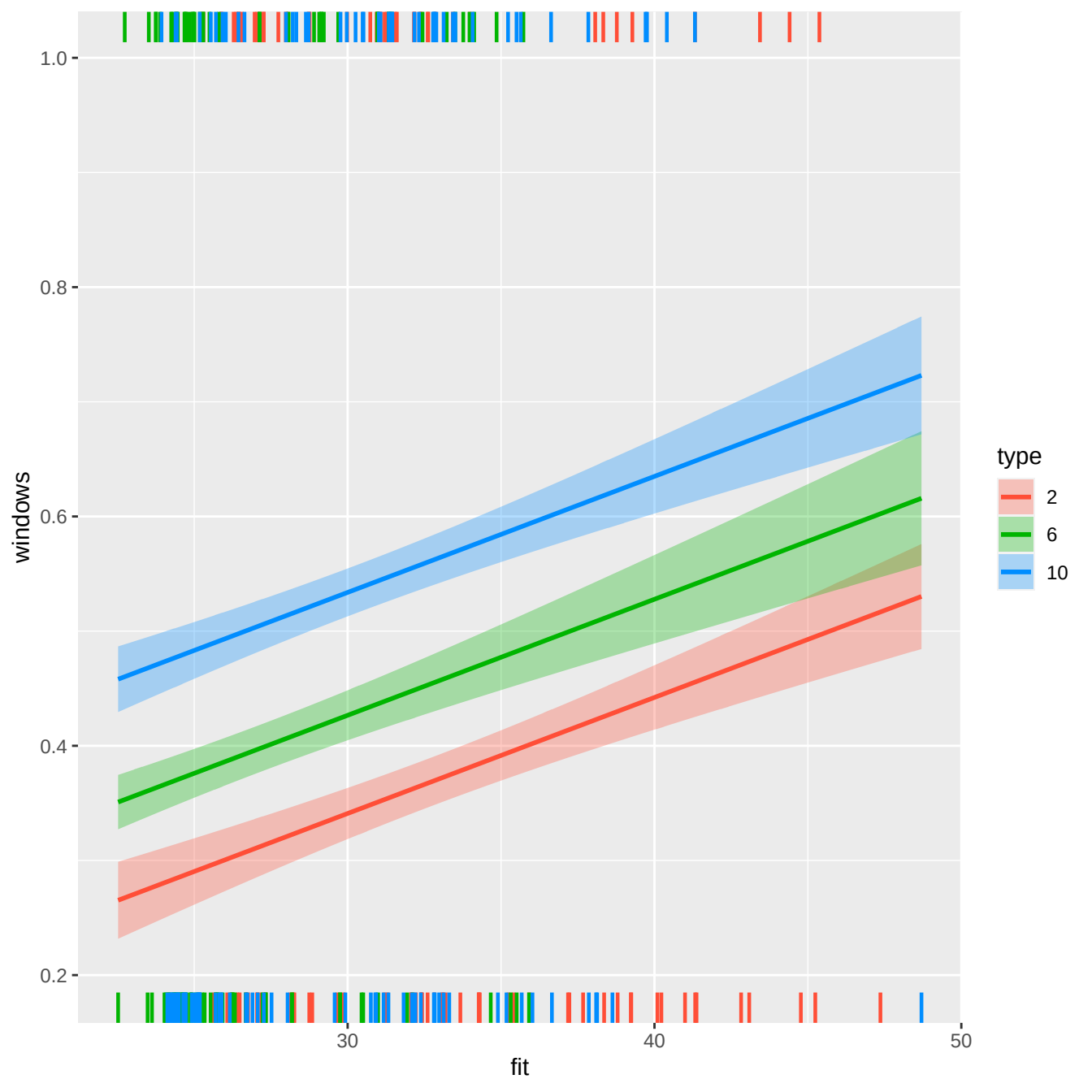

Abi_Anterior_bias_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

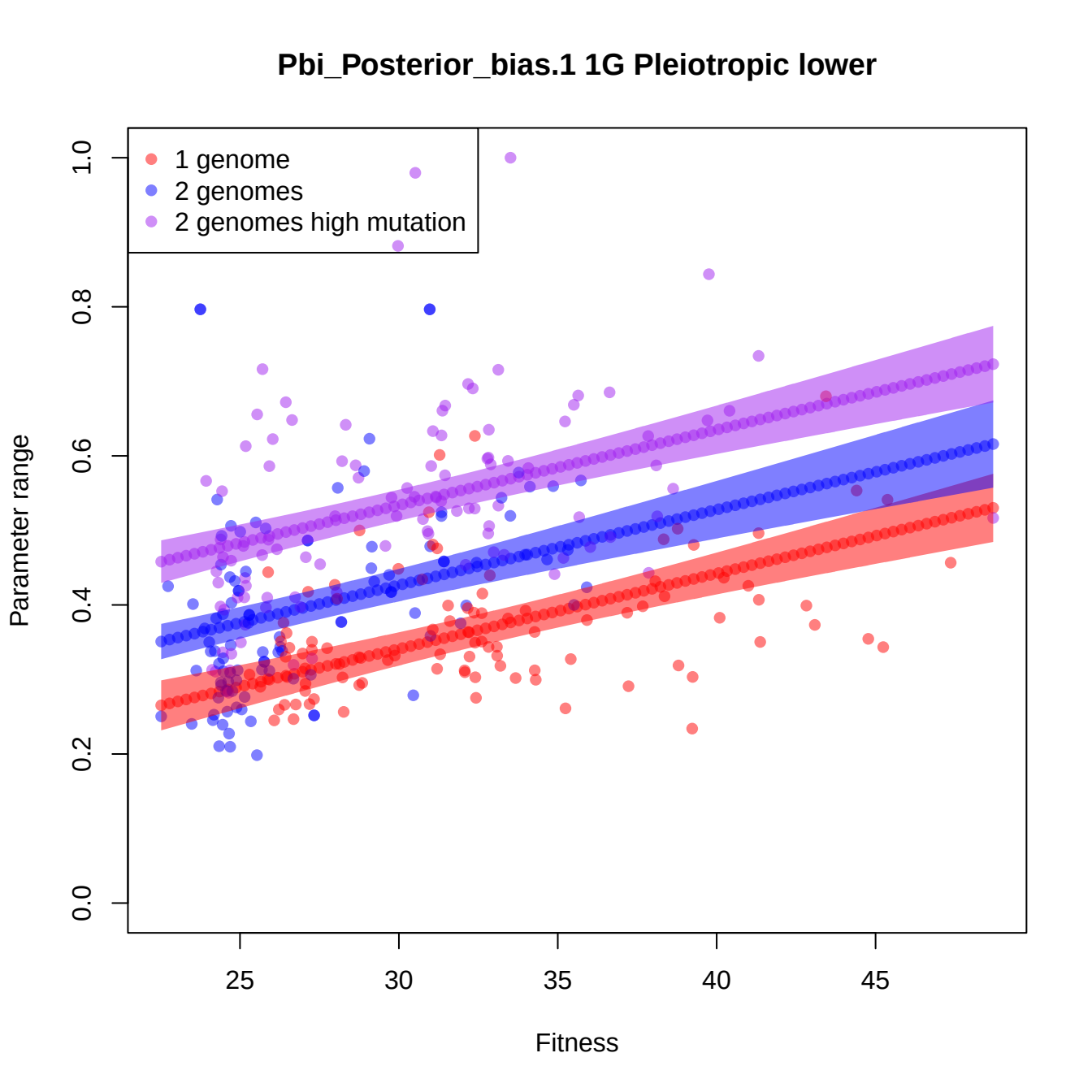

Abi_Anterior_bias_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.0601038803224912
0.00210167780607736
0.405357176201133
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

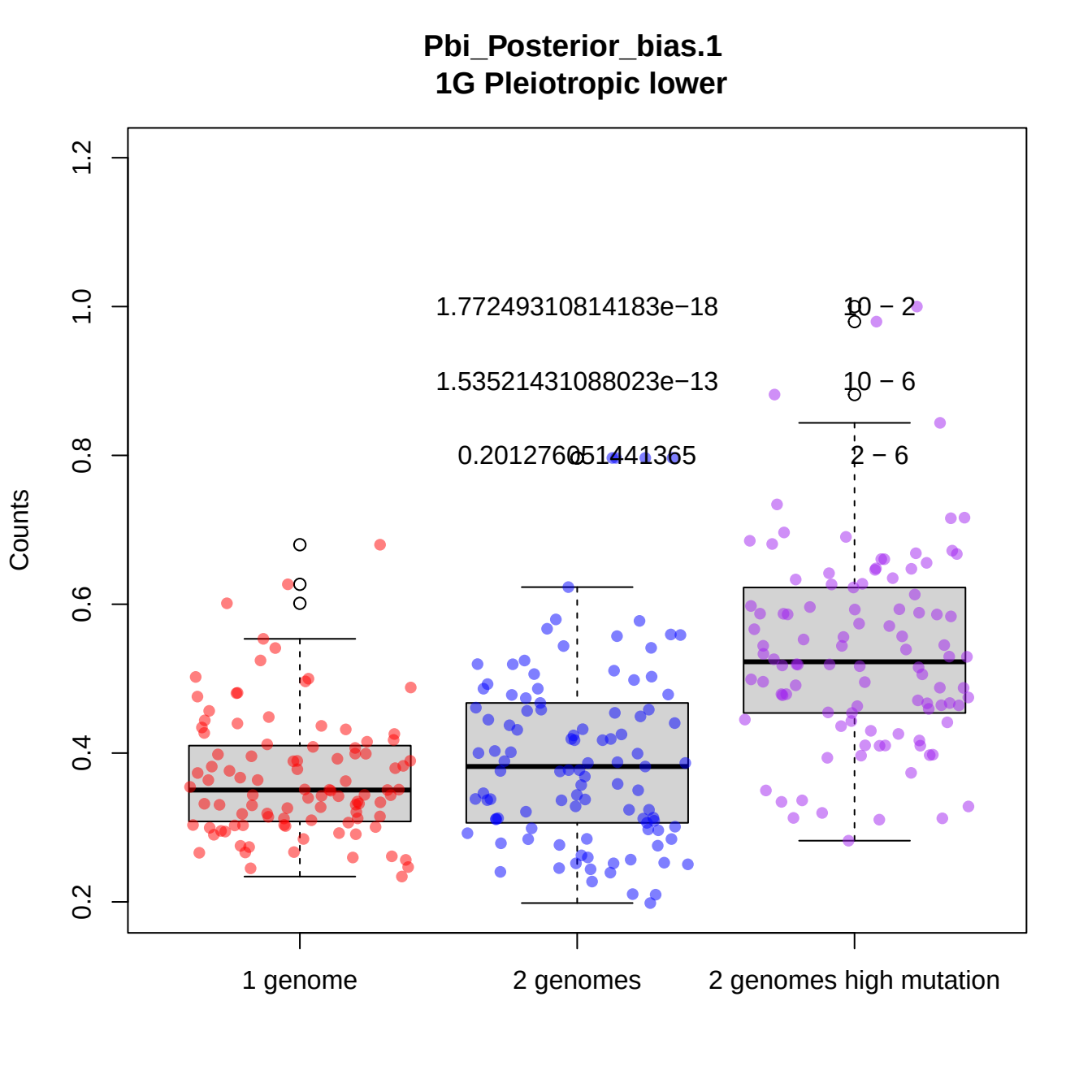

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
1
5
9

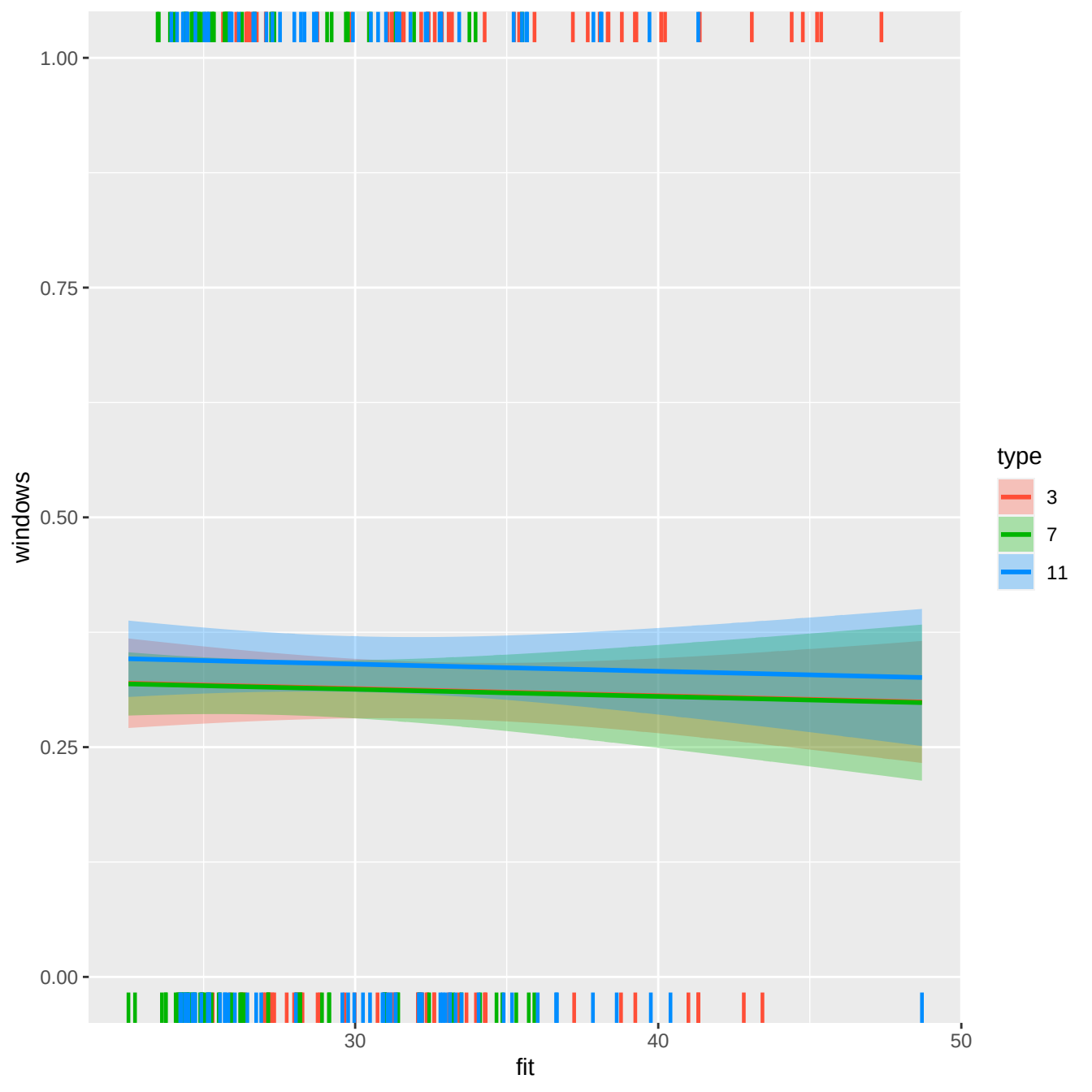

Pbi_Posterior_bias.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

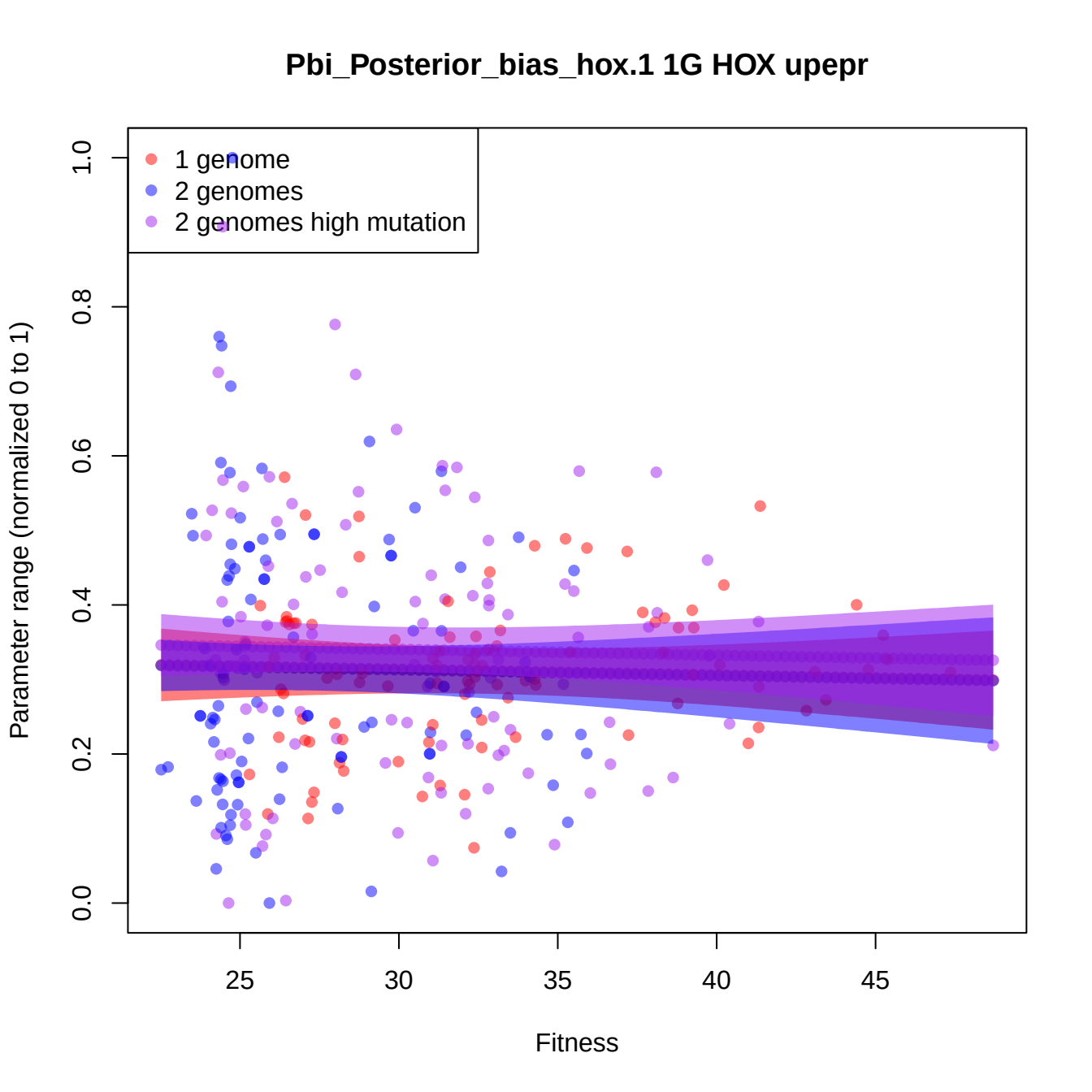

Pbi_Posterior_bias.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
0.0033742650475745
1 − 5
4.60273541060557e−21
1 − 9
1.04024002052688e−10
5 − 9
1 genome
2 genomes
2 genomes high mutation

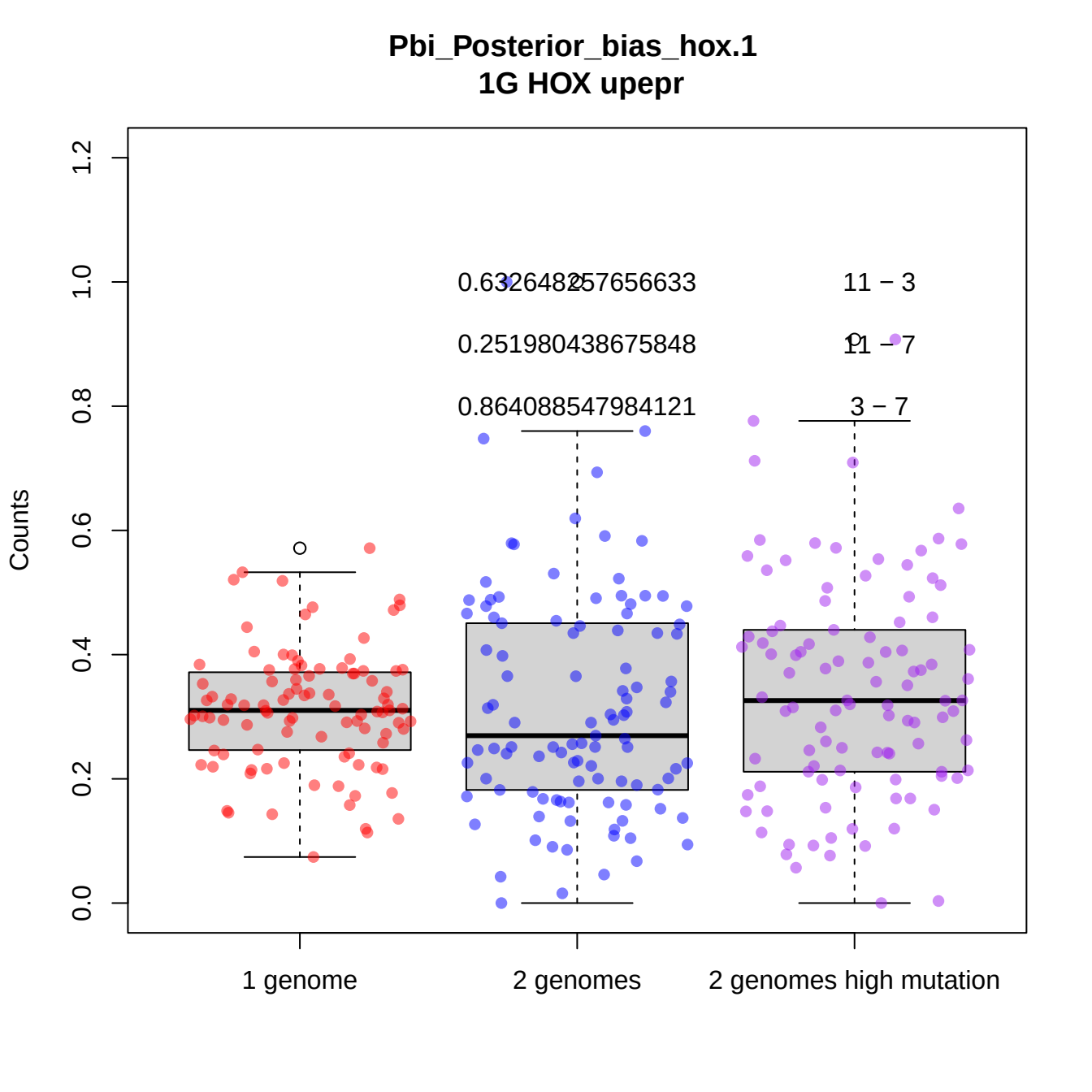

1.0
0.8
0.6
0.4
0.2
windows
30
40
50
fit
type
2
6
10

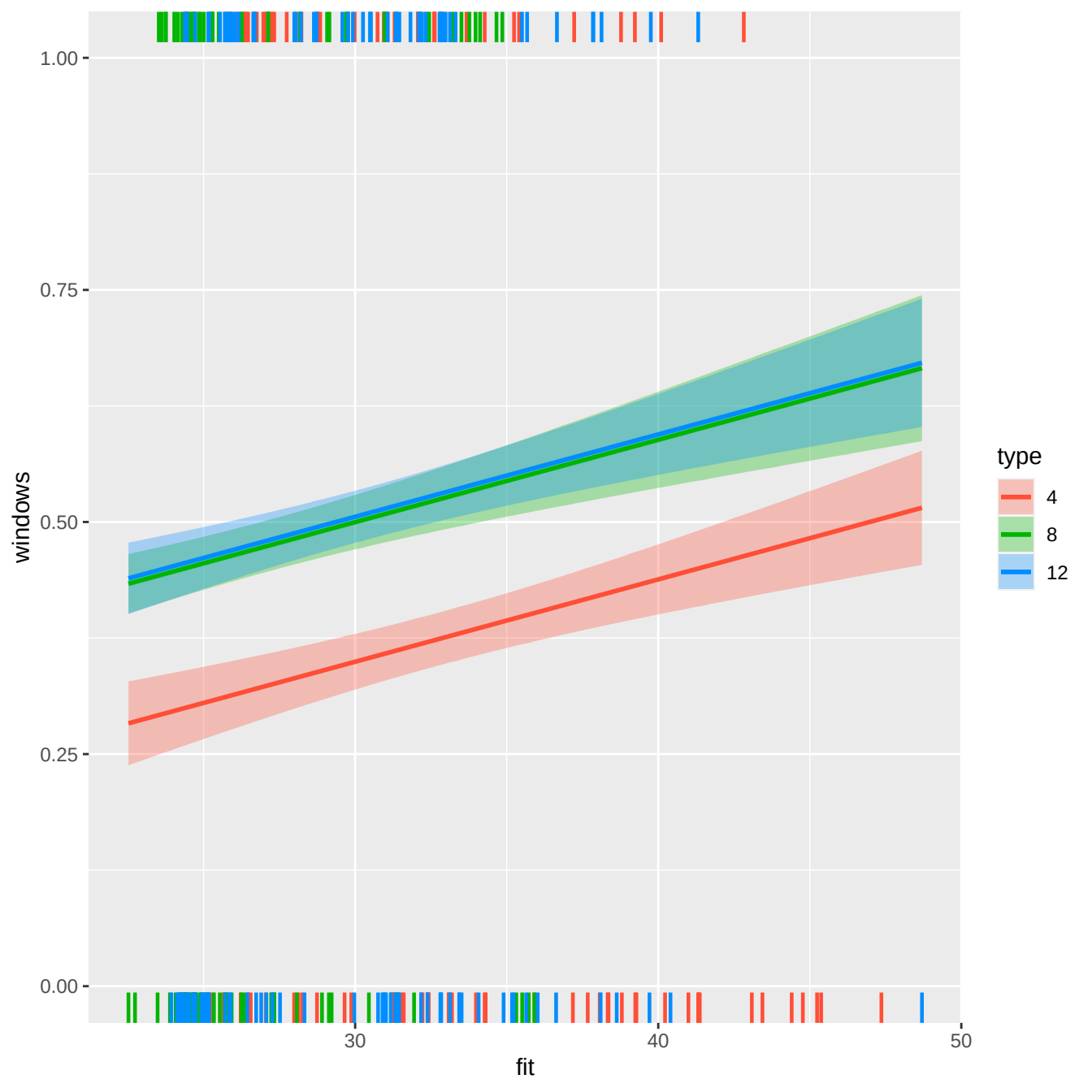

Pbi_Posterior_bias.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

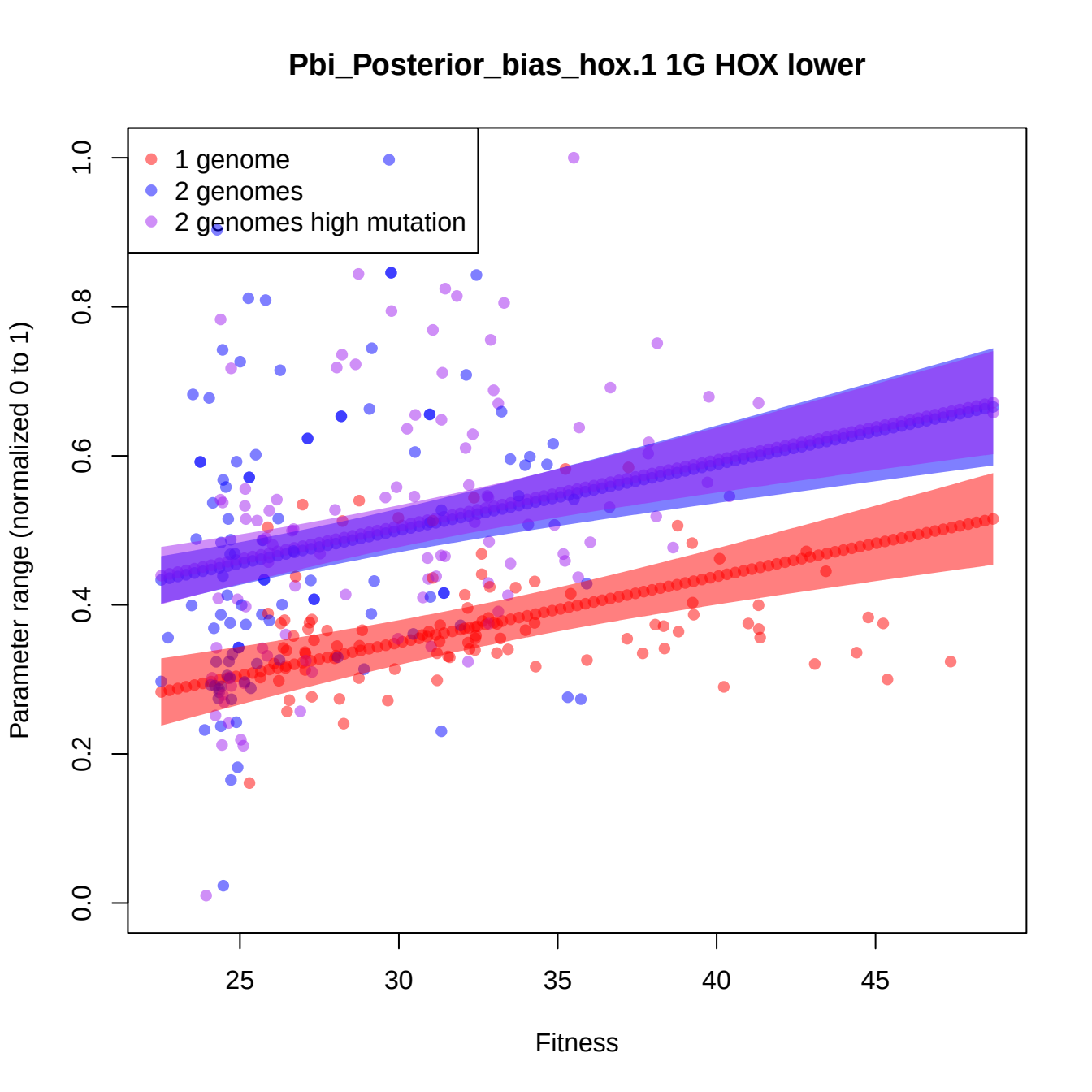

Pbi_Posterior_bias.1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
1.77249310814183e−18
1.53521431088023e−13
0.201276051441365
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

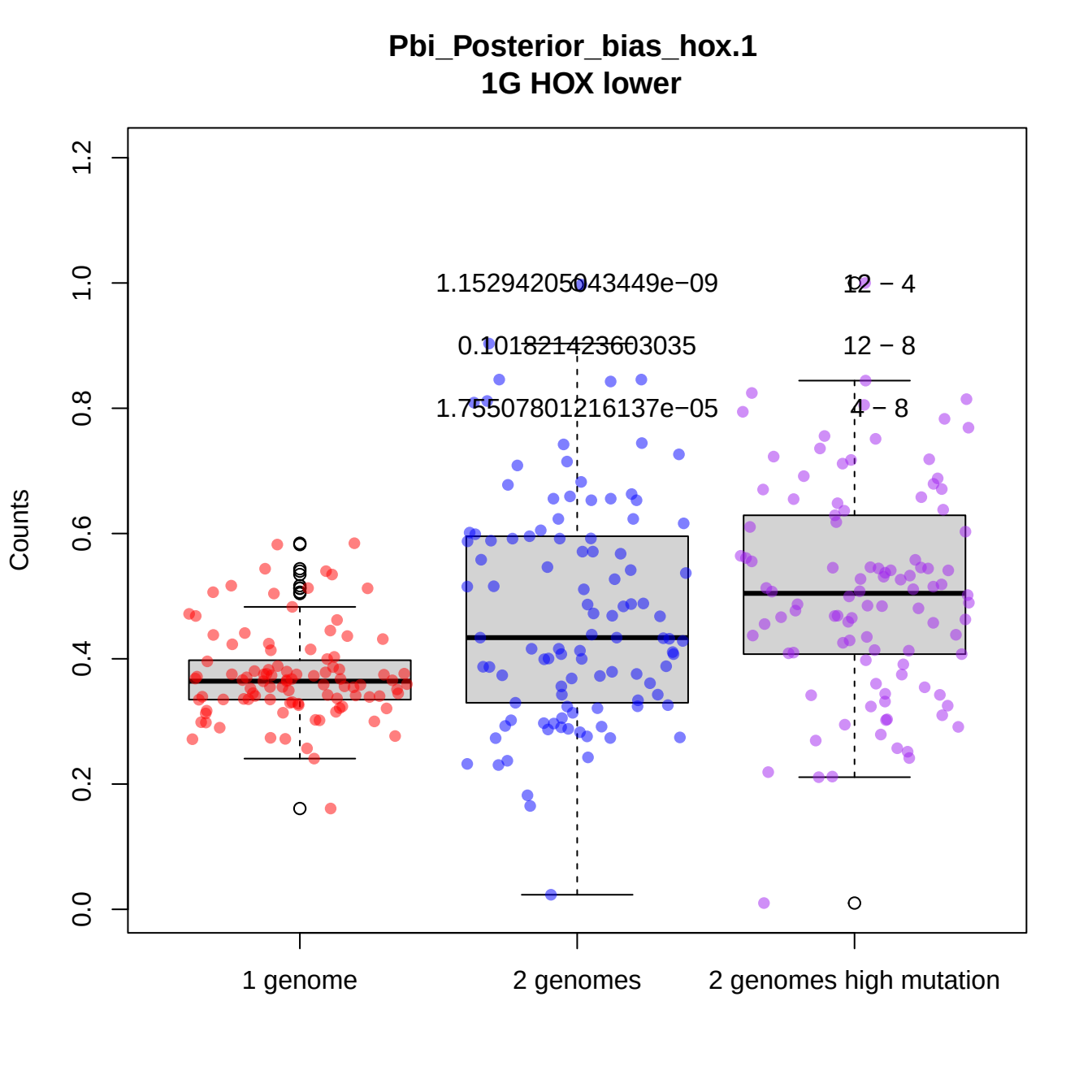

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11

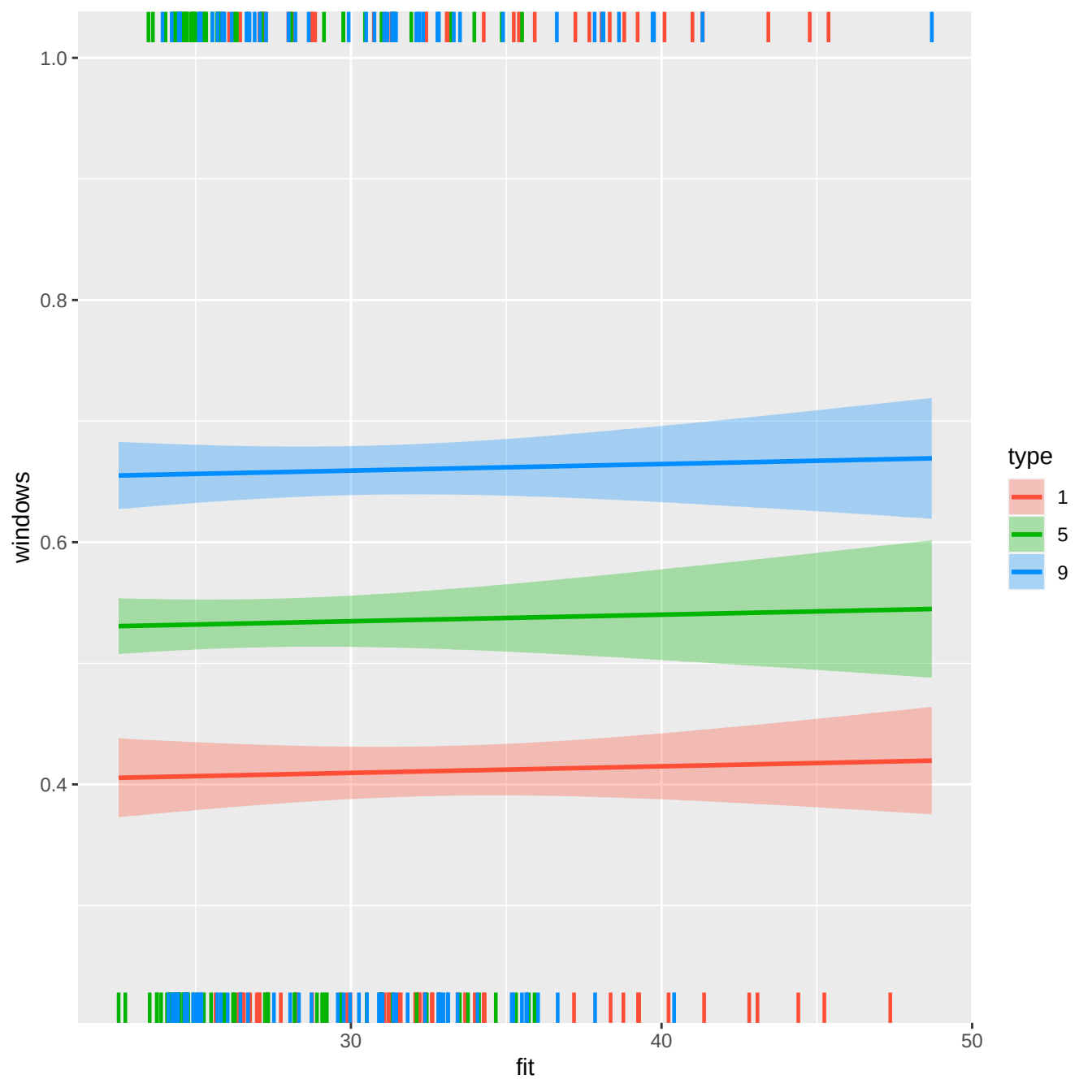

Pbi_Posterior_bias_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

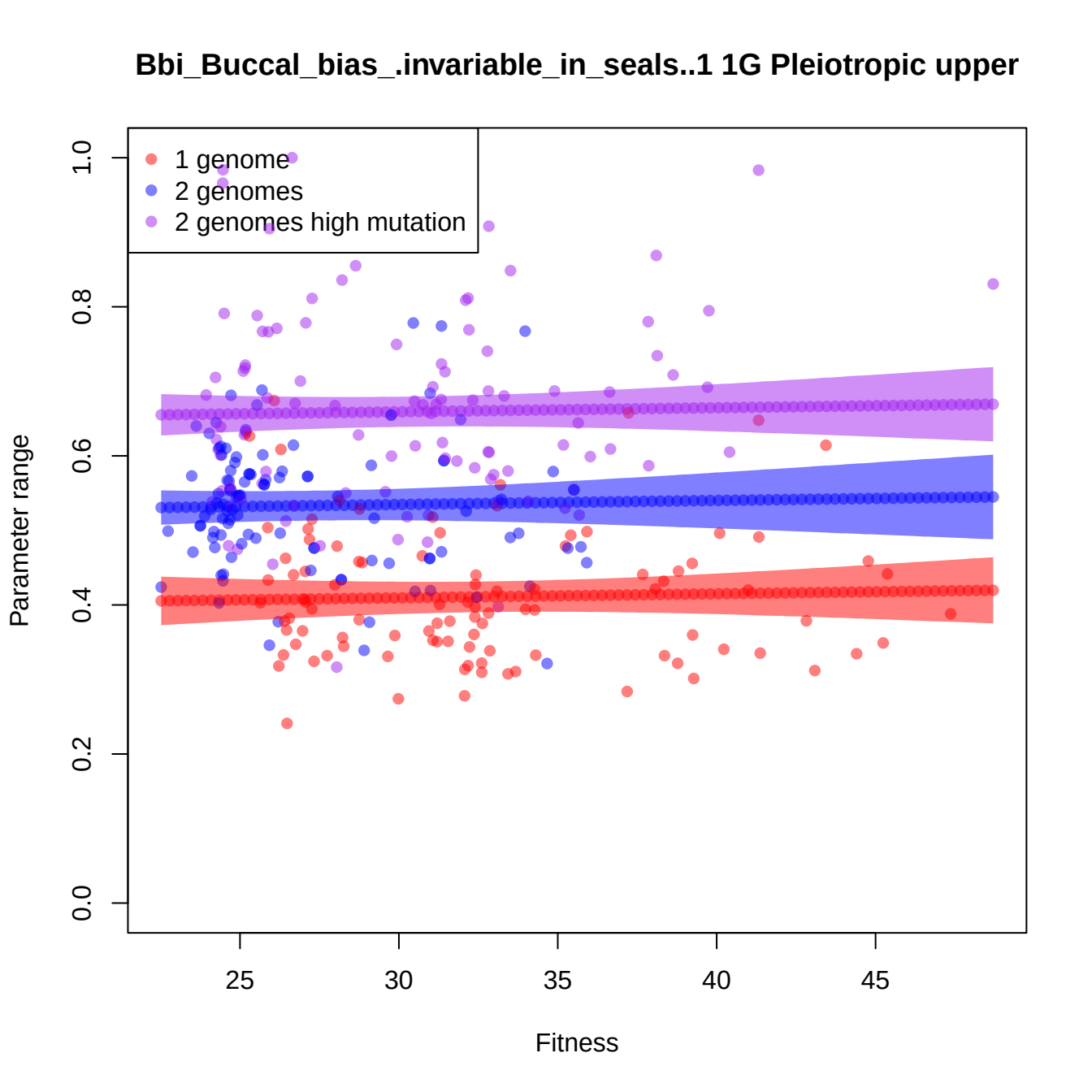

Pbi_Posterior_bias_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.6326482257656633
0.251980438675848
0.864088547984121
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

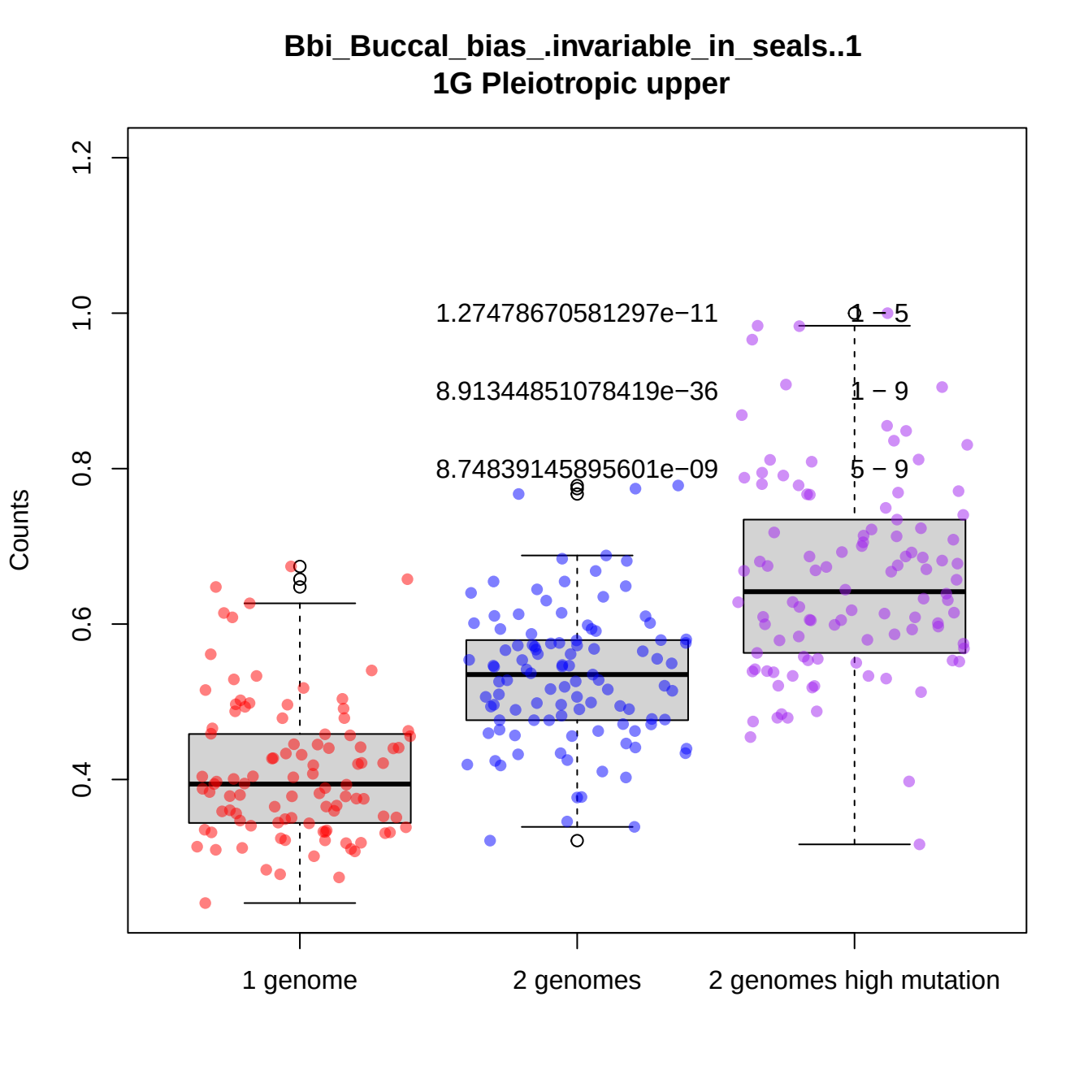

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

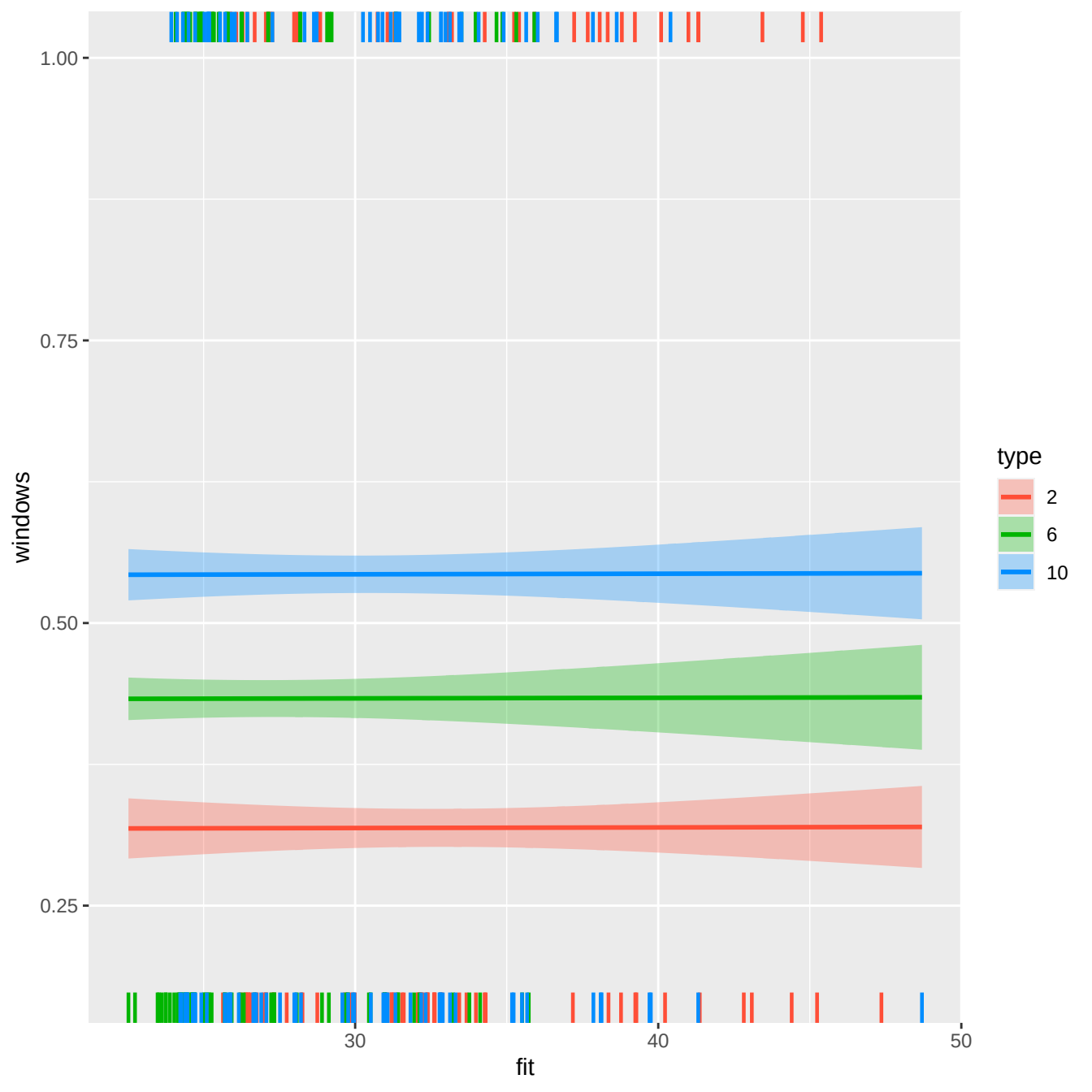

Pbi_Posterior_bias_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

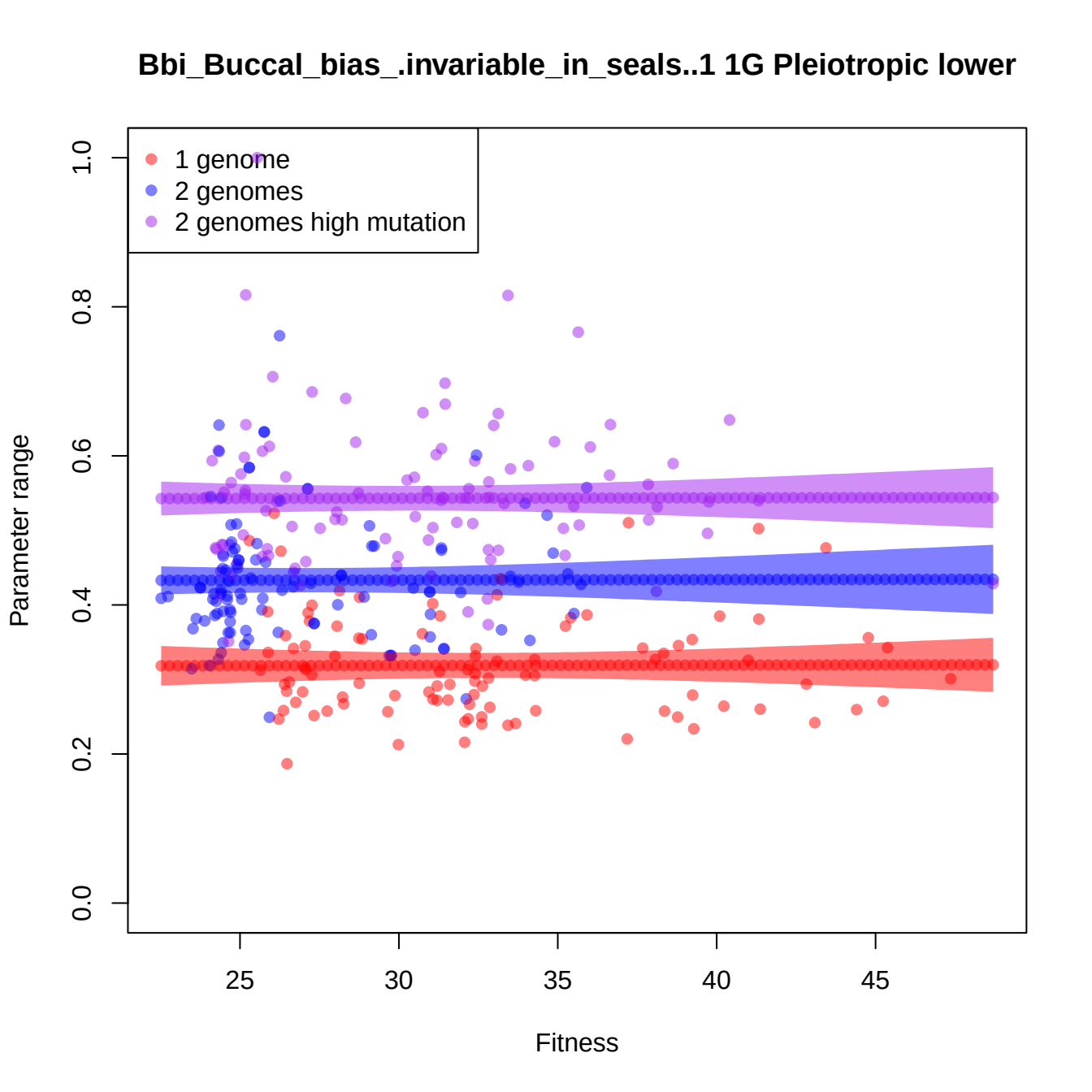

Pbi_Posterior_bias_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.15294205043449e−09
0.101821423603035
1.75507801216137e−05
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

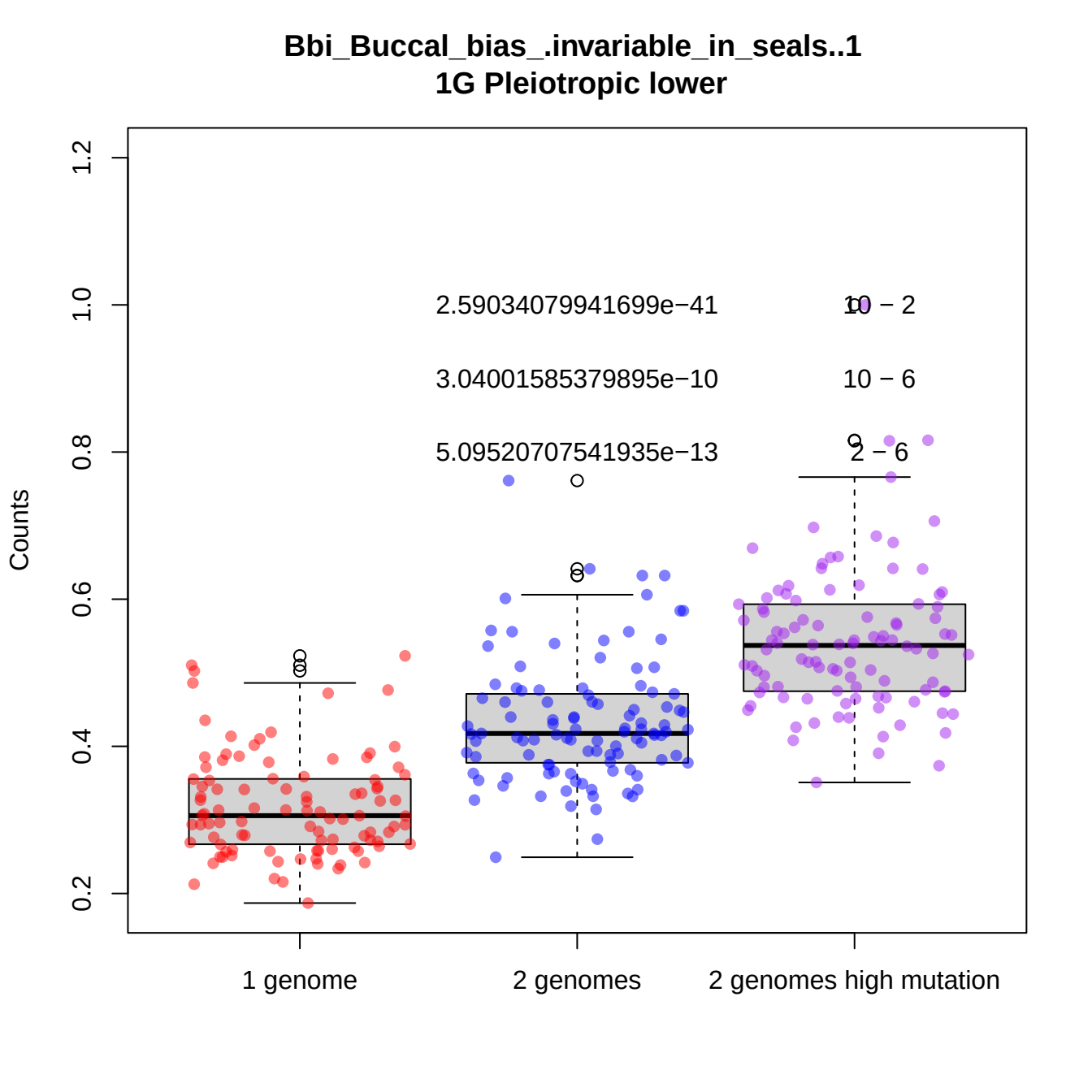

1.0
0.8
0.6
0.4
windows
30
40
50
fit
type
1
5
9

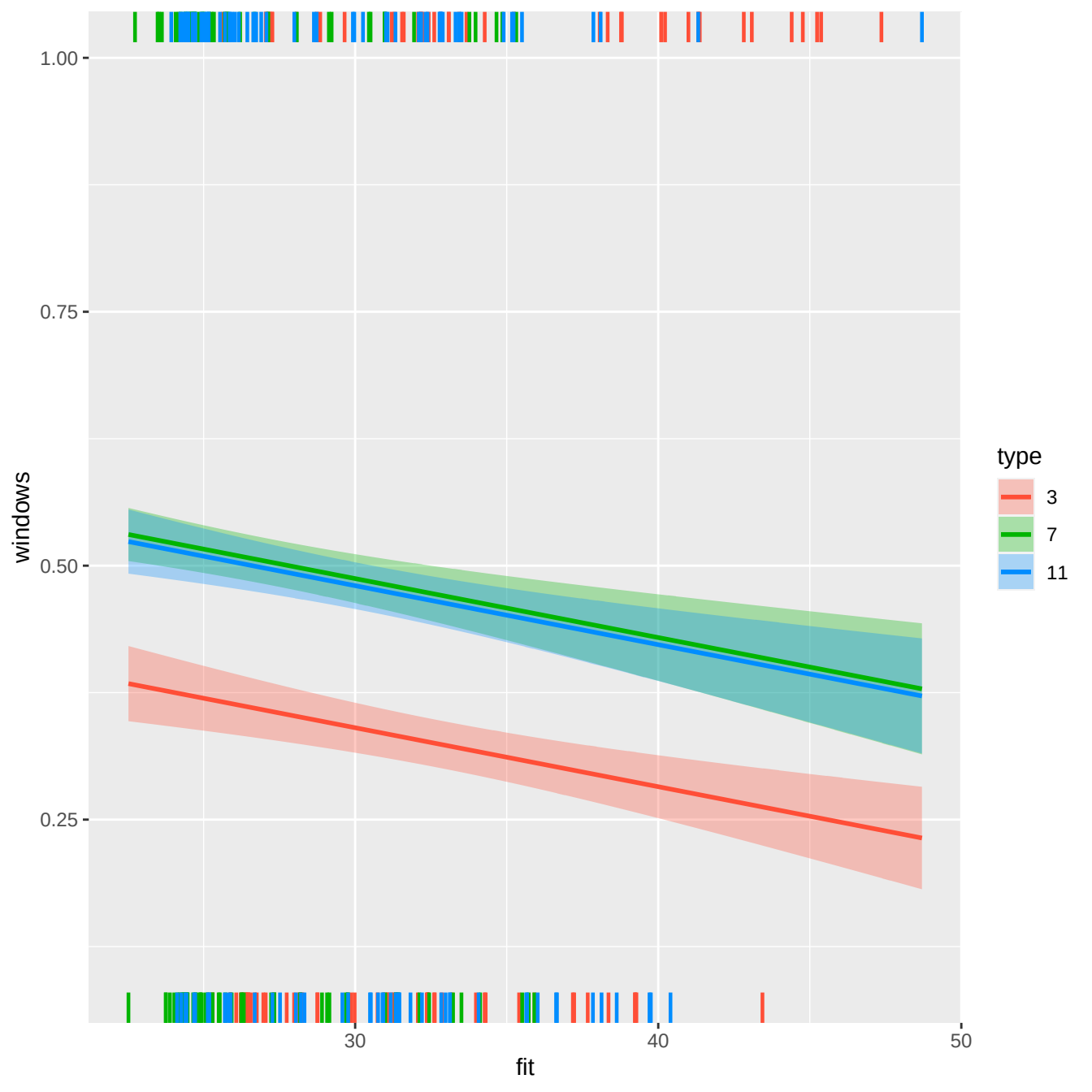

Bbi_Buccal_bias_.invariable_in_seals..1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

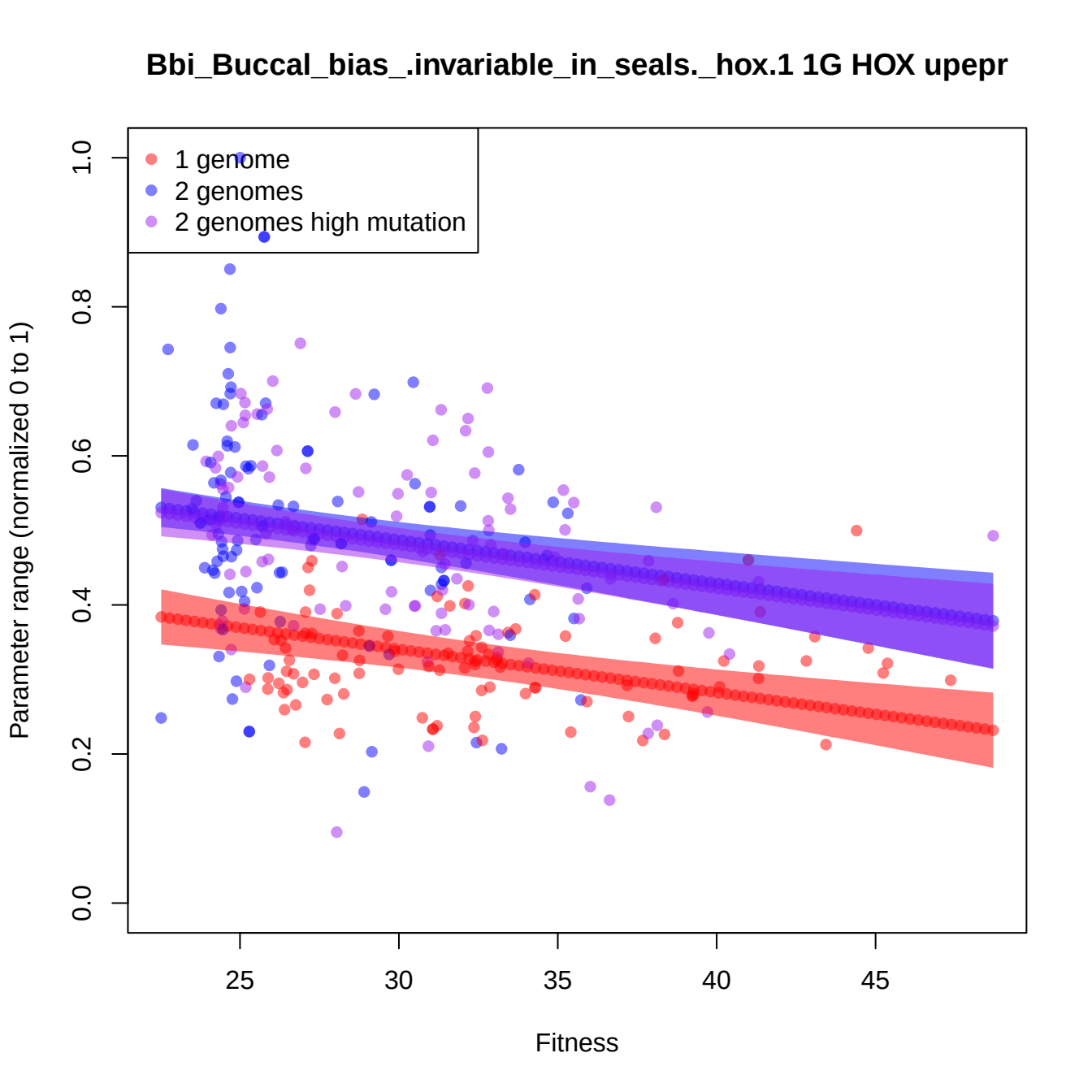

Bbi_Buccal_bias_.invariable_in_seals..1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
1.27478670581297e−11
1 − 5
8.91344851078419e−36
1 − 9
8.74839145895601e−09
5 − 9
1 genome
2 genomes
2 genomes high mutation

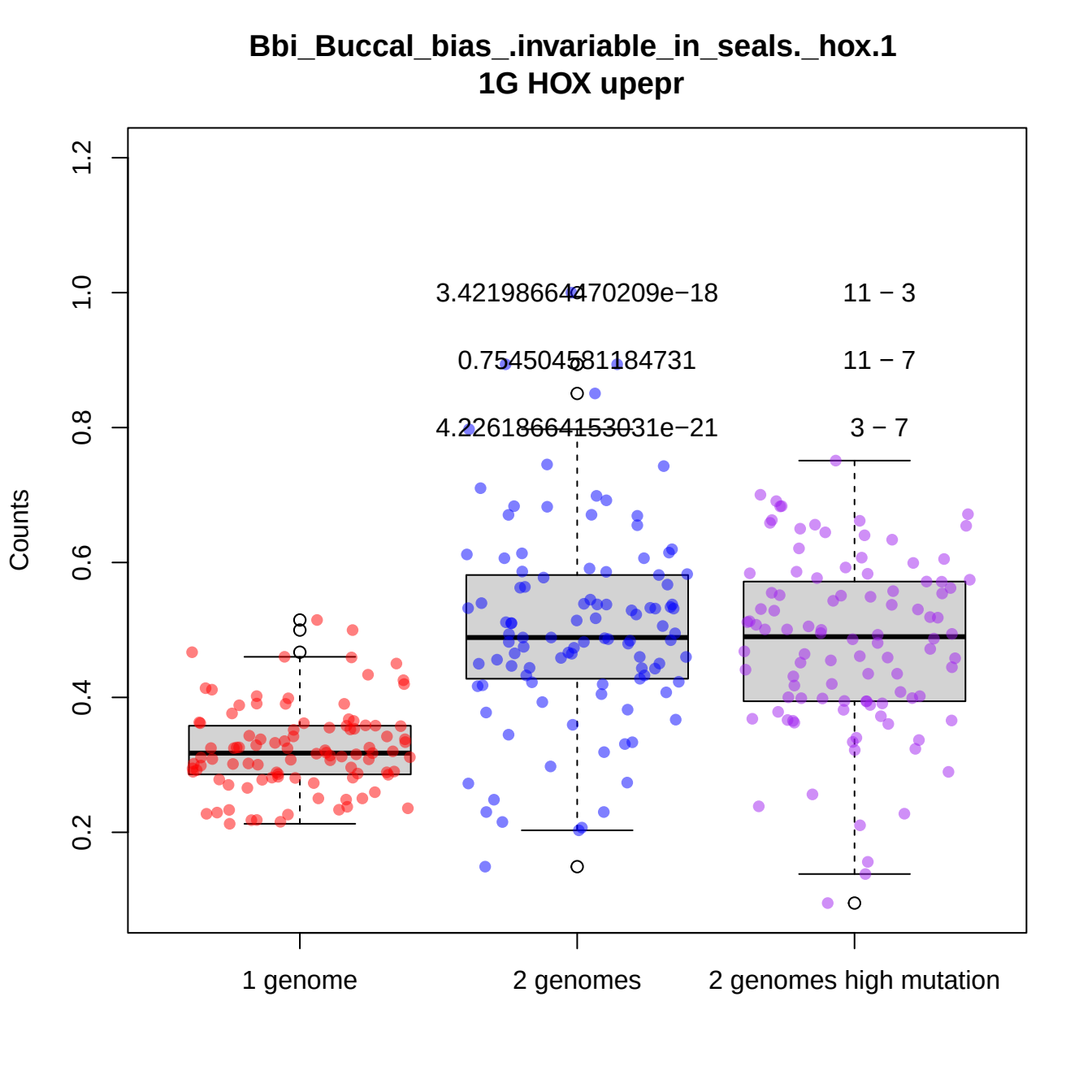

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

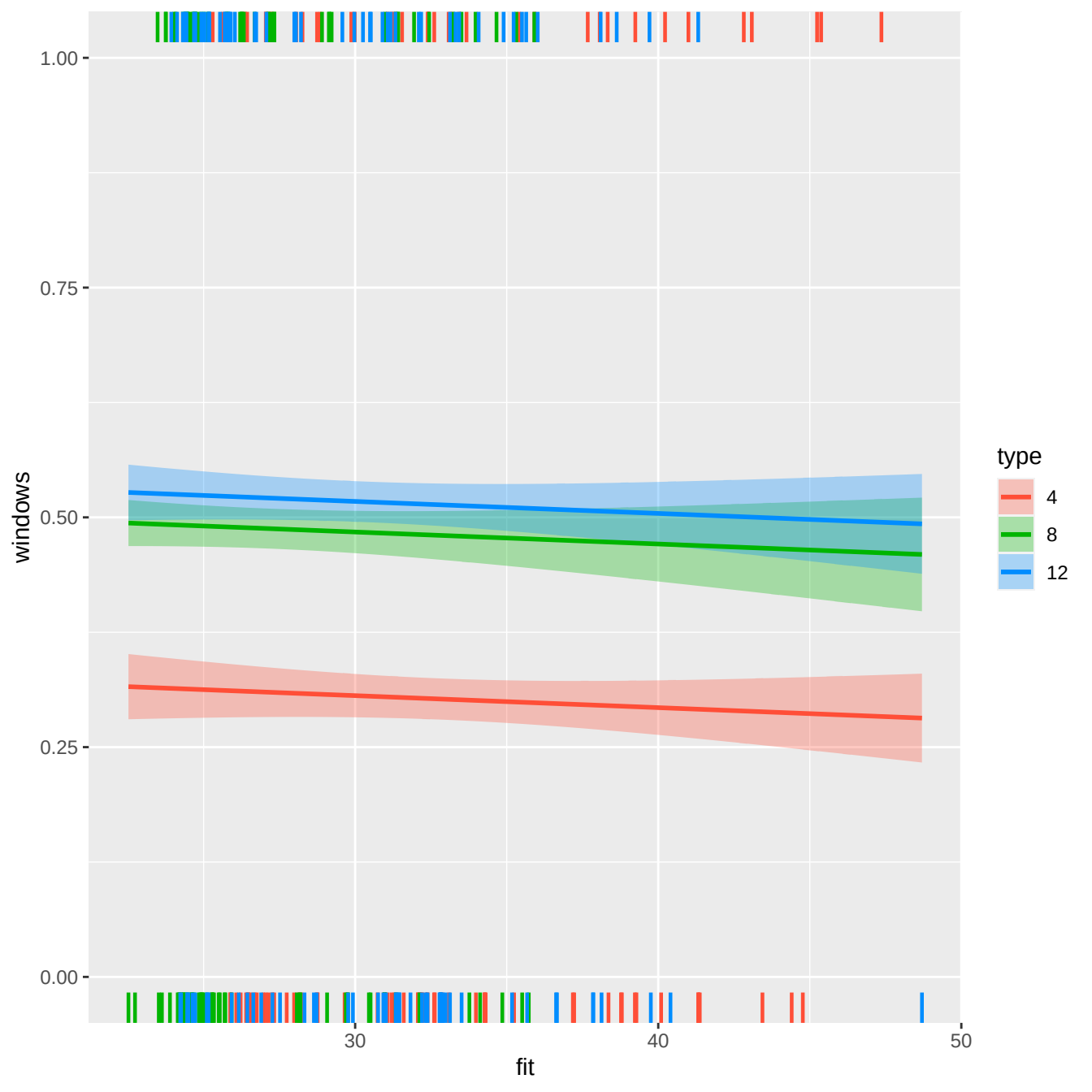

Bbi_Buccal_bias_.invariable_in_seals..1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

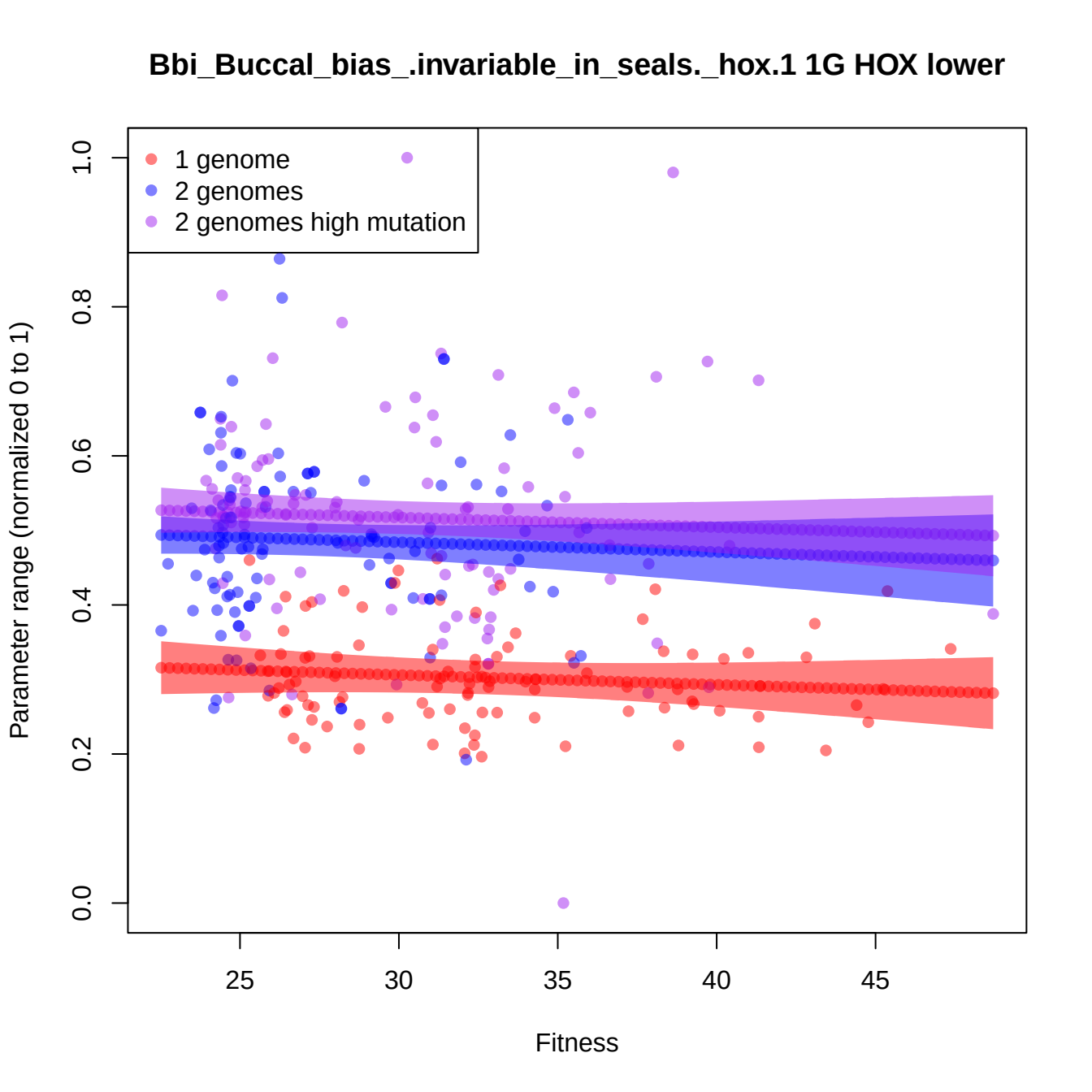

Bbi_Buccal_bias_.invariable_in_seals..1
1G Pleiotropic lower
Counts
1.2
1.0
0.8
0.6
0.4
0.2
2.59034079941699e−41
3.04001585379895e−10
5.09520707541935e−13
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

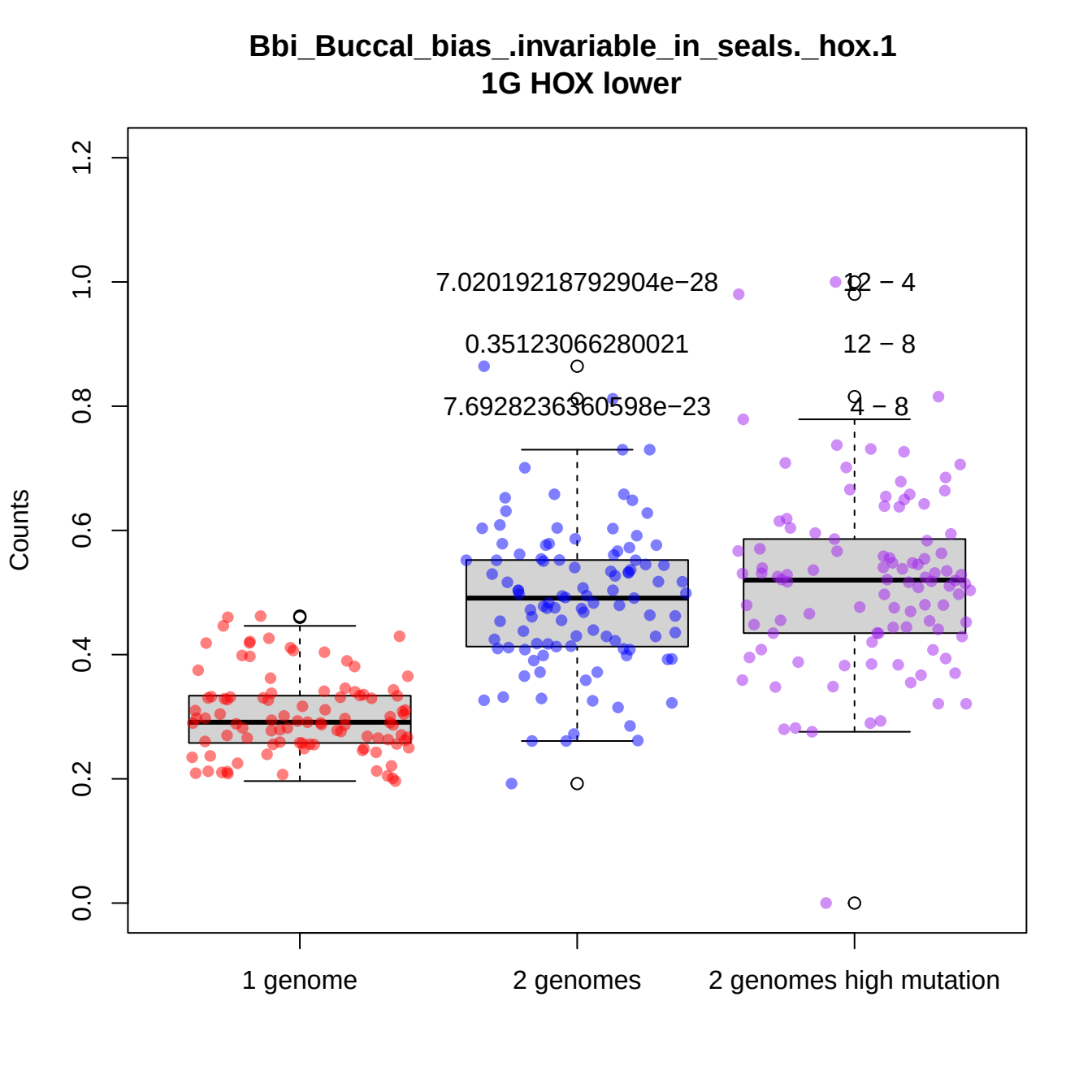

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
3
7
11

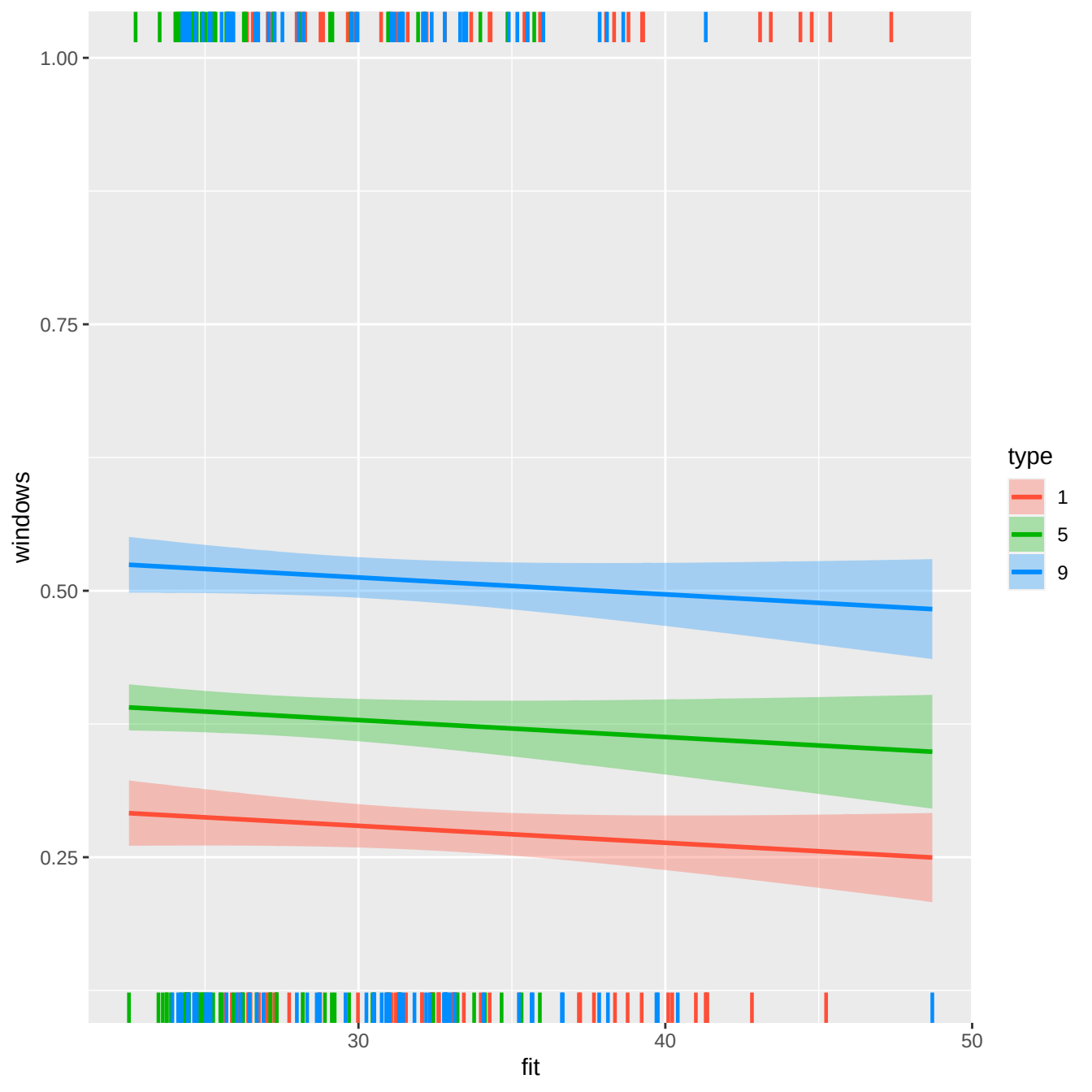

Bbi_Buccal_bias_.invariable_in_seals._hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

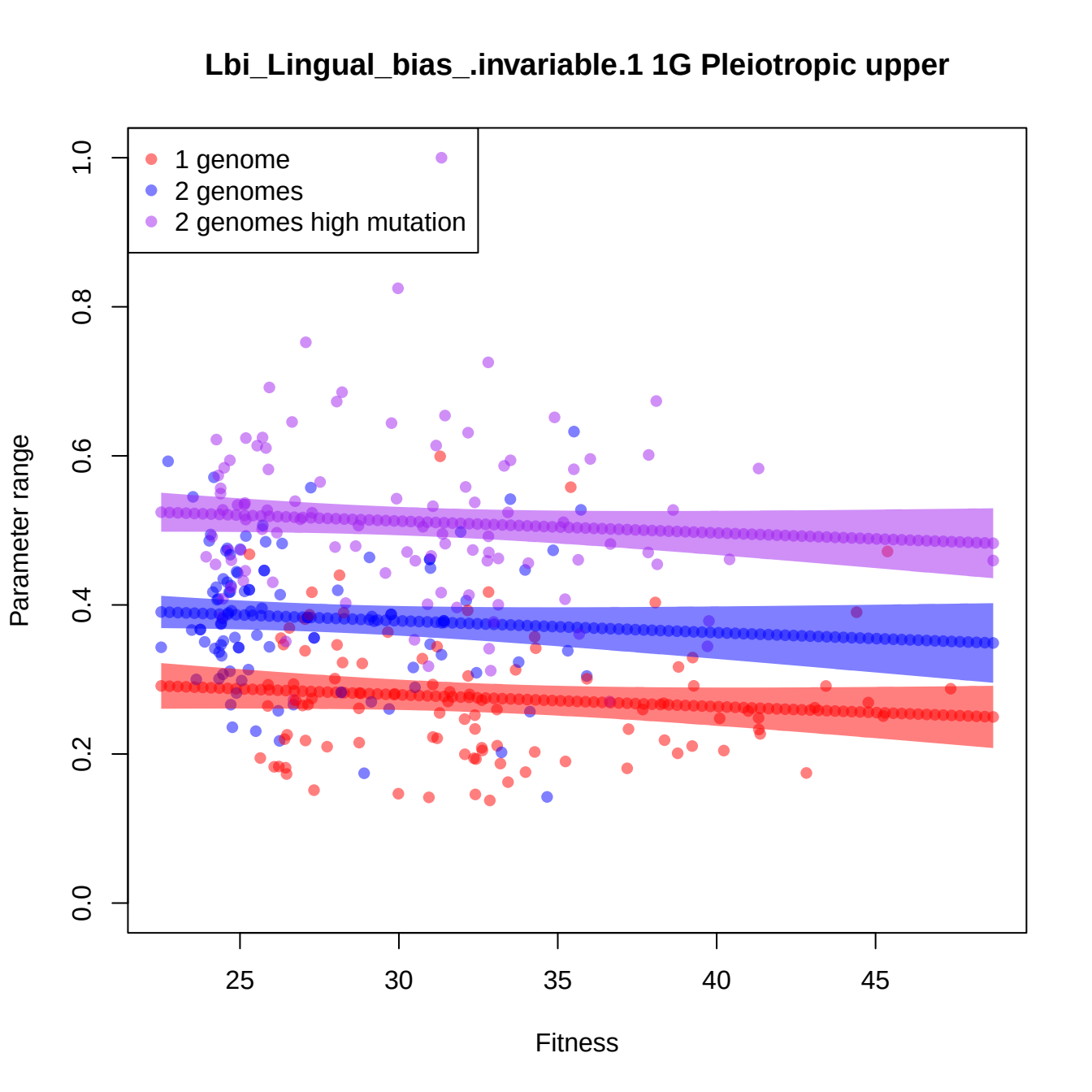

Bbi_Buccal_bias_.invariable_in_seals._hox.1
1G HOX upepr
Counts
1.2
1.0
0.8
0.6
0.4
0.2
3.421986644702O9e−18
0.7545045811847 31
4.2261866415303 1e−21
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation

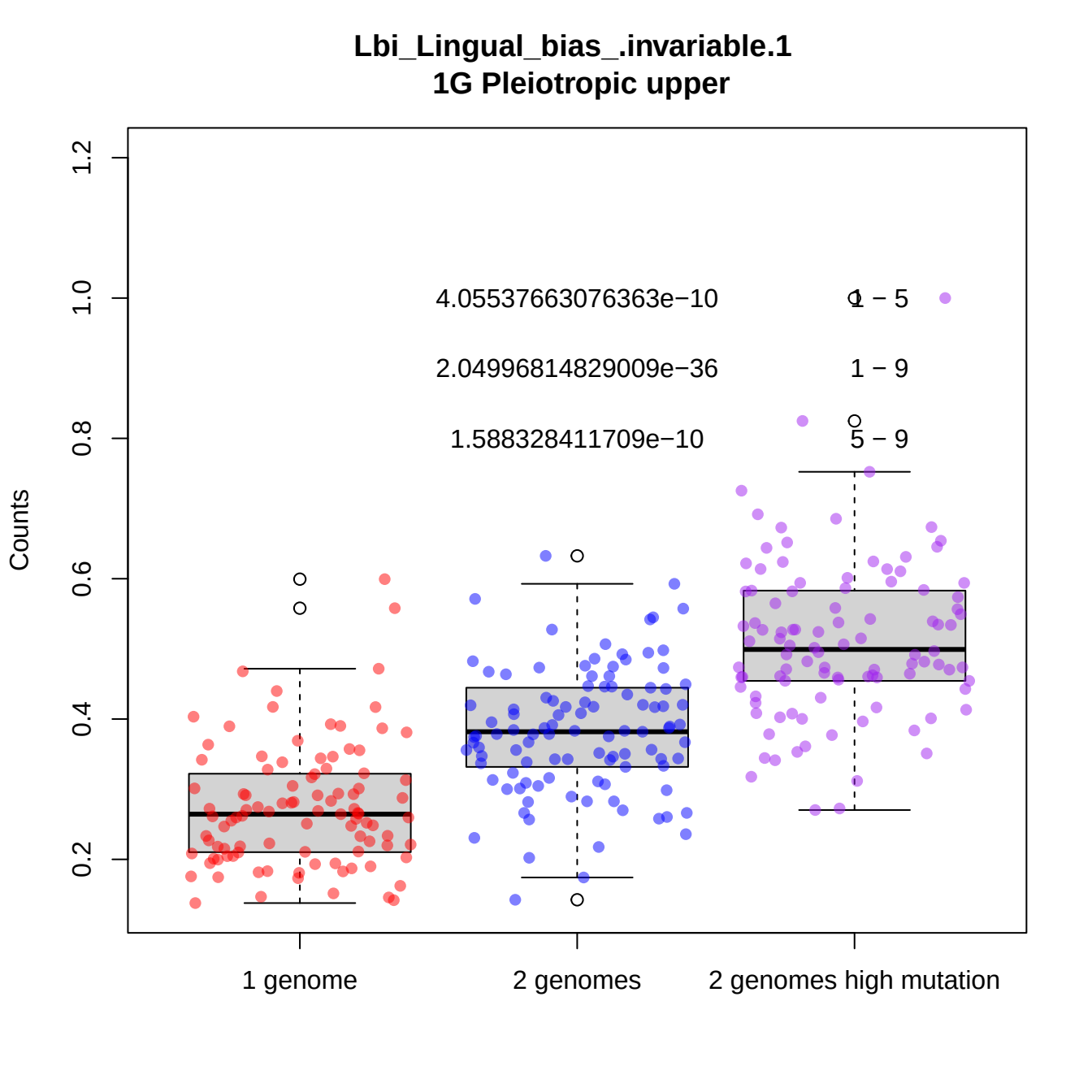

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12

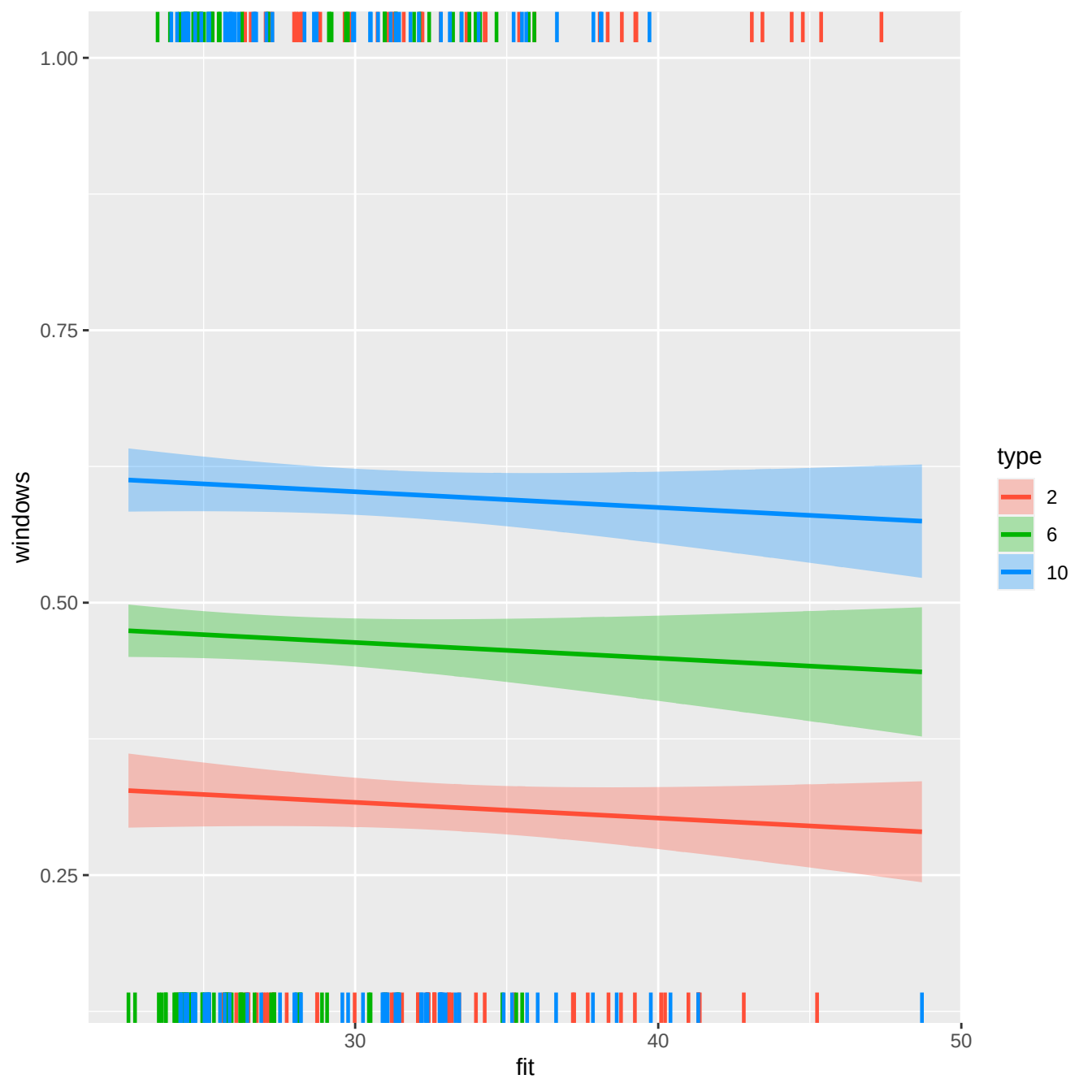

Bbi_Buccal_bias_.invariable_in_seals._hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

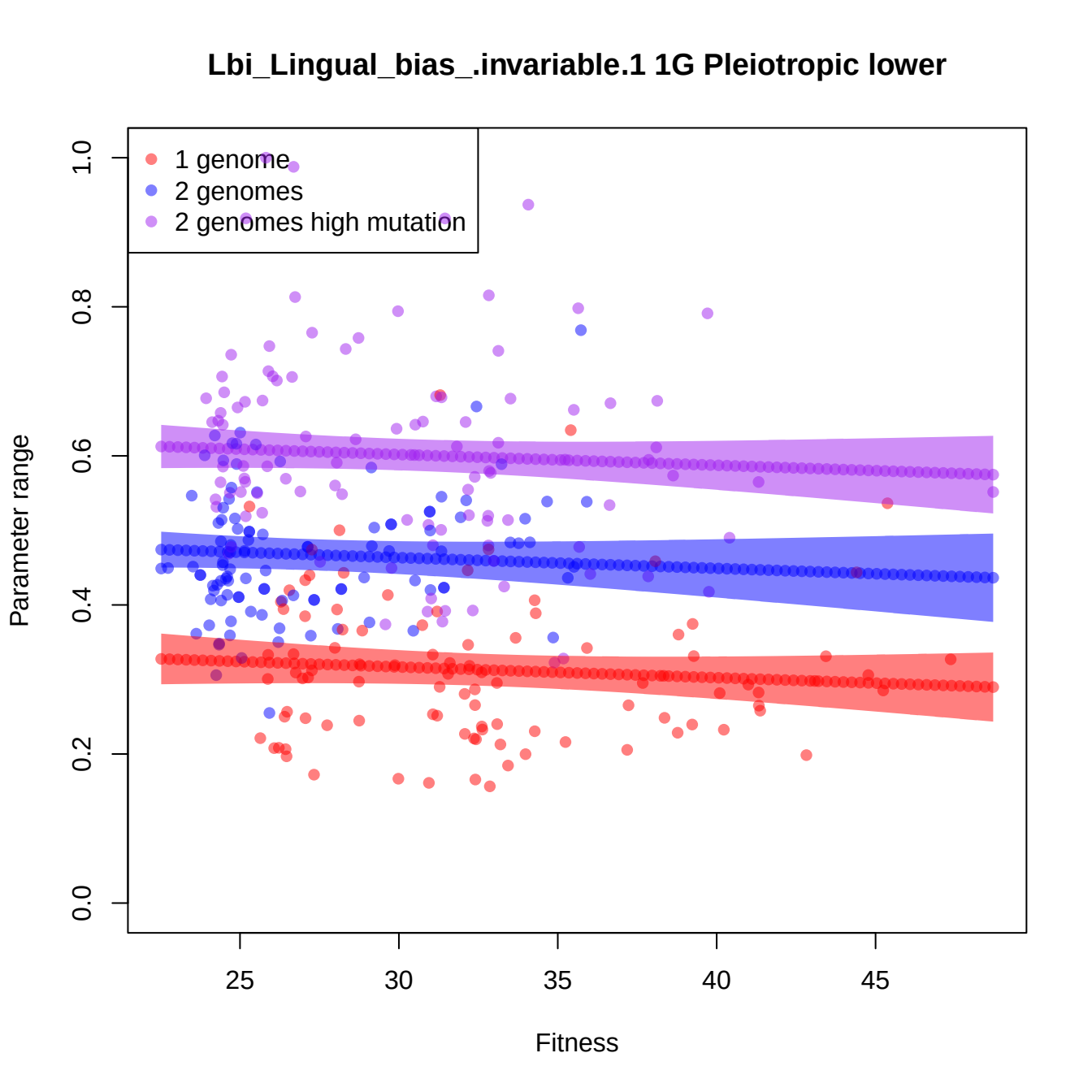

Bbi_Buccal_bias_.invariable_in_seals._hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
7.02019218792904e−28
0.35123066280021
7.69282360598e−23
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation

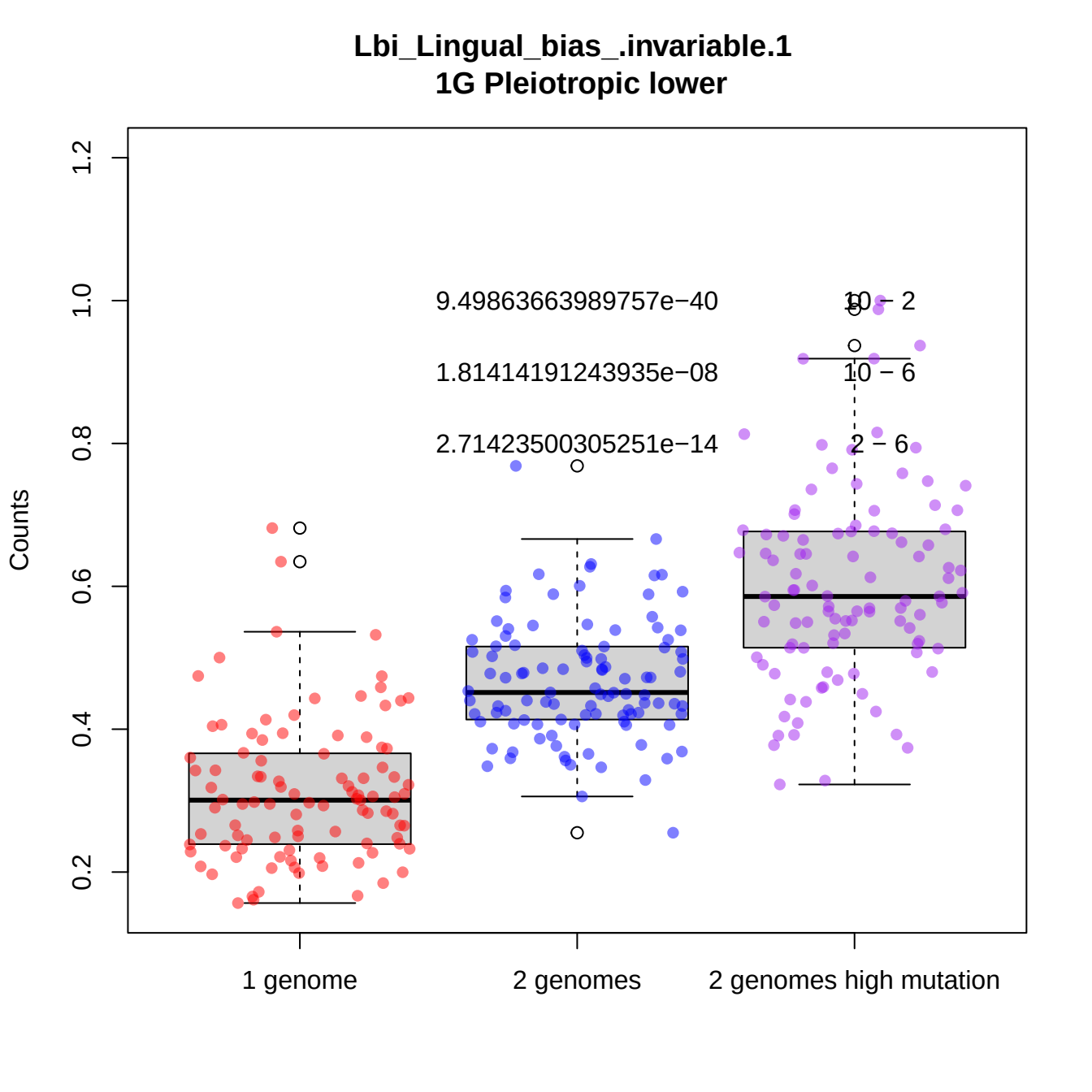

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
1
5
9

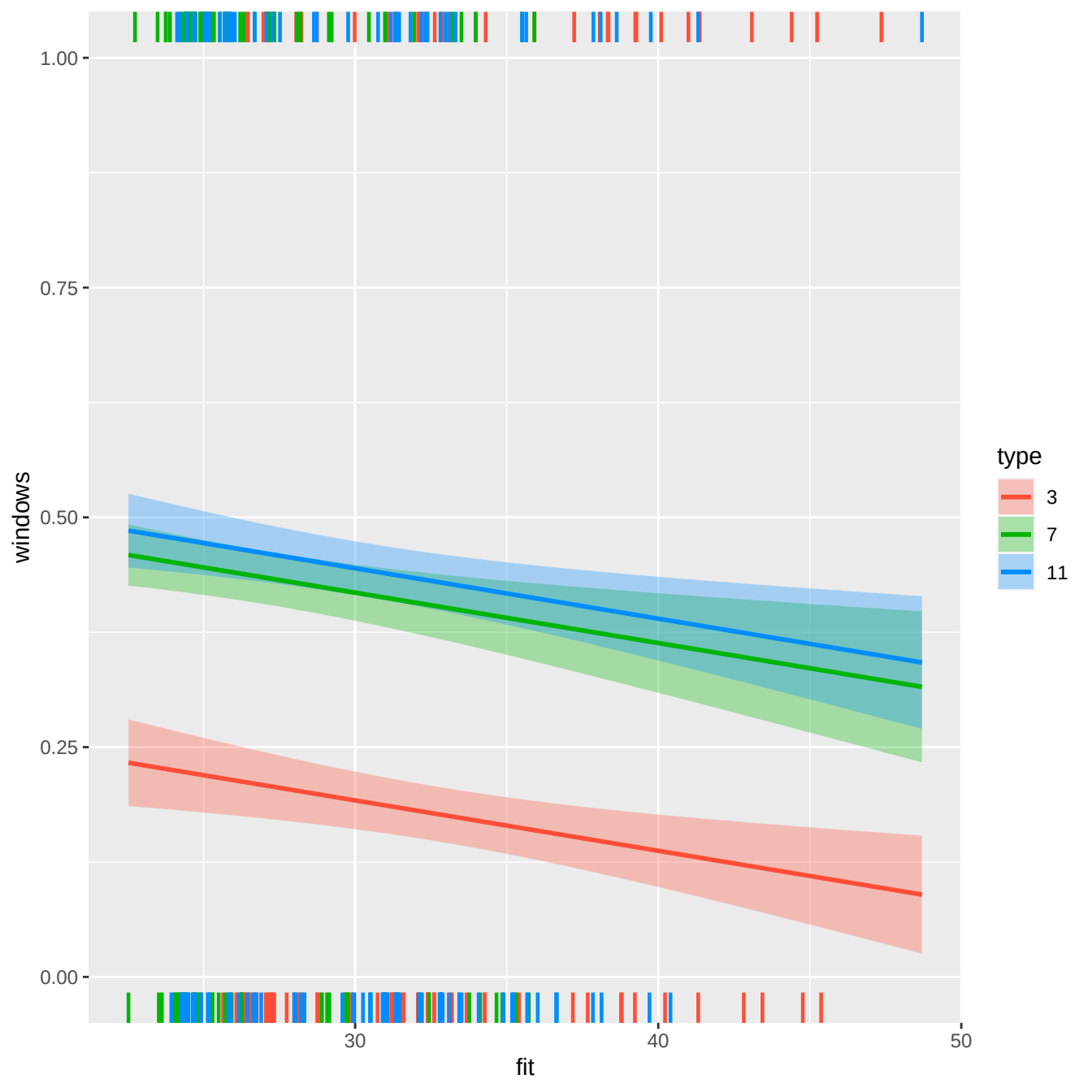

Lbi_Lingual_bias_.invariable.1 1G Pleiotropic upper
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

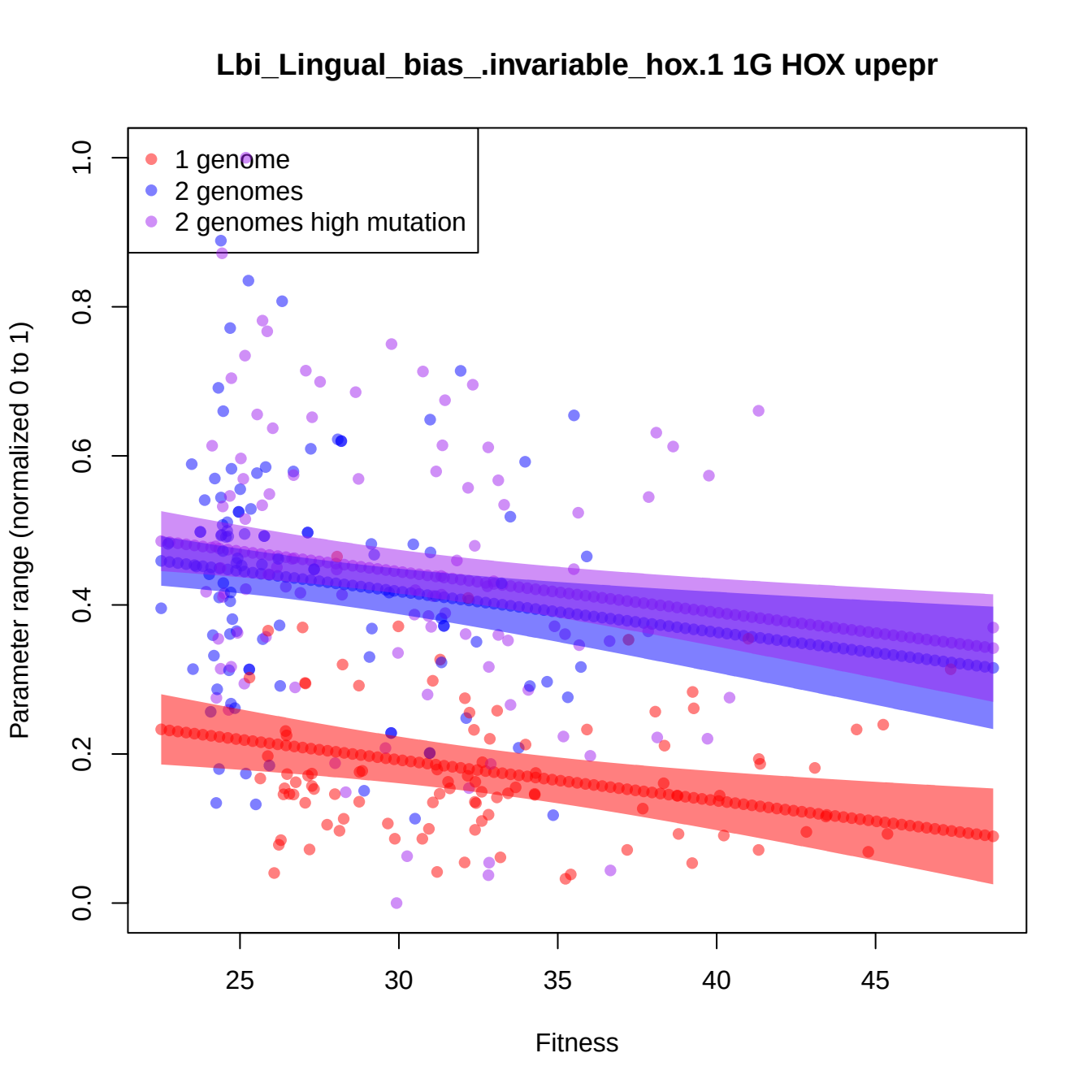

Lbi_Lingual_bias_.invariable.1
1G Pleiotropic upper
Counts
1.2
1.0
0.8
0.6
0.4
0.2
4.05537663076363e−10
2.04996814829009e−36
1.588328411709e−10
1 − 5
1 − 9
5 − 9
1 genome
2 genomes
2 genomes high mutation

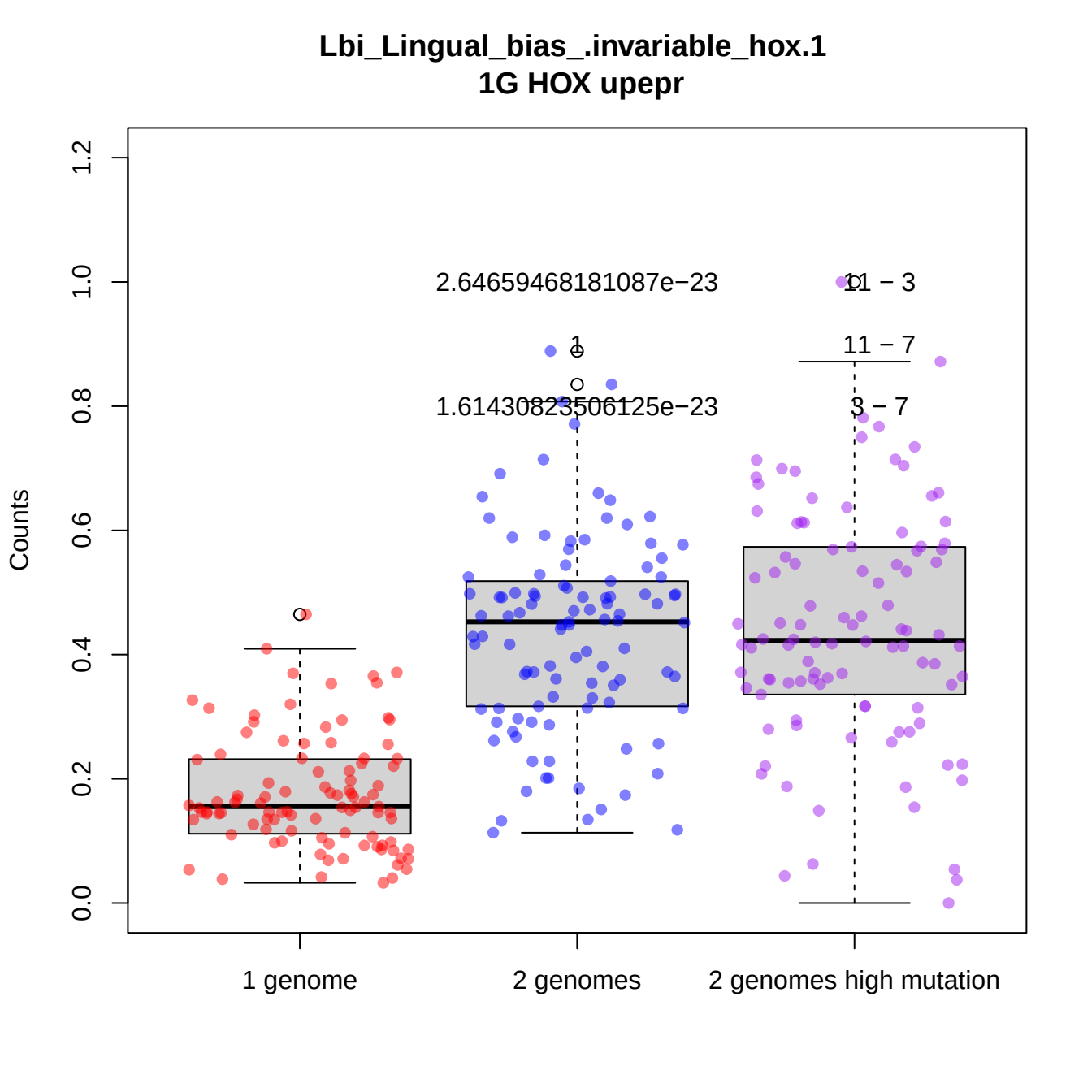

1.00
0.75
0.50
0.25
windows
30
40
50
fit
type
2
6
10

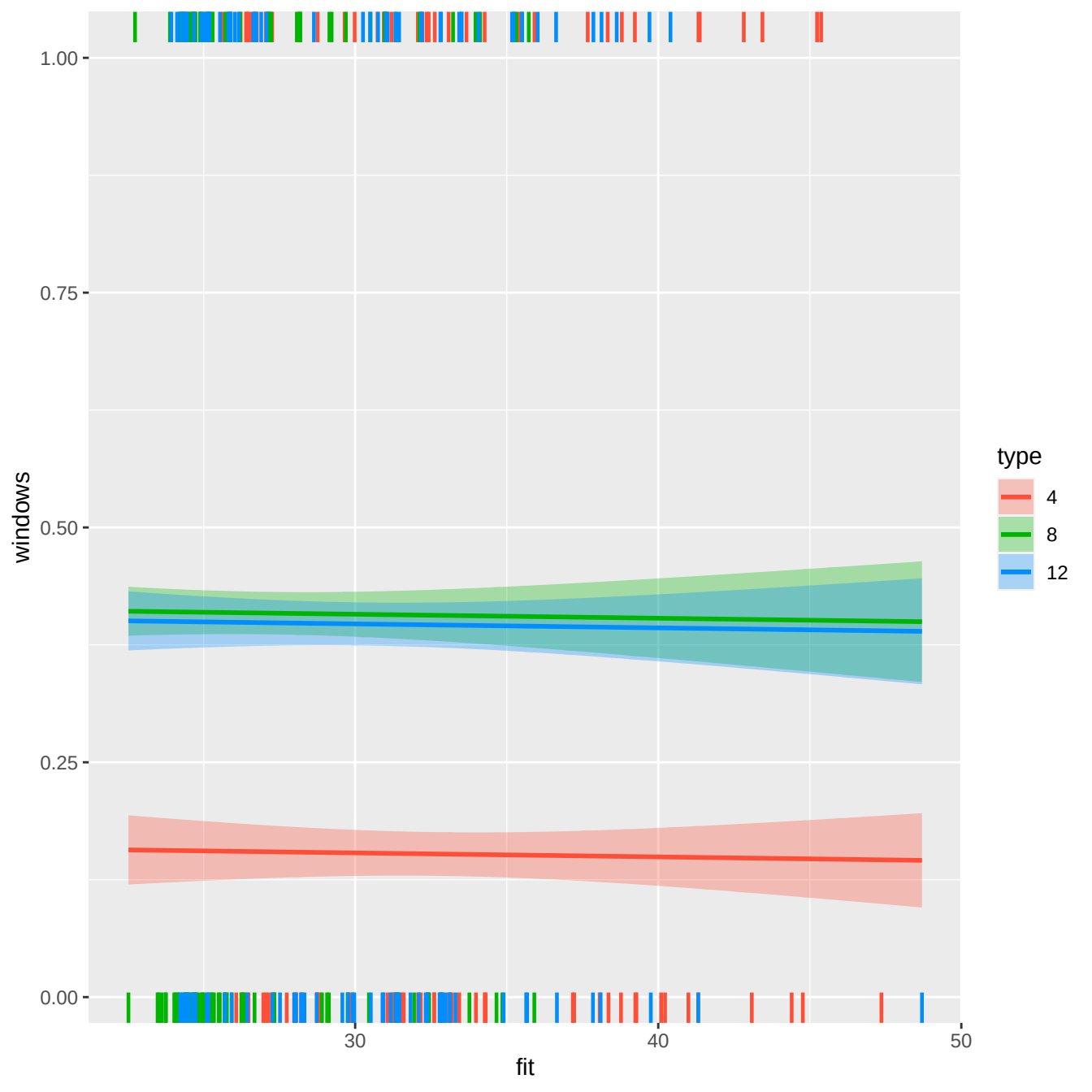

Lbi_Lingual_bias_.invariable.1 1G Pleiotropic lower
1 genome
2 genomes
2 genomes high mutation
Parameter range
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness

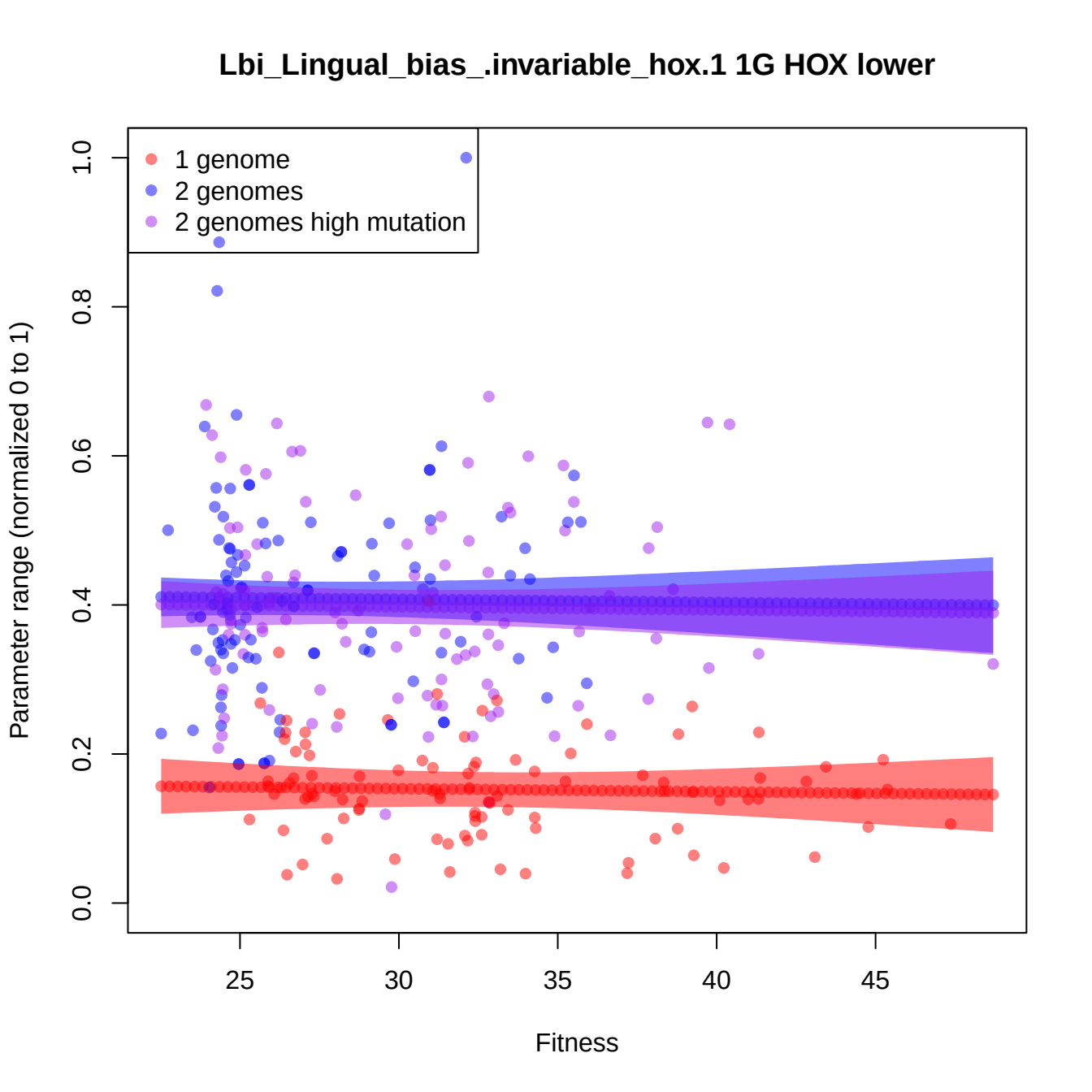

Lbi_Lingual_bias_.invariable.1
1G Pleiotropic lower
Counts
0.2
0.4
0.6
0.8
1.0
1.2
9.49863663989757e−40
1.81414191243935e−08
2.71423500305251e−14
10 − 2
10 − 6
2 − 6
1 genome
2 genomes
2 genomes high mutation

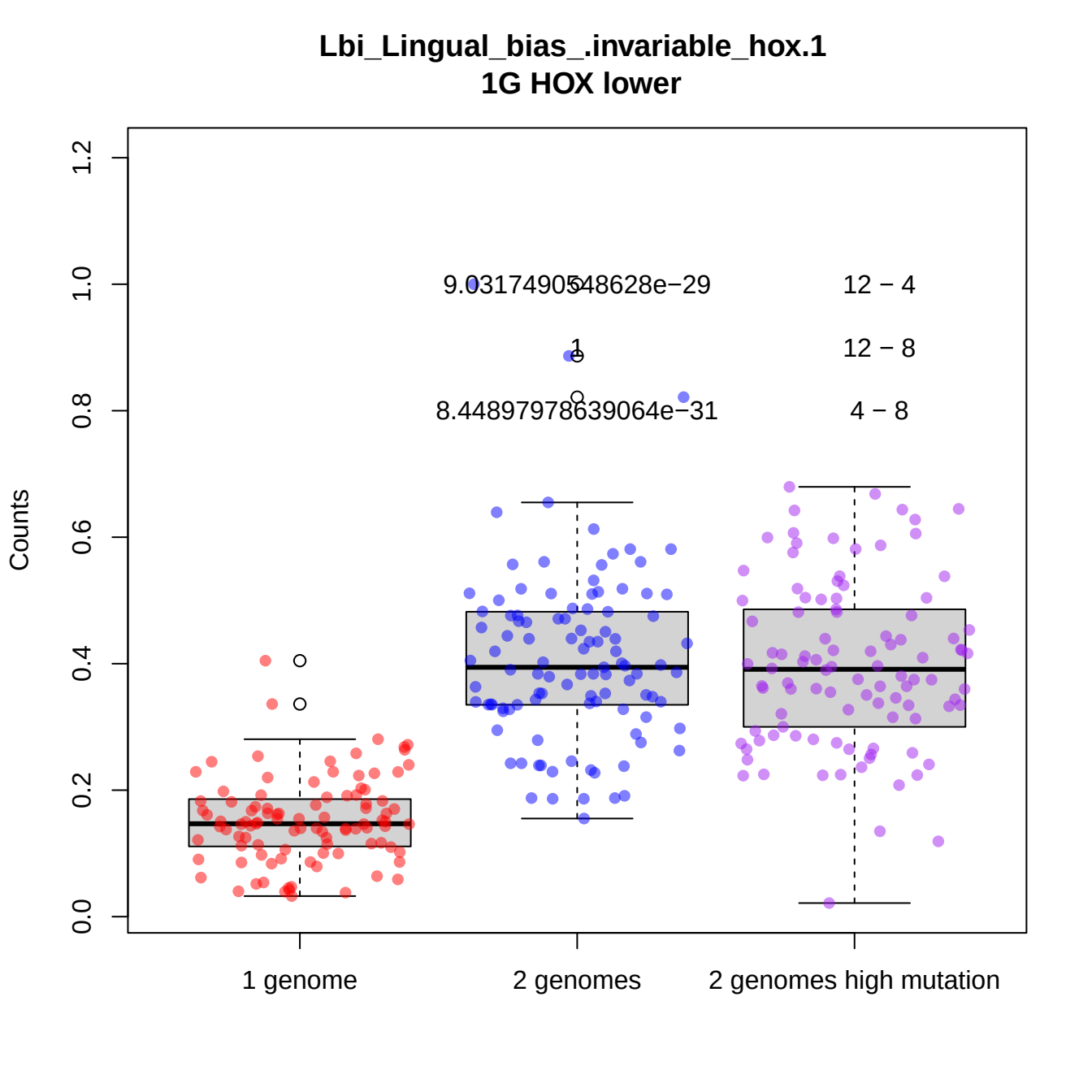

1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
3
7
11


Lbi_Lingual_bias_.invariable_hox.1 1G HOX upepr
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness


Lbi_Lingual_bias_.invariable_hox.1
1G HOX upepr
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.64659468181087e−23
1.61430823506125e−23
11 − 3
11 − 7
3 − 7
1 genome
2 genomes
2 genomes high mutation


1.00
0.75
0.50
0.25
0.00
windows
30
40
50
fit
type
4
8
12


Lbi_Lingual_bias_.invariable_hox.1 1G HOX lower
1 genome
2 genomes
2 genomes high mutation
Parameter range (normalized 0 to 1)
0.0
0.2
0.4
0.6
0.8
1.0
25
30
35
40
45
Fitness


Lbi_Lingual_bias_.invariable_hox.1
1G HOX lower
Counts
0.0
0.2
0.4
0.6
0.8
1.0
1.2
9.031749054628e−29
1
8.448979786390 64e−31
12 − 4
12 − 8
4 − 8
1 genome
2 genomes
2 genomes high mutation